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Applications:
Word, Excel, PowerPoint, Outlook, OneNote, etc.
The essential productivity suite for business and personal use.

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Key Functionality:
Spreadsheets, Data Analysis, Charts & Graphs, Formulas, Financial Modeling.
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Primary Uses:
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MS Office
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Contents: **Pages**

Microsoft Office

Word 2016 for Windows

INTRODUCTION TO MS-WORD

Learning Objectives

After completing the instructions in this booklet, you will be able to:

- Identify the components of the Word 2016 interface.
- Use the Tell Me feature to enter words and phrases related to what you want to do next to quickly
- Access features or actions.
- Create a new document.
- Set document margins.
- Set paragraph alignment, indentation, and spacing.
- Set tabs.
- Add headers and footers to a document.
- Apply a theme to a document.
- Format text.
- Check the document for spelling and grammar.

The Word 2016 Interface

The Word 2016 interface is very similar to the Word 2013 interface, with a few minor changes. The following describes the Word 2016 interface.

The Backstage View

When first opening the program, the user will be presented with options to open recent documents, start a new blank document, or select from a number of templates. The following explains how to enter the *Backstage View* after creating your document:

1. Click the **File** tab.

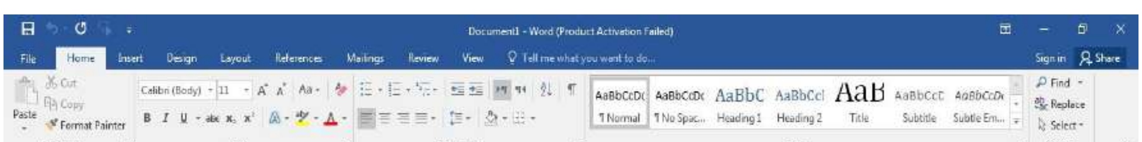


Figure 1 - File Tab (Backstage View)

2. From the *Backstage View*, you can perform the following actions:
   a. **Back** - Takes you back to edit your document (See Figure 2).
   b. **Info** - Obtain information about your documents (See Figure 2).

c. **New** - Create a new document from a blank or pre-formatted template (See Figure 2).

d. **Open** - Open a document (See Figure 2).

e. **Save** - Save the document to keep your edits (See Figure 2).

f. **Save As** - Resave a saved document as a different filename or file type (See Figure 2).

g. **Print** - Print documents and see a preview of your document (See Figure 2).

h. **Close** - Close the document (See Figure 2).

Figure 2 - Backstage View

The Ribbon

The Ribbon is a panel that contains functional groupings of buttons and drop-down lists organized by tabs. Each product in the Office Suite has a set of tabs that pertain to the functionality of that application. Each tab is further divided into *groups* such as the *Font* and *Paragraph*.

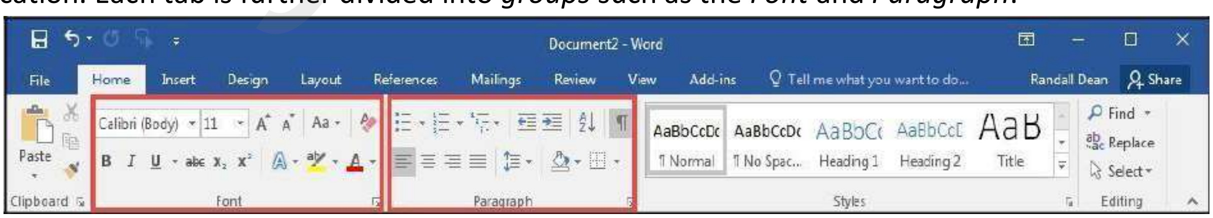


Figure 3 - The Ribbon

At the bottom right-hand corner of some groups, there is a diagonal arrow called a *Dialog Box Launcher* (See Figure 4). Clicking this button opens a dialog box for that group containing further option selections for the group.

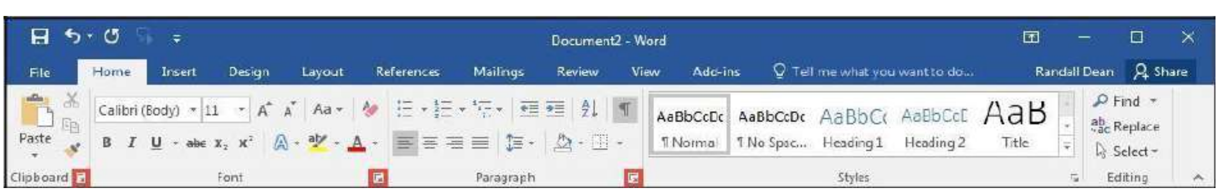


Figure 4 - Dialog Box Launcher

Contextual tabs will appear depending on what you are working on. For example, if you've inserted pictures, the tab appears whenever a picture is selected.

Figure 5 - Contextual Tab

Customizing the Ribbon

You can hide/unhide tabs that you do not use, or create your own tab of favorite tools. The following explains how to customize the *Ribbon* to build your own tabs and groups.

1. Click the **File** tab.

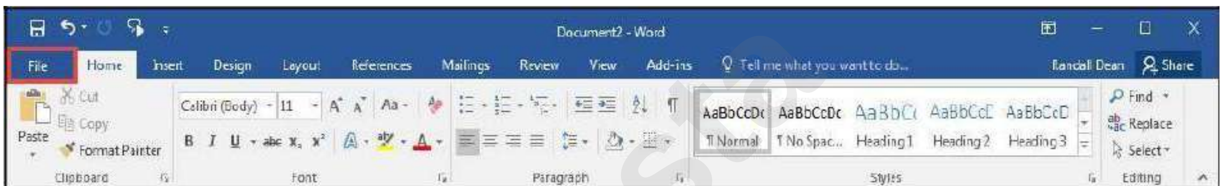


Figure 6 - File Tab (Backstage View)

2. In the *Backstage View*, click **Options**.
3. In the *Word Options* dialog box, click **Customize Ribbon** (See Figure 7).
4. Click the **New Tab** button (See Figure 7).
5. A *New Tab* has been added in the list of *Main Tabs*. Under *New Tab*, you will see *New Group* is already selected for you (See Figure 7).
6. From the column on the left, select a **command** from the list under *Choose commands from* to add to the *New Group* (See Figure 7).
7. Click the **Add** button (See Figure 7).

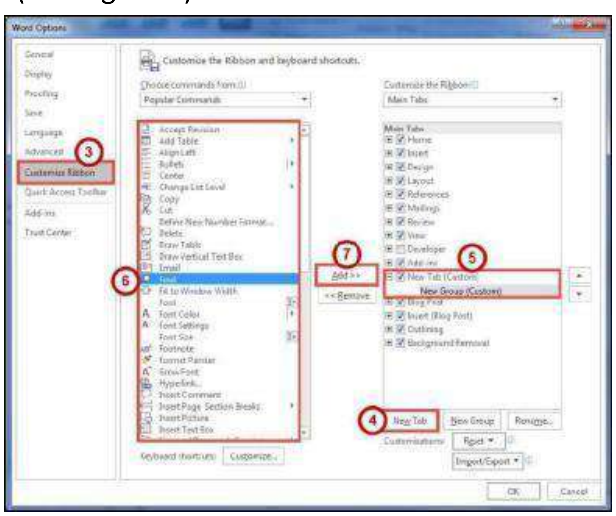


Figure 7 - Creating a New Tab

8. The command will be added to your *New Group* tab.
9. To rename the tab or group, right-click on the **New Tab** or **New Group** (See Figure 8).
10. Click **Rename** (See Figure 8).
11. To hide a tab, remove the **check mark** next to the name of the tab (See Figure 8).

Figure 8 - Rename New Tab or New Group

The Quick Access Toolbar

The *Quick Access Toolbar* is located in the upper-left part of the main Word window, above the *File* and*Home* tabs. It provides easy access to commands that you may use often and be customized to your preferences. The following explains how to customize the *Quick Access Toolbar*:

1. Click the **drop-down arrow** in the *Quick Access Toolbar* (See Figure 9).
2. In the *Customize Quick Access Toolbar drop-down* menu, click the **command(s)** you wish to add or remove from your *Quick Access Toolbar* (See Figure 9).
3. Click **More Commands** (See Figure 9).

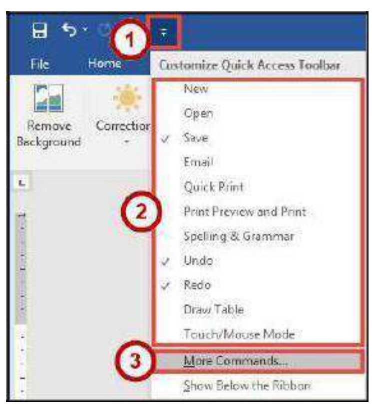


Figure 9 - Customize Quick Access Toolbar

4. In the *Customize Quick Access Toolbar* window, from the column on the left, select a **command** from the list under *Choose commands from* to add to your *Quick Access Toolbar* (See Figure 10).
5. Click the **Add** button (See Figure 10).
6. Click the **Ok** button (See Figure 10).

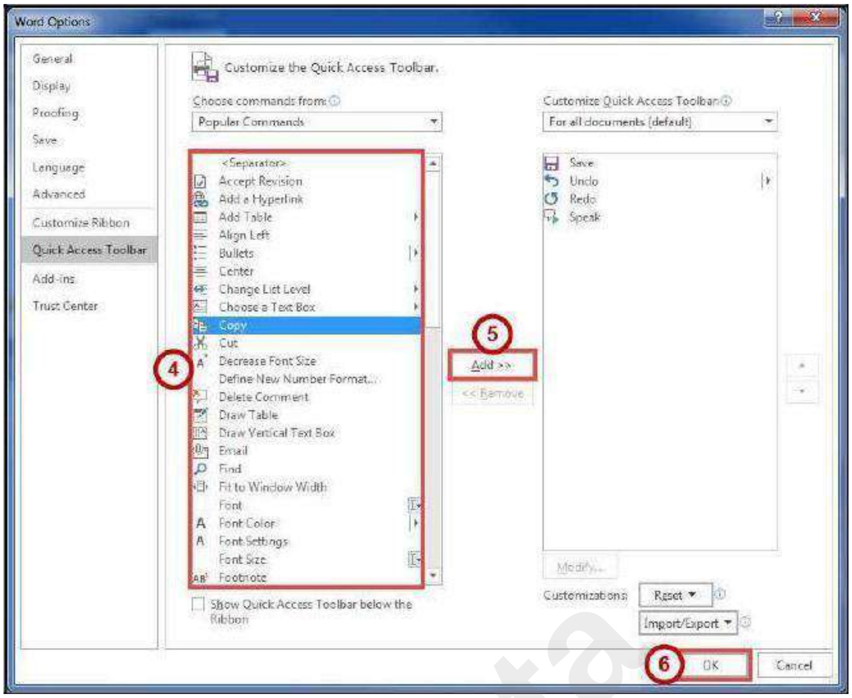


Figure 10 - Adding Commands to the Quick Access Toolbar

Tell Me

The Tell Me feature allows you to enter words and phrases related to what you want to do next to quickly access features or actions. It can also be used to look up helpful information related to the topic. It is located on the *Menu bar*, above the *ribbon*.

Search for Features

1. Click the **Tell Me** box.

Figure 11 - Tell Me

2. Type the **feature** you are looking for (See Figure 12).
3. In the *Tell Me* drop-down, you will receive a list of *features* based on your search. Click the **Feature** you were looking for (See Figure 12).

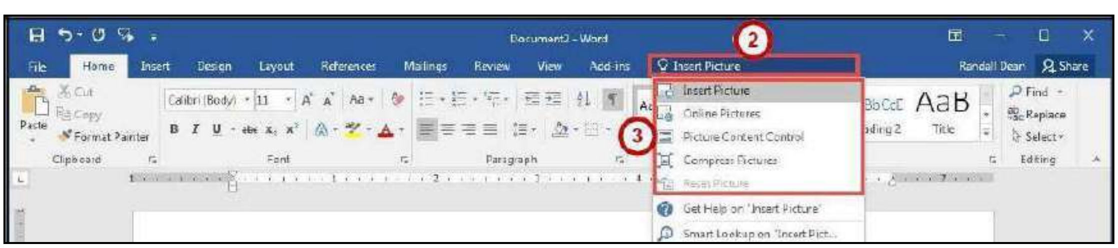


Figure 12 - Select a Feature

4. You will either be taken to the *feature* or a dialog box of that feature will *open*.

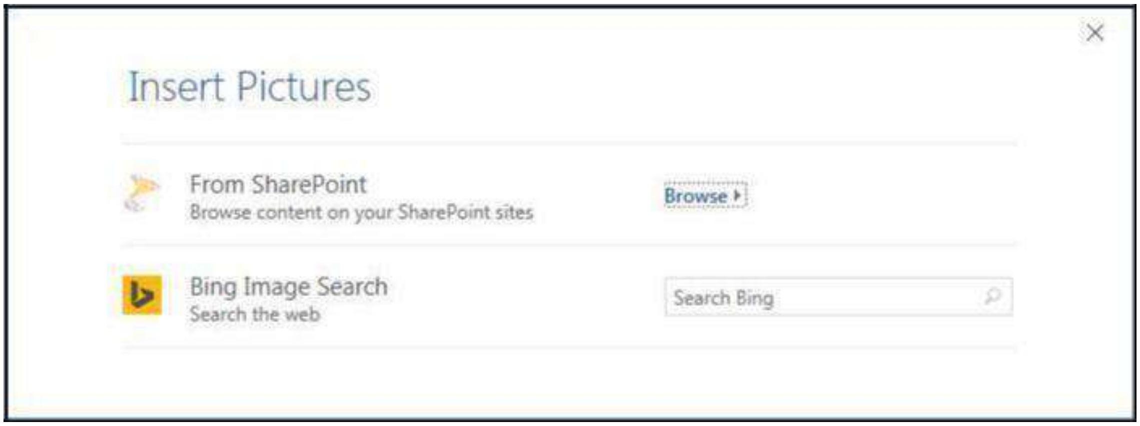


Figure 13 - Insert Pictures Online Dialog Box

Get Help with Word

To receive Microsoft Word Help, either type in the *Tell Me* box or press the *F1* key on the keyboard.

1. Click in the **Tell Me** box.

Figure 14 - Tell Me

2. Type your **question** you want help with (See Figure 15).
3. In the *Tell Me* drop-down, click **Get Help on question** (See Figure 15).

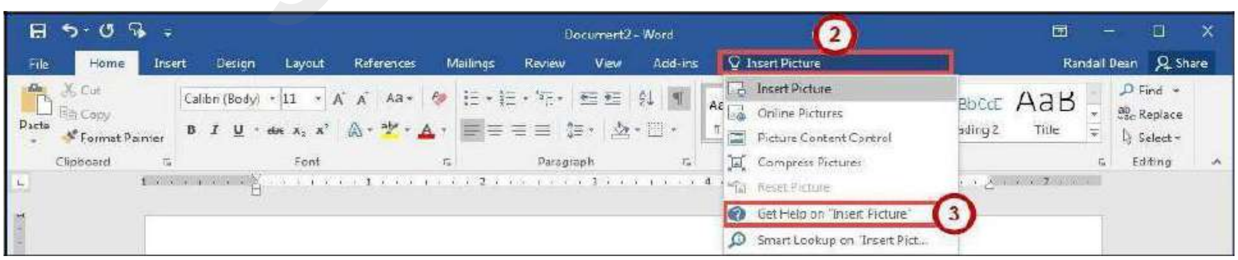


Figure 15 - Get Help on Feature

4. In the *Word 2016 Help* dialog box, you will get a list of help topics based on your search. Click the **Topic** you wanted help with.

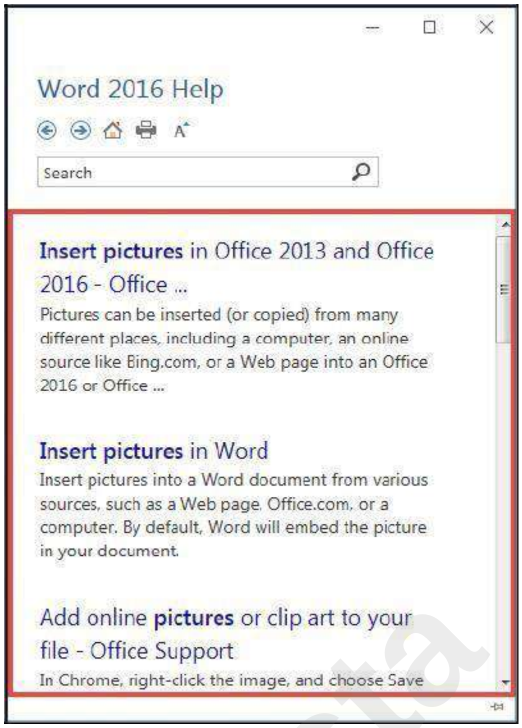


Figure 16 - Word 2016 Help

Smart Lookup

Use *Smart Lookup* to search *Bing* **Mi rosoft's** internet search engine) to provide you with search results for a word or phrase.

1. Click the **Tell Me** box.

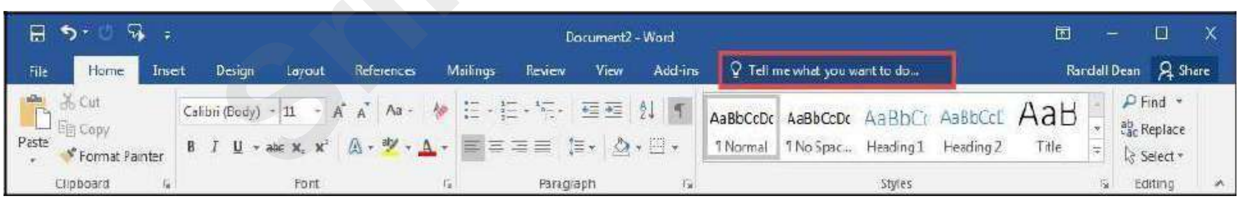


Figure 17 - Tell Me

2. Type a **word or phrase** you are looking up information for (See Figure 18).
3. In the *Tell Me* drop-down, click **Smart Lookup on word** (See Figure 18).

Figure 18 - Smart Lookup on Word

4. In the *Insights* pane, you will receive the following information:
   a. **Explore** - Wiki articles, image search, and related searches from the internet (See Figure 19).

b. Define - A list of definitions (See Figure 19).

Note: The *Insights* pane uses the Microsoft search engine Bing. For *Smart Lookup* to work you have to be connected to the internet.

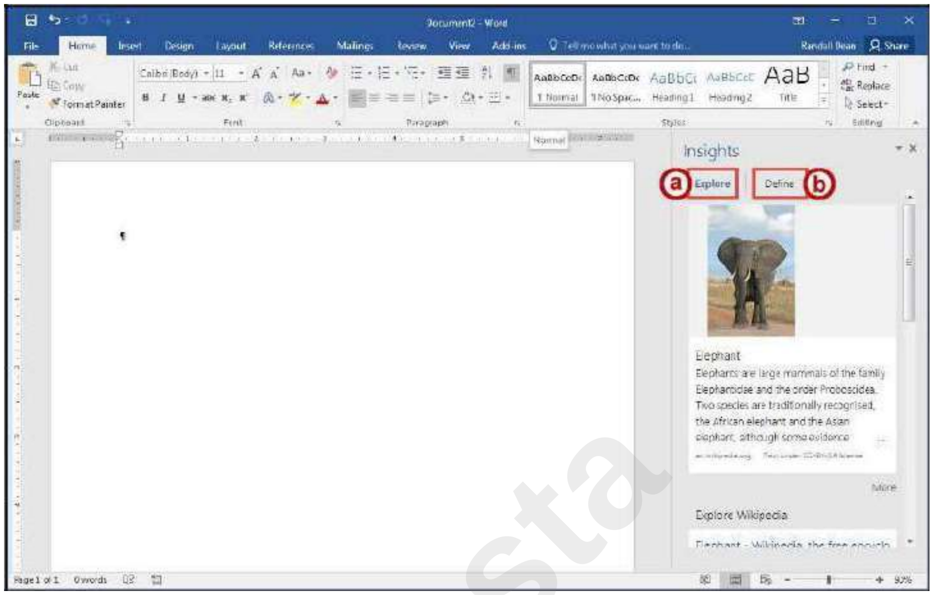


Figure 19 - Insights Pane

The Mini Toolbar

The *Mini Toolbar* is a toolbar that appears when you select text. The *Mini Toolbar* provides quick access to some commonly used formatting tools, such as font, font size, bold, italics, and more.

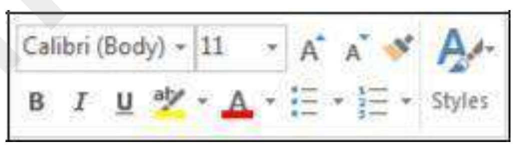


Figure 20 - Mini Toolbar

Disable the Mini Toolbar

1. Click the **File** tab.
2. In the *Backstage View*, click **Options**.
3. In the *Word Options* dialog box, click **General** (See Figure 21).
4. Under the *User Interface options* section, click the **checkbox** for *Show Mini Toolbar on selection* (See Figure 21).
5. Click the **OK** button (See Figure 21).

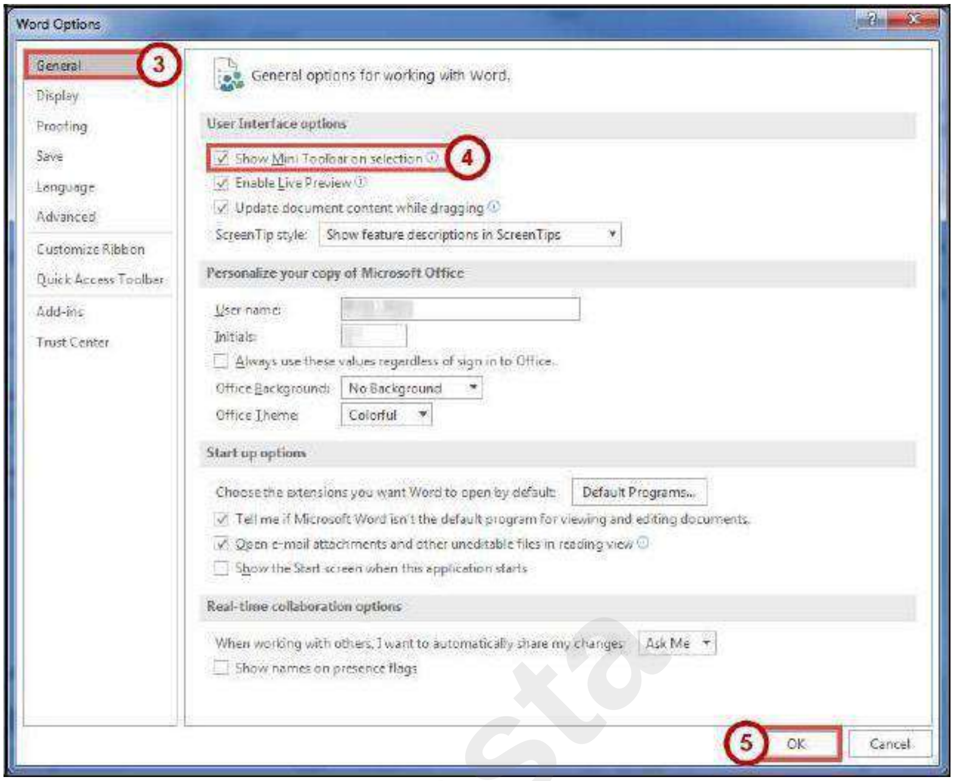


Figure 21 - Turn off the Mini Toolbar

The Status Bar

The *Status Bar* is located at the bottom of the Word **i do a d gi es ou a at a gla e s apshot of** important information regarding your current document (e.g. number of pages, number of words, proofing errors, etc.).

Figure 22 - Status Bar

Customize the Status Bar

1. Right-click the **Status Bar** (See Figure 23).
2. In the *Customize Status Bar* drop-down menu, click an **option** to add it (See Figure 23).
3. When finished, click anywhere **outside** the *Customize Status Bar* drop-down.

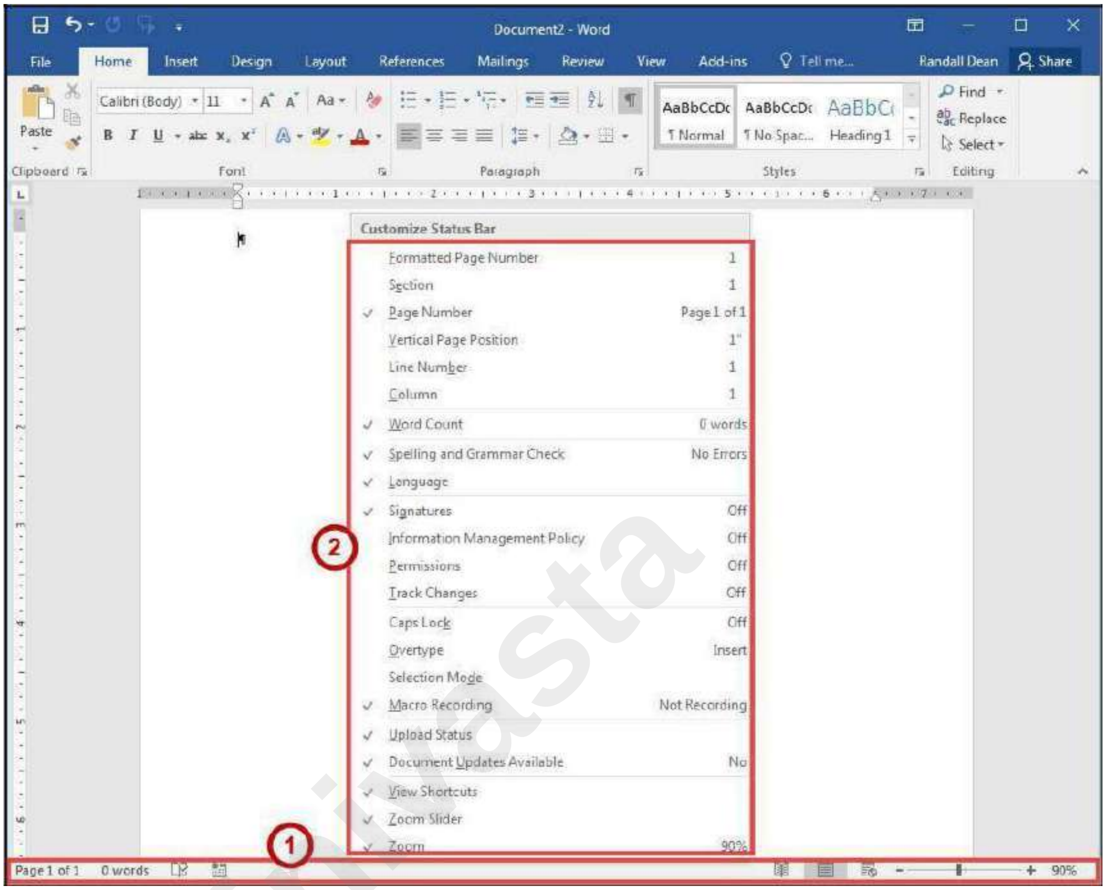


Figure 23 - Customize Status Bar

Creating a New Document

The following shows how to create a *Blank Document* in Word:

1. Click the **File** tab.

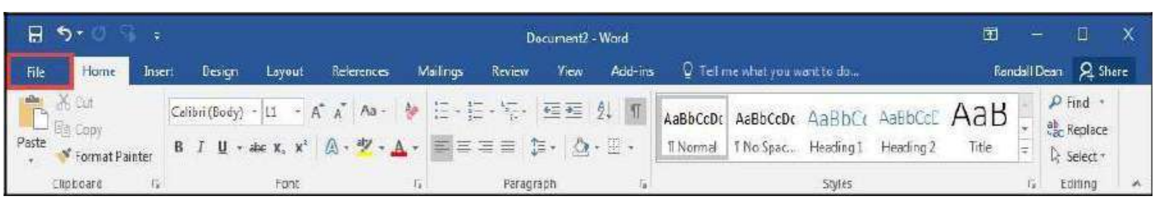


Figure 24 - File Tab (Backstage View)

2. In the *Backstage View*, click **New** (See Figure 25).
3. Click **Blank Document** (See Figure 25).

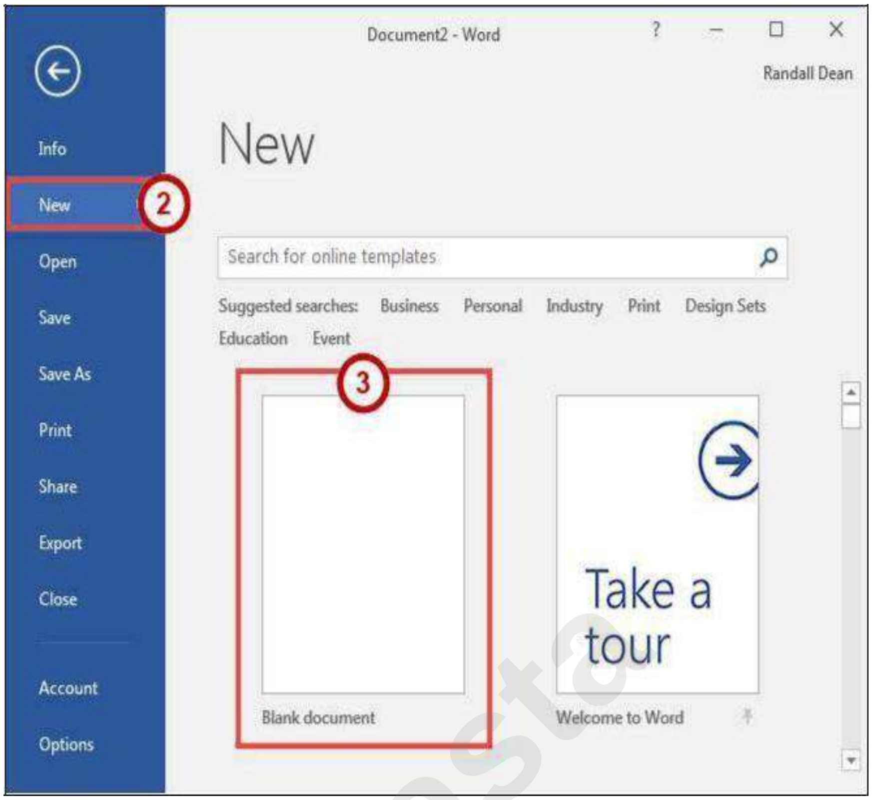


Figure 25 - Blank Document

Setting up Your Document

When you begin creating a new Word document, there are certain aspects of the document that you should consider, such as margin settings, fonts and styles, and line spacing. Making these choices before you begin typing could save you time editing your document later on.

Setting Margins

Page margins, the blank space around the edges of the page, can contribute to the impression your document makes, and even how easy it is to read. A few clicks set the margins for a page or an entire document. The following instructions explain how to change the margin settings:

1. Click the **Layout** tab (See Figure 26).
2. Click the **Margins** button (See Figure 26).
3. Click one of the **present margin options** (See Figure 26).

Note: Clicking on **Custom Margins** at the bottom of the *Margin* options will open the *Page Setup* window and allow you to enter the specific margin values that you want.

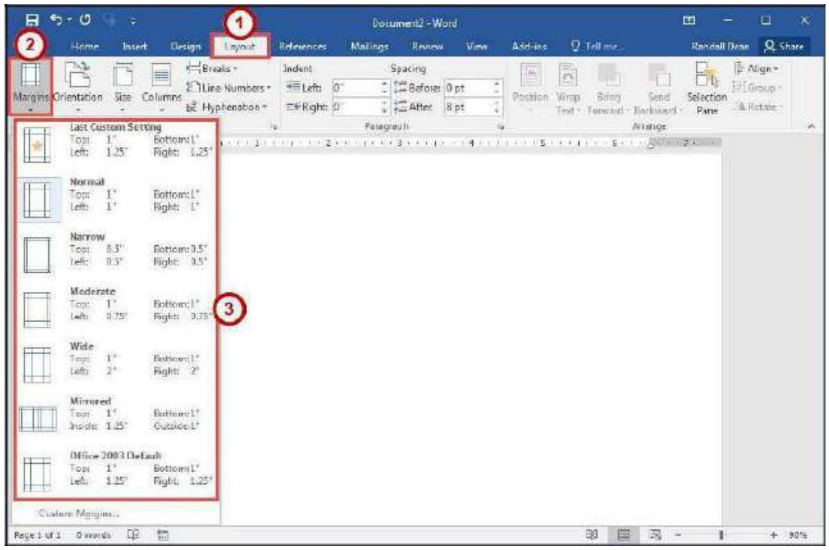


Figure 26 - Margins

Setting the Default Font

You can change the default font options so your favourite font is always selected in Word. The following explains how to change the default font options:

1. Click the **Home** tab (See Figure 27).
2. In the *Font* group, click the **Font Dialog Box Launcher** (See Figure 27).

Figure 27 - Font Dialog Box Launcher

3. In the *Font* dialog box, you can change the *Font, Font style, Size, Font color, Underline style,* and *Effects* (See Figure 28).
4. Click **Set As Default** (See Figure 28).

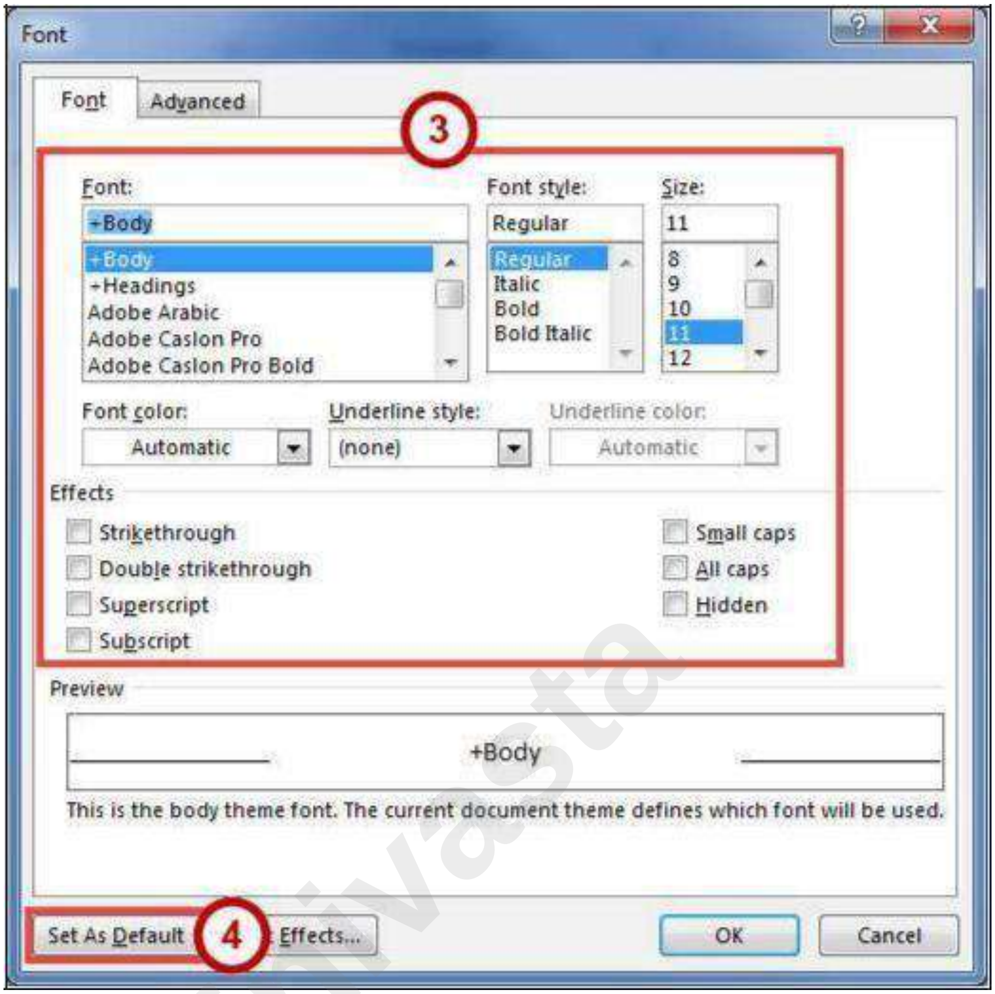


Figure 28 - Font Dialog Box

5. In the *Set as Default* dialog box, make a **selection** based on your preference (See Figure 29).
6. Click the **OK** button (See Figure 29).

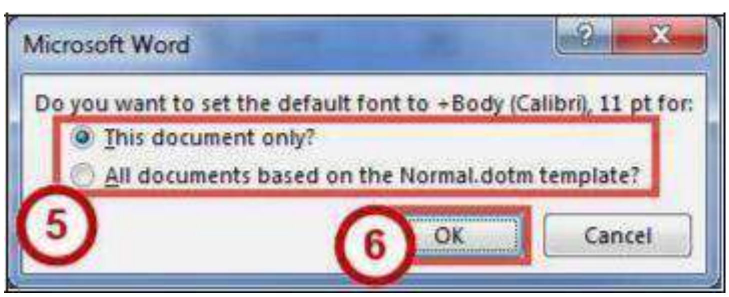


Figure 29 - Set as Default Dialog Box for Font

Setting Default Paragraph Alignment

You can change the default paragraph alignment options so is always spaced how you want it to be in Word. Alignment, indentation, and line spacing are all set from the Paragraph dialog box. The following explains how to change the default paragraph options:

1. Click the **Home** tab (See Figure 30).
2. In the *Paragraph* group, click the **Paragraph Dialog Box Launcher** (See Figure 30).

Figure 30 - Paragraph Dialog Box Launcher

3. In the *Paragraph* dialog box, you can change the *Alignment, Outline level, Indentation,* and *Spacing* (See Figure 31).
4. Click **Set As Default** (See Figure 31).

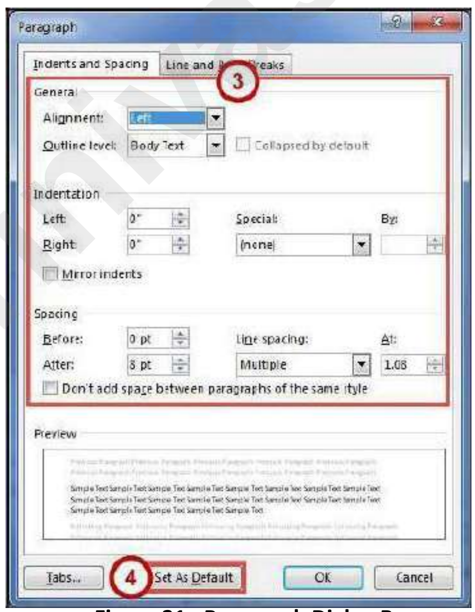


Figure 31 - Paragraph Dialog Box

5. In the *Set As Default* dialog box, make a **selection** based on your preference (See Figure 32).
6. Click the **OK** button (See Figure 32).

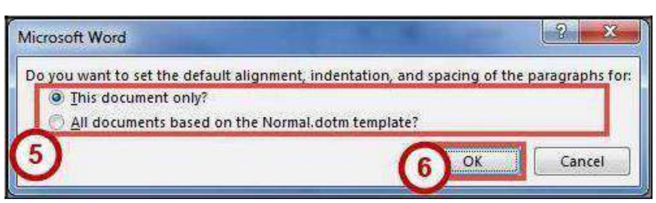


Figure 32 - Set as Default Dialog Box for Paragraph

Adding Styles

The *Styles Gallery* is a combination of text formatting options which are saved under a single name. Using styles can make formatting text faster and easier than applying individual formatting options. Styles can also help with navigating your document, and marking sections for later use in a table of contents.

The choices in the *Styles Gallery* incorporate a feature called *Live Preview*. When you hover your mouse over a selection in a Gallery, your document takes on the formatting attributes of that selection in order to give you a preview of how that selection will look when applied to your document.

Add a Style to Your Document

1. Select the **text** you wish to add a style to (See Figure 33).
2. Click the **Home** tab (See Figure 33).
3. Click one of the **preset styles** (See Figure 33).

Note: Only the most recently used *Styles* in the *Styles Gallery* are displayed on the ribbon. The entire *Styles Gallery* can be displayed by clicking the more button at the right of the displayed *Styles*.

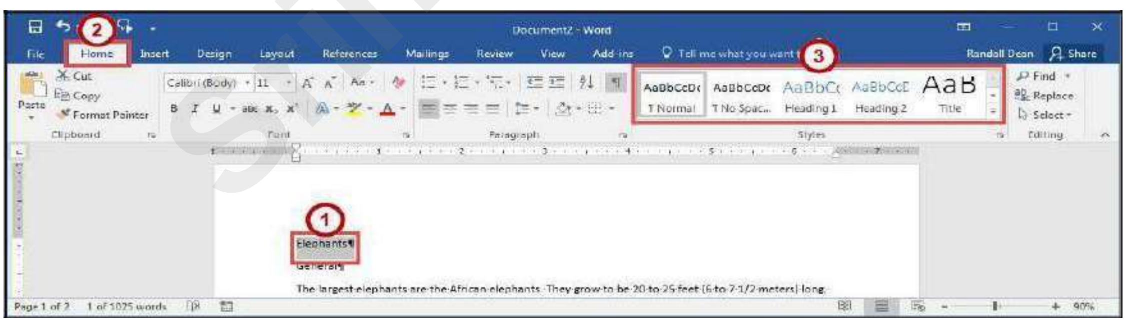


Figure 33 - Styles

4. The style will be added to your selected text.

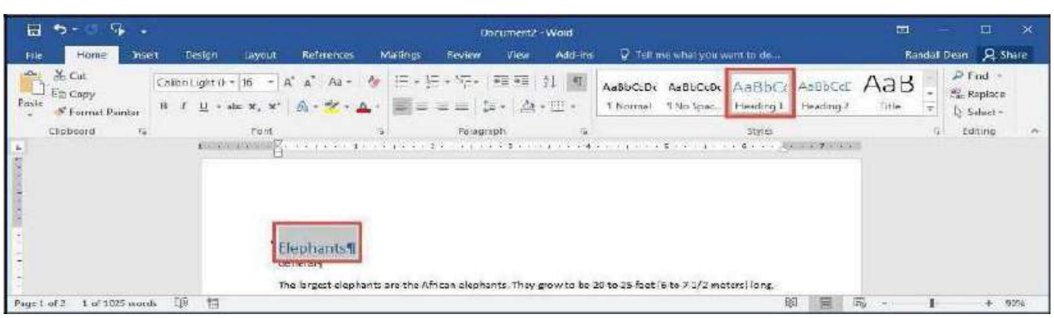


Figure 34 - Style Added

Adding Themes

A *Theme* is a set of formatting options that is applied to an entire document. A theme includes a set of colors, a set of fonts, and a set of effects. Using themes shortens formatting time and provides a unified, professional appearance.

Themes can be accessed from the *Design* tab. From here you can select a theme from the *Document Formatting* group, as well as customize the colors, fonts, and effects of a theme. The default theme that is applied to every new document is the *Office Theme*. The following shows how to apply a different theme to a document:

1. Click the **Design** tab (See Figure 35).
2. Click the **Themes** button (See Figure 35).
3. Click the **Theme** to apply it to your document (See Figure 35).

Note: Hover the mouse over the selections in the *Themes Gallery* to preview how each *theme* will look when applied to your document.

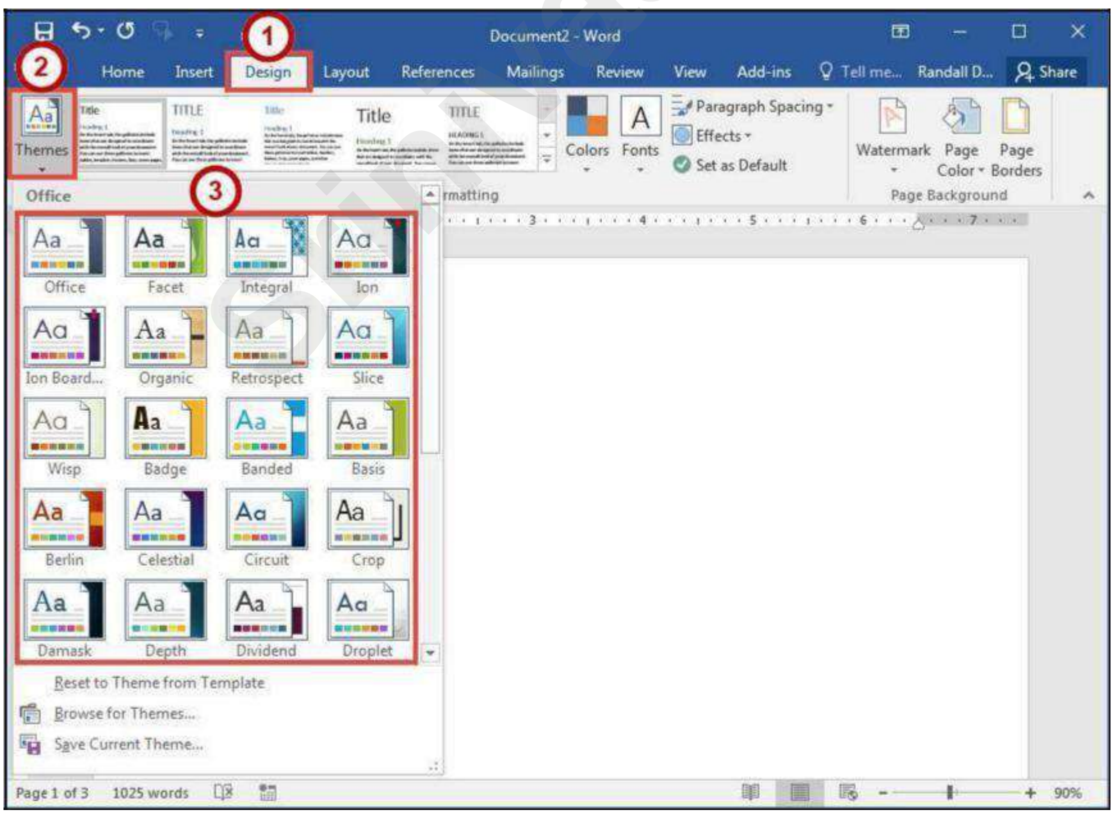


Figure 35 - Themes

Editing a Document

Entering Text

Word will automatically wrap text when the cursor reaches the right margin. There is no need to press the ENTER key unless you want to start a new paragraph or add space between lines of text.

Formatting Text

Character and paragraph formatting commands are found on the Home tab of the Ribbon. To apply any of the formatting options, first select the text and then click the button or check box for the option(s) that you want to apply.

Character Formatting

Character formatting involves changing the font, size, color, and spacing of characters, as well as applying bold, italics, and various other effects. Commonly used commands are found in the *Font group*, under the *Home* tab, while more commands can be found in the *Font Dialog Box*. The *Font Dialog Box* can be accessed by clicking the **Font Dialog Box Launcher**.

Figure 36 - Font Dialog Box Launcher

Paragraph Formatting

Paragraph formatting involves changing the alignment, line spacing, or indentations of paragraphs. Commonly used commands are located in the *Paragraph group*, under the *Home* tab, while more commands can be found in the *Paragraph Dialog Box.* The *Paragraph Dialog Box* can be accessed by clicking the **Paragraph Dialog Box Launcher**.

Figure 37 - Paragraph Dialog Box Launcher

Enabling the Ruler

Word has a ruler that fits along the top and left side of your document and provides a point of reference when aligning text and images within your document. The following explains how to enable the ruler:

1. Click the **View** tab (See Figure 38).
2. Click the **checkbox** next to *Ruler* (See Figure 38).
3. The ruler will appear along the top and left side of your document (See Figure 38).

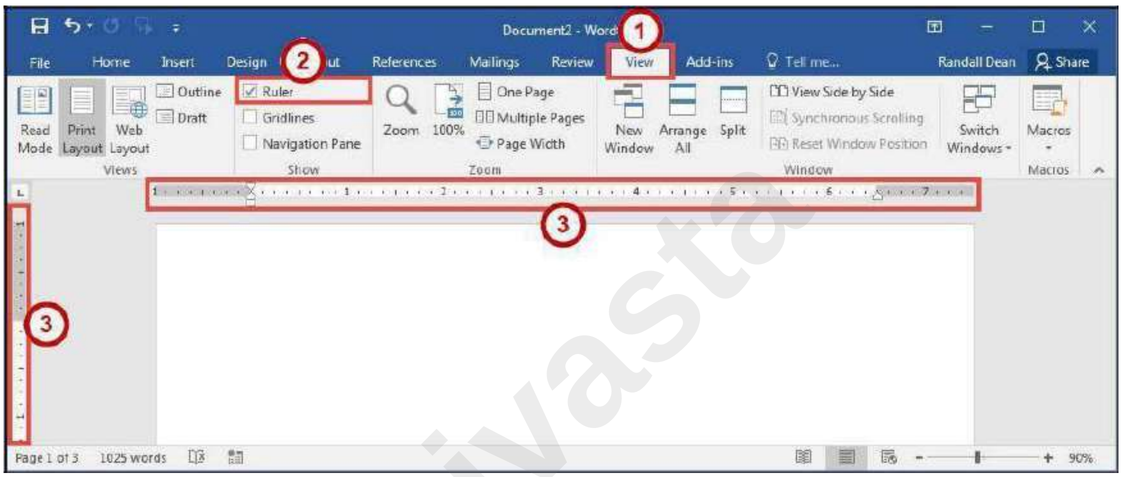


Figure 38 - Ruler

Setting Tab Stops

Tab stops provide a position for placing and aligning text on your document, and are most easily set from the Ruler (See Enabling the Ruler). The following explains how to insert a tab stop:

1. Click within the paragraph or text that you want to insert your *tab stop*.
2. Position your mouse pointer over the *Ruler* at the location you want to add the *tab stop*.
3. Left-click the **mouse button**. A *tab stop* will be placed on the ruler.

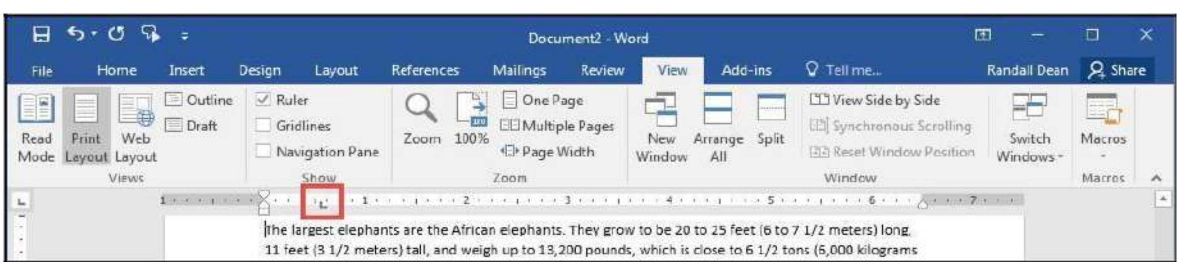


Figure 39 - Tab Stop

Adding a Leader to Tab Stops

After adding a *tab stop* (See Setting Tab Stops), a leader can be added to make text easier to read. For example, on a menu the leaders make it easier to read the food and the corresponding price. The following explains how to add *leaders* to your *tab stops*:

1. Double-click on the **tab stop** in your *Ruler*.

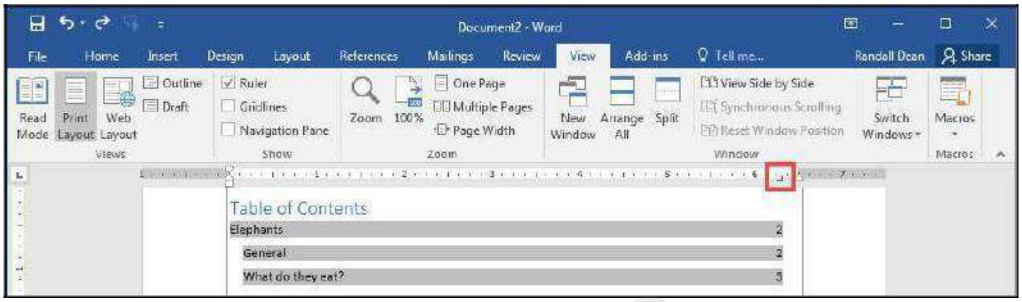


Figure 40 - Tab Stop

2. In the *Tabs* dialog box, click on one of the **Leader** options (See Figure 41).
3. Click the **OK** button (See Figure 41).

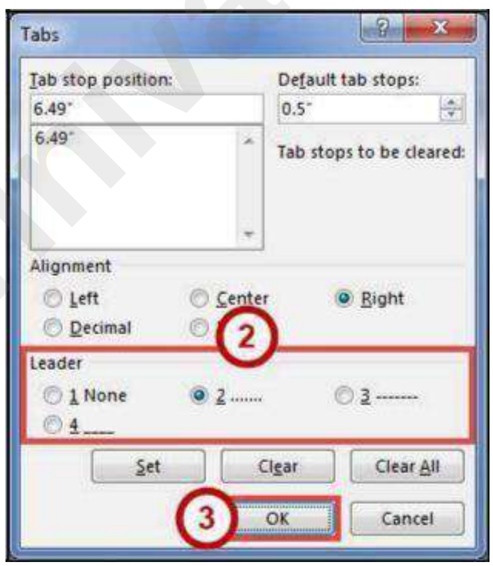


Figure 41 - Tabs Dialog Box

4. The *leader* will appear when you tab to the *tab stop*.

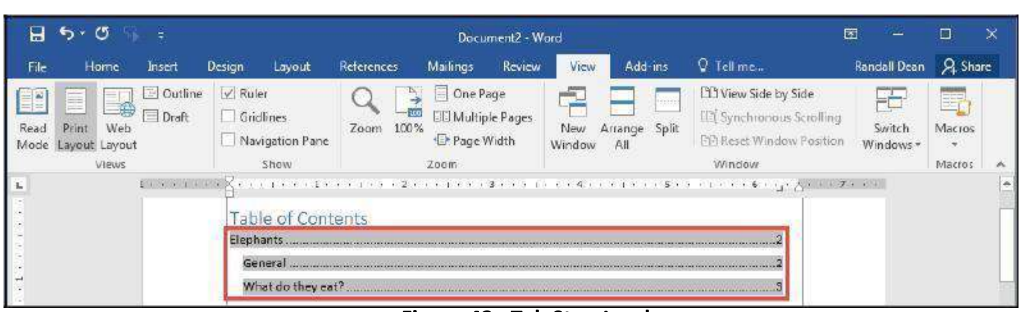


Figure 42 - Tab Stop Leader

Selecting Different Tab Stops

At the far left of the horizontal ruler is the tab selector that lets you choose from different types of tab stops. The most commonly used tab stop is a Left Tab and more can be selected by clicking the **tab selector**.

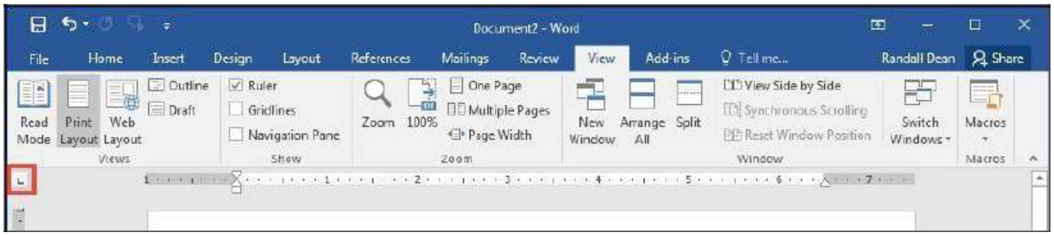


Figure 43 - Tab Selector

The following is an explanation of the *tab stops* available:

- **Left Tab** - Left aligns text to the left of the tab stop.
- **Center Tab** - Centers the text on the tab stop.
- **Right Tab** - Right aligns the text to the right of the tab stop.
- **Decimal Tab** - Aligns decimal numbers by their decimal point.
- **Bar Tab** - Draws a vertical line on the document.
- **First Line Indent** -
  Inserts the indent marker on the ruler and indents the first line of text in a paragraph.
- **Hanging Indent** -
  Inserts the hanging indent marker and indents all lines other than the first line.

Headers and Footers

Headers and footers are areas containing text that will be displayed on every page. The header is located along the top of every page, while the footer is located at the bottom of every page. The following explains how to add a header to a document:

1. Click the **Insert** tab (See Figure 44).
2. Click **Header** (See Figure 44).
3. In the *Header Gallery* drop-down menu, click on a **header design** (See Figure 44).

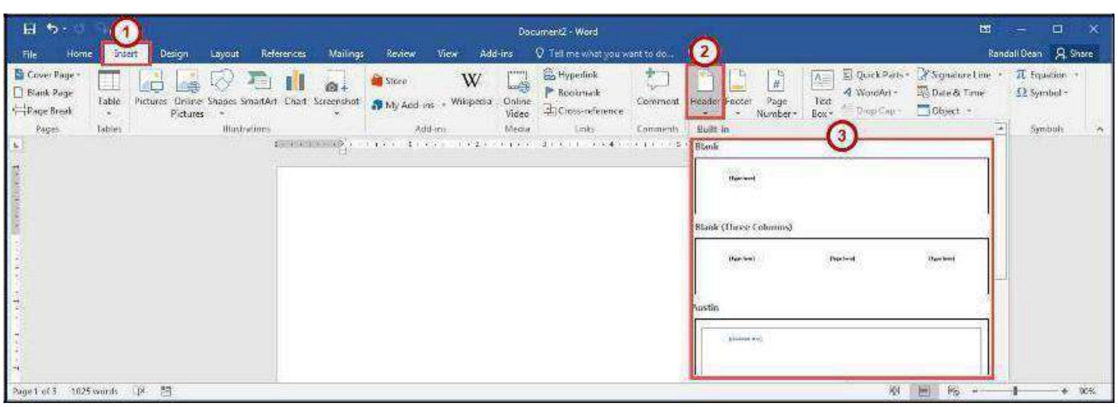


Figure 44 - Header

4. To edit the header, **type** the text that you want to appear in the indicated areas (See Figure 45).
5. When you are finished, click **Close Header and Footer** in the *Header & Footer Design Tools - Design* tab (See Figure 45).

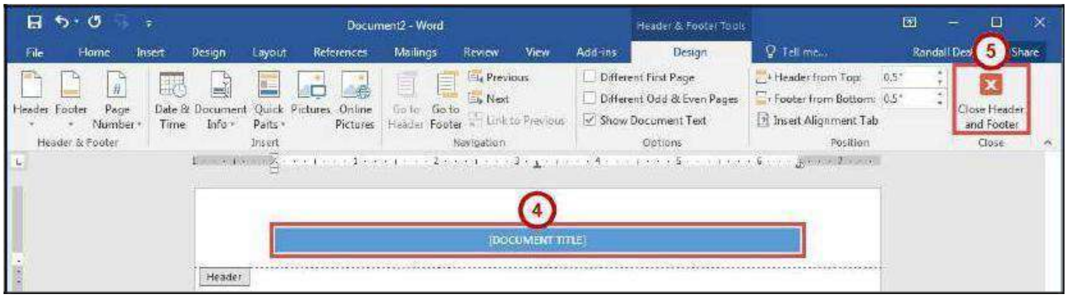


Figure 45 - Close Header and Footer

Editing Headers & Footers

1. Click the **Insert** tab (See Figure 46).
2. Click **Header** (See Figure 46).
3. In the *Header Gallery* drop-down menu, click **Edit Header** (See Figure 46).
4. The *Header & Footer Tools - Design* tab will open. You will be able to make your edits to the *header* and/or *footer*.

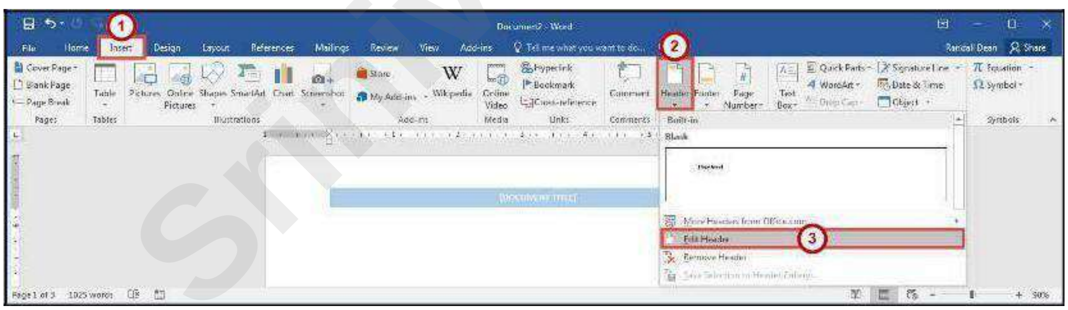


Figure 46 - Edit Header

Removing Headers & Footers

1. Click the **Insert** tab (See Figure 47).
2. Click **Header** (See Figure 47).
3. In the *Header Gallery* drop-down menu, click **Remove Header** (See Figure 47).

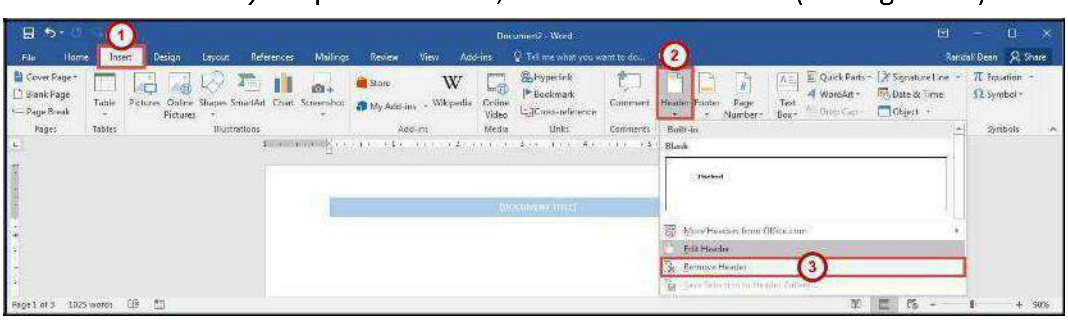


Figure 47 - Remove Header

Adding Page Numbers to Your Document

1. Click the **Insert** tab (See Figure 48).
2. Click **Page Number** (See Figure 48).
3. In the *Page Number* drop-down menu, click **Bottom of Page** (See Figure 48).
4. In the *Bottom of Page* drop-down menu, click on a **page number design** (See Figure 48).

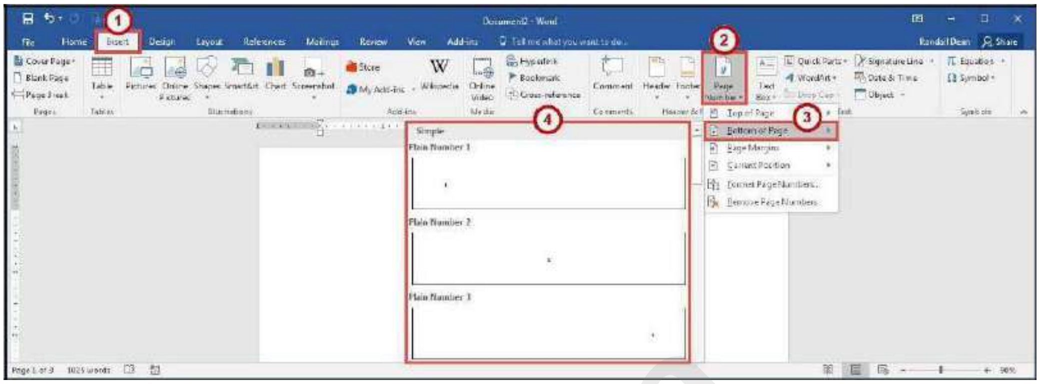


Figure 48 - Adding Page Numbers to Your Document

Cut, Copy, and Paste Text

To remove text from one place in the document and put it in another you *cut and paste* the text. You can also *copy and paste* text if you want to add copies of your selection elsewhere in your document.

The Cut, Copy, and Paste commands, along with the Format Painter, are located in the *Clipboard* group on the *Home* tab.

Cutting and Pasting Text

1. **Select** the text to be cut.
2. Click the **Home** tab (See Figure 49).
3. Click the **Cut** button (See Figure 49).

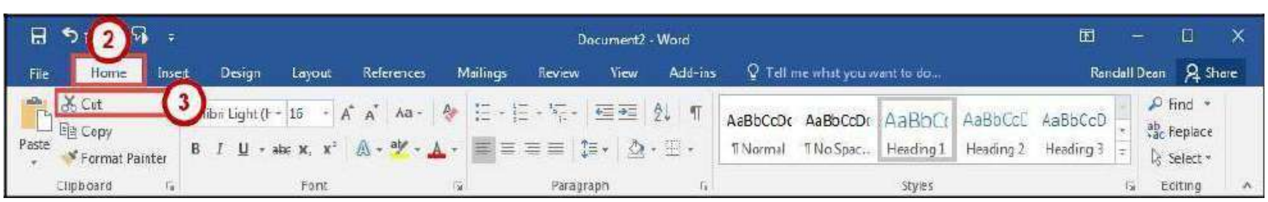


Figure 49 - Cut Tool

4. Click within the document where you want to paste the text.
5. Click the **Paste** button.

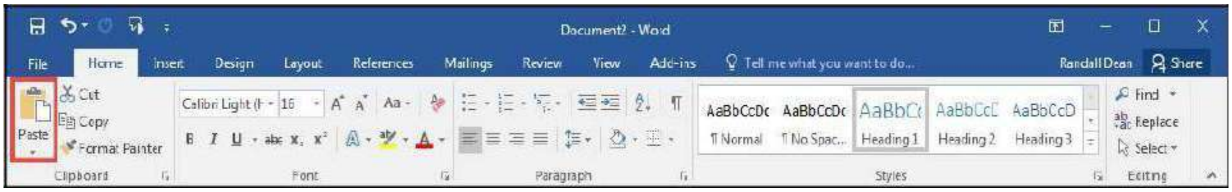


Figure 50 - Paste Tool

6. Your text will be reinserted into the location you specified within your document.

Note: You can move text from one place to another by selecting the text and then clicking and dragging the text to the new location.

Copying and Pasting Text

1. **Select** the text to be copied.
2. Click the **Home** tab (See Figure 51).
3. Click the **Copy** button (See Figure 51).

Figure 51 - Copy Tool

4. Click within the document where you want to paste the text.
5. Click the **Paste** button.

Figure 52 - Paste Tool

6. Your text will be reinserted into the location you specified within your document.

Navigating the Document

By using the *Find* and *Go To* features, you can search for key words in your document, or move to a specific page, section, comment, etc. If the heading styles have been added to your document, you can also use the *Navigation Pane* to move to sections of your document.

Using Find

The *Find* feature is useful for finding one or all instances of a specific word, as well as replacing the word with another.

1. Click the **Home** tab (See Figure 53).
2. Click the **Find** tool (See Figure 53).
3. The *Navigation Pane* will open to the left of the document. In the *search document* field, **type** the word you wish to search for (See Figure 53).
4. As you type your word in the search bar, your results will auto populate below. Click on one of the **search results** to jump to that word in your document (See Figure 53).

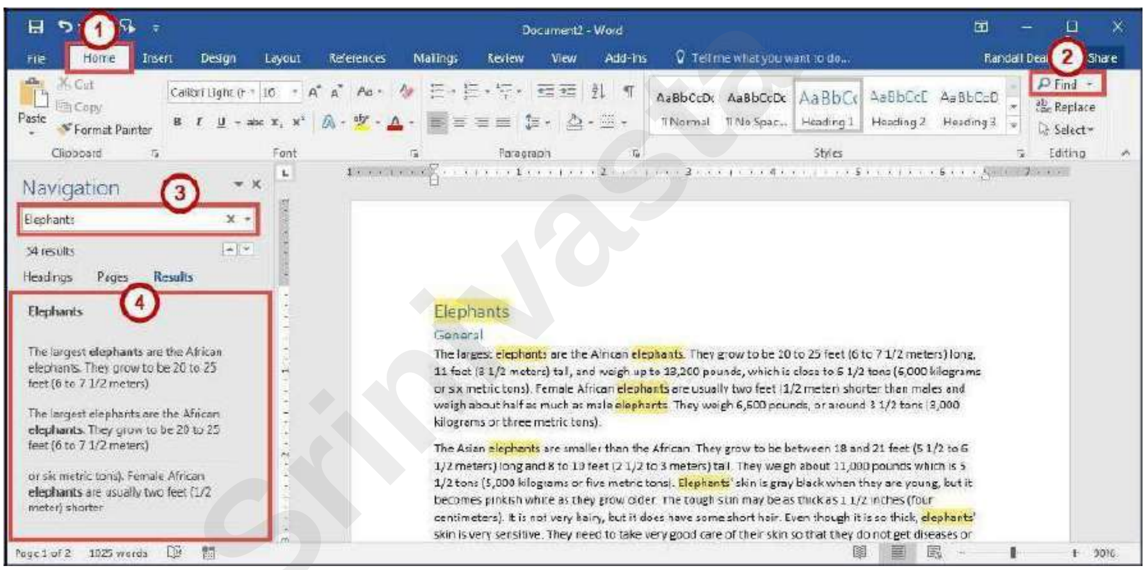


Figure 53 - Find Tool

Using Go To

The *Go To* feature is helpful for moving to a specific page, section, comment, etc. within your document.

1. Click the **Home** tab (See Figure 54).
2. Click the **drop-down arrow** next to the *Find* tool (See Figure 54).
3. In the drop-down menu, click the **Go To** tool (See Figure 54).

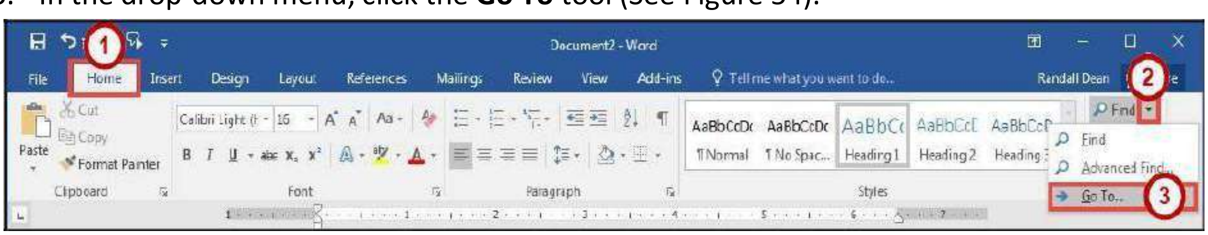


Figure 54 - Go To Tool

4. In the *Find and Replace* dialog box*,* make a **selection** under *Go to what* (See Figure 55).
5. Type your **search parameters** in the *search field* (See Figure 55).
6. Click the **Go To** button (See Figure 55).

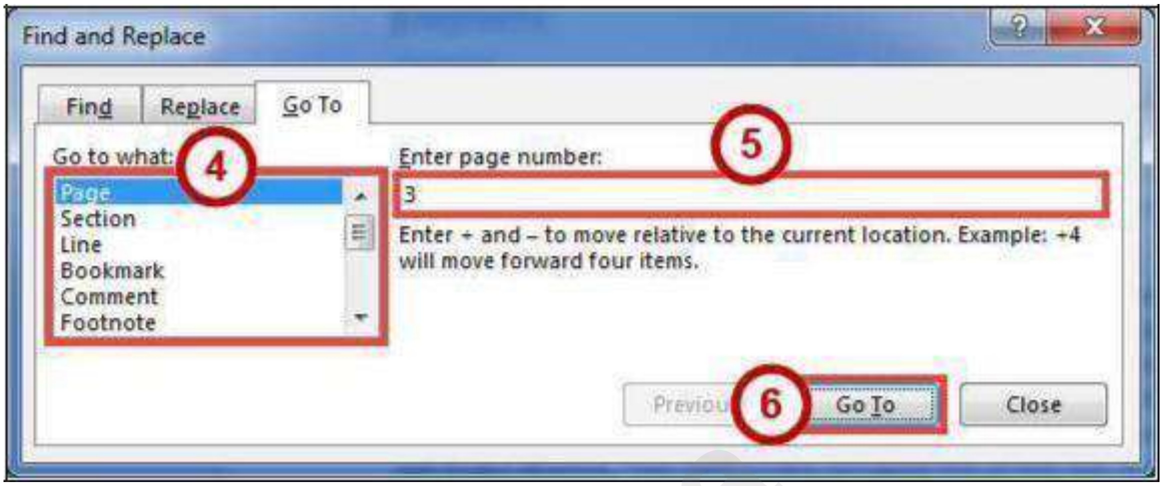


Figure 55 - Go To Search

Navigation Pane

The *Navigation Pane* provides an easy way to move throughout your document, without having to scroll. You can navigate quickly to areas of your document that have headings, or move to other pages. You can also use the *Results* tab to find text within your document. The following explains how to enable the *Navigation Pane*:

1. Click the **View** tab (See Figure 56).
2. Click the **checkbox** next to *Navigation Pane* (See Figure 56).
3. The *Navigation Pane* will display to the left of your document (See Figure 56).

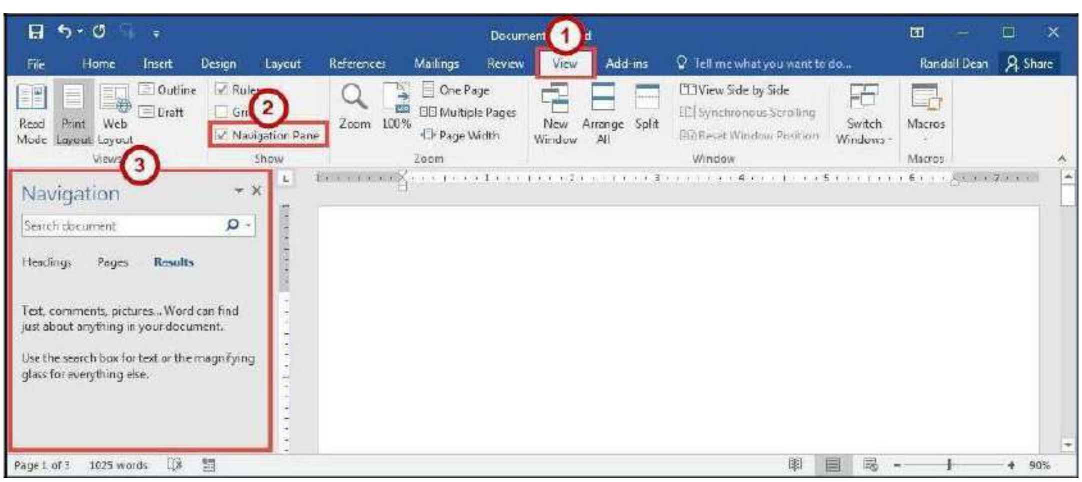


Figure 56 - Navigation Pane

Proofing and AutoCorrect Options

Word is configured to check your spelling and grammar as you type. As a result, you may notice that certain markings appear on the page as you type. These markings indicate possible mistakes that Word has identified in your document.

A jagged red line underneath a word indicates a spelling error (or an unrecognized word), while a jagged blue line indicates a grammatical error.

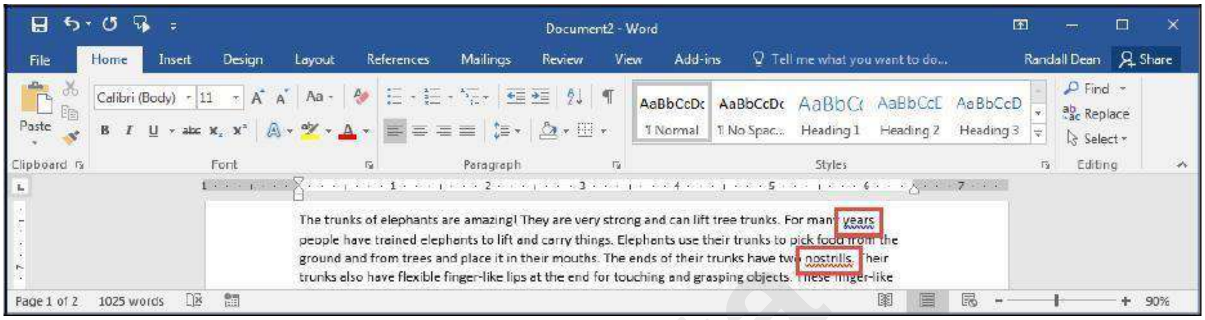


Figure 57 - Proofing Results

AutoCorrect Options

Word will automatically correct misspelled words that are contained in the AutoCorrect list (e.g. typing **"the"** will automatically be changed to "the"**).** The following explains how to customize the AutoCorrect options:

1. Click the **File** tab.
2. In the B*ackstage View,* click **Options**.
3. In the *Word Options* dialog box, click **Proofing** (See Figure 58)**.**
4. In the *AutoCorrect options* section, click the **AutoCorrect Options** button (See Figure 58).

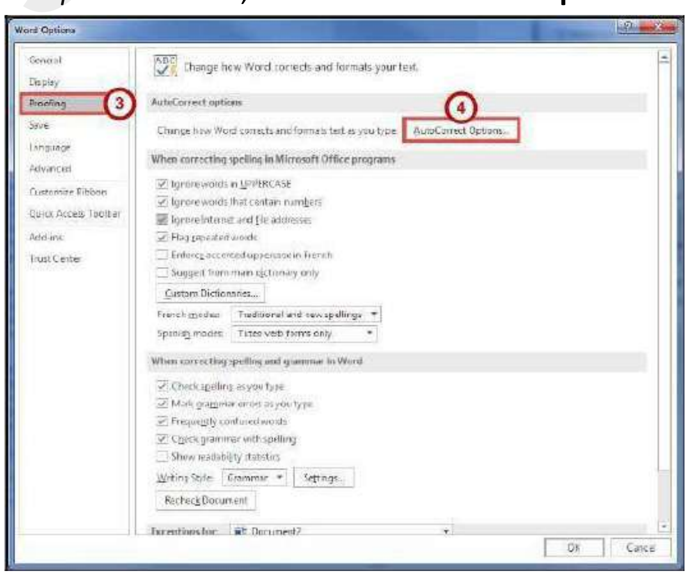


Figure 58 - AutoCorrect Options

5. From the *AutoCorrect Options* dialog box you can change the following actions:
    a. To disable any of the default options, click the **check box** next to the option (See Figure 59).
    b. To add your own *Replace text as you type* entry (See Figure 59):
        i. Type the **word** that you want to correct in the *Replace* box (See Figure 59).
        ii. Type the **word** with which you wish to replace it in the *With* box (See Figure 59).
        iii. Click the **Add** button (See Figure 59).
    c. Click the **OK** button (See Figure 59).
6. In the *Word Options* dialog box, click the **OK** button.

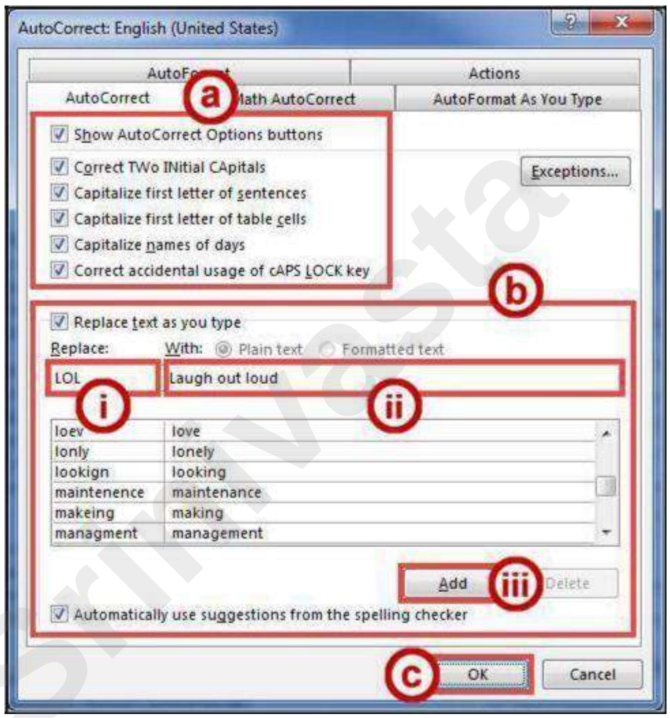


Figure 59 - AutoCorrect Dialog Box

Spelling & Grammar Check

Word automatically checks for spelling and grammar mistakes as you type. However, it’s a good idea to run the *Spelling & Grammar* tool as the final step when finishing your document. When run, the *Spelling & Grammar* tool will check your entire document for spelling & grammar errors, and allow you to insert the corrections. The following explains how to run the *Spelling & Grammar* tool:

1. Click the **Review** tab (See Figure 60).
2. Click **Spelling & Grammar** (See Figure 60).

Figure 60 - Spelling & Grammar Tool

3. The *Spelling & Grammar* checker will open to the right side of the document.
4. From the *Spelling & Grammar* tool you can choose the following options:
   a. **Ignore** - Ignore the currently selected misspelled word (See Figure 61).
   b. **Ignore All** - Ignore the currently selected misspelled word and all instances of the misspelled word in the document (See Figure 61).
   c. **Add** - Add the selected misspelled word to the dictionary so it will not be identified as a mistake (See Figure 61).
   d. **Suggested Word List** - A list of suggested words for the misspelled word (See Figure 61).
   e. **Change** - Apply the currently selected suggestion to the misspelled word (See Figure 61).
   f. **Change All** - Apply the currently selected suggestion to all instances of the misspelled word in the document (See Figure 61).

Note: The *Spelling & Grammar* tool will continue to check your document for any misspelled words, or grammar errors. The *Spelling and Grammar* check will notify you when it has completed checking your document.

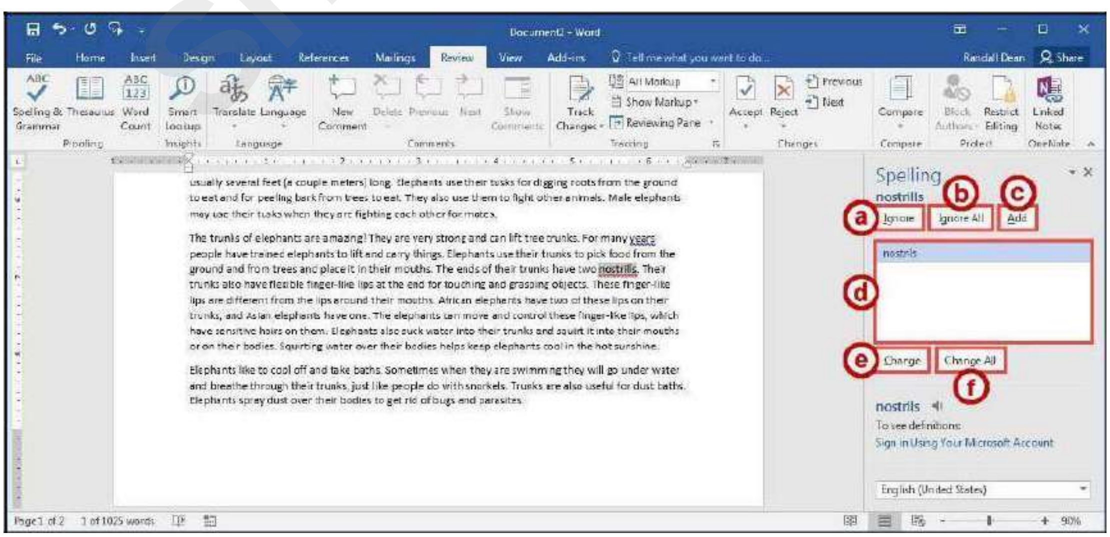


Figure 61 - Spelling Results

Thesaurus

The *Thesaurus* tool can help you find synonyms for words, and insert the new word into your document.

1. Select a **word** in your document that you wish to find a synonym for.
2. Click the **Review** tab (See Figure 62).
3. Click **Thesaurus** (See Figure 62).

Figure 62 - Thesaurus Tool

4. The *Thesaurus* tool will open to the right side of the document, with a list of synonyms for your selected word (See Figure 63).
5. Hover over the word you wish to insert, and click the **drop-down** arrow (See Figure 63).
6. In the *drop-down* menu, click **Insert** (See Figure 63).

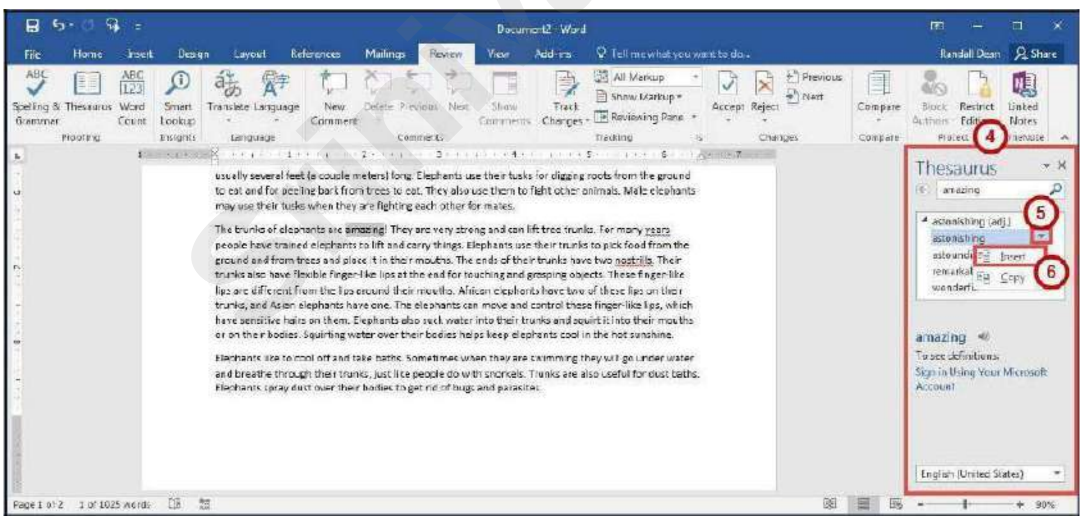


Figure 63 - Thesaurus Results

7. The chosen word from the *Thesaurus* will replace the selected word in your document.

Saving a Document

Saving your document will create a file that will allow you to access the document at a later time for editing. You can also save your work to share the file with others. The following explains how to save your document in the Word format to your Desktop:

1. Click the **File** tab.

Figure 64 - File Tab (Backstage View)

2. In the B*ackstage View*, click **Save As** (See Figure 65).
3. Click **This PC** (See Figure 65).
4. Click **Desktop** (See Figure 65).

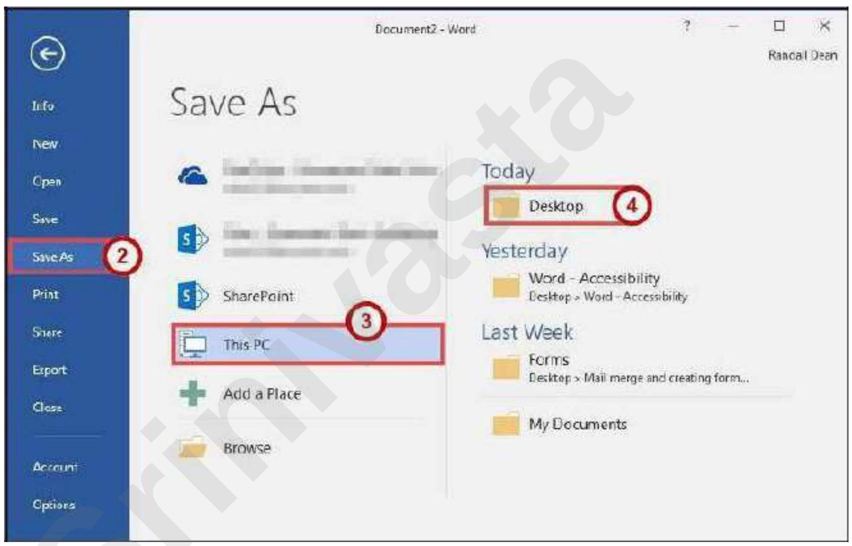


Figure 65 - Backstage View

5. In the *Save As* dialog box, type a **File name** for your document (See Figure 66).

Note: Make sure *Save as type* is *Word Document (*.docx)*. This will save the document as a Word 2016 document.

6. Click the **Save** button (See Figure 66).

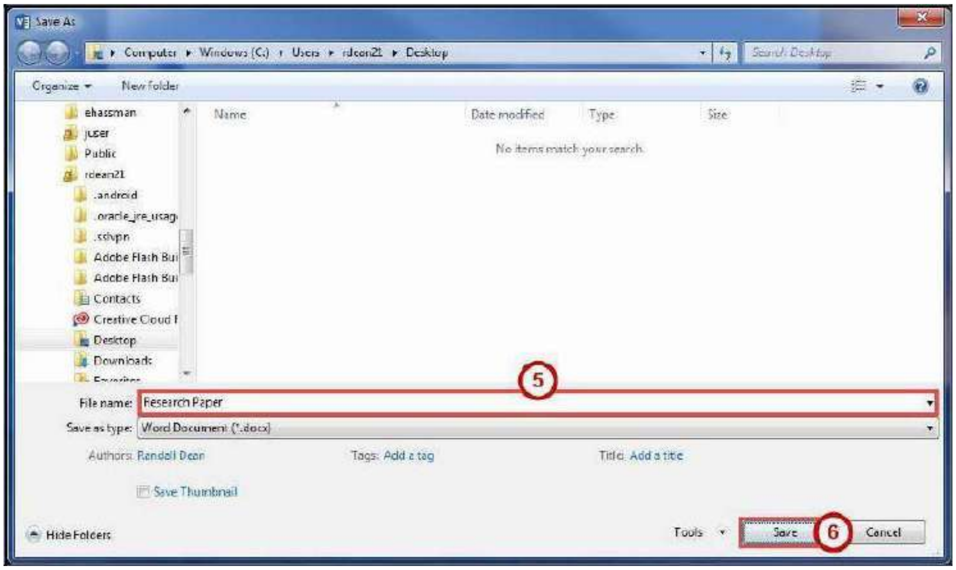


Figure 66 - Save As Dialog Box

Compatibility Mode

When opening a document created in a previous version of Word, Word 2016 will begin operating in *Compatibility Mode*. At the top of the window, the words *[Compatibility Mode]* will be displayed next to the name of the document.

Compatibility Mode means that some of the newer features of Word 2016 will not be available to you because the document was created with an earlier version of Word. If you save the document as a Word 2016 document, you will exit *Compatibility Mode* and all of the new features will be available to you. The following explains how to save a document so that you can exit *Compatibility Mode*:

1. Click the **File** tab.

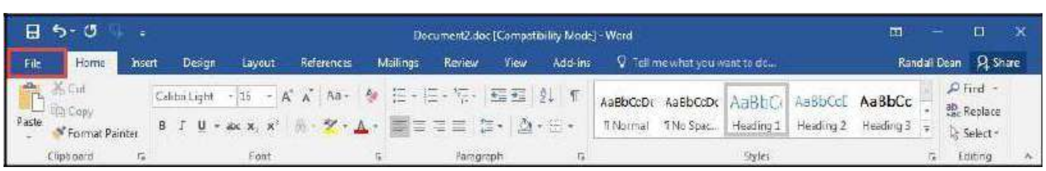


Figure 67 - File Tab (Backstage View)

2. In the B*ackstage View*, click **Info** (See Figure 68).

3. Click **Convert** (See Figure 68).

Figure 68 - Convert

4. In the *confirm your conversion* dialog box, click the **OK** button.

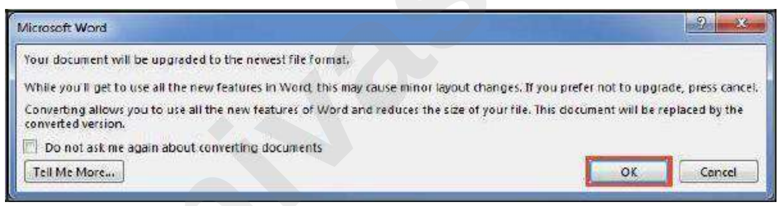


Figure 69 - Confirm Your Conversion

5. The document will be converted to a Word 2016 document. The *[Compatibility Mode]* label will be removed from the document's title.

Printing a Document

If your computer is connected to a printer, you will be able to print your document to share a hard copy with others.

1. Click the **File** tab.

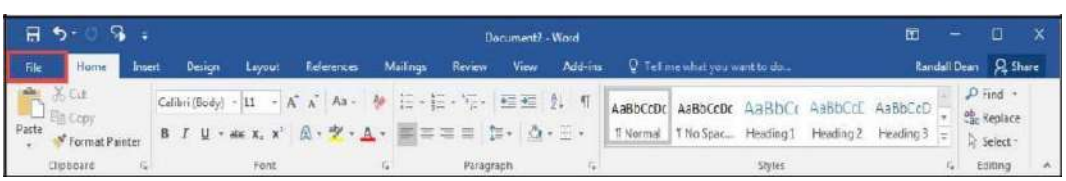


Figure 70 - File Tab (Backstage View)

2. In the B*ackstage View*, click **Print** (See Figure 71).
3. From *Print* you can choose the following options:
   a. Copies - Set the number of copies you want to print (See Figure 71).

b. **Printer** - Select a printer (See Figure 71).

Note: Your list of available printers will be determined by the printers you have installed on your computer.

c. **Settings** - Configure how and what you want to print (See Figure 71).

4. Click the **Print** button (See Figure 71).

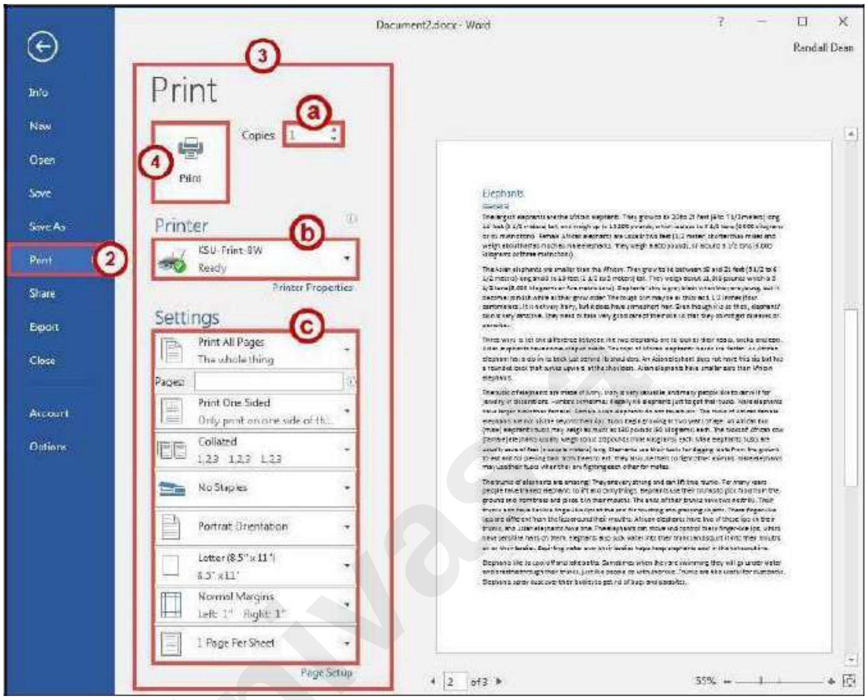


Figure 71 - Print

Microsoft Office

Word 2016 for Windows

FORMATTING YOUR DOCUMENT

Learning Objectives

After completing the instructions in this booklet, you will be able to:

- Create styles and use them to format document text.
- Create and modify tables.
- Insert section breaks in a document.
- Format the document text as columns.

Using Styles

A Style is a predefined combination of font style, color, and size that you can use to format the text in your document. Using styles can help you create documents that have a more professional, and consistent, appearance. Some styles (like the built-in heading styles) can be used to easily navigate your document, or insert a table of contents.

You can use the styles available in Word, modify them, or create your own style and save it to use every time you need it.

Applying a Style

1. Select the **text** that you want to format.
2. Click the **Home** tab (See Figure 1).
3. In the *Home* tab, scroll through the styles with the **up and down arrows** (See Figure 1).
4. Click the desired **style** to apply (See Figure 1).

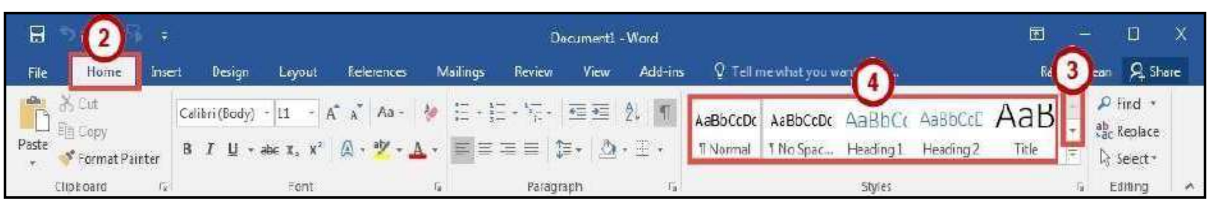


Figure 1 - Styles on Ribbon

Note: You can also access your styles from the *Mini Toolbar*. After selecting your text, the *Mini Toolbar* will appear. Click the **Styles** button on the right.

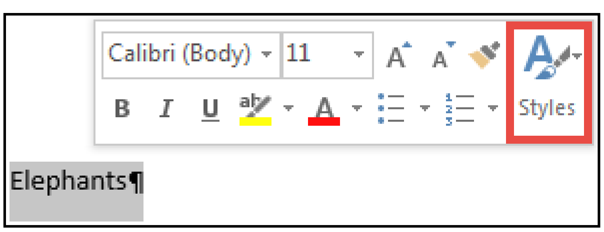


Figure 2 - Styles on Mini Toolbar

Modifying a Style Using the Ribbon

An easy way to modify your styles is to apply any formatting changes to your selected text, then to apply the change to a preset style. The following explains how to modify a style using the ribbon:

1. Select the **text** that you want to format.
2. Format the selected text with the new attributes that you want (e.g. Bold, 10pt, Times New Roman font, and Red lettering).
3. Click the **Home** tab (See Figure 3).
4. On the *Home* tab, right-click the **style** you wish to modify (See Figure 3).
5. In the drop-down menu, click **Update (Style to modify) to Match Selection** (See Figure 3).
6. Your style will be updated to match your selection.

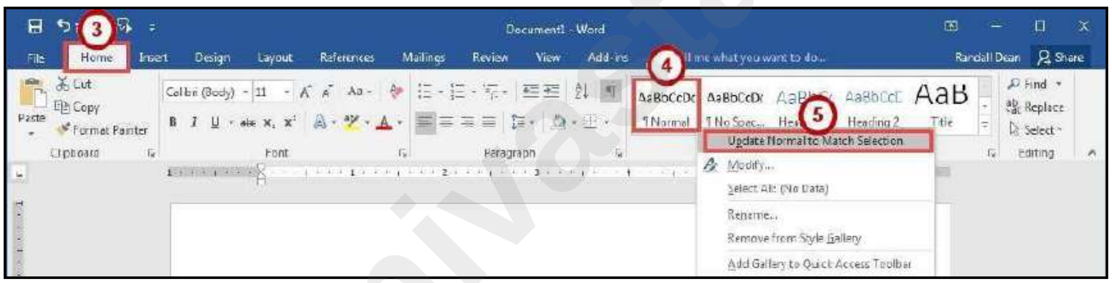


Figure 3 - Update Style to Match Selection

Modifying a Style Using the Styles Dialog Box

1. Click the **Home** tab (See Figure 4).
2. On the *Home* tab, right-click the **style** you wish to modify (See Figure 4).
3. In the drop-down menu, click **Modify** (See Figure 4).

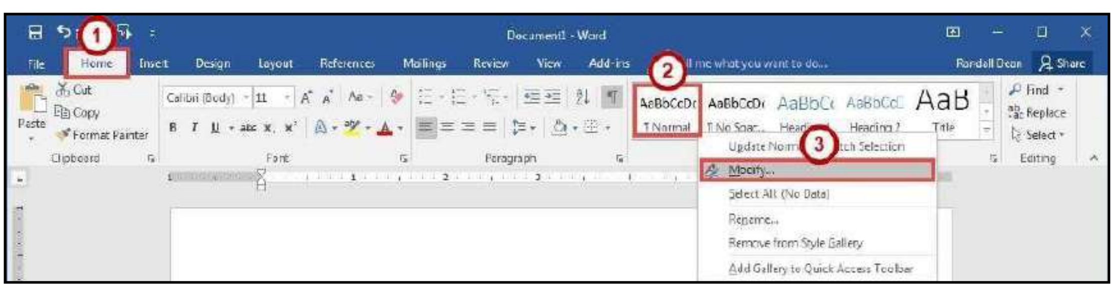


Figure 4 - Modify Style

4. In the *Modify Style* dialog box, select the **new attributes** that you want to apply to the style (See Figure 5).
5. For further options, click the **Format** button (See Figure 5).

6. Click the **OK** button (See Figure 5).

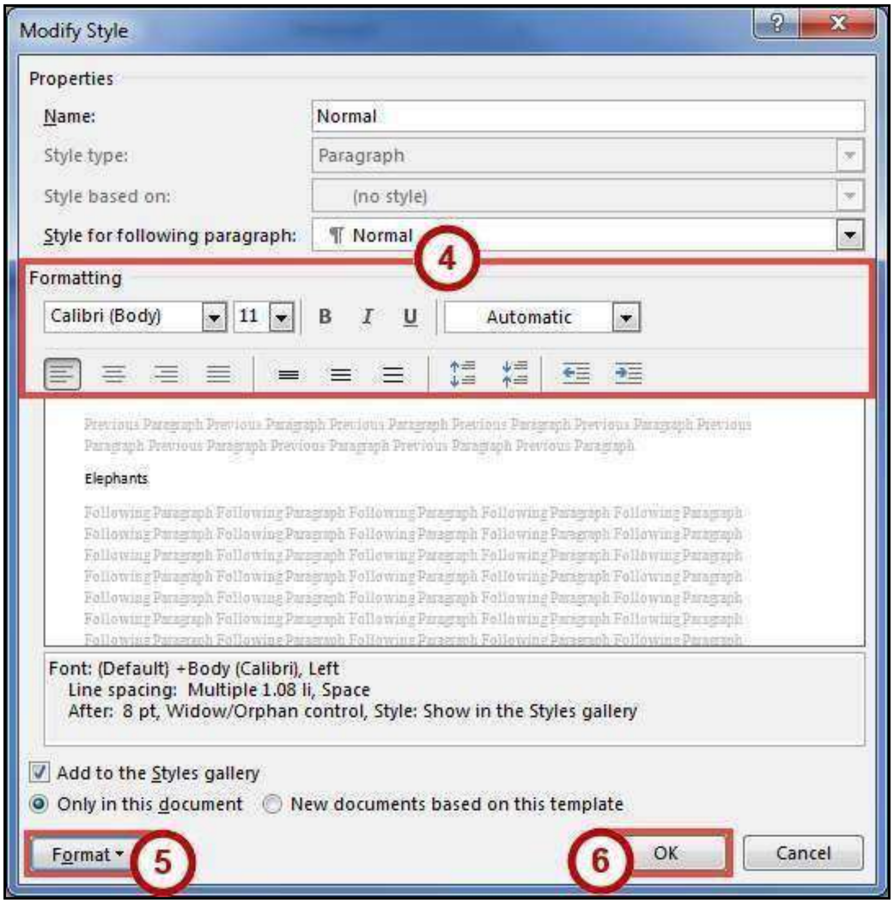


Figure 5 - Modify Style Dialog Box

Creating Your Own Custom Style

In addition to modifying existing styles, you can also create your own custom styles. The following explains how to create a custom style:

1. Select the **text** that you want to format as a new style.
2. Format the selected text with the new attributes that you want (e.g. Bold, 10pt, Times New Roman font, and Red lettering).
3. Click the **Home** tab (See Figure 6).
4. In the *Home* tab, click the **Style drop-down** arrow (See Figure 6).

Figure 6 - Style Drop-down

5. In the *Style* drop-down menu, click **Create a Style**.

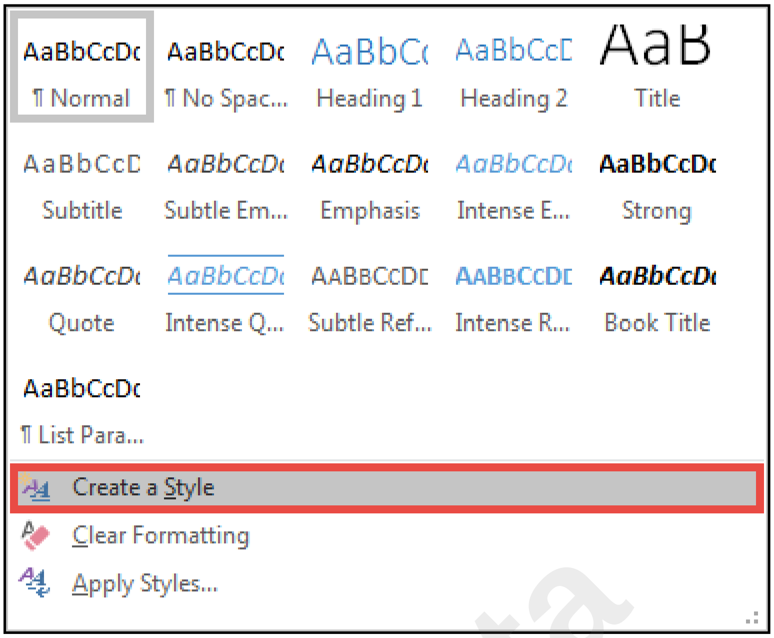


Figure 7 - Create a Style

6. In the *Create New Style from Formatting* dialog box, type a **Name** for your style (See Figure 8).

7. Click the **OK** button (See Figure 8).

Figure 8 - Create New Style from Formatting Dialog Box

8. Your new style can be selected from within the *Styles* on the *Home* tab.

Figure 9 - My New Style

Tables

Tables are useful for presenting text information and numerical data in a neat and orderly fashion. A Table consists of rows and columns that intersect to form boxes called cells, which you can then fill with text, numbers, or graphics. You can also format your table for added effect (e.g. make the lines within the table visible or invisible).

Creating a Table Using the Table Menu

1. Click the **Insert** tab (See Figure 10).
2. Click the **Table** icon (See Figure 10).
3. In the *Insert Table* drop-down menu, move your mouse pointer over the **boxes** until you have the number of rows and columns that you want in the table (See Figure 10).

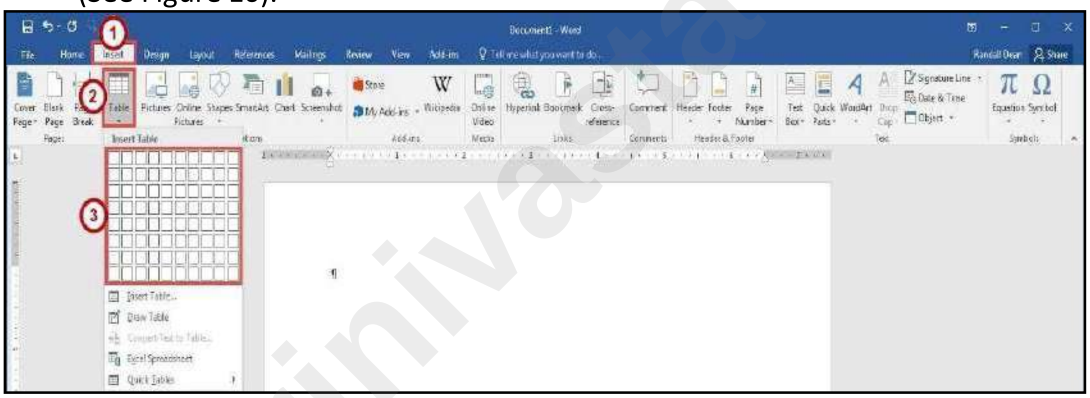


Figure 10 - Table Icon

4. The *Live Preview* feature will show what the table will look like in your document. Click to **confirm** your table.

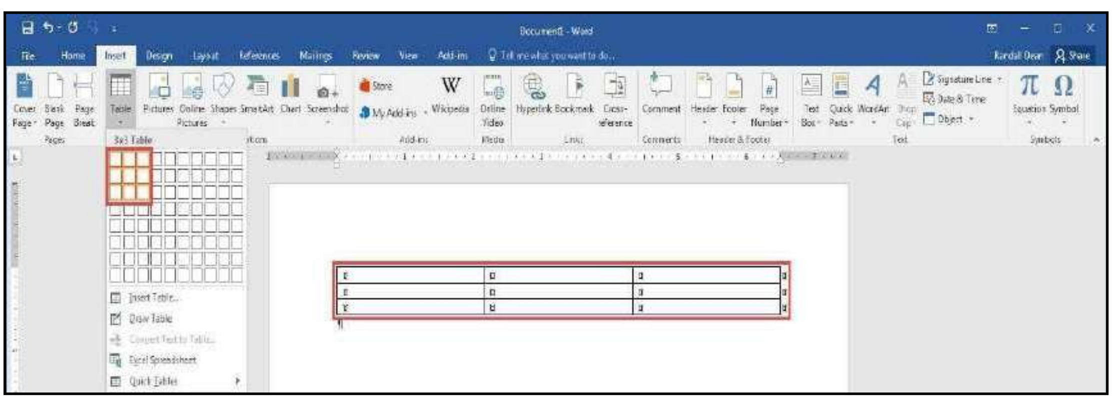


Figure 11 - Click to Confirm Your Table

5. Your table will be placed within your document.

Note: This method will only allow you to create a table up to 10x8 in size. To create a table with more rows and columns, see creating a Table Using the Insert Table Command.

Creating a Table Using the Insert Table Command

If you need to create a table that contains more than 10 columns and/or 8 rows, then you can use the *Insert Table* command to designate how many columns and rows to enter into your table. The following instructions explain how to create a table using the Insert Table Command:

1. Click the **Insert** tab (See Figure 12).
2. Click the **Table** icon (See Figure 12).
3. In the *Insert Table* drop-down, click **Insert Table** (See Figure 12).

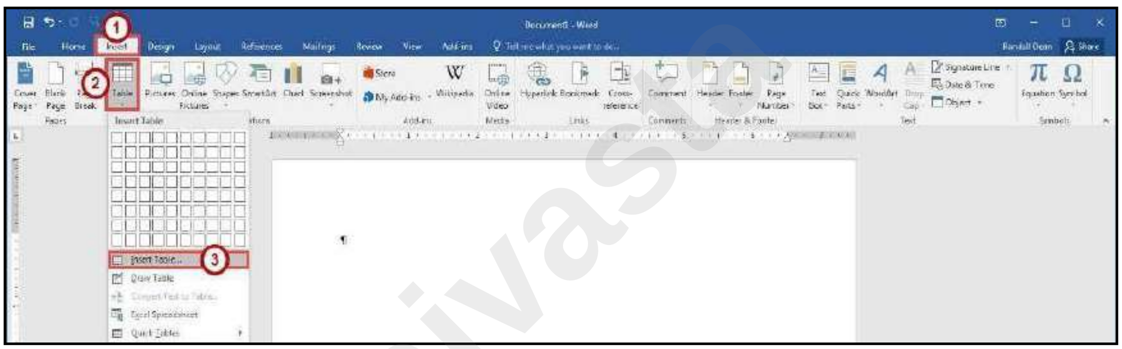


Figure 12 - Insert Table

4. In the *Insert Table* dialog box, enter the **Number of columns and rows** for the table (See Figure 13).
5. Click the **OK** button (See Figure 13).

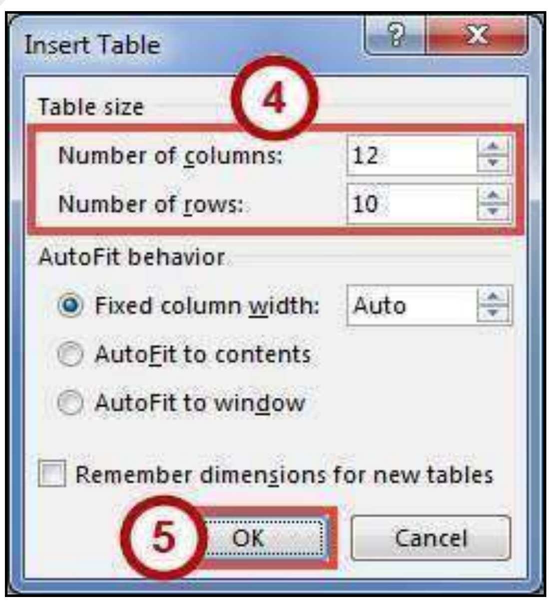


Figure 13 - Insert Table Dialog Box

Creating a Table Using the Drawing Tools

If you want more control over the shape of your table's columns and rows, you can also draw your own table. The following explains how to create a table using the *drawing tools*.

1. Click the **Insert** tab (See Figure 14).
2. Click the **Table** icon (See Figure 14).
3. In the *Insert Table* drop-down menu, click **Draw Table** (See Figure 14).

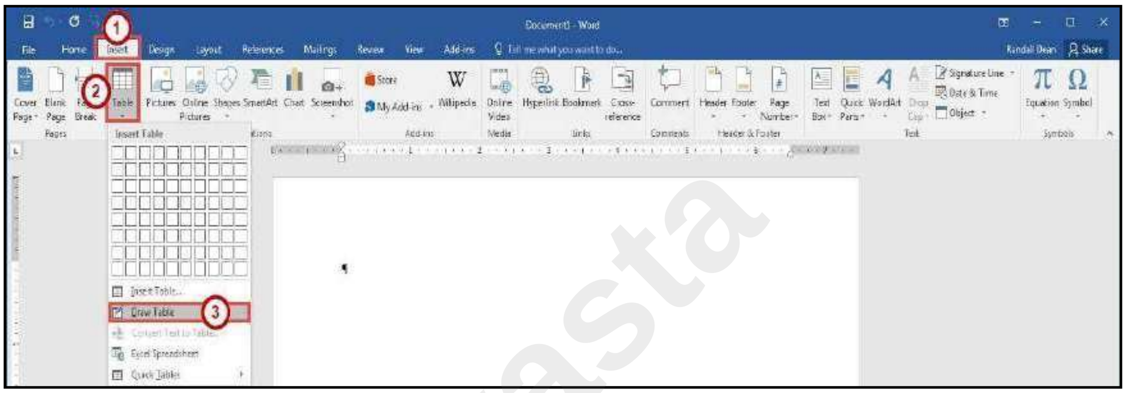


Figure 14 - Draw Table

4. The mouse pointer will change to a pencil icon. Click and drag the **pencil** to draw the outer border of the table.

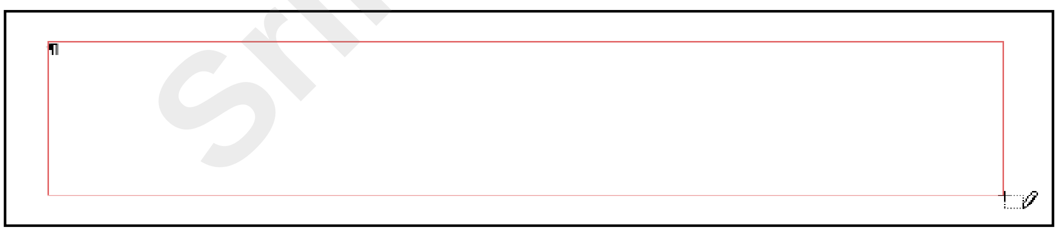


Figure 15 - Drawing a Table

5. Click and drag the **pencil** to draw lines within the border to create cells.

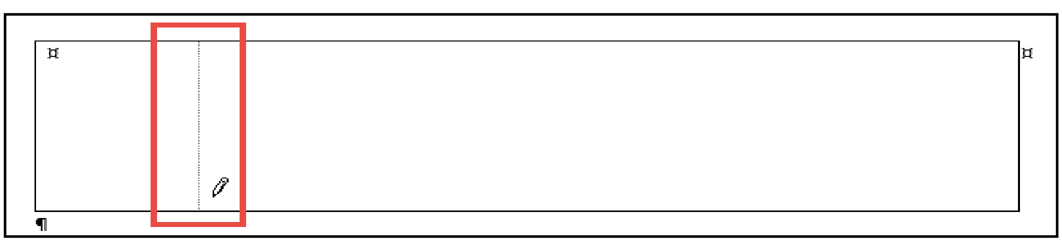


Figure 16 - Drawing Table Lines

6. Press the **ESC** key to return to the regular mouse pointer.

Accessing the Table Tools

Tables have their own set of editing tools accessible by a context sensitive tab on the *Ribbon*. To access this tab, click a table in your document and the *Table Tools* tab will appear in the *Ribbon*. From these tabs, you can modify the *Design* or the *Layout* of your tables.

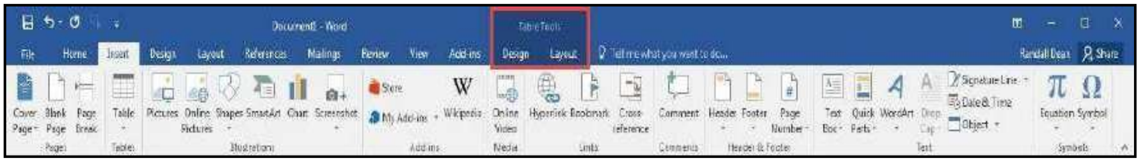


Figure 17 - Table Tools

Splitting Cells

1. Click in the **cell(s)** that you want to split.
2. Click the **Layout** tab for *Table Tools* (See Figure 18).
3. In the *Table Tools - Layout* tab, click **Split Cells** (See Figure 18).

Figure 18 - Split Cells

4. In the *Split Cells* dialog box, enter the N**umber of rows and columns** into which you want to split the cells (See Figure 19).
   Click **OK** button (See Figure 19).

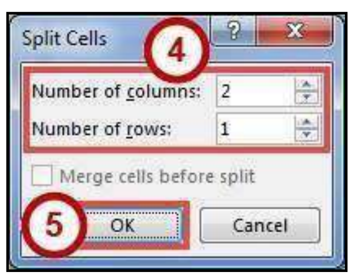


Figure 19 - Split Cells Dialog Box

Merging Cells

1. Select the **cells** that you want to merge.
2. Click the **Layout** tab for *Table Tools* (See Figure 20).
3. In the *Table Tools - Layout* tab, click **Merge Cells** (See Figure 20).

Figure 20 - Merge Cells

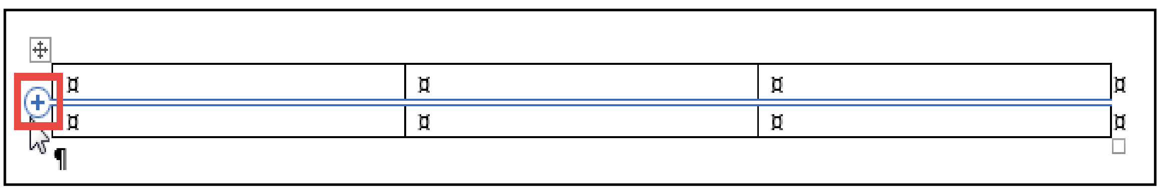
Inserting Rows and Columns

The following section explains how to insert additional rows and columns into a table.

Using Insert Controls to quickly add Rows and Columns

The *Insert Controls* allow you to quickly add rows and columns to your table using a mouse. The *Insert Controls* will appear right outside your table when you move your cursor above or the left of two existing columns or rows. The following explains how to use the *Insert Controls* to add a row to your table:

1. Hover your cursor just to the left of two existing rows (See Figure 21).
2. The *Insert Control* option will appear. Click the **Insert Control** to add a row (See Figure 21). **Note**: For inserting columns using *Insert Controls*, hover just above two



existing columns and click the **Insert Control**.

Figure 21 - Insert Controls

Using the Table Tools Layout Tab to Insert Rows and Columns

1. Click in the **cell** that you want to insert a row or column around.
2. Click the **Layout** tab for *Table Tools* (See Figure 22).
3. In the *Table Tools - Layout* tab, click **Insert (Above, Below, Left, Right)** to insert a row or column (See Figure 22).

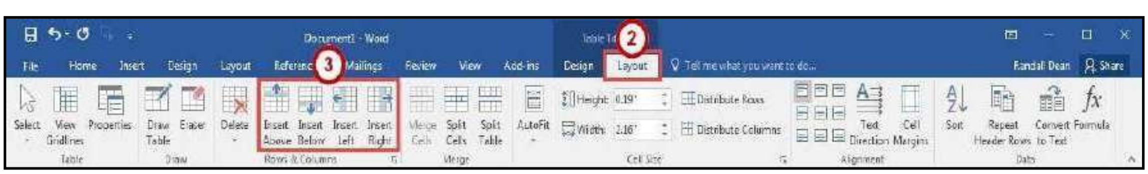


Figure 22 - Insert Rows or Columns

Using the Table Tools Layout Tab Rows & Columns Dialog Box

1. Click in the **cell** that you want to insert a row or column around.
2. Click the **Layout** tab for *Table Tools* (See Figure 23).
3. In the *Table Tools - Layout* tab, click the **Dialog Box launcher** for the *Rows & Columns* group (See Figure 23).

Figure 23 - Rows & Columns Dialog Box Launcher

4. In the *Insert Cells* dialog box, click one of the **last two options** to insert a row or column (See Figure 24).
5. Click the **OK** button (See Figure 24).

Note: You can also select *Shift cells right* or *Shift cells down* to insert a single cell.

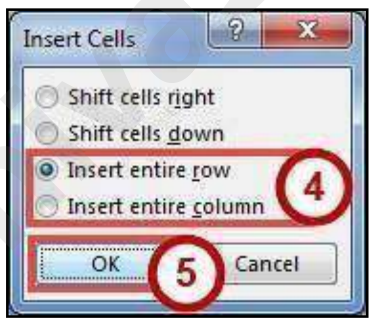


Figure 24 - Insert Cells Dialog Box

Deleting Cells, Rows, Columns, and Tables

1. Select the **rows, columns, cells, or whole table** that you want to delete.
2. Click the **Layout** tab for *Table Tools* (See Figure 25).
3. In the *Table Tools - Layout* tab, click **Delete** (See Figure 25).
4. In the *Delete* drop-down menu, click **Delete (Cells, Columns, Rows, or Table)** (See Figure 25).

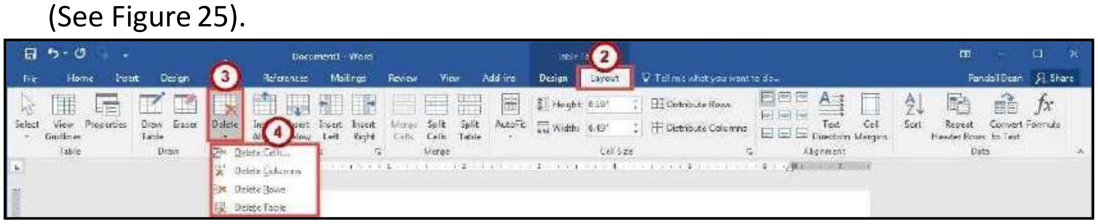


Figure 25 – Table Delete Options

Changing Cell Size

There are three ways to change the size of rows and columns in a table; either by setting the dimensions, dragging the cell borders, or by distributing your rows and/or columns.

Setting the Dimensions

1. Click in the **cell** inside the row or column where the size needs to be adjusted.
2. Click the **Layout** tab for *Table Tools* (See Figure 26).
3. In the *Table Tools - Layout* tab, change the **measurement** in the height and/or width fields (See Figure 26).

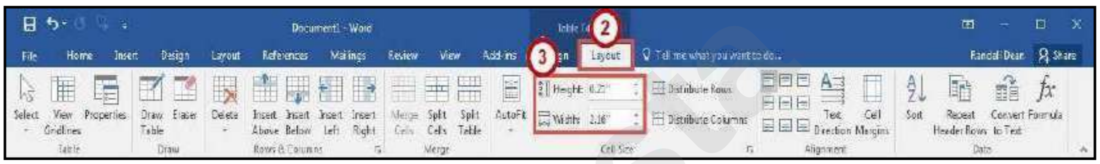


Figure 26 - Cell Size

Distribute Rows and Columns Evenly

1. Select the **rows** or **columns** you want to distribute evenly.
2. Click the **Layout** tab for *Table Tools* (See Figure 27).
3. In the *Table Tools - Layout* tab, click **Distribute Rows** or **Distribute Columns** (See Figure 27).

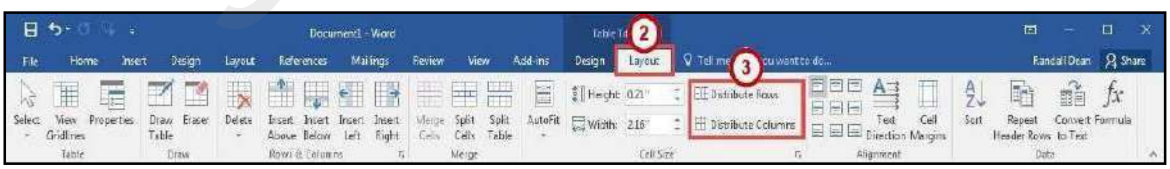


Figure 27 - Distribute Rows or Columns

Dragging the Cell Border

To change the cell size by dragging, hover your mouse cursor over the border of the column or row you wish to change. The cursor will change to a resize cursor. **Left-click and drag** to resize your column or row.

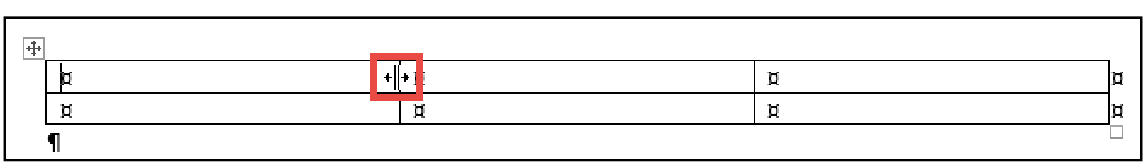


Figure 28 - Dragging Cell Borders

Add Text Wrapping to a Cell

As you type text into a cell, the column will expand to accommodate your entry. If you want the text to wrap inside the cell (move down when it hits the column border), the following steps will explain how to enable text wrapping:

1. Select the **cell(s)** in your table to add text wrapping.
2. Click the **Layout** tab for *Table Tools* (See Figure 29).
3. In the *Table Tools - Layout* tab, click **Dialog Box launcher** for the *Cell Size* group (See Figure 29).

Figure 29 - Cell Size Dialog Box Launcher

4. In the *Table Properties* dialog box, click the **Cell** tab (See Figure 30).
5. Click the **Options** button (See Figure 30).

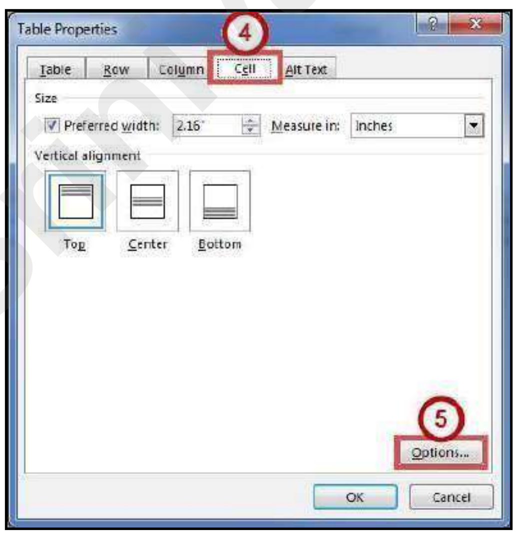


Figure 30 - Table Properties Dialog Box

6. The *Cell Options* window will open. Click the **box** next to *Wrap text* (See Figure 31).
7. Click the **OK** button (See Figure 31).

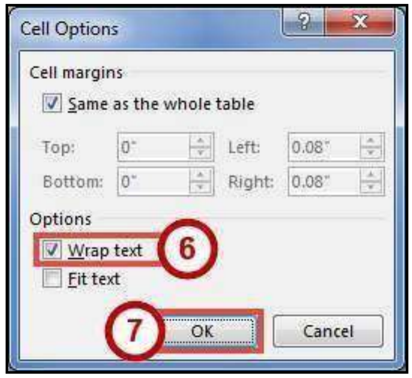


Figure 31 - Cell Options

8. In the *Table Properties* dialog box, click the **OK** button.

Moving a Table

1. Click in the **table**.
2. Move the mouse over the **Table Selector** in the top left corner of the table.

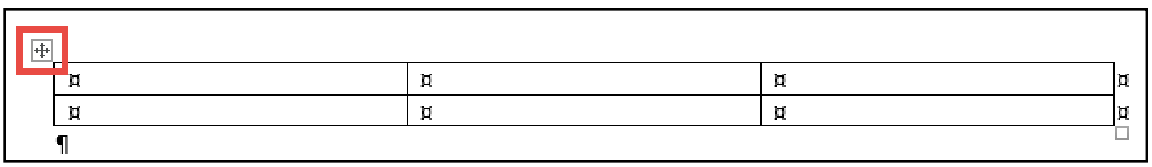


Figure 32 - Table Selector

3. Left-click and drag the **table** to the new location within the document.

Altering the Design of Your Table

The *Table Tools - Design* tab contains tools for altering the design of your table by adding preset table styles (e.g. banded columns, highlighted total row, etc.), shading options, and changing the color and style of your borders. The *Table Tools - Design* tab is shown in the *Ribbon* by first selecting atable.

Figure 33 - Table Tools Design Tab

Adding Table Styles

1. Click in the **table**.
2. Click the **Design** tab for *Table Tools* (See Figure 34).
3. In the *Table Tools - Design* tab, scroll through the styles with the **up and down arrows** (See Figure 34).
4. Click the desired **table style** to apply (See Figure 34).

Note: You can hover over a *Table Style* to see a live preview.

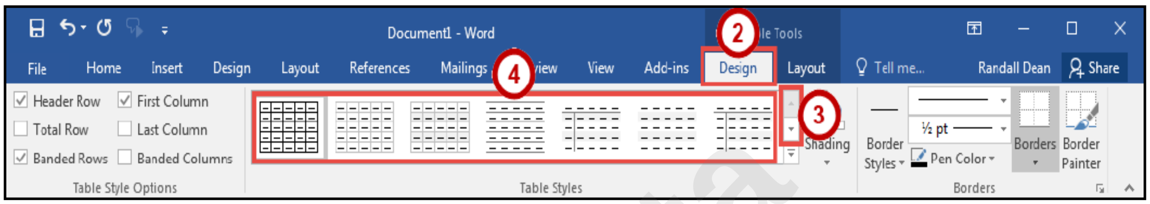


Figure 34 - Table Styles

5. You can further customize the *Table Style* by selecting **options** under the *Table Style Options*.

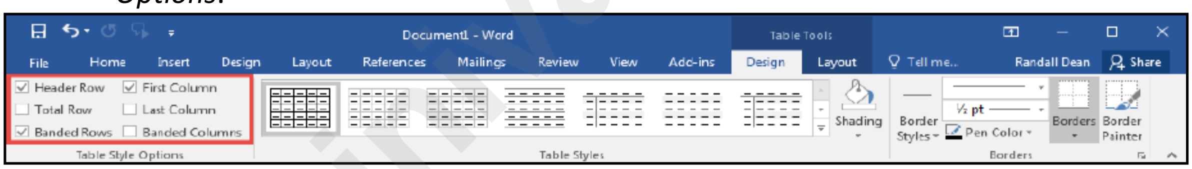


Figure 35 - Table Style Options

Headers and Footers

Headers and Footers allow you to add information (e.g. name, title of document, etc.) within the top or bottom margins of your document, and will repeat on every page for your document. The following explains how to insert a simple header:

1. Click the **Insert** tab (See Figure 36).
2. In the *Insert* tab, click **Header** (See Figure 36).
3. In the *Header* drop-down, select a **Header** from the list (See Figure 36).

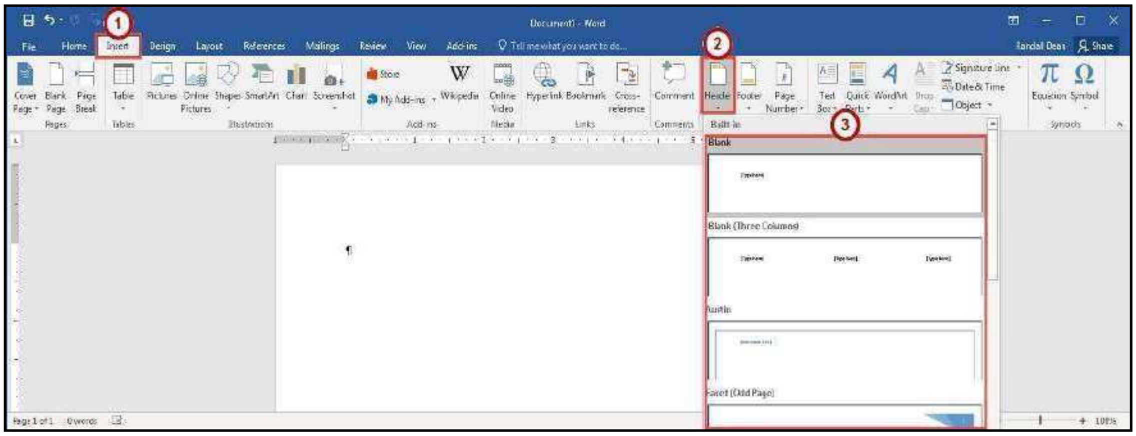


Figure 36 - Insert Header

4. The *Header* will be inserted into your document. Type your **information** into the header.

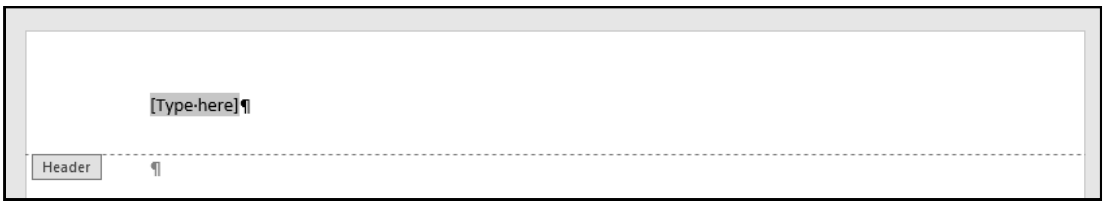


Figure 37 - Header Inserted

5. While editing your header, the *Header & Footer Tools - Design* tab will open. You will notbe able to return to the body of your document until you close the *Header & Footer Tools* (See Figure 38).
6. To close the *Header & Footer Tools* tab and return to editing your document, click the **Close Header and Footer** button (See Figure 38).

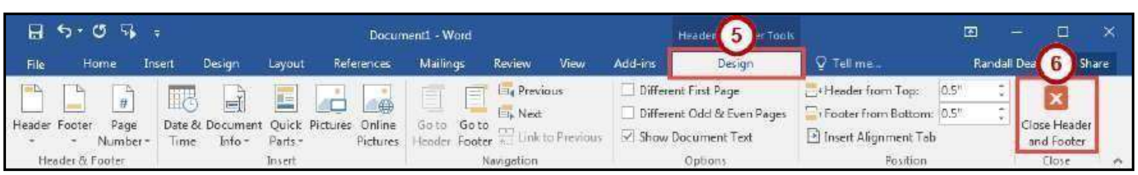


Figure 38 - Close Header & Footer Tools

Note: To return to editing your *Header*, double-click within the **Header** to select it.

Note: The steps for inserting a footer into your document are the same. Follow steps 1-6 above; selecting the **Footer** option instead.

Inserting Page Numbers

Word provides a simple tool for adding page numbers to your document. The *insert page number* tool also provides a selection of simple, and colorful options for your page numbers. The following explains how to insert page numbers into the bottom page of your document:

1. Click the **Insert** tab (See Figure 39).
2. In the *Insert* tab, click **Page Number** (See Figure 39).
3. In the *Page Number* drop-down menu, click **Bottom of Page** (See Figure 39).
4. In the *Bottom of Page* drop-down menu, select a **Page Number** from the list (See Figure 39).

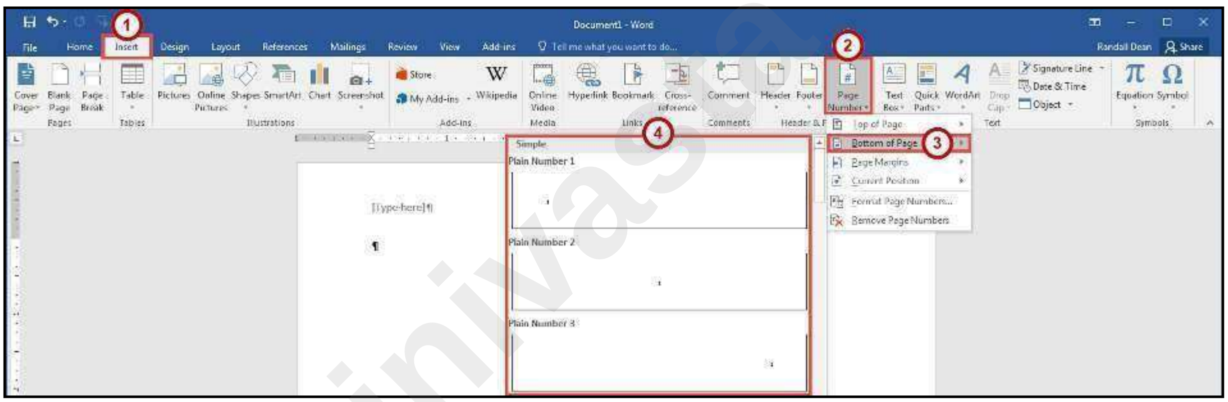


Figure 39 - Page Number

Section Breaks

A *section break* is a partition in a document that allows you to apply different layout and formatting options to different sections of the document, allowing for more control over the document's format and style. For example, section breaks can be used to start a new section on the next page, allowing you to maintain your spacing between sections. There are four types of section breaks: next page; continuous; even page; and odd page.

Inserting a Section Break

1. Click at the **end of a page** in the document.
2. Click the **Layout** tab (See Figure 40).
3. In the *Page Layout* tab, click **Breaks** (See Figure 40).
4. In the *Breaks* drop-down menu, select **Next Page** from the list under *Section Breaks* (See Figure 40).

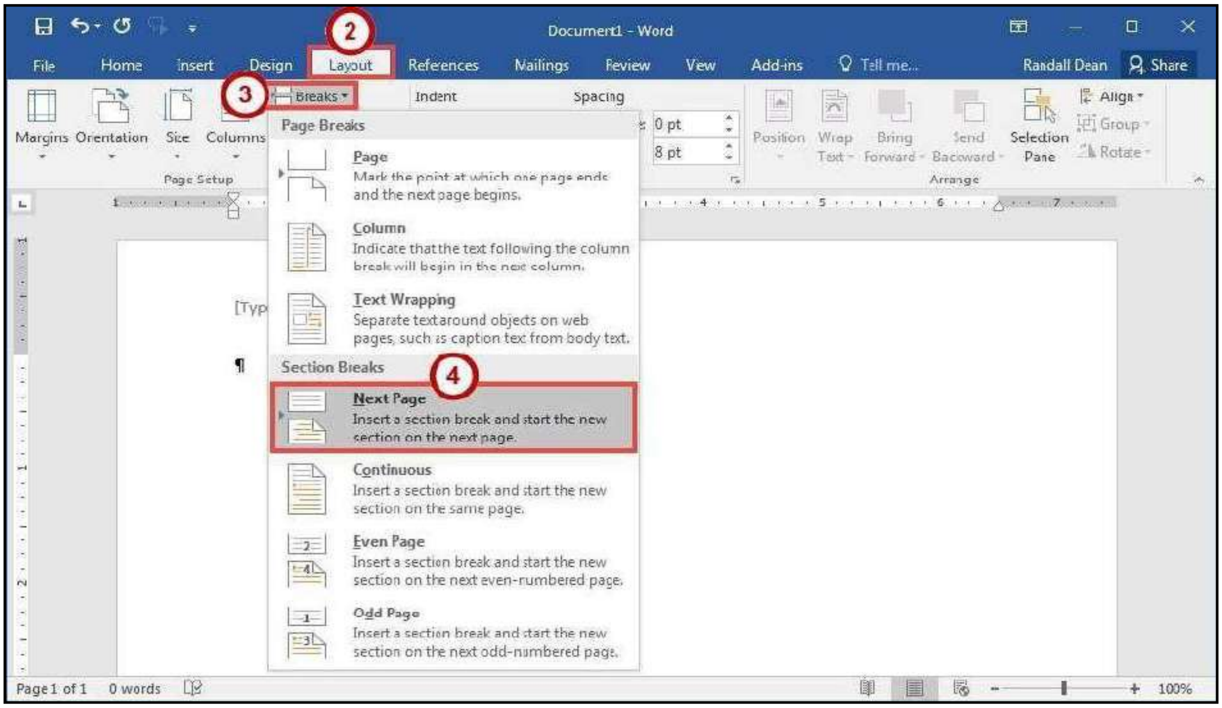


Figure 40 - Next Page Section Break

Note: Breaks inserted into your document will be hidden from view. In the *Home* tab, click the **Show/Hide** button under the *Paragraph* group.

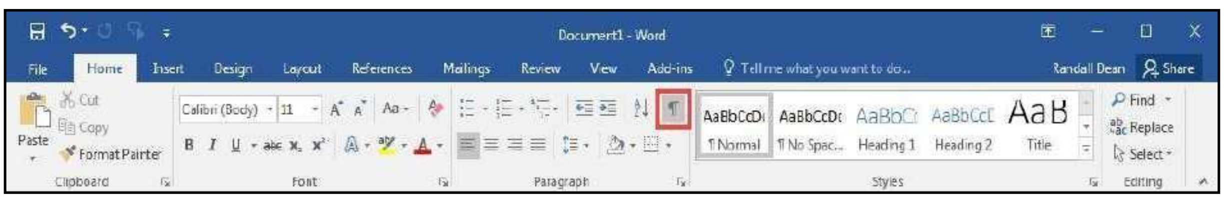


Figure 41 - Show/Hide

Using Section Breaks to Alter Page Numbering

The following example describes how to use a section break to display different page numberings in separate sections of a document. For example, you are creating a research paper with a title page and table of contents. You would like to have page numbers on the pages of the paper itself, but not on the title page or table of contents.

1. Click at the **end** of the table of contents page.
2. Click the **Layout** tab (See Figure 42).
3. In the *Page Layout* tab, click **Breaks** (See Figure 42).
4. In the *Breaks* drop-down menu, select **Continuous** from the list under *Section Breaks* (See Figure 42).

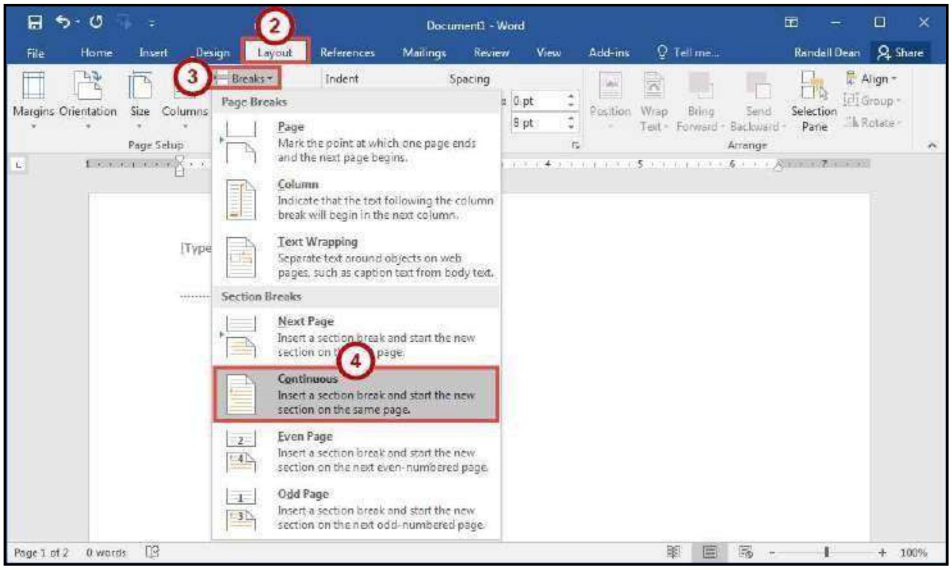


Figure 42 - Continuous Section Break

5. Click the **Insert** tab (See Figure 43).
6. In the *Insert* tab, click **Page Number** (See Figure 43).
7. In the *Page Number* drop-down menu, click **Bottom of Page** (See Figure 43).
8. In the *Bottom of Page* drop-down menu, select a **Page Number** from the list (See Figure 43).

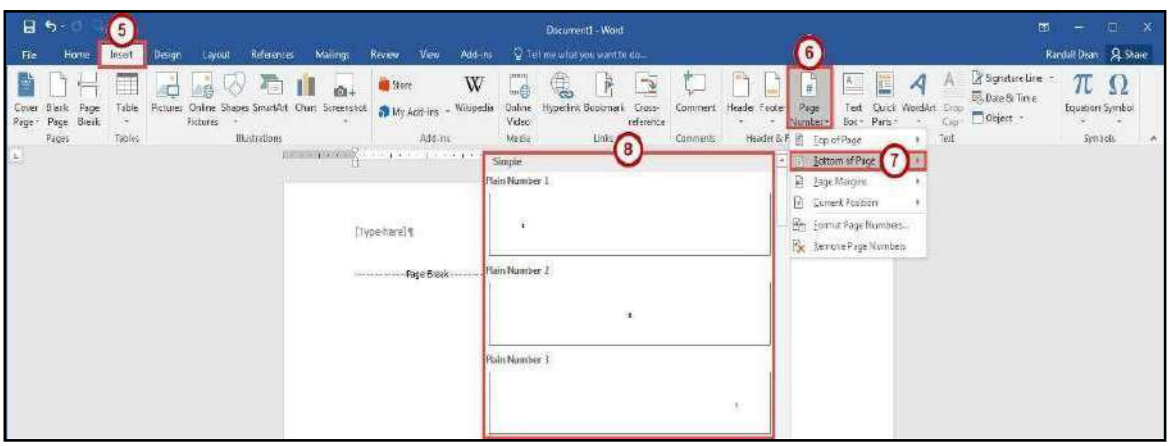


Figure 43 - Insert Page Number

9. You should now see a *Section 1* header and footer on the *table of contents* page, and a *Section 2* header and footer on the following page. The *Section 2* header and footer should also be labelled *same as Previous* (See Figure 44).
10. Click in the **Footer -Section 2-** to select it (See Figure 44).
11. In the *Header & Footer Tools - Design* tab, click the **Link to Previous** button (this will break the link between the two sections) (See Figure 44).

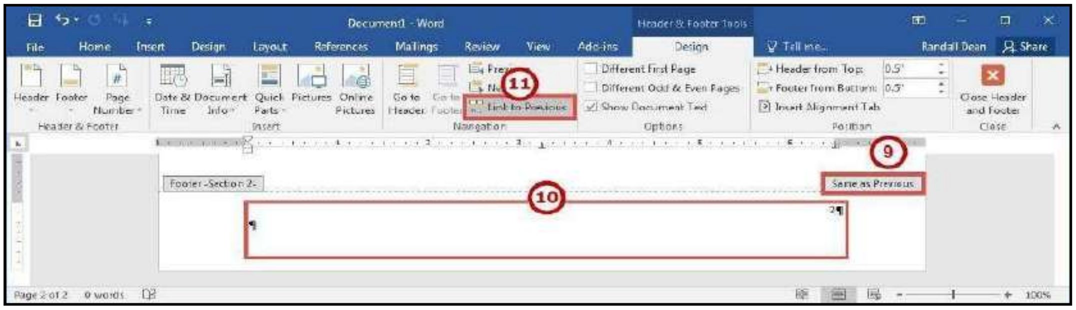


Figure 44 - Link to Previous

12. Select the page number in the *Footer -Section 1-* and **delete** it.

13. Click in the **Footer -Section 2-** to select it.

14. In the *Header & Footer Tools - Design* tab, click **Page Number** (See Figure 45).

15. Click **Format Page Numbers** (See Figure 45).

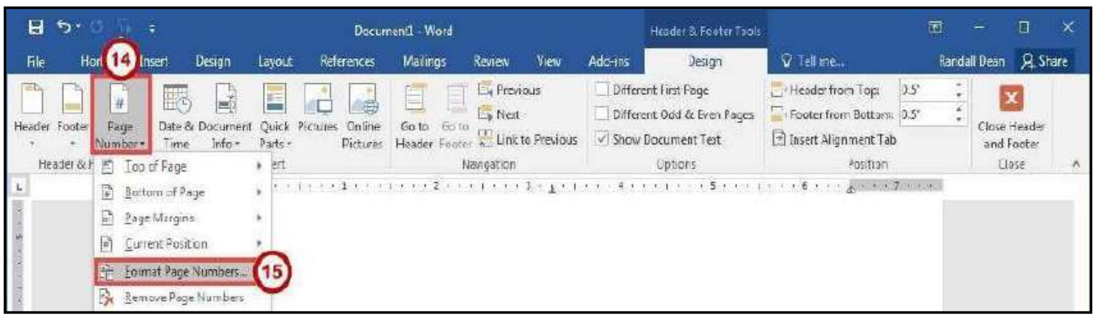


Figure 45 - Format Page Numbers

16. In the *Page Number Format* dialog box, in the *Start at* box **enter 1** (See Figure 46).

17. Click the **OK** button (See Figure 46).

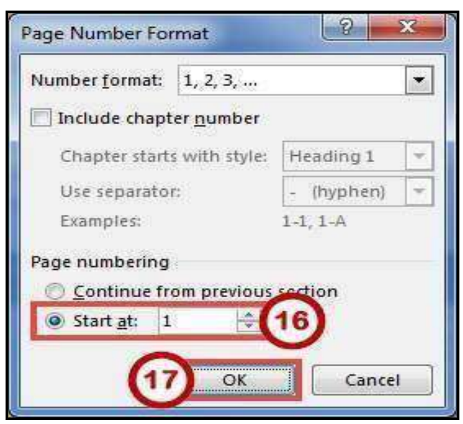


Figure 46 - Page Number Format Dialog Box

Removing a Section Break

1. Click the **Home** tab (See Figure 47).

2. In the *Home* tab, click the **Show/Hide** button (See Figure 47).

Figure 47 - Show/Hide

3. The section break will be revealed. Select the **section break** in your document.

4. Press the **Delete** key on your keyboard.

Columns

You can use Word's "columns" feature to create a newsletter-style layout for a document. You can create the column structure before you start typing, and then enter the text; however, you may find it easier to type the text in paragraph format, and then apply the paragraph structure. Either way, the text always flows from one column to the next.

Creating Columns

The following explains how to create columns within your document:

Note: If you don't want to apply columns to specific areas of your document, insert section breaks to separate that text from area of the document that will include columns. Insert section breaks before and after the text that you want to format as columns (See Section Breaks for moreinformation).

1. Click **inside the section** that you want to format into columns.

2. Click the **Layout** tab (See Figure 48).

3. In the *Layout* tab, click **Columns** (See Figure 48).

4. In the *Columns* drop-down menu, select the **number of columns** to create (See Figure 48).

Figure 48 - Choose Number of Columns

Accessing Additional Column Options

Additional column options are available to further customize your columns (e.g. separate columns by a line, varying lengths for columns, multiple columns). The following shows how to access the column options.

1. Click the **Layout** tab (See Figure 49).
2. In the *Layout* tab, click **Columns** (See Figure 49).
3. In the *Columns* drop-down menu, click **More Columns** (See Figure 49).

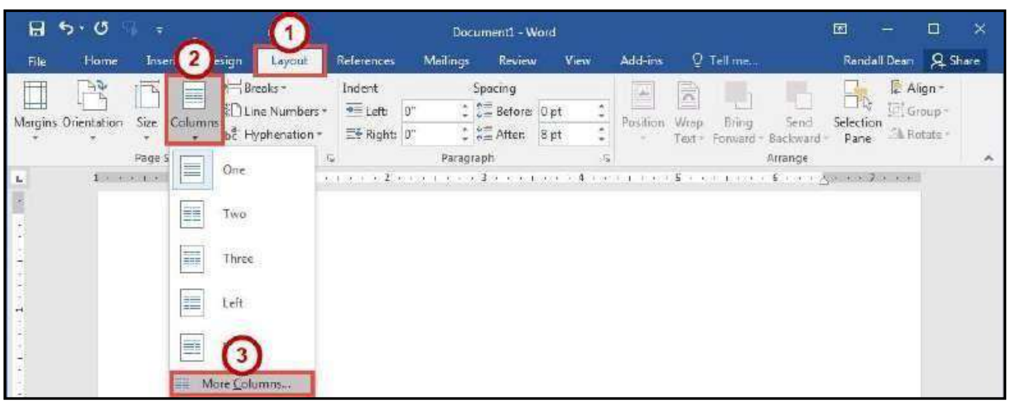


Figure 49 - More Columns

4. In the *Columns* dialog box, make your changes as necessary (See Figure 50).
5. Click the **OK** button (See Figure 50).

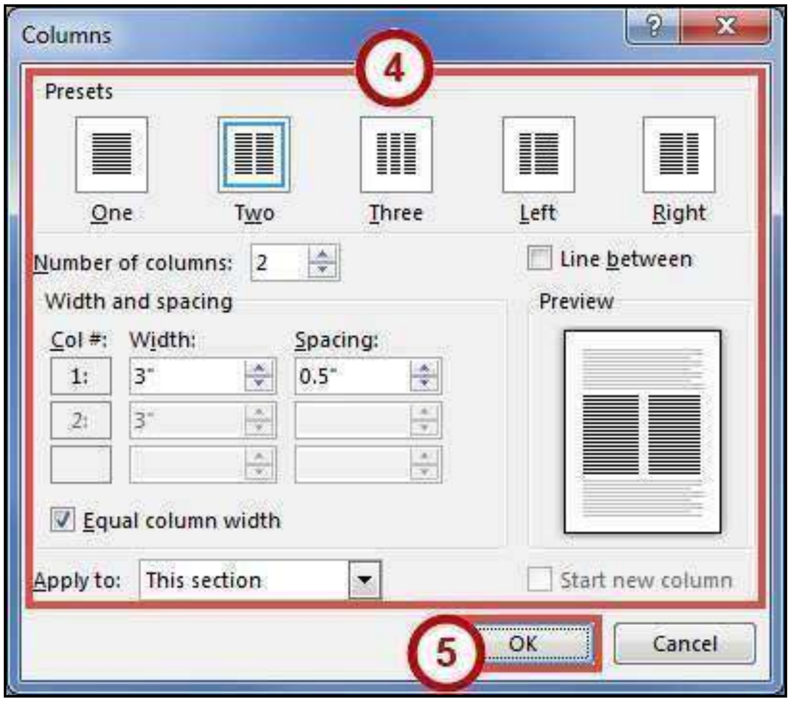


Figure 50 - Columns Dialog Box

Microsoft Office

Word 2016 for Windows

WORKING WITH GRAPHICS

Learning Objectives

After completing the instructions in this booklet, you will be able to:

- Insert graphics in the form of Pictures, Clipart, Shapes, Video, and Screenshots.
- Modify and format graphics.
- Transform and format shapes.
- Insert video into your document.
- Discover how Text Wrapping works.
- Insert text boxes and link them together.
- Insert text into shapes.

Working with Graphics

The use of graphics will enhance your documents and allow you to provide the reader with additional information in the form of a visual aid. The following section explains the various graphics features in Word 2016.

Inserting Pictures

The following explains how to insert an existing picture saved to your computer into your Word document:

1. Click the **Insert** tab (See Figure 1).
2. In the *Illustrations* group, click the **Pictures** button (See Figure 1).

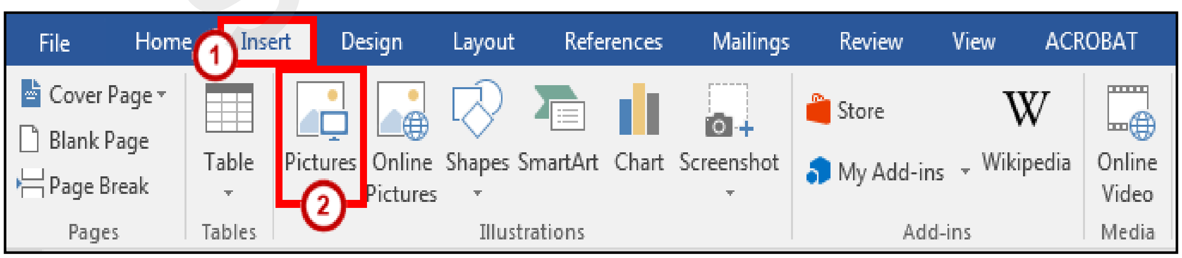


Figure 1 - *Illustrations:* Pictures

3. The *Insert Picture* dialog box will appear. Select the **picture** you wish to insert from your computer.

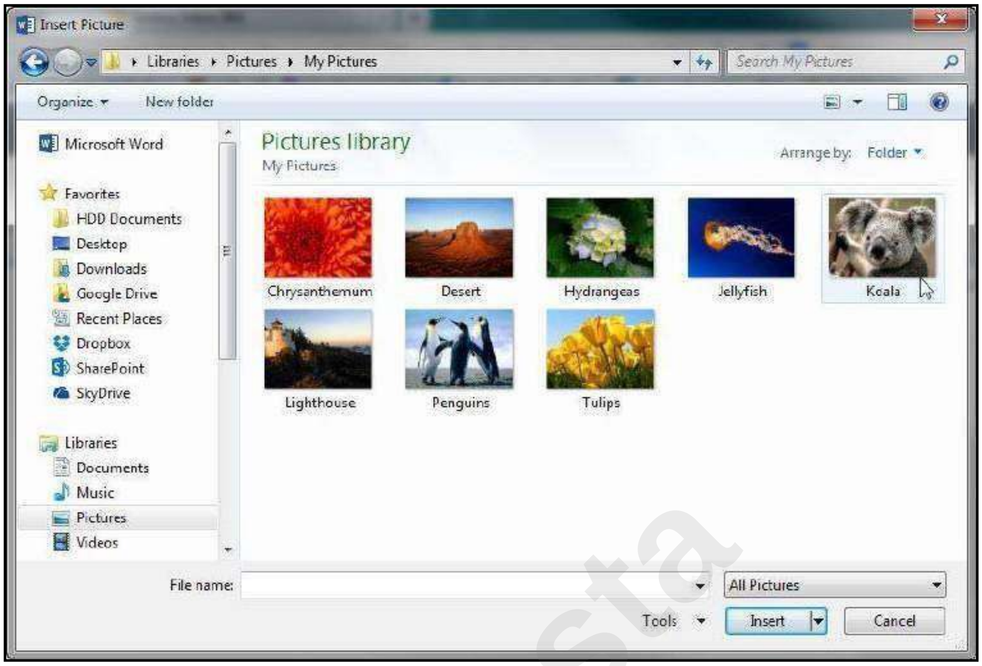


Figure 2 - Insert Picture Dialog Box

4. Click the **Insert** button. 

Inserting Online Pictures

The *Online Pictures* tool provides access to online picture resources such as *Microsoft Clipart* and *Bing Image Search*. The following explains how to insert online pictures into your Word document from a variety of internet sources.

Note: The Online Pictures icon has replaced the *Clip Art* icon seen in previous versions of Microsoft Word.

1. Click the **Insert** tab (See Figure 3).
2. In the *Illustrations* group, click the **Online Pictures** button (See Figure 3).

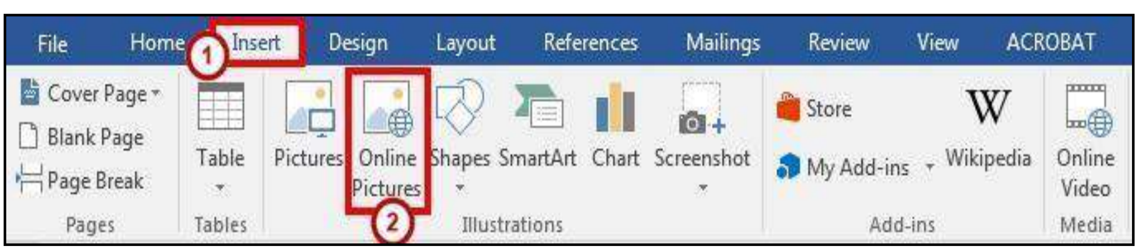


Figure 3 - *Illustrations*: Online Pictures

3. The *Insert Pictures* window will open.
4. In the search box next to *Bing Image Search*, type a **word or phrase** that describes the desired image (See Figure 4).
5. Press **Enter**.

Figure 4 - Insert Online Pictures

6. A list of your search results will appear (See Figure 5).
7. Click the desired picture you wish to add and click the **insert** button (See Figure 5).

Figure 5 - Clip Art Search Results

Note: The procedure is the same for inserting images using *Bing Image Search*.

Modifying Graphics with Picture Tools

After inserting a picture, you can make changes to the size, brightness, shading, etc. by accessing the

Picture Tools. The following explains how to access the *Picture Tools*:

1. Click the **graphic** to select it. The *Picture Tools - Format* contextual tab appears.

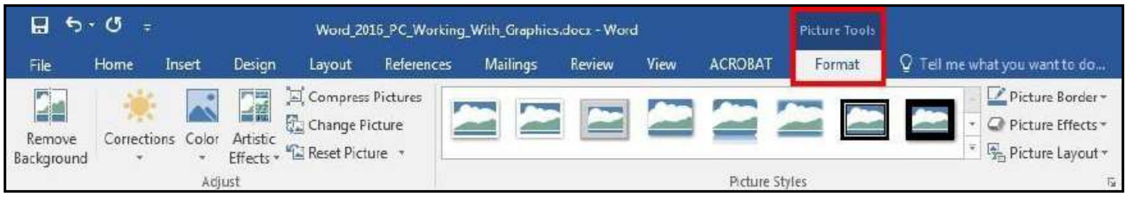


Figure 6 - Picture Tools

2. Click the **Picture Tools - Format** contextual tab. The *Picture Tools - Format groups* will appear in the ribbon.

Figure 7 - Picture Tools Groups

Adding Styles Your Graphic

Graphics have preset styles under the *Picture Tools* – *Format* tab that you can use to alter the color, border, and any special effects to the shape. The following explains how to apply a style to a shape:

1. In the *Picture Styles* group, click on a **style** of your choice.

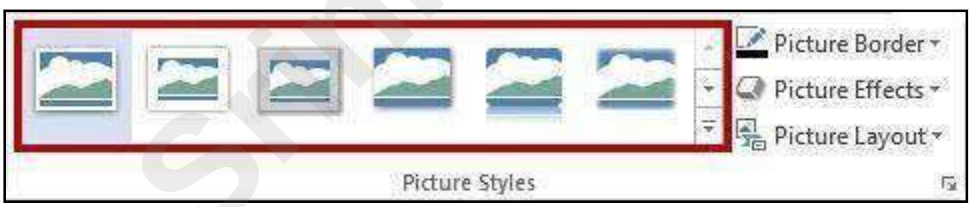


Figure 8 - Picture Styles

2. Click the **drop-down arrow** to access additional styles.

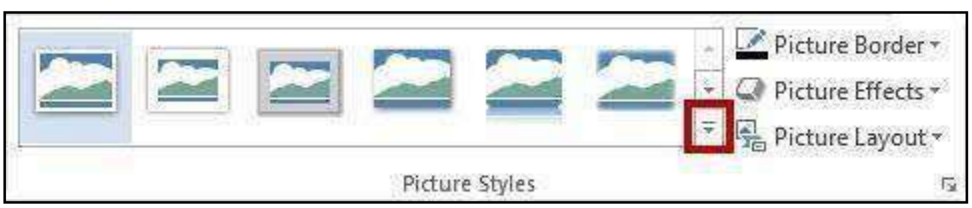


Figure 9 - Accessing Additional Picture Styles

3. To access additional formatting options, click the **dialog launcher** in the lower-right corner of the *Picture Styles* grouping.

Figure 10 - Picture Styles Dialog Box

4. The *Format Picture* options will appear to the right of your document (See Figure 11).

Re-Sizing a Graphic

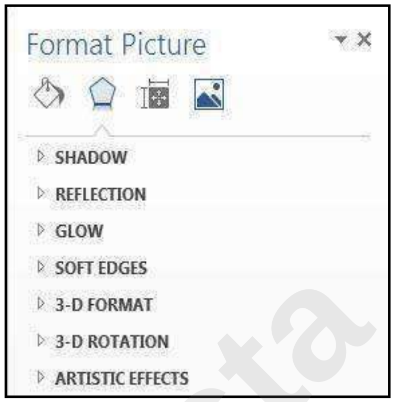


Figure 11 - Format Picture Options

The following explains how to re-size graphics:

1. Click the **graphic** to select it. The sizing handles will appear around the border of the picture.

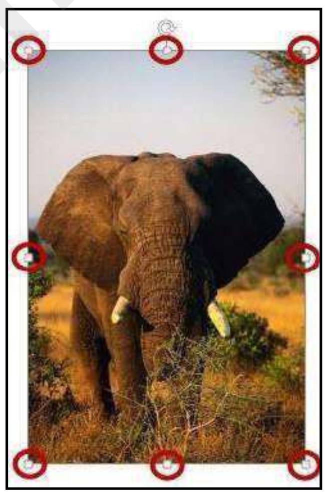


Figure 12 - Sizing Handles

2. To resize the picture so the dimensions remain proportional, place your mouse over one of the corner sizing handles, click, and drag. If you drag a side sizing handle you will change one dimension only.

3. The curved arrow at the top of the picture allows you to rotate the picture.

Figure 13 - Rotational Arrow

4. To resize the picture to a specific value, enter number for the **height** and **width** in the *Size* grouping under *Picture Tools – Format* (See Figure 14).

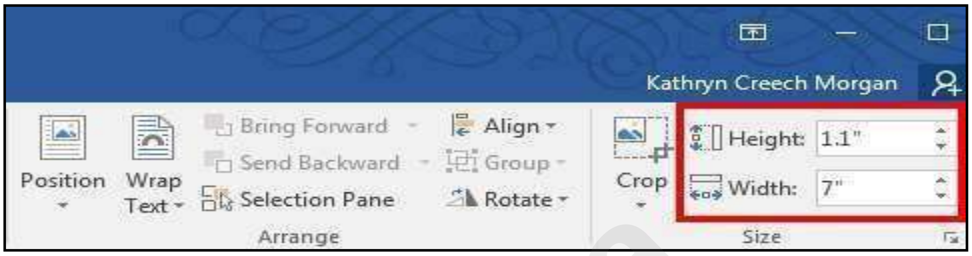


Figure 14 - Size Grouping

5. You can also enter a specific size value by clicking the **Size dialog box**.

Figure 15 - Size Dialog Box Launcher

6. The *Layout* window will appear. Set the dimensions by entering the height and width or set the scale by entering the height and width in the Scale section.

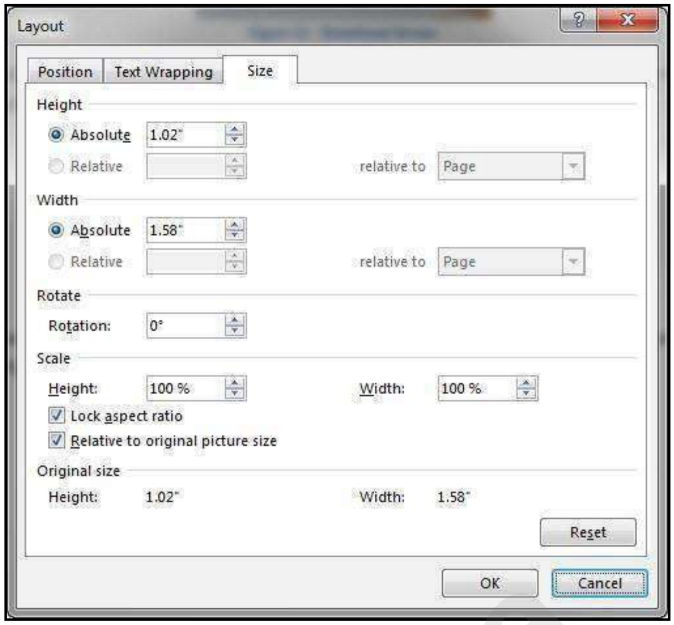


Figure 16 - Size Dialog Box

Note: If the *Lock aspect ratio box* is checked, changing either the height or width in either section will also change the other dimension by the same amount.

Repositioning a Picture

Pictures can be repositioned on the page either by using the Object Position feature, or by dragging the picture to the new location. Before you can drag a picture, text wrapping must be set to something other than *In Line with Text*. See Text Wrapping for more information.

The following explains how to reposition a picture:

1. Click on the **picture** to be repositioned.
2. Under the *Picture Tools – Format* tab, click the **Position** icon.

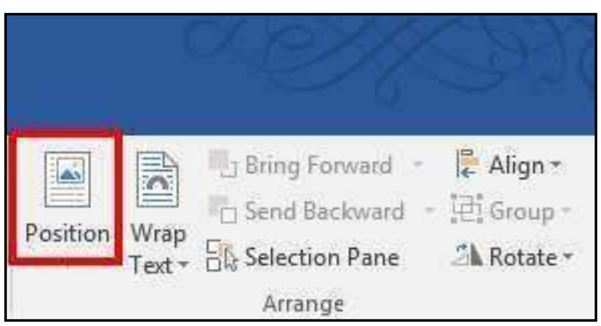


Figure 17 - Position Tool

3. A drop-down will appear. Select the **position layout** of your choice.

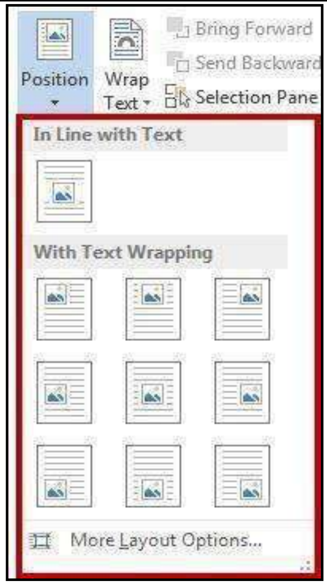


Figure 18 - Position Options

Note: The position selected also determines the text wrapping for the picture.

4. To manually move your picture within the document, left-click on your **picture** and drag it to its new location.

Text Wrapping

Sometimes you may need to have text wrap around a picture. The following explains how to apply text wrapping to your document:

1. Click on the **picture** to apply text wrapping.
2. Under the *Picture Tools – Format* tab, click the **Wrap Text** icon.

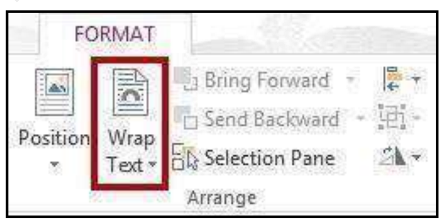


Figure 19 - Wrap Text

3. A drop-down will appear. Select the type of **text wrapping** that you want to apply to your picture.

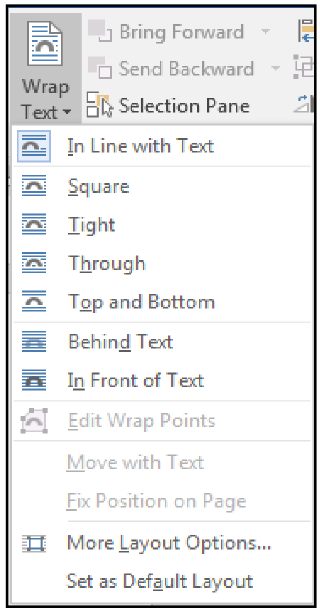


Figure 20 - Text Wrapping Options

Note: You can also access the text wrapping options by clicking the **layout button** that appears when clicking on a picture.

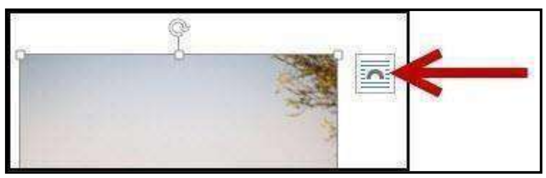


Figure 21 - Layout Button

Inserting Shapes

The following explains how to insert a variety of predefined shapes into your Word document:

1. Click the **Insert** tab (See Figure 22).
2. In the *Illustrations* group, click on the **Shapes** button (See Figure 22).

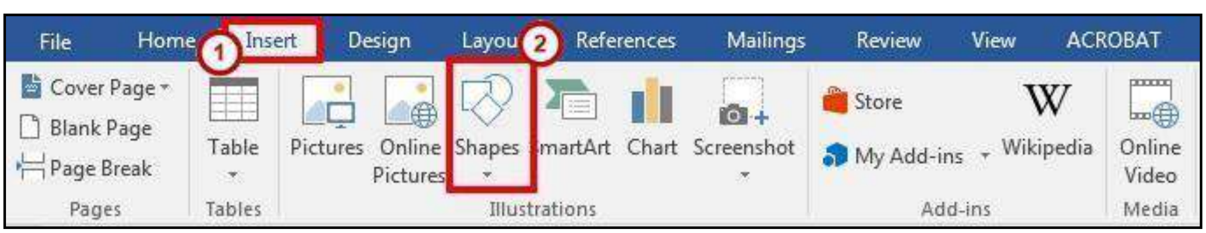


Figure 22 - *Illustrations*: Shapes

3. A drop-down window will appear with a library of shapes to choose. Click on a **shape** to select it.

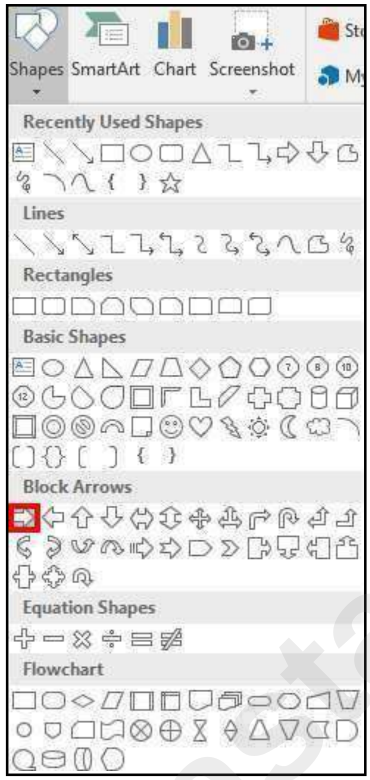


Figure 23 - Select a Shape

4. Your cursor will change to a crosshair and you will be ready to insert your selected shape.

5. To draw your selected shape within your document, hold the left mouse button and **drag the cursor** to draw your shape.

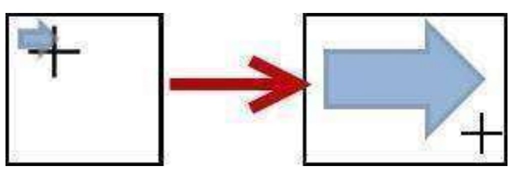


Figure 24 - Click and Drag to Draw Shape

6. Release the left mouse button when you are finished drawing your shape. The shape will be added to your document.

Modifying Shapes with Drawing Tools

You can modify shapes in a number of ways. The following describes some of the ways that shapes can be changed:

Re-sizing a Shape

Re-sizing a shape is similar to re-sizing a graphic (picture or clip art). See the section in this booklet on Re-Sizing a Graphic for more information.

Re-Shaping

After clicking on a shape, the *shaping handles* will appear (along with the sizing handles) as yellow squares. A two-dimensional shape can be altered by clicking and dragging the yellow squares to alter a certain aspect of the shape.

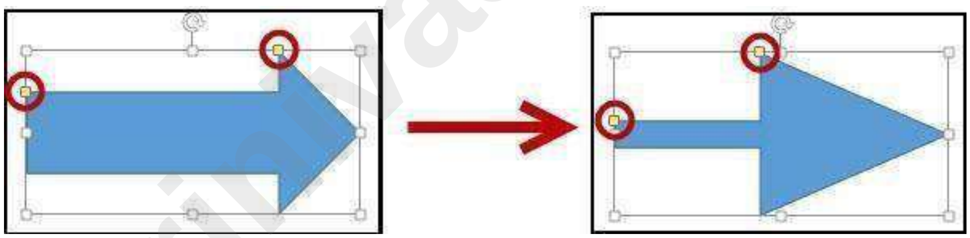


Figure 25 - Re-shaping a 2D Shape

Accessing the Drawing Tools – Format Tab

Shapes will have their own set of editing tools accessible by a context sensitive tab on the *Ribbon*. To access this tab, click on a **shape** in your document and the *Drawing Tools – Format* tab will appear in the *Ribbon*.

Figure 26 - Drawing Tools - Format Tab

Adding Styles to your Shapes

Shapes have preset styles under the *Drawing Tools – Format* tab that you can use to alter the color, border, and any special effects to the shape. The following explains how to apply a style to a shape:

1. In the *Shape Styles* group, click on a **Style** of your choice.

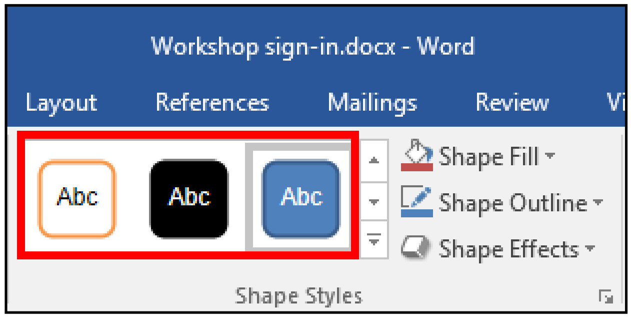


Figure 27 - Shape Styles

2. Click the **drop-down arrow** to access additional styles.

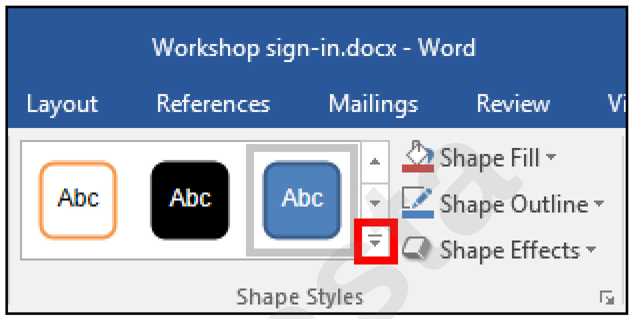


Figure 28 - Accessing Additional Shape Styles

3. To access additional formatting options, click the **dialog launcher** in the lower-right corner of the *Shape Styles* grouping.

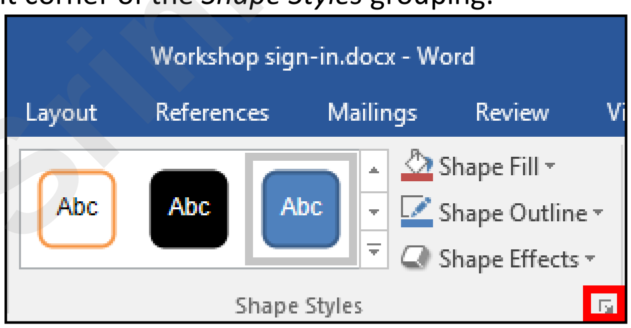


Figure 29 - Shape Styles Dialog Box

4. The *Format Shape* options will appear to the right of your document.

Figure 30 - Format Shape Options

Applying Color to Your Shape

The following explains how to add/change the color of a shape:

1. In the *Shape Styles* group, click the **drop-down arrow** for Shape Fill.

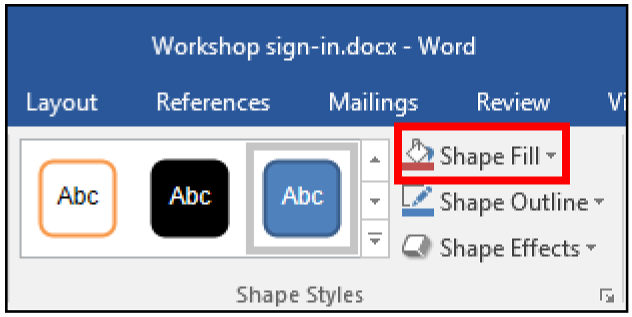


Figure 31 - Shape Fill

2. Select a **color** to apply it to your shape.

Changing the Shape Outline

The following explains how to change the outline for a shape:

1. In the *Shape Styles* group, click the **drop-down arrow** for Shape Outline.

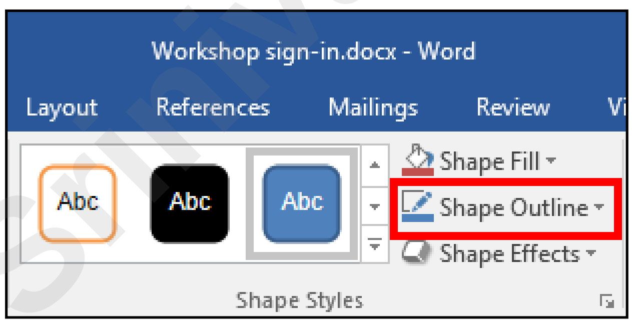


Figure 32 - Shape Outline

2. A drop-down menu will appear. From here you can alter the color, thickness (weight) of the outline, and add dashes.

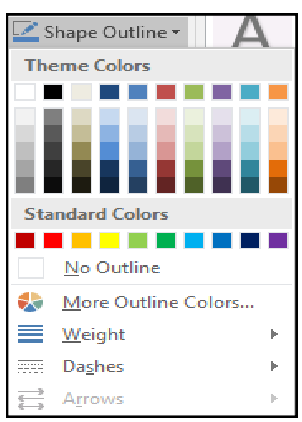


Figure 33 - Shape Outline Drop-down

Switching Shapes

The following explains how to switch shapes in the document:

1. In the *Insert Shapes* group, click the **Edit Shape** icon. A drop-down menu will appear.

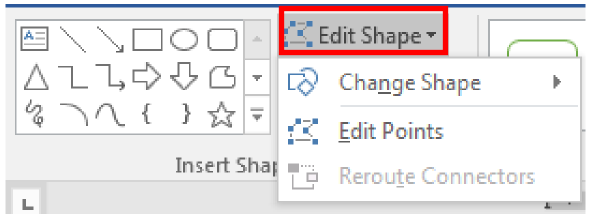


Figure 34 - Edit Shape Icon

2. Click **Change Shape**. A list of shapes will appear.
3. Select a shape from the options available to switch it with your selected shape.

Add Text to a Shape

To add text to your shape, simply click on your shape to select it and begin typing. Your text will automatically fill into the shape. Adjust your shape as needed to make additional space for your text. Text can be formatted like regular text in the document.

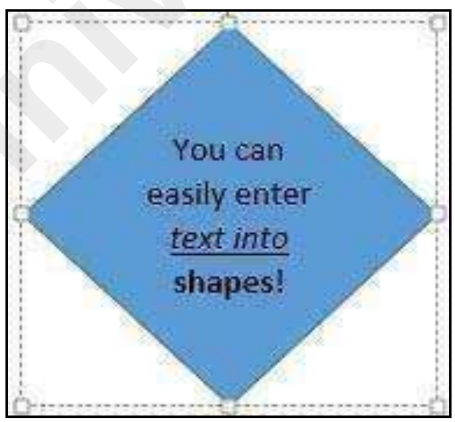


Figure 35 - Entering Text into Shapes

Inserting SmartArt

SmartArt graphics provide a visual representation of information or ideas. The following explains how to insert SmartArt graphics into your Word document.

1. Click the **Insert** tab (See Figure 36).
2. In the *Illustrations* group, click on the **SmartArt** button (See Figure 36).

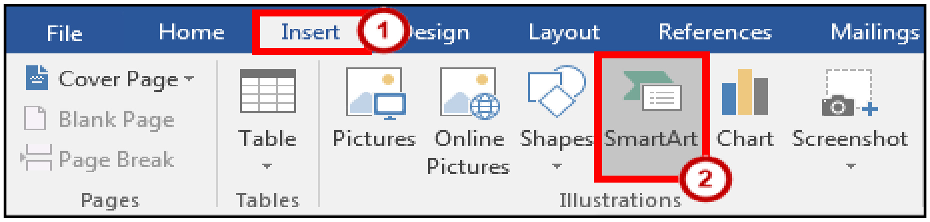


Figure 36 - *Illustrations*: SmartArt

3. The *Choose a SmartArt Graphic* window will open. Select a **SmartArt** graphic from the list (See Figure 37).

4. Click the **OK** button (See Figure 37).

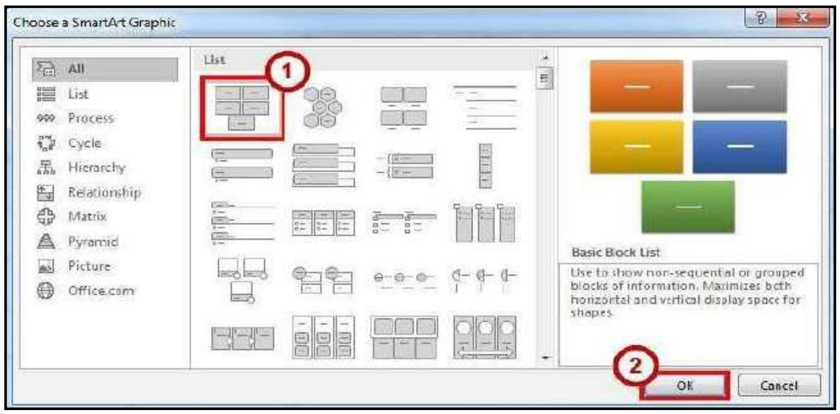


Figure 37 - Choose a SmartArt Graphic

5. The selected **SmartArt** graphic will be inserted into your Word document.

Editing SmartArt Graphics

Once inserted into your Word document, your **SmartArt** can be edited and customized by accessing the *SmartArt Tools* tab. The following explains how to access the *Design* and *Format* tabs:

1. Click on the **image** of your *SmartArt* graphic. The *SmartArt Tools* contextual tab will appear at the top of the screen.

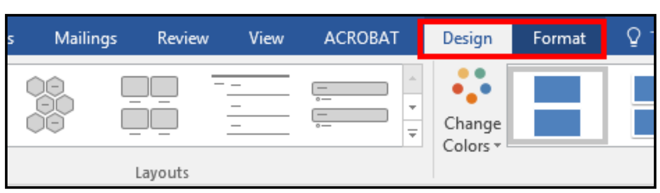


Figure 38 - SmartArt Tools

To alter aspects of the SmartArt design, click the **Design** tab.

Figure 39 - SmartArt Tools: Design

Note: *The SmartArt Tools: Design* tab contains tools that will allow you to alter the layout of the selected design, change colors, and select from preset styles.

a. To format aspects of the SmartArt graphic, click the **Format** tab.

Figure 40 - SmartArt Tools: Format

Note: *The SmartArt Tools*: *Format* tab contains tools that will allow you to change the shape of your SmartArt, add preset styles to the shape and words, and choose individual aspects of the SmartArt graphic to modify (e.g. changing the shape fill or outline).

Inserting Charts

Adding charts to your word document can help readers visualize a relationship among sets of data. The following explains how to add Charts to your word document.

1. Click the **Insert** tab (See Figure 41).
2. In the *Illustrations* group, click on the **Chart** button (See Figure 41).

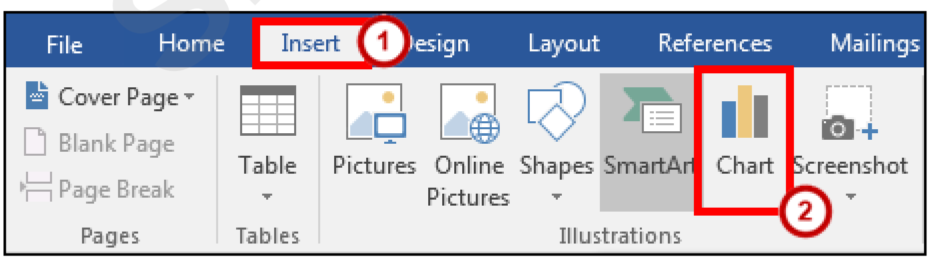


Figure 41 - *Illustrations*: Chart

3. The *Insert Chart* window will appear. Click on the **chart** you wish to use (See Figure 42).

4. Click the **OK** button (See Figure 42).

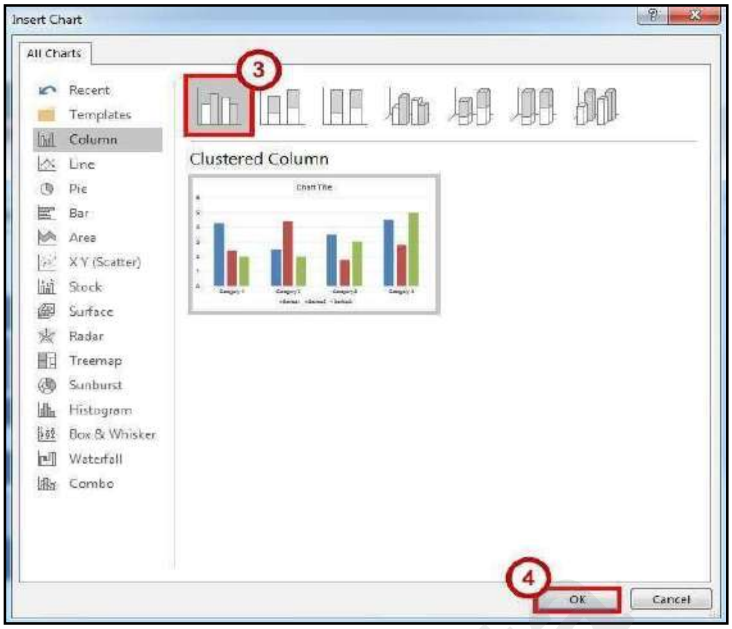


Figure 42 - Insert Chart

Note: Hover your mouse over the picture of a graph for an enlarged preview of the chart.

5. The selected chart will be inserted into your document.

Editing your Chart

Once inserted into your Word document, your Chart can be edited and customized by accessing the *Chart Tools* tab. The following explains how to access the *Design* and *Format* tabs:

1. Click on your **chart**. The *Chart Tools* contextual tab will appear at the top of the screen.

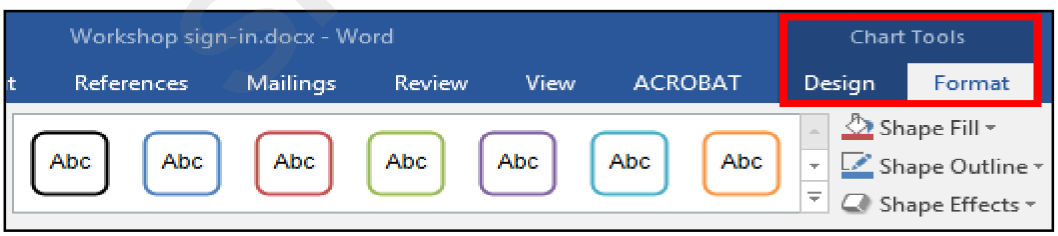


Figure 43 - Chart Tools

a. To alter aspects of the Chart's design, click the **Design** tab.

Figure 44 - Chart Tools: Design

Note: *The Chart Tools – Design* tab contains tools that will allow you to alter the layout of the selected chart, change colors, select and edit your data, and select from preset styles.

b. To format aspects of the SmartArt graphic, click the **Format** tab.

Figure 45 - Chart Tools - Format

Note: *The Chart Tools – Format* tab contains tools that will allow you to add preset styles to the shape and words, arrange and position your chart within your document, and format specific areas of your chart (e.g. axis, chart title, values, legend, etc.).

Inserting Screenshots

The *screenshot* feature in word will allow you to capture a designated window or area of your screen, and insert it as a picture in your document. The following explains how to add screenshots to your word document:

To take a screenshot of an active window:

1. Open the program or website you wish to take a screenshot of (make sure the window is not minimized).
2. Click the **Insert** tab (See Figure 46).
3. In the *Illustrations* group, click on the **Screenshot** button (See Figure 46).

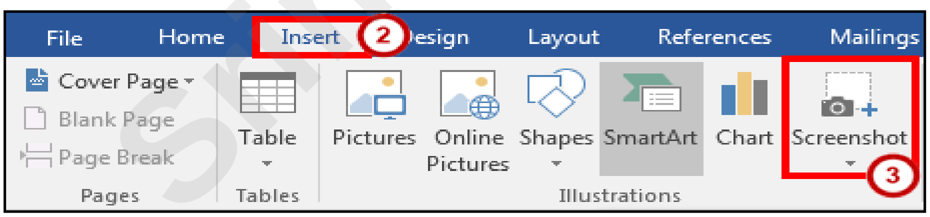


Figure 46 - *Illustrations*: Screenshot

4. A drop-down window will appear. Under *Available Windows*, you will see a thumbnail preview of all open windows.

Figure 47 - Screenshot: Available Windows

5. Click the thumbnail preview to insert the screenshot into your document.

To take a screenshot of a section of your screen:

1. Open the program or website you wish to take a screenshot of (make sure the window is not minimized).
2. Click the **Insert** tab (See Figure 48).
3. In the *Illustrations* group, click on the **Screenshot** button (See Figure 48).

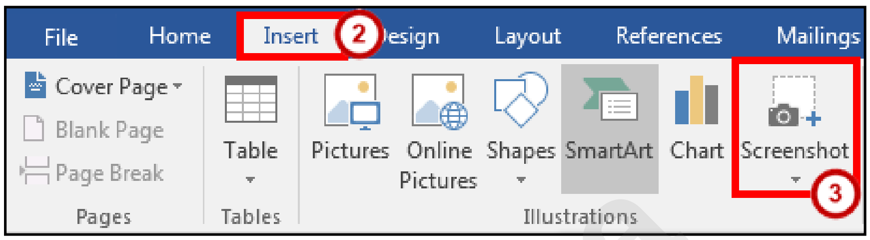


Figure 48 - *Illustrations*: Screenshot

4. A drop-down window will appear. Click on **Screen Clipping**.

Figure 49 - Screenshot: Screen Clipping

5. Your screen will turn grey and a black crosshair will appear. Hold down the **left mouse button** and drag the crosshair over the area of your screen you wish to capture.
6. Release the **left mouse button** when you are finished capturing. Your screenshot will automatically be inserted into your document.

Inserting Video into your Document

New to Word 2016 is the ability to insert online videos into your document. When inserted into your document, the video will show a still image of the video. When clicked, the video will open in a separate window and will play. When printing out your document, the video will look like a picture.

Note: To use this feature, you will need an internet connection. Videos will not play if the computer accessing the document is not connected to the internet.

To add video to your document:

1. Click the **Insert** tab (See Figure 50).
2. In the *Media* group, click on the **Online Video** button (See Figure 50).

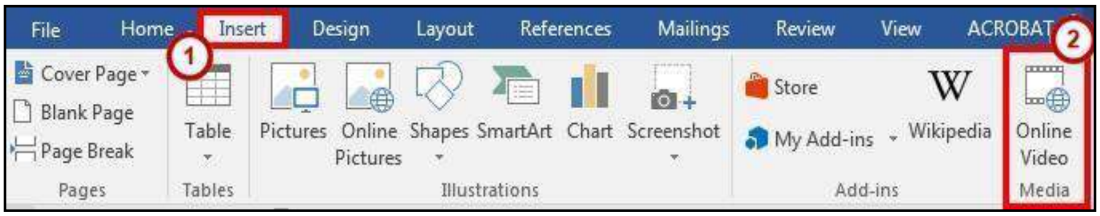


Figure 50 - Online Video

3. The *Insert Video* window will appear. A list of search options are presented.

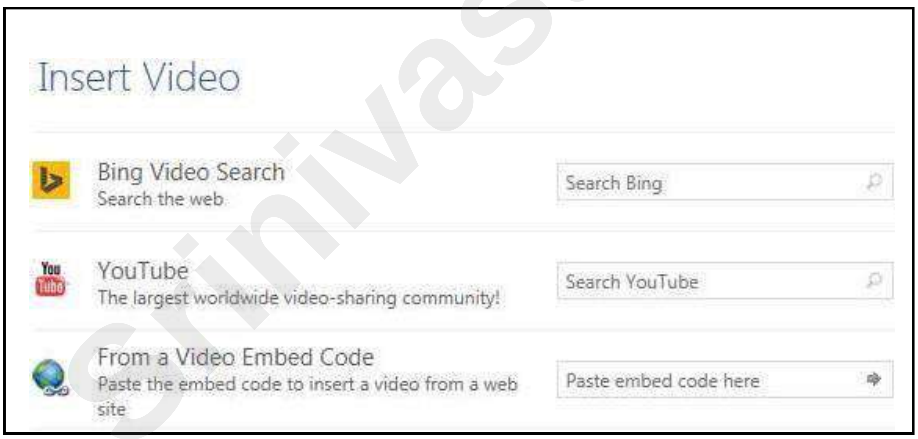


Figure 51 - Insert Video

4. In one of the search boxes, type a **word or phrase** that describes the desired video you wish to search for (Figure 52 shows a YouTube search for Elephants).

Figure 52 - Insert Video: YouTube Search

5. Press **Enter**.
6. A list of your search results will appear.

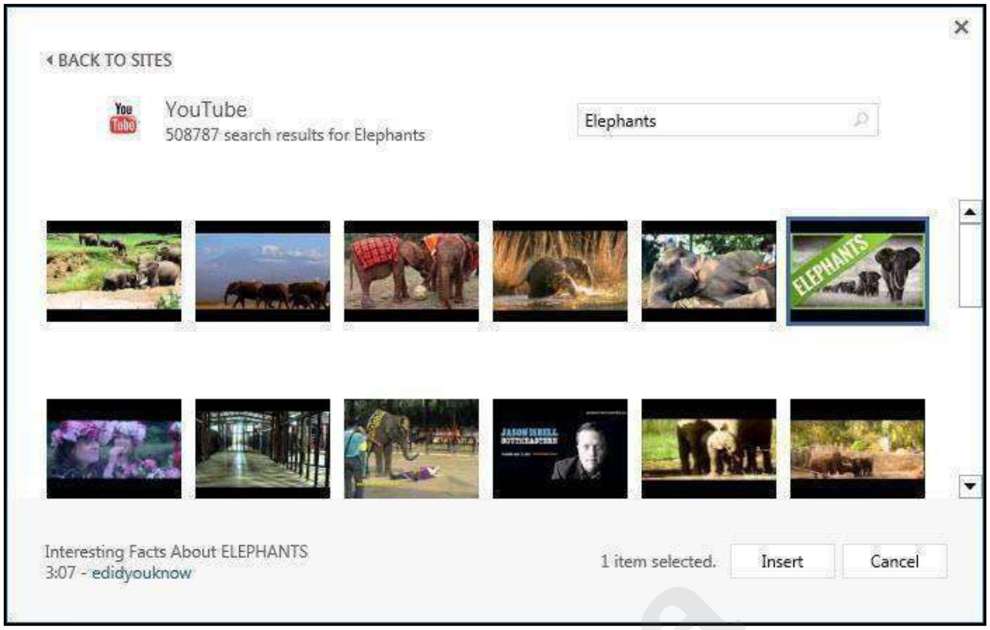


Figure 53 - Video Search Results

7. To preview the video, mouse over the image and click the **magnifying glass**.

Figure 54 - Preview Video

Click the desired video you wish to add and click the **Insert** button.

The video will be added to your document. It will show a still image from the video that will display in the document as a picture (this is the image that will also appear if you print the document). To play the video, click the **play** button on top of the still image (See Figure 55).

Figure 55 - Video in Document

Inserting Text Boxes

A text box brings focus to the content inside and is helpful for showcasing important text (e.g. headings or quotes). You can use text boxes to place text at specific locations in a document, and format the text box with a border, shading, etc.

Inserting a Text Box

The following instructions explain how to insert a text box into your document:

1. Click the **Insert** tab (See Figure 56).
2. In the *Text* group, click the **Text Box** icon (See Figure 56).

Figure 56 - Text Box Icon

3. A list of pre-defined text boxes will appear. Click on a **text box**.

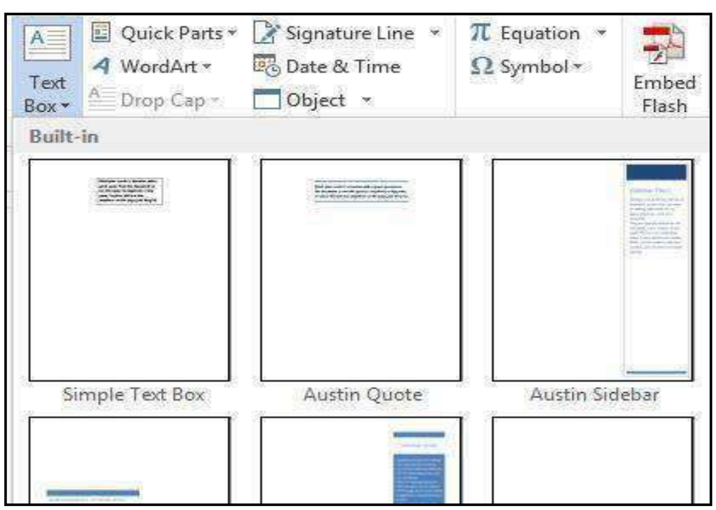


Figure 57 - Text Box Gallery

4. The *Text Box* will be inserted into your document. Click on the **Text Box** and begin typing to add your text.
5. To move the text box, select the **text box**, then drag the border of the text box the new location.

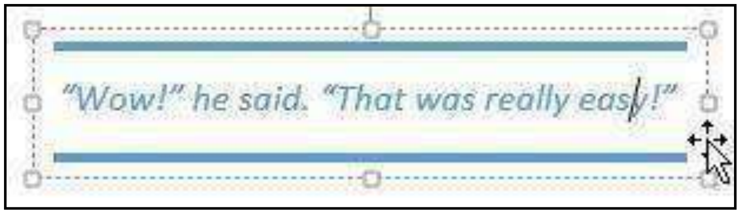


Figure 58 - Move a Text Box

6. You can re-size your text box the same way you re-size pictures. Refer to the section in this booklet on Re-Sizing a Graphic for more information.

Drawing a Text Box

In addition to selecting a preset text box, you can also draw your own text box and insert it into your document. The following explains how to draw a text box:

1. From the *Insert* tab, click the **Text Box** icon.

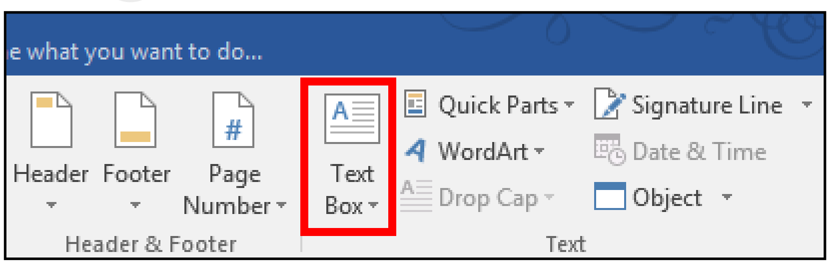


Figure 59 - Text Box Icon

2. A list of pre-defined text boxes will appear. At the bottom of the window, click on **Draw Text Box**.

Figure 60 - Draw Text Box

3. Your cursor will change to a crosshair + and you will be ready to draw your text box.
4. To draw your text box within your document, hold the left mouse button and **drag the cursor** to draw the text box.

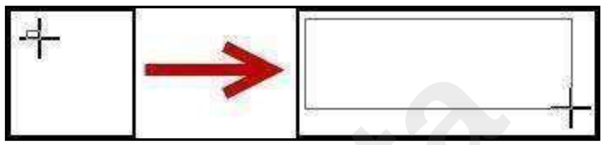


Figure 61 - Click and Drag to Draw Text Box

5. Release the left mouse button when you are finished drawing your text box. The text box will be added to your document and you can begin entering text.

Flowing Text between Text Boxes

It is possible to connect two empty text boxes so your text can flow from one box into the other. The following explains how to allow text to flow between text boxes:

1. Delete any text in the second text box (Text will only flow into an empty text box).
2. Select the **Text Box** to display the *Drawing Tools – Format* tab in the *Ribbon*.
3. In the Text group, click the **Create Link** button.

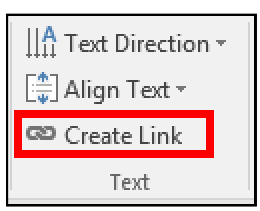


Figure 62 - Create Link

4. The mouse pointer will change to a pitcher. 

5. Move the mouse pointer over the second text box (the pitcher will change to a pouring pitcher) and left click the **inside of the second text box**.

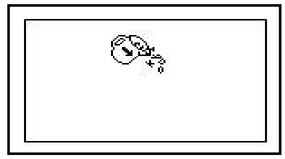


Figure 63 - Link to the Second Text Box

1. When the first text box is full, the text will begin to flow into the second text box.

Figure 64 - Linked Text Boxes

Microsoft Office

Word 2016 for Windows

REVIEWING YOUR DOCUMENT

Learning Objectives

After completing the instructions in this booklet, you will be able to:

- Turn track changes on and off.
- Understand the different review display settings and how to apply them.
- Accept/reject changes to the document.
- Add/delete/reply to comments in the document.
- Lock tracking changes for your document.
- Combine changed documents.
- Compare changed documents.

Collaborating on Documents

Word contains features that make it easy for several people to work on a document together. Rather than passing a hard copy of the document containing manual changes back and forth, you can have Word automatically track the changes, and then you can pass the document electronically.

Track Changes

By using the *Track Changes* tool, you can easily see what changes have been made to the existing document. This feature is very useful if you are collaborating with others, or wish to make suggestions that can be later accepted or rejected. Furthermore, you can quickly switch between different views to see the extent of the changes, or if you wish to view the original document in its entirety.

If you wish to make changes to a document that you want to share with others, you must enable *Track Changes* first before making any changes to your document.

1. Click on the **Review** tab (See Figure 1).
2. In the *Review* tab, click on **Track Changes** (See Figure 1).

Figure 1 - Track Changes

3. Word will now begin to track changes to your document (e.g. inserting text, deleting text, etc.). The *Track Changes* button will be shaded to indicate track changes has been activated.
4. To turn off T*rack Changes,* click the **Track Changes** button.

Note: *Track Changes* will remain on unless it is deactivated; even if you save your document. Be sure to turn off Track Changes if you don't want others to track changes. If you want to prevent others from turning off Track Changes, see Locking Track Changes.

Making Changes to Your Document with Track Changes

Once *Track Changes* has been activated, Word will make note of all changes made to your document. To make changes to your document, simply edit the document as you normally would. The default settings for changes will appear as red lettering for

The largest elephants are the ~~African~~ Asian elephants. They grow to be 20 to 25 feet (6 to 7 1/2 meters) long, ~~11~~ 10 feet (about 3 1/2 meters) tall~~, and~~. These elephants weigh up to 13,200

insertions, and red lettering with a strikethrough for deletions.

Figure 2 - Changes to the Document

Leaving Comments

You can leave notes in your document for others to read that ask for clarification, explain a revision, etc. When your review settings are set to *Simple Markup*, all comments will be hidden and areas that have had a comment added will display a speech bubble.

New Comment

1. Click within your **document**, or select a **section of text** that you want to add the comment to.
2. Click on the **Review** tab (See Figure 3).

3. In the *Review* tab, click on **New Comment** (See Figure 3).

Figure 3 - New Comment

4. A *comment textbox* will be added to your document. Type your **message** within the *comment textbox* to leave your comment.

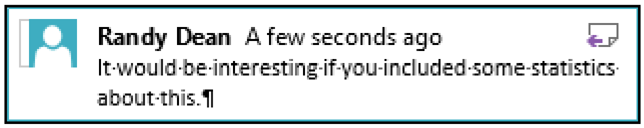


Figure 4 - Comment Textbox

5. Once finished, click **anywhere** inside your document to leave the *comment textbox*.

Edit a Comment

1. Search within your **document** for the comment to edit.
2. Click on the comment **speech bubble**.
3. Once you are finished editing, click **anywhere** inside your document to leave the *comment textbox*.

Reply to a Comment

The following shows how to reply to a comment while in *Simple Markup* view:

1. Click on the comment **speech bubble** to display your comment.
2. In the *comment window*, click on the **Reply** icon.

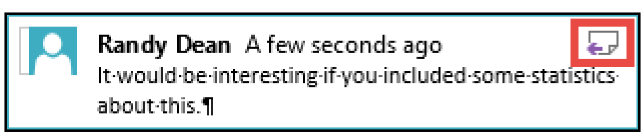


Figure 5 - Reply to a Comment

3. Your *username* will be added to the *comment window*. Type your **response** to the comment.

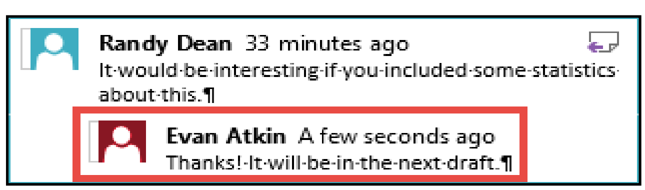


Figure 6 - Response to a Comment

Delete a Comment

The following shows how to delete a comment while in *Simple Markup* view:

1. Click on the **speech bubble** to display your comment.
2. Click on the **Review** tab (See Figure 7).
3. In the *Review* tab, click on **Delete** (See Figure 7).

Figure 7 - Delete a Comment

Changing Your Review Display Settings

While *Track Changes* is active, you can alter your display settings to show how changes appear in your document. To alter your settings:

1. Click on the **Review** tab (See Figure 8).
2. In the *Review* tab, click on the **Display for Review drop-down** (See Figure 8).
3. A list of display options will appear. Select one of the following to apply:
   a. **Simple Markup** - Provides a clean, uncomplicated view of your document. You will see indicators where tracked changes have been made as a red line. It will also show comments as a speech bubble (See Figure 8).
   b. **All Markup** - Will show all changes and comments made to your document (See Figure 8).
   c. **No Markup** - Will show how the final version of the document will look with changes.No comments will be shown (See Figure 8).
   d. **Original** - Will show how the original version of the document looks without changesand comments (See Figure 8).

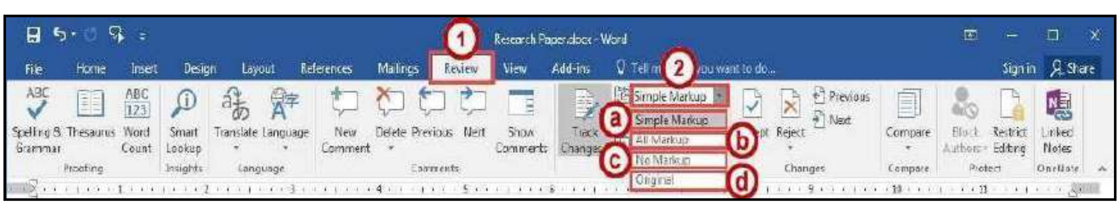
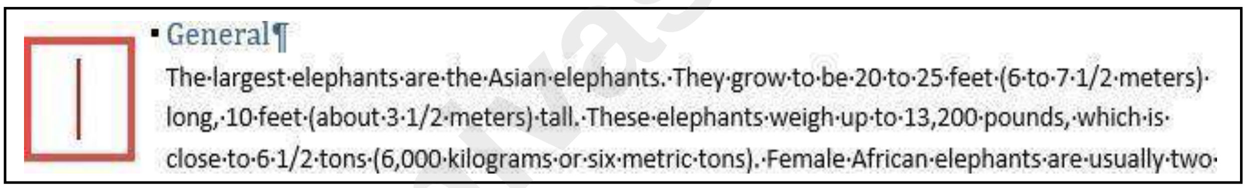


Figure 8 - Display for Review

Using the Simple Markup Display Settings

The *Simple Markup* display setting will show you sections of your document that have had changes made to them as a red line in the left side margin. Comments will also be shown as a speech bubble in the right margin of your document. The changes will remain hidden until you click the red line, upon which the changes and comments will be made visible.

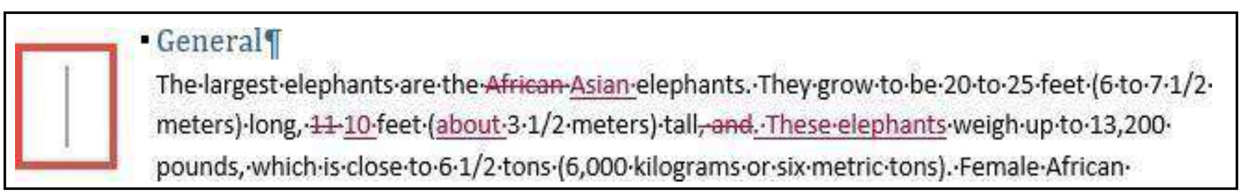
1. Ensure that *Simple Markup* has been selected as your *Display for Review* on the *Review* tab.
2. *Red lines* will appear in the left margin of your document, indicating that a change hasbeen made on this line (See Figure 9).
3. Click the **red line** to show the changes made in your document (See Figure 9).


General¶

The largest elephants are the Asian elephants. They grow to be 20 to 25 feet (6 to 7 1/2 meters) long, 10 feet (about 3 1/2 meters) tall. These elephants weigh up to 13,200 pounds, which is close to 6 1/2 tons (6,000 kilograms or six metric tons). Female African elephants are usually two

Figure 9 - Red Line Indicating a Change

4. The red line will become gray and the changes in your document will become visible.
5. To hide the changes again, click the **Gray Line** (See Figure 10).


General¶

The largest elephants are the ~~African~~ Asian elephants. They grow to be 20 to 25 feet (6 to 7 1/2 meters) long, ~~11~~ 10 feet (about 3 1/2 meters) tall~~, and~~. These elephants weigh up to 13,200 pounds, which is close to 6 1/2 tons (6,000 kilograms or six metric tons). Female African

Figure 10 - Gray Line Indicating Changes Visible

Activating the Reviewing Pane

When active, the *Reviewing Pane* will display the number of revisions in the document, the type of change made, and what was changed. You can also use the *Reviewing Pane* to quickly jump to revisions within your document by selecting them. The following shows how to activate the *Reviewing Pane*:

1. Click on the **Review** tab (See Figure 11).

2. In the *Review* tab, click on **Reviewing Pane** (See Figure 11).

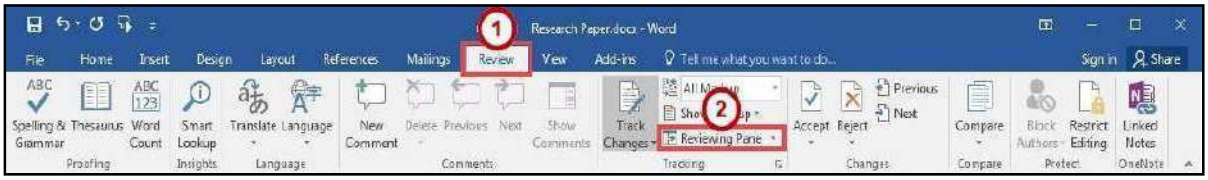


Figure 11 - Enable Reviewing Pane

3. The *Reviewing Pane* will appear on the left side of your document.

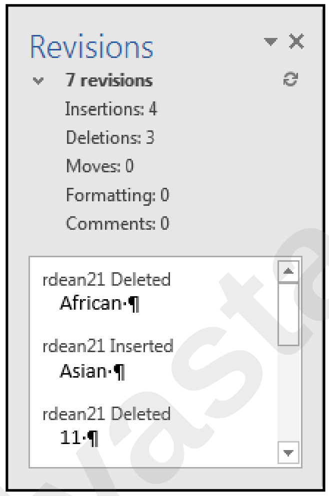


Figure 12 - Reviewing Pane

Locking Track Changes

If you are collaborating on a document with users, and want to ensure that *Track Changes* is used, you can *Lock Tracking* so the tracking option cannot be turned off.

Enable Lock Tracking

1. Click on the **Review** tab (See Figure 13).
2. In the *Review* tab, click on the **lower-half of the Track Changes** icon (See Figure 13).
3. In the *Track Changes* drop-down, click **Lock Tracking** (See Figure 13).

Figure 13 - Lock Tracking

4. In the *Lock Tracking* window, set a password (if desired) and click **OK**.

Figure 14 - Lock Tracking

Disable Lock Tracking

1. Click on the **Review** tab (See Figure 15).
2. In the *Review* tab, click on the **lower-half of the Track Changes** icon (See Figure 15).
3. In the *Track Changes* drop-down, click **Lock Tracking** (See Figure 15).

Figure 15 - Disable Lock Tracking

Accepting and Rejecting Changes

If you receive a document that has had changes made, you can move through the document to accept or reject the changes in the document. Once the changes have been accepted/rejected, the track markings will disappear.

The easiest way to accept/reject changes is to start from the beginning of your document. The following explains how to accept/reject changes:

1. Click to **place your cursor** at the beginning of the document.
2. Click on the **Review** tab (See Figure 16).
3. In the *Changes* group on the *Review* tab, you can do the following:
   a. **Accept** - Accept the change and move to the next change in your document (See Figure16).
   b. **Reject** - Reject the change and move to the next change in your document (See Figure 16).
   c. **Previous/Next** - Navigate between changes without accepting/rejecting (See Figure 16).

4. Continue this process until you have moved through the entire document.

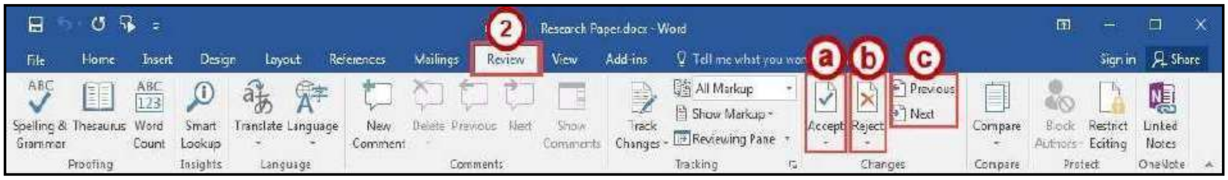


Figure 16 - Accepting and Rejecting Changes

Personalize Your Copy of Word

By personalizing your copy of Word, changes made to the document will show your "User name" and make it easier for others to identify your revisions when multiple reviewers are involved. This is helpful if you plan to collaborate with other users. The following explains how to modify your user information:

1. Click the **File** tab.
2. In the *Backstage View*, click **Options**.
3. The *Word Options* window will appear, click **General** (See Figure 17).
4. In the *Personalize your copy of Microsoft Office* section, *e*nter your **name** and **initials** (See Figure 17).
5. Click the **OK** button (See Figure 17).

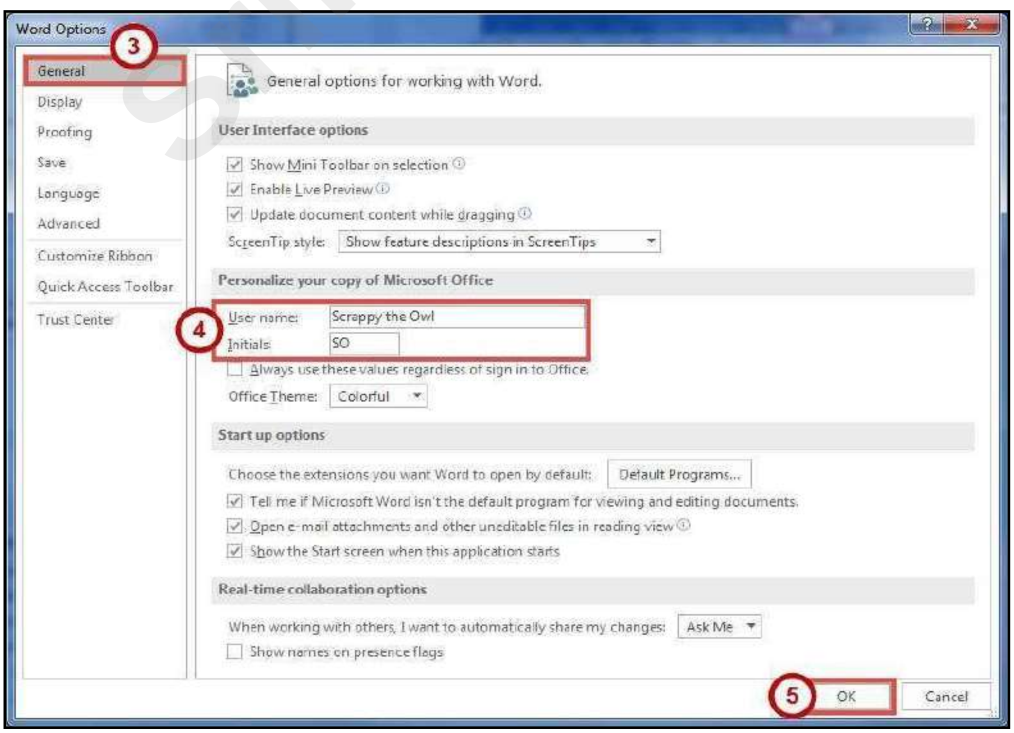


Figure 17 - Personalize your copy of Microsoft Office

Compare Changed Documents

If you receive a document that has been revised, but *track changes* were not enabled, then you can use the *Compare* tool in Word to determine what changes were made between the original document and the revised one. Please note, that to use this tool, you will need the original document and the revised document in order to compare.

1. Click on the **Review** tab (See Figure 18).
2. In the *Review* tab, click on **Compare** (See Figure 18).
3. In the *Compare* drop-down, click **Compare** (See Figure 18).

Figure 18 - Compare Changes

4. In the *Compare Documents* window, click on the **Folders** next to O*riginal document* and R*evised document* to browse through your computer and select the appropriate documents
   (See Figure 19).
5. Click the **OK** button (See Figure 19).

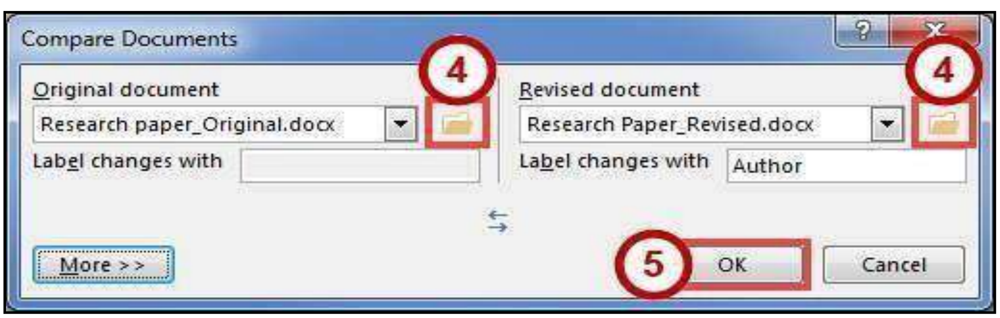


Figure 19 - Compare Documents Window

A screen will appear that shows three documents (from left to right): *Compared Document*, *Original Document*, and *Revised Document*.

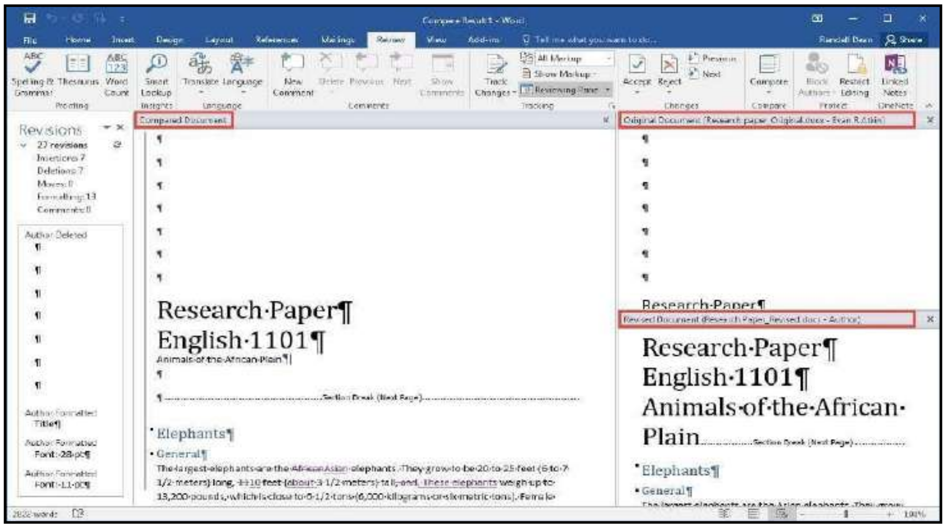


Figure 20 - Documents Compared

6. You can go through the changes in the *Compared Document* section and accept/reject changes as needed.
7. When finished reviewing the *Compared Document*, you can save it as its own version of the document.

Combine Changed Documents

If you send a document for review to several reviewers, and each reviewer returns the document, you can *combine* the documents two at a time until all the reviewer changes have been incorporated into a single document. The following explains how to combine multiple documents:

1. Click on the **Review** tab (See Figure 21).
2. In the *Review* tab, click on **Compare** (See Figure 21).
3. In the *Compare* drop-down, click **Combine** (See Figure 21).

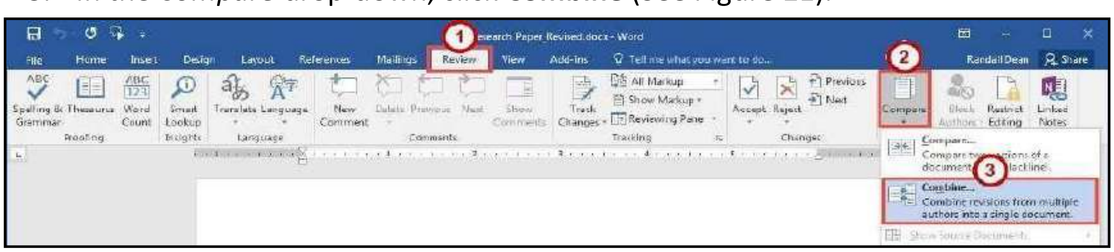


Figure 21 - Combine Changes

4. In the *Combine Documents* window, click on the **Folder** next to *Original document* and *Revised document* to browse through your computer and select

the appropriate documents (See Figure 22).

5. Click the **OK** button (See Figure 22).

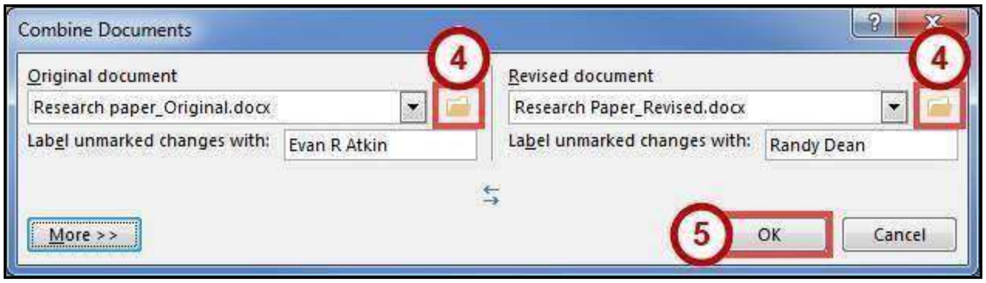


Figure 22 - Combine Documents Window

6. In the *Microsoft Word* window, select the document you want to **Keep formattingchanges from** (See Figure 23).
7. Click the **Continue with Merge** button (See Figure 23).

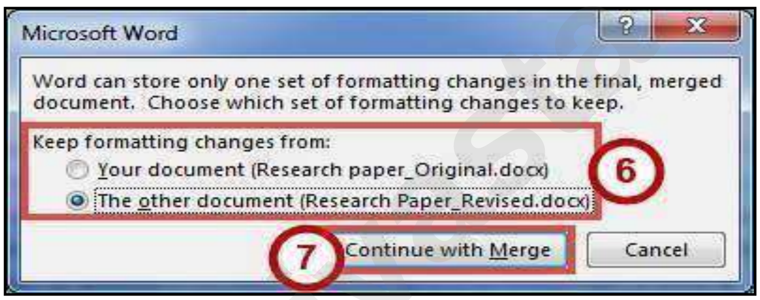


Figure 23 - Keep Formatting Changes From

A screen will appear that shows three documents (from left to right): *Combined Document*, *Original Document*, and *Revised Document*.

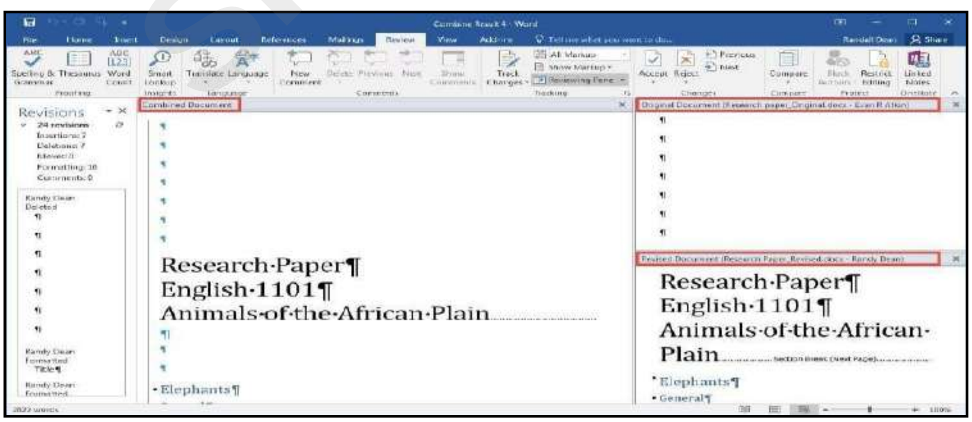


Figure 24 - Documents Combined

8. You can go through the changes in the *Combined Document* section and accept/reject changes as needed.
9. When finished reviewing the *Combined Document*, you can save it as its own version of the document.

Microsoft Office

Word 2016 for Windows

MAIL MERGE

Learning Objectives

After completing the instructions in this booklet, you will be able to:

- What is Mail Merge
- Use of Mail Merge .
- Reassign fields using the matching fields tool.
- Create and Print Labels

What Is Mail Merge?

Mail Merge is most often used to print or email form letters to multiple recipients. Using Mail Merge, you can easily customize form letters for individual recipients. Mail merge is also used to create envelopes or labels in bulk.

A mail merge in Word will combine a preprepared letter with a mailing list, so that bulk mail is personalised before it is sent out. For example, you might be part of an organisation that has a list of members and you want to let them know about an upcoming Annual General Meeting. Your mailing list would be the list of members' names and addresses, and the preprepared letter would be a letter informing them of the AGM. Each letter produced will be identical, apart from the personalised portions.

The three files involved in the mail merge process are:

1. Your main document
2. Your mailing list
3. The merged document

The Main Document

You should prepare your document before you start the mail merge, so type that out and save it now. Once it's ready, you will then tell Word the type of mail merge you are about to start: go to the Mailings tab and click Start Mail Merge > Letters.

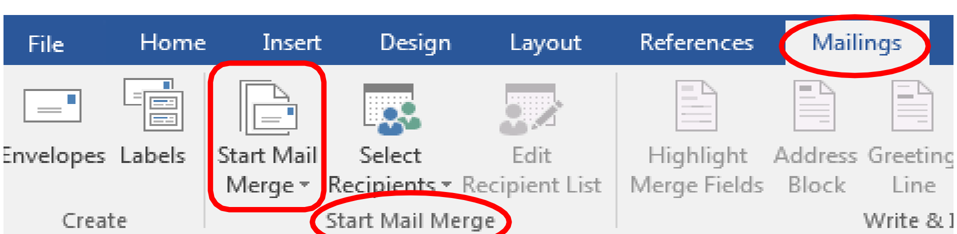


Figure 1: Mail Tab

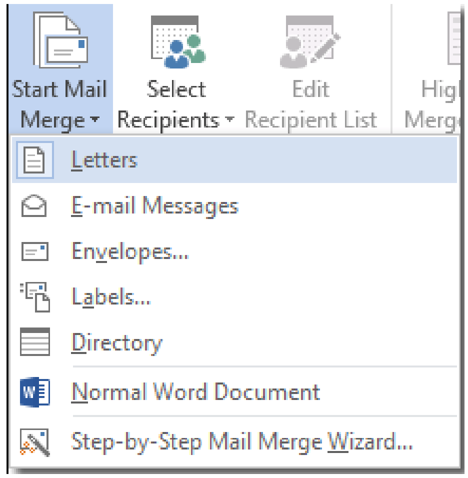


Figure 2: Selecting Letter

Now we will link the letter to your mailing list.

The Mailing List

Your mailing list can be stored in a variety of different locations, such as an Excel spreadsheet, an Access database, a directory of Outlook contacts, or an Office address list. It contains the personalised details that will be combined with the document.

If you don't have a mailing list when you begin the mail merge, you can get Word 2016 to create one during the merge. We recommend that you create the mailing list before you commence the merge though, so for this example we'll assume you have an Excel spreadsheet that contains names and address that you want to mail the letter to.

Link the Mailing List To The Document

Go to the Mailings tab and click Select Recipients in the Start Mail Merge Group. We're going to assume you have a mailing list ready to use, although you can create one on the fly. Select Use an Existing List.

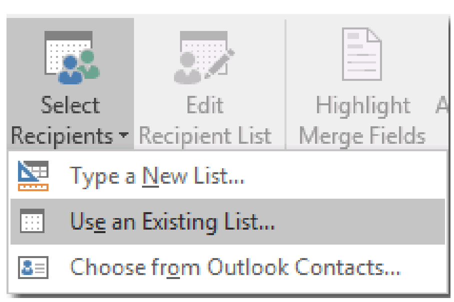


Figure 3: Linking Mailing List

In our example, we have an Excel Spreadsheet, so navigate to where that is and select it. Select the sheet that contains your data and click OK. If the first row in your spreadsheet is a header row, make sure that you check the box to say so.

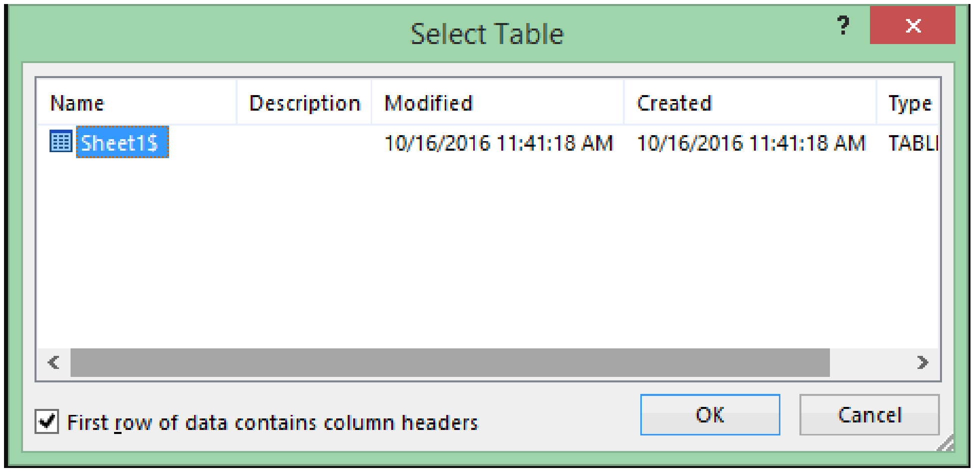


Figure 4: Linking Excel Workbook

Now that you have established the link between your document and mailing list, save the document.

We're going to assume that you want to send your letter to everyone on your list, but if you want to, you can select only certain entries from the list.

Insert Merge Fields

We now need to tell Word what personal details to add to the letter and where. To do this, we will insert merge fields in the main document. We're going to keep things very simple and we're just going to insert the most basic information. In our example we will insert member names and addresses.

First of all position the cursor where you want the address to appear on your letter. Then, on the Mailings tab, in the Write & Insert Fields group, choose Address Block.

In the window that opens, you get the chance to review and amend the format of the address that will be inserted when the merge is actually run.

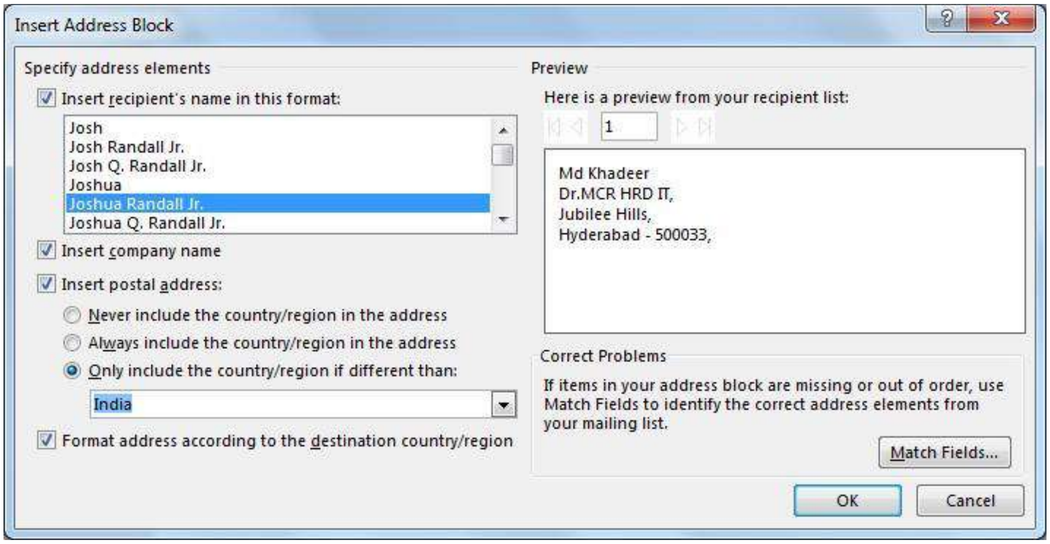


Figure 5: Inserting Address Block

Make any changes you need and click OK. You will ten see the <<AddressBlock>> placeholder appear where you inserted it.

Let's add a greeting line: position the cursor where you want the greeting and in the Write & Insert Fields group, click Greeting Line. Again, you get the chance to review and amend teh greeting, so make your changes and click OK. You will see the <<GreetingLine>> placeholder where you inserted it.

The address and greeting are standard items that most people will want to use in their mail merges, so they get their own special buttons in the ribbon. What if you want to insert data that is not in the Write & Insert Fields group? You will need the Insert Merge Fields button for that.

In our example, we have an email address column in the spreadsheet. We will add that now: click Insert Merge Fields > email.

Figure 6: Insert Merge Field

Word gets all the column headings in the spreadsheet and lists them for you to select from. An <<email>> placeholder is inserted.

All the <<>> placeholders inserted will get populated with real data from your spreadsheet when the merge is run.

Run The Mail Merge

The preparation is complete! All that is left to do is preview what the merged letter will look like, and then run it. Click Preview Results, and then choose the Next record button or Previous record button to make sure the names and addresses in the body of your letter look right.

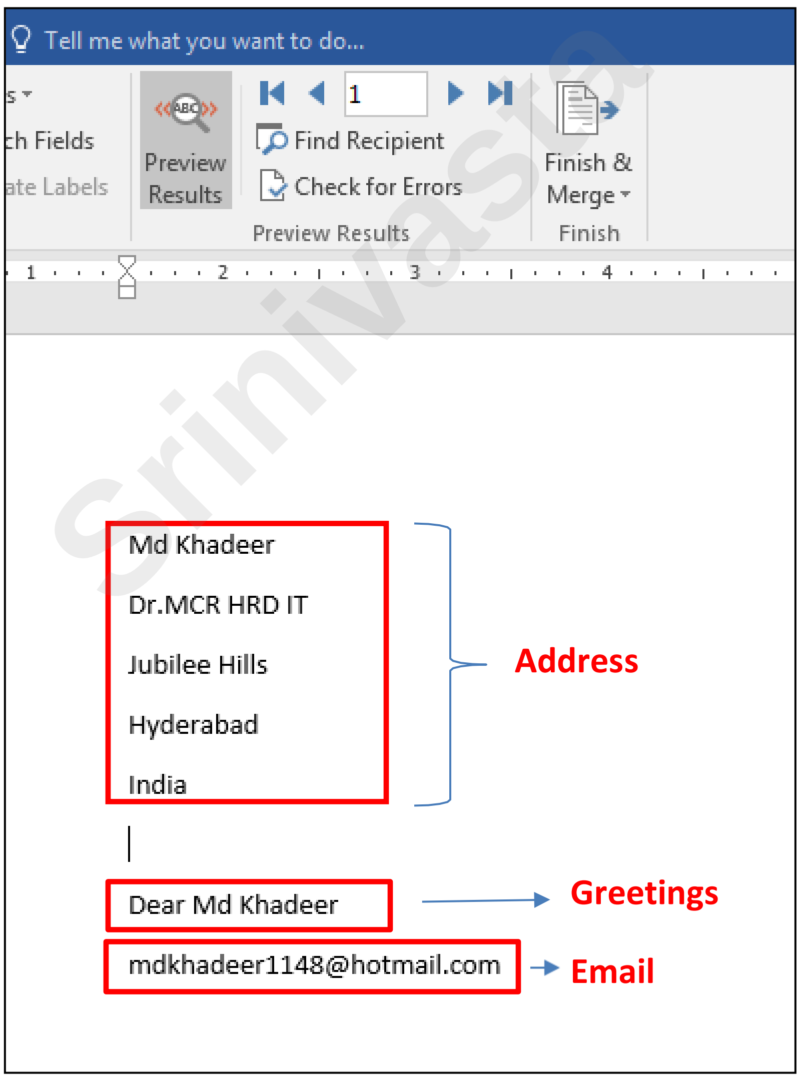


Figure 7 : Preview Merged Field

Make any changes you feel are necessary, and then, once happy, click Finish & Merge > Print Documents to run the merge and print.

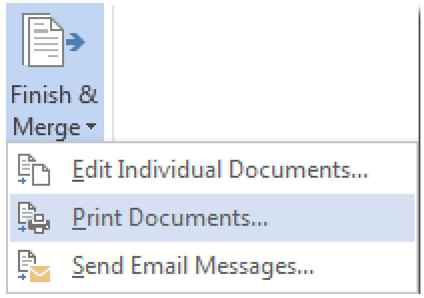


Figure 8: Merge and Print Document

Don't forget to save your document.

Create and Print labels in Microsoft Word

Start a new document to create new labels, or open an existing document that you used previously to merge labels.

Note:
Before labels can be made, the names and addresses must be stored in CSV file or another format that can be imported into the Mail Merge.

Once a CSV file is prepared, the steps below can be followed to create labels using a mail merge.

Figure 9: Mailings Tab

Step one and two

1. In Microsoft Word, on the Office Ribbon, click Mailings, Start Mail Merge, and then labels.
2. In the Label Options window, select the type of paper you want to use. If you plan on printing one page of labels at a time, keep the tray on *Manual Feed*; otherwise, select *Default*.
3. In the *Label vendors* drop-down list, select the type of labels you are using. In our example, we are using *Avery labels*.
4. Select the product number of the labels. The product number is often shown in one of the corners of the label package.
5. Once everything is selected, click OK.

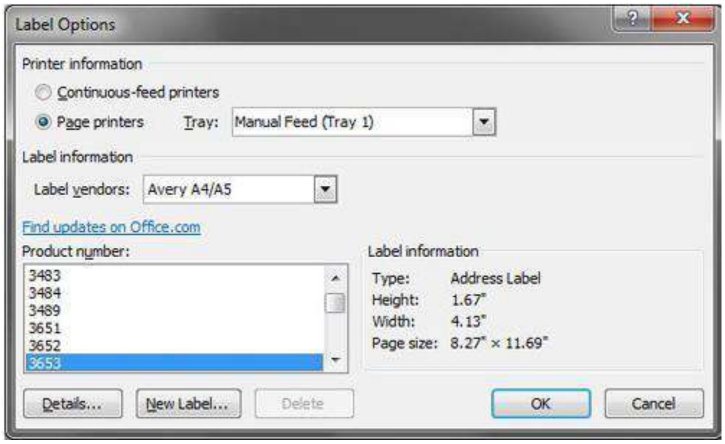


Figure 10: Labels Option

Step three

6. After the labels have been created, click the Select Recipients button in Mailings tab, click Use Existing List and then click Browse under the Use an existing list heading. The Select Data Source dialog box appears.

7. Click the appropriate drive and folder, select the file that you want to use, and then click Open. Your data is probably in Sheet 1, so click OK. (If you renamed your sheet, it would have that name instead of Sheet1$.) Also make sure to have the box checked if your first row contains headers.

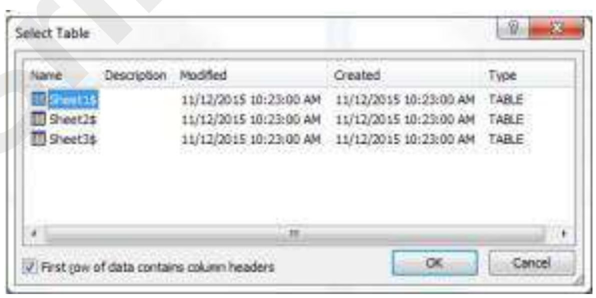


Figure 11: selecting sheet name

Word displays the Mail Merge Recipients dialog box for you to select specific recipients. When you finish, click OK.

Your labels are displayed. You will see <>. This is good!

Step four

8. Click the Address Block option in the Ribbon and verify the address is properly formatted.

9. If the address is not being displayed properly, click the Match Fields button and change how the fields are being matched.

10. Once the address is being displayed properly, click OK to place the "<<Address Block>>" field into the first label.

Step five

11. Click the Update Labels button to update all fields.

12. The first label should have only the "<<AddressBlock>>" field and all other labels should have "<<Next Record>><<AddressBlock>>" to step through each address field and print the address block in each label.

Step six

13. To make sure everything looks ok, click the Preview Results button, which should show each label and a different address for each label.

14. If you want to preview more than the first page, click the arrow pointing to the right while still in preview mode to show other pages.

Step seven

15. If everything looks ok, click the Finish & Merge button.

16. To merge on the screen, click Edit individual labels, The Merge to new document dialog box appears, so that you can select which records to merge. Select All and Click OK to merge the labels. You can view the labels before you print them.

After the merged document appears on the screen, you can save it as a separate document. Maybe call it something like “address labels final.” Note that in this saved version, you can’t toggle back and forth between the Preview Results and data fields. You can edit further if you want, but remember that you will be editing the final document, and not the label template.

Printing

1. Do a test print! This is a must, even if you think they will print fine. Print on regular paper first, and hold it up to your labels to see if they will fit.
2. Insert the labels in your printer. Make sure you have the correct orientation (face up/face down). Look for a picture on your printer.

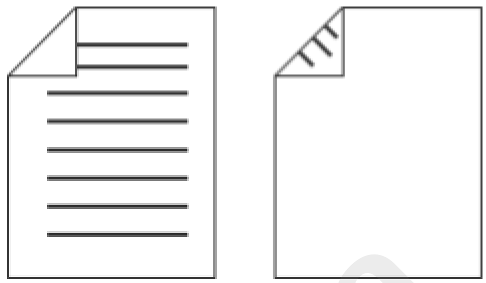


Figure 12: Paper Orientation

3. Print the merged document by clicking File and then print.

Microsoft Office Excel 2016

for Windows

INTRODUCTION TO MS-EXCEL

Learning Objectives

- The Ribbon, Customizing the Ribbon
- The Quick Access Toolbar, Tell Me and Smart Lookup
- The Smart Lookup Tool, The File Tab
- Columns and Rows, Entering Text
- Long Words and Numbers, Completing a Series
- Selecting Multiple Cells, Moving Text and Numbers
- Copying Data
- Insert a Row or Column, Delete a Row or Column
- Changing Column Width and Row Height, Formatting Numbers
- Formatting Text and Numbers, Changing the Font, Borders
- Graphics Clip Art, Images from a File Printing, Saving

Intro to Excel spreadsheets What is a Spreadsheet?

A spreadsheet is the computerized equivalent of a general ledger. It has taken the place of the pencil, paper, and calculator. Spreadsheet programs were first developed for accountants but have now been adopted by anyone wanting to prepare a budget, forecast sales data, create profit and loss statements, compare financial alternatives and any other mathematical applications requiring calculations.

The electronic spreadsheet is laid out similar to the paper ledger sheet in that it is divided into columns and rows. Any task that can be done on paper can be performed on an electronic spreadsheet faster and more accurately.

The problem with manual sheets is that if any error is found within the data, all answers must be erased and recalculated manually. With the computerized spreadsheet, formulas can be written that are automatically updated whenever the data are changed.

What can a Spreadsheet do?

In contrast to a word processor, which manipulates text, a spreadsheet manipulates numerical data and text. Using a spreadsheet, one can create budgets, analyze data, produce financial plans, and perform various other simple and complex numerical applications.

By having formulas that automatically recalculate, either built by you, the user, or the built-in math functions, you can play with the numbers to see how the result is affected. Using this “what-if?” analysis, you can see what affect changing a data value or calculation can have on your monitoring program.

Spreadsheets can also be used for graphing data points, reporting data analyses, and organizing and storing data.

Starting Excel

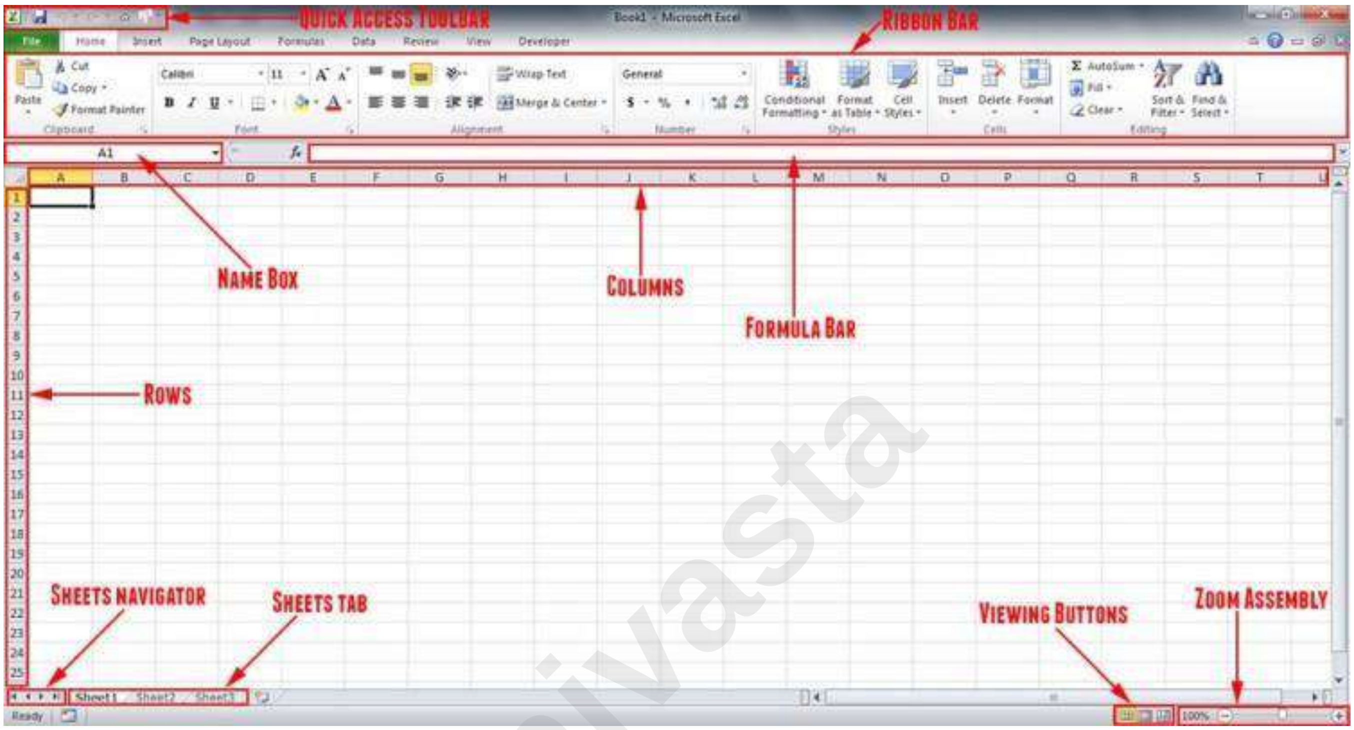
You are encouraged to start using MS Excel as you read through the following materials to familiarize yourself with the topics and procedures.

1. Click the Start button on the Windows taskbar.
    - The Start menu opens
2. Point to Programs
    - The Programs menu opens
3. Click Microsoft Excel
    - Excel opens a new workbook

Note: An icon for MS Excel may be located either on the desktop or on the Office toolbar.

The Excel Screen

The screen in Excel looks different than those used in other types of applications.



Columns

The large window, labeled "Microsoft Excel" may take up the entire screen. This is referred to as the Application Window. The top line is called the Title Bar and has three buttons (Minimize, Restore, and Close) to the right. These buttons are used to size the window and close it. This title bar is standard in all Windows programs.

The second line is called the Menu Bar. Notice that one character of each selection is highlighted or underlined. This menu bar is also standard in all Windows programs.

The next two lines contain buttons with text or images and are referred to as the Standard and Formatting Toolbars. If you have a mouse, these toolbars allow you to enhance your worksheet without accessing the menu. Keep in mind that these may not be in the exact same place as on the illustration above. All toolbars can be customized to display any buttons you desire.

The next line is the Formula Bar and displays the current cell address (see below) and contents. As you move from cell to cell, Excel will keep track of the current cell address for you. The Formula Bar can also be used to edit the text (contents) or formulas contained in the cell.

Columns and Cells and Row

The horizontal bar across the top of the worksheet area is filled with letters, beginning with A. Each letter represents a column while the vertical bar on the left side of the worksheet filled with numbers refers to rows.

The intersection between a column and a row is referred to as a cell. A cell is similar to a box that can be used to store pieces of information. Each piece of information could be a word or group of words, a number or a mathematical formula.

Each cell has its own address. This address is used in formulas for referencing different parts of the worksheet. The address of a cell is defined by the letter of the column in which it is located and the number of the row. For example, the address of a cell in column B, row 5 would be referred to as B5. The column is always listed first followed by the row without any spaces between the two.

The outlined cell (the one with the dark borders) within the worksheet is referred to as the active cell. Each cell may contain text, numbers, or dates. You can enter up to 32,000 characters in each cell (Equivalent to a 44 page report!).

These cell addresses are useful when entering formulas. Instead of typing actual values in your equations, you simply type the cell address where the value is stored. Then, if you need to go back and change one of the values the spreadsheet automatically updates the result of the formula based on the new data.

For example, instead of typing 67*5.4 you could enter C5*D5. The number 67 is stored in cell C5 and the number 5.4 is stored in cell D5. If these numbers change next month or next year, the formula remains correct because it references the cells - not the actual values. With the second formula, you can change the numbers stored in cells C5 and D5 as often as required and see the result recalculated immediately.

The next section of the screen lists the columns and rows within the current worksheet. As mentioned, columns are lettered and rows are numbered. The first 26 columns are lettered A through Z. Excel then begins lettering the 27th column with AA and so on. In a single Excel worksheet there are 256 columns (lettered A-IV) and 65,536 rows (numbered 1-65,536), totaling 16,777,216 individual cells.

Sheets and Workbooks

Towards the bottom of the worksheet is a set of small Tabs that identify each sheet in the workbook (file). If there are multiple sheets, you can use the tabs to easily identify what data is stored on each sheet. For example, the top sheet could be "Expenses" and the second sheet could be called "Income". When you begin a new workbook, the tabs default to being labeled Sheet1, Sheet2, etc.

At the bottom of the screen is another bar called the Status Bar. This bar is used to display various information about the system and current workbook.

The left- hand corner of this line lists the Mode Indicator, which tells you what mode you are currently working in.
The Zoom button (located on the toolbar at the top of the screen) allows you to change the size of the viewing area. This does not affect the actual printing of the file. Click on the down arrow located to the right side of the current zoom factor. Scroll through the available zoom choices. When you select a zoom factor, Excel will zoom in or out of the worksheet area - as specified in the Zoom. You can also access the View Æ Zoom menu. In addition, you can hide everything except the worksheet and the menu (which will increase your working area) by accessing the View Æ Full Screen menu.

The Ribbon

The Ribbon is designed to help you quickly find the commands that you need to complete a task. Commands are organized in logical groups, which are collected together under Tabs. Each Tab relates to a type of activity, such as formatting or laying out a page. To reduce clutter, some Tabs are shown only when needed. For example, the Picture Tools tab is shown only when a picture is selected.

File Menu

Here you will find the basic commands such as open, save, print, etc.

Quick Access Toolbar

The place to keep the items that you not only need to access quickly, but want to be immediately available regardless of which of the Ribbon's tabs you're working on. If you put so many items on the Quick Access Toolbar that it becomes too big to fit on the title bar, you can move it onto its own line.

Tell Me

This is a text field where you can enter words and phrases about what you want to do next and quickly get to features you want to use or actions you want to perform. You can also use Tell Me to find help about what you're looking for, or to use Smart Lookup to research or define the term you entered.

Formula Bar

A place where you can enter or view formulas or text. Expand Formula Bar Button

This button allows you to expand the formula bar. This is helpful when you have either a long formula or large piece of text in a cell.

Worksheet Navigation Tabs

By default, every workbook starts with 1 sheet.

Insert Worksheet Button

Click the Insert New Worksheet button to insert a new worksheet in your workbook.

Horizontal/Vertical Scroll

Allows you to scroll vertically/horizontally in the worksheet. Normal View

This is the "normal view" for working on a spreadsheet in Excel.

Page Layout View

View the document as it will appear on the printed page. Page Break Preview

View a preview of where pages will break when the document is printed. Zoom Level

Allows you to quickly zoom in or zoom out of the worksheet. Navigating in the Excel Environment

Below is a table that will assist you with navigating/moving around in the Excel environment.

Key	**Description**
ARROW KEYS	**Move one cell up, down, left, or right in a worksheet.** **SHIFT+ARROW KEY extends the selection of cells by one cell.**
BACKSPACE	**Deletes one character to the left in the Formula Bar.** **Also clears the content of the active cell.** **In cell editing mode, it deletes the character to the left of the insertion point.**
DELETE	**Removes the cell contents (data and formulas) from selected cells without affecting cell formats or comments.** **In cell editing mode, it deletes the character to the right of the insertion point.**
END	**Moves to the cell in the lower-right corner of the window when SCROLL LOCK is turned on.**

Also selects the last command on the menu when a menu or submenu is visible.

CTRL+END moves to the last cell on a worksheet, in the lowest used row of the rightmost used column. If the cursor is in the formula bar, CTRL+END moves the cursor to the end of the text. CTRL+SHIFT+END extends the selection of cells to the last used

Cell on the worksheet (lower-right corner). If the cursor is in the formula bar, CTRL+SHIFT+END selects all text in the formula bar

ENTER Completes a cell entry from the cell or the Formula Bar, and selects the cell below (by default).

ESC Cancels an entry in the cell or Formula Bar. Closes an open menu or submenu, dialog box, or message window.

HOME Moves to the beginning of a row in a worksheet.

CTRL+HOME moves to the beginning of a worksheet.

PAGE DOWN Moves one screen down in a worksheet. PAGE UP Moves one screen up in a worksheet.

SPACEBAR In a dialog box, performs the action for the selected button, or selects or clears a check box.

CTRL+SPACEBAR selects an entire column in a worksheet.

SHIFT+SPACEBAR selects an entire row in a worksheet.

CTRL+SHIFT+SPACEBAR selects the entire worksheet.

TAB **Moves one cell to the right in a worksheet.**

Highlighting/Selecting Areas Using the Mouse Select cells:

Moves a cell's contents:

Activate the Autofill feature:

To Select a Column: Click on the column letter To Select a Row: Click on the row number

To Select the Entire Worksheet: Click above row 1 and to the left of column A or hit CTRL A on the keyboard

Entering Text

Any items that are not to be used in calculations are considered, in Excel's terminology, labels. This includes numerical information, such as phone numbers and zip codes. Labels usually include the title, column and row headings.

To Enter Text/Labels:

1) Click in a cell

2) Type text

3) Press Enter

NOTE: By default, pressing the Enter key will move you to the cell below the active cell.

The label actually "lives" in the cell you typed it into. If you type long text it might appear to be in multiple columns. It is important to understand this concept when trying to apply formatting to a cell. Using the formula bar will confirm where the label actually "lives."

Autofills

Frequently, it is necessary to enter lists of information. For example, column headings are often the months of the year or the days of the week. To simplify entering repetitive or sequential lists of information, Excel has a tool called Autofill. This tool allows preprogrammed lists, as well as custom lists, to be easily added to a spreadsheet.

Entering Values

Numerical pieces of information that will be used for calculations are called values. They are entered the same way as labels. It is important NOT to type values with characters such as "," or "$".

To Enter Values:

1) **Navigate to a cell**

2) **Type a value**

3) **Press Enter**

Creating Formulas

Formulas perform calculations or other actions on the data in your worksheet. A formula starts with an equal sign (=). It is possible to create formulas in Excel using the actual values, such as "4000*.4" but it is more beneficial to refer to the cell address in the formula, for example "D1*.4". One of the benefits of using a spreadsheet program is the ability to create a formula in one cell and copy it to other cells. Most spreadsheet formulas use a concept called relative referencing.

This is the explanation of relative referencing from Excel's help file:

"A relative cell reference in a formula, such as A1, is based on the relative position of the cell that contains the formula and the cell the reference refers to. If the position of the cell that contains the formula changes, the reference is changed. If you copy the formula across rows or down columns, the reference automatically adjusts. By default, new formulas use relative references. For example, if you copy a relative reference in cell B2 to cell B3, it automatically adjusts."

It is also important to know the operators Excel uses for formulas:

Operator (Key)	Function
=	Begins all Excel functions and formulas
+	Addition
-	Subtraction
*	Multiplication
/	Division

To Create a Formula:

1) Click in a cell
2) Press the = key
3) Type the formula
4) Press Enter

Copying Formulas

Like many things in Excel, there is more than one way to copy formulas. Feel free to choose what works best for you.

To Copy Formulas Using Autofill:

1) Click in the cell that contains the formula
2) Position the mouse on the Autofill handle (a thin black cross will appear)
3) Click and drag to copy the formula

To Copy Formulas Using Copy and Paste:

1) Click in the cell that contains a formula

2) Select Copy on the Home Ribbon in the Editing group

3) Highlight the cell where you would like to paste the formula

4) Select Paste on the Home Ribbon in the Editing group

ALTERNATE METHODS

Keyboard:	**Ribbon**:	**Mouse**:
Press CTRL + C	Copy	Right-click and choose Copy

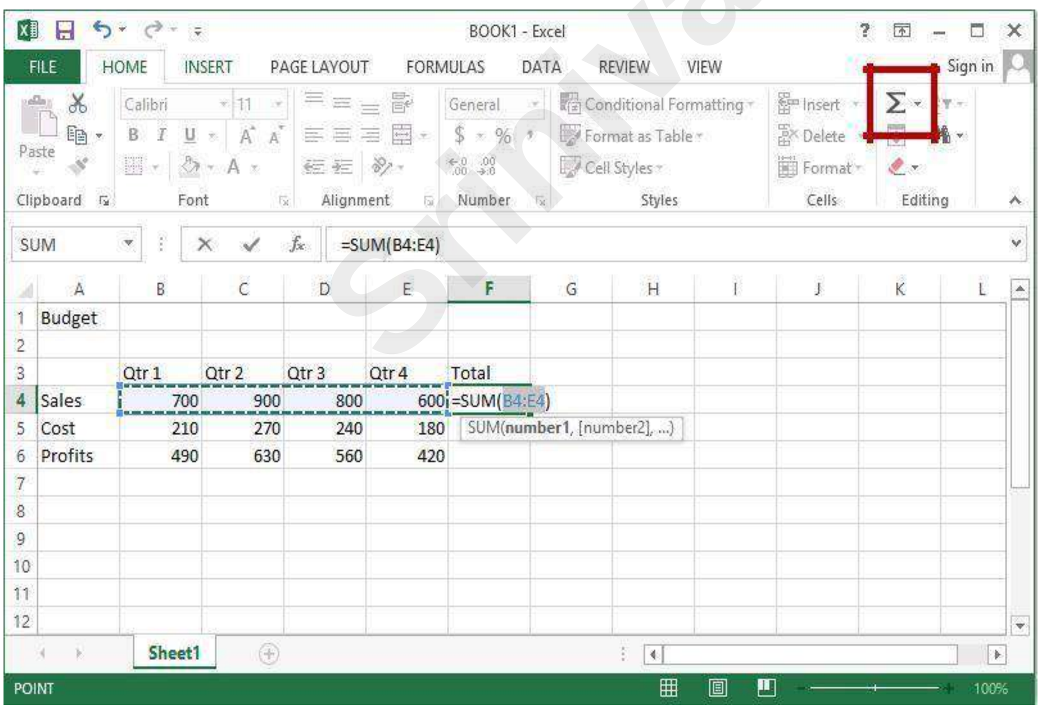
AutoSum Function

The most common formula in Excel is SUM, or the addition of multiple values. In this example, we could create a formula that reads =C6+D6+E6+F6+G6+H6. That's a lot of typing! Instead, we can use the SUM function and specify a range of cells.

Functions are more complex formulas that are invoked by typing their name. In this example, we will use the SUM function. Excel has over 200 functions that can be used. Because SUM is the most common function, it is the only one with its own toolbar button.

When working with functions, the cells used in the formula are referred to as the range. A range is a group of cells that are specified by naming the first cell in the group and the last cell. For example, A1:D1 is a range that includes cells A1, B1, C1 and D1.

To Create the Total Column's Values Using AutoSum:



1) Click in the cell where you would like the Total to be located

2) Press the AutoSum button on the Home Ribbon

The AutoSum function automatically looks for cells that have values in them. It will read values until it finds the first blank cell. AutoSum will always look for values in the cells above it first, then to the left. This means that you need to be aware of what cells will be in the formula. AutoSum will select the range of cells to use in the formula by highlighting the range.

3) Press Enter

Saving a Worksheet

When working in Excel it is necessary to save your files. It is also very important that while working, your file is saved frequently. When naming a file, you are restricted to 255 characters. Avoid most punctuation; spaces are acceptable.

To Save the Files:

1) Click on the File tab

2) Click Save

3) Choose the destination

4) Type a file name

5) Click Save

Editing Cells

Excel provides a major enhancement over earlier spreadsheet products in its ability to edit cells easily. There are various methods for cell editing, including double-clicking in the cell, using the F2 key, and typing in the formula bar.

To Edit a Cell in the Worksheet:

1) Position yourself in the cell you would like to edit

2) Press the F2 key on the keyboard or double-click in the cell

3) Use the backspace or delete keys to edit the cell

4) Press Enter when you have finished editing the cell

5) Click in the cell you would like to edit

6) Click in the formula bar and make any necessary changes

7) Press Enter when you have finished editing the cell

Undo

Excel and other Windows applications have a convenient method of correcting mistakes known as Undo. In many applications, including Excel, you can undo an almost limitless number of commands. The Undo button has a small down-pointing arrow next to it. When pressed, it will display a list of actions that can be undone. Redo works in the same way, allowing you to repeat actions.

Excel will undo actions in reverse chronological order, meaning that the most recent command is reversed first, then the one prior to that, and so on. You cannot reverse an earlier action using Undo without first undoing the actions that were performed after it.

NOTE: The list of commands to undo is reset after the file is saved. You cannot use Undo to fix an error after the file is saved.

To Undo a Command:

Click Undo

Clearing Cells

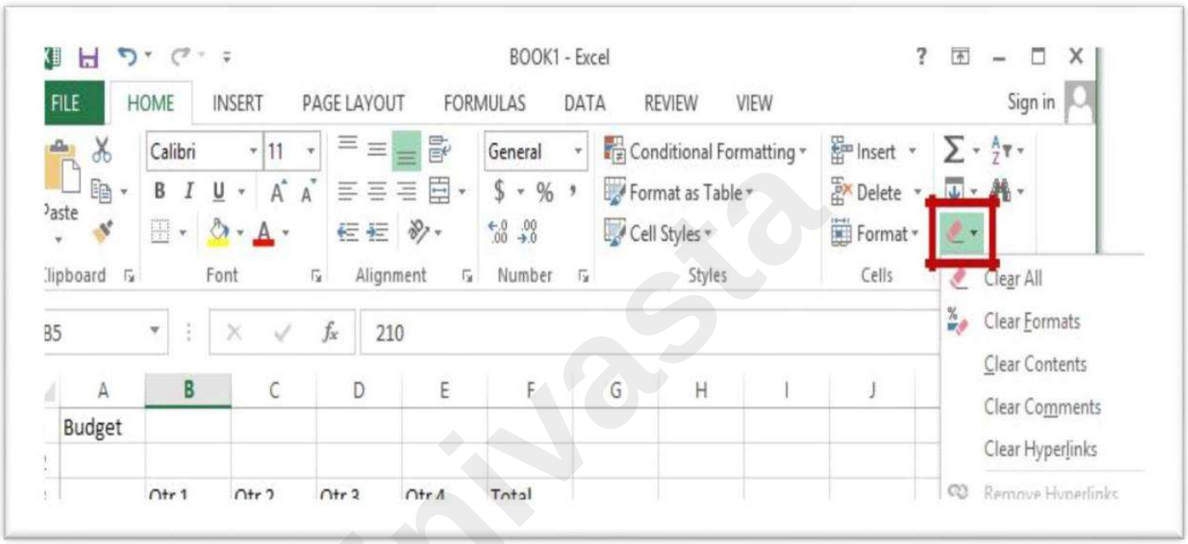
As we begin to look at formatting, it is important to understand what makes up the contents of a cell. There are three distinct items that can be in a cell:

- Contents
- Formats
- Comments

These allow items to be formatted properly, even if the values change. However, when trying to delete or clear a cell, it can be a bit tricky. Excel stores formats and contents separately, simply deleting the contents does not delete the format.

To Clear a Cell Format:

1) Click in the cell that contains formatting



Click the drop-down arrow next to the Clear button on the Home tab in the Editing group Click Clear

Formats

Formatting Values

Applying formats to any cell(s) can be done either using the Font, Alignment and Number groups or using the dialog box which will include all the formatting options.

To Apply the Currency Format:

1) Highlight the cell(s)

2) Click on the Currency Style button on the Home tab in the Number group

3) If necessary, click on the Increase or Decrease Decimal button on the Number group

To Apply the Comma Format:

1) Highlight cells

2) Click on the Comma Style button on the Number group

3) If necessary, click on the Increase or Decrease Decimal button on the Number group

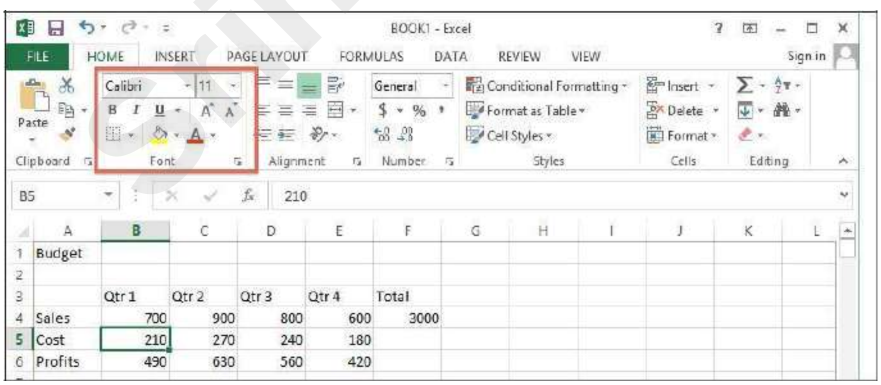
Formatting Labels

A Label, or text formatting is applied virtually the same way it is done in word processing programs. To Format the Title Labels:

1) Highlight the cell(s)

2) Select a font from the Font group

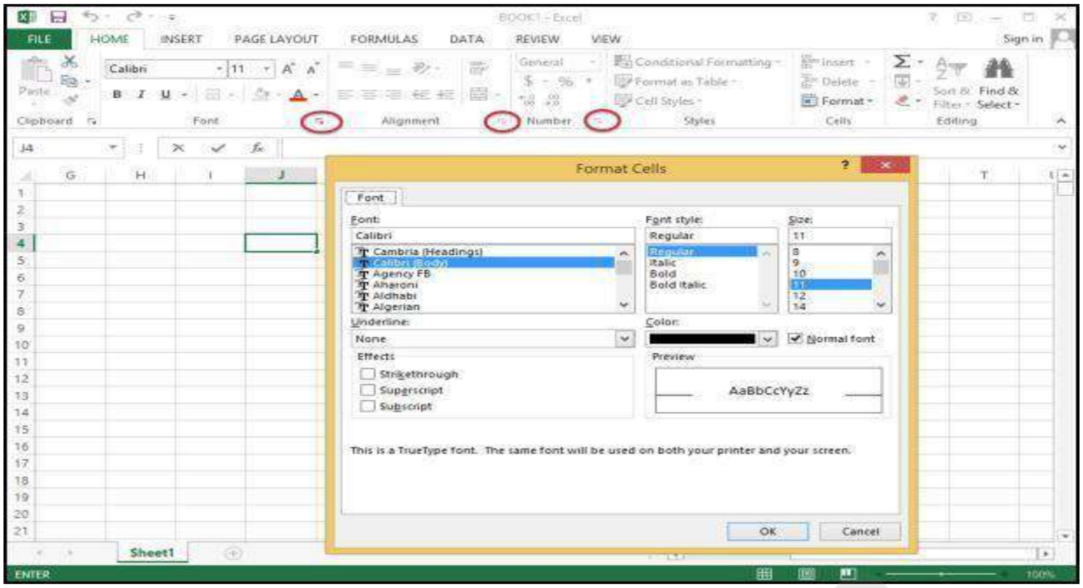
3) Select a point size from the Font group



Using the Dialog Box:

1) Highlight the cells

- Click on the arrow in the corner of one of the formatting groups (Font, Alignment, Number) to open the Format Cells dialog box and click on one of the tabs



Format Painter

Frequently, you will need to take a format that is applied to one cell and apply it to other cells. A quick way to do this is by using the Format Painter.

To Apply a Format to Cells:

1) Highlight cell(s)

2) Format the cell(s) to the desired format

3) Select the formatted cell(s)

4) Click the Format Painter from the Clipboard group of the Home tab

5) Highlight the cells you wish to format

Tips and Tricks: If you would like the Format Painter to remain active, double-click the Format Painter. It will remain active until you press the Esc key.

Cantering Text Across Columns

When it comes to titles, it may be preferable to have the information cantered across the document, rather than in only one cell. Excel uses the feature Merge Cells to accomplish this.

To Centre the Title Across Columns:

1) Highlight cell(s)

2) Click the Merge and Centre button on the Alignment group

NOTE: Each cell must be done individually. Excel will delete the contents of all but the top most cell if multiple cells are selected.

This option basically takes all the cells in the highlighted range and merges them into one large cell. For example, the range A1:F1 became cell A1 after the Merge Cells button was selected. There is no cell B1, C1, etc. any longer.

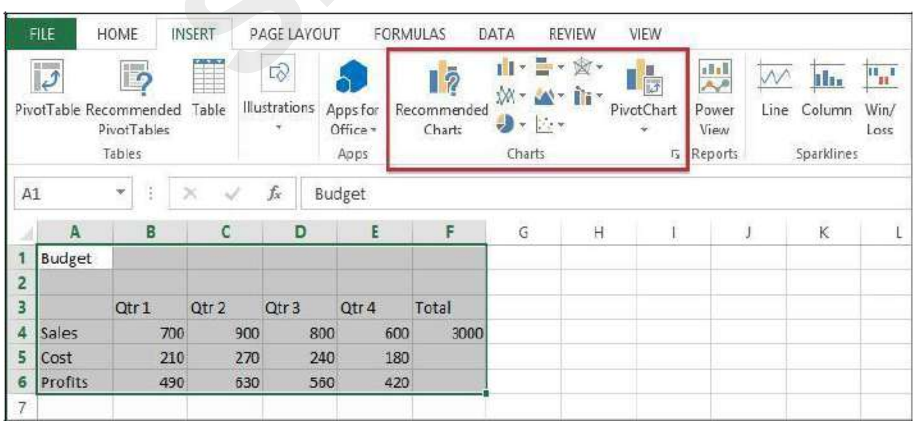
Creating a Basic Chart

1) Highlight the data to be charted

2) Click on the Insert tab

3) Click on a Chart Type in the Charts group

4) Click on a Chart Style

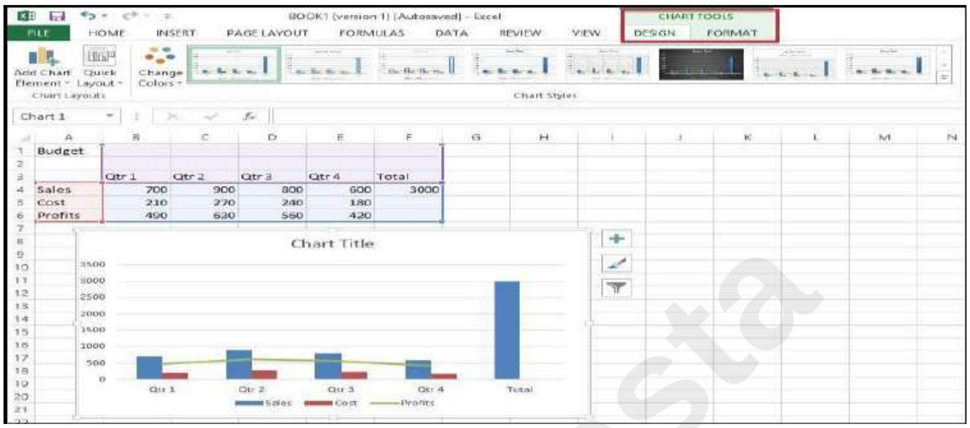


To Move your Chart:

Click and drag the chart to a new location on the worksheet.

When the chart is selected you will notice a new tab "Chart Tools" on the Ribbon. If you do not see the Chart Tools, click on the chart to select it. Under Chart Tools you will find 2 tabs:

- Design
- Format`



Excel Functions

As we have previously seen, the power of Excel lies in its ability to perform calculations. The real strength of this is shown in Functions. Functions are more complex formulas that are executed by using the name of a function and stating whatever parameters the function requires.

To Enter the SUM Function:

1) Click in a cell
2) Click on the AutoSum button in the Editing group
3) Highlight the range of cells that are to be added (The colon means "through")
4) Press ENTER

To Insert the Average Function into the Worksheet:

1) Click in a cell
2) Click on the drop-down arrow next to the AutoSum button
3) Click on Average

4) Highlight the range of cells be calculated

5) Press ENTER

To Insert the MAX Function into the Worksheet:

1) Click in a cell

2) Click on the drop-down arrow next to the AutoSum button

3) Click on Max

4) Highlight the range of cells be calculated

5) Press ENTER

To Insert the MIN Function into the Worksheet:

1) Click in a cell

2) Click on the drop-down arrow next to the AutoSum button

3) Click on Min

4) Highlight the range of cells be calculated

5) Press ENTER

To Insert the COUNT NUMBERS Function into the Worksheet:

1) Click in a cell

2) Click on the drop-down arrow next to the AutoSum button

3) Click on Count Numbers

4) Highlight the range of cells be calculated

5) Press ENTER

Printing a Worksheet

To Print, Preview and Modify Page Setup

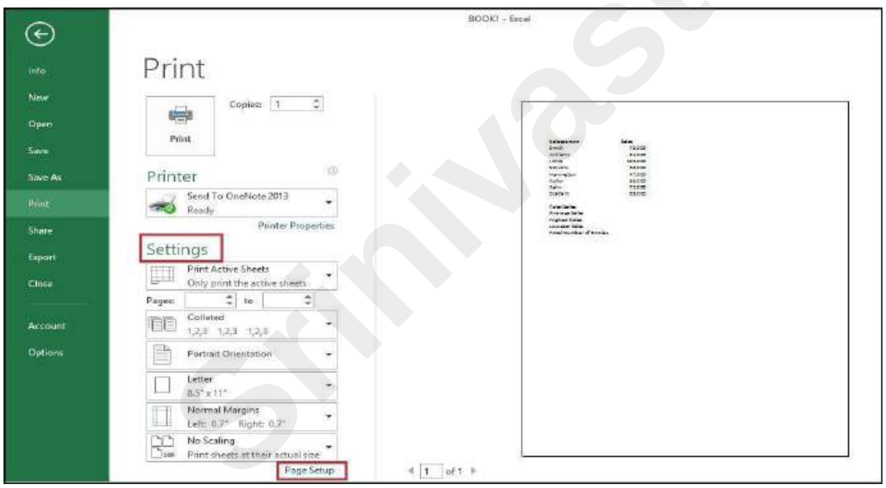
1) Click on the File tab

2) Click on Print

The spreadsheet shows as it will be printed. You can proceed to print the document from here, or you can change things to make the printed output look different.

Page Setup

You can change options under Settings or you can click on Page Setup.



Clicking on Page Setup will open a dialog box with four tabs:

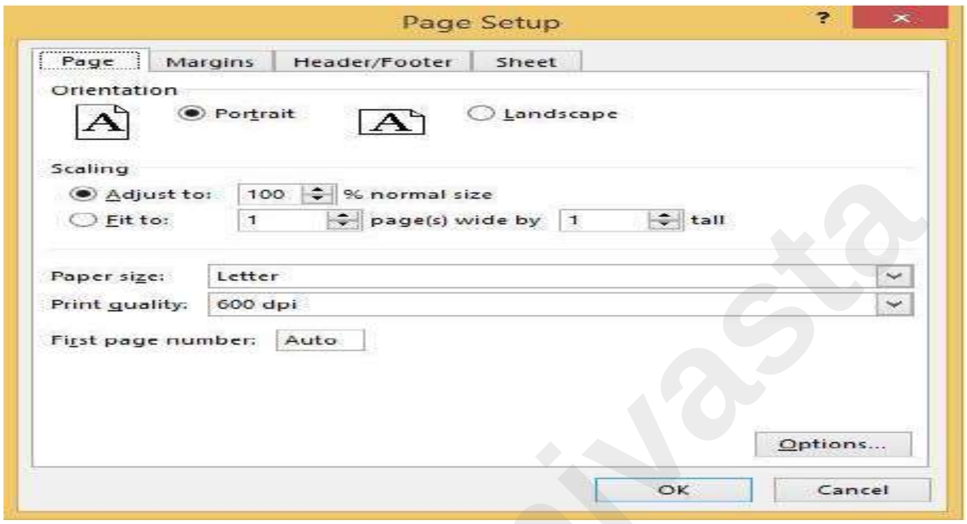
- Page
- Margins
- Header/Footer
- Sheet

Page:

1) Change the Orientation

2) Adjust the Scaling

3) Change the Paper Size



Margins:

1) Change the margins

2) Center on the page either horizontally, vertically or select both

Header/Footer:

1) To select from one of the already created headers/footers, click on the drop-down arrow for Header and also for Footer and choose from the list

2) To create a custom header and/or footer, click on Custom Header and Custom Footer

This area is made of three sections – left, center and right. Any information added in these sections will appear in that area (left, center or right) in the header or footer. You will also see a row of buttons in this dialog box. Following are their functions:

Click in a section to position your cursor

3) Enter text/fields

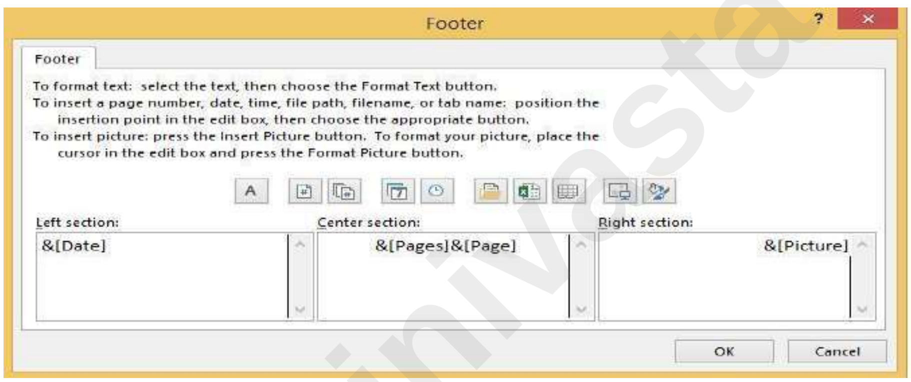
4) Click OK when finished

Sheet Tab:

1. Repeat Rows and Columns under Print Titles

2. Check off what to print under Print

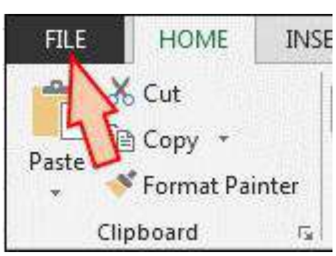
3. Change the Page Order

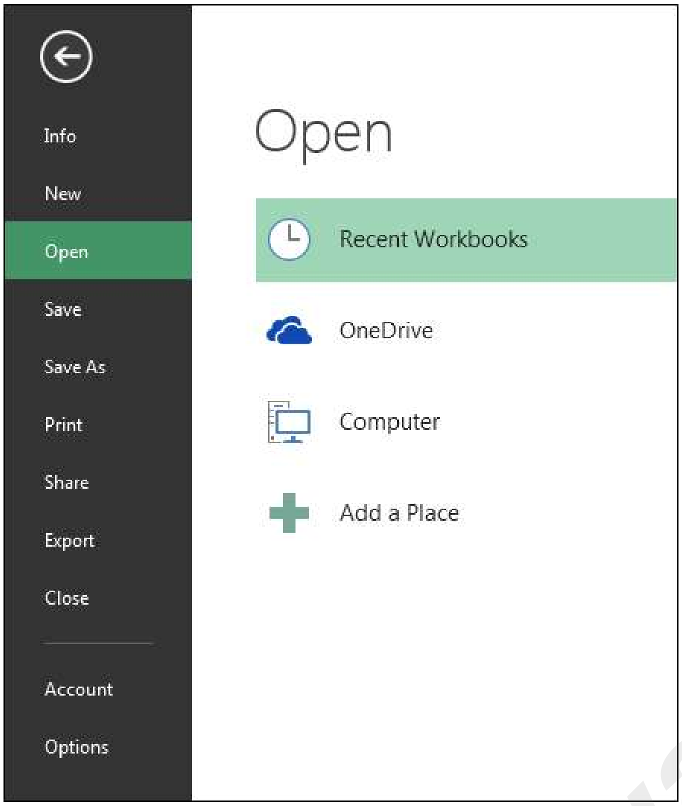


File Tab in Microsoft Excel

What is File tab and its uses?

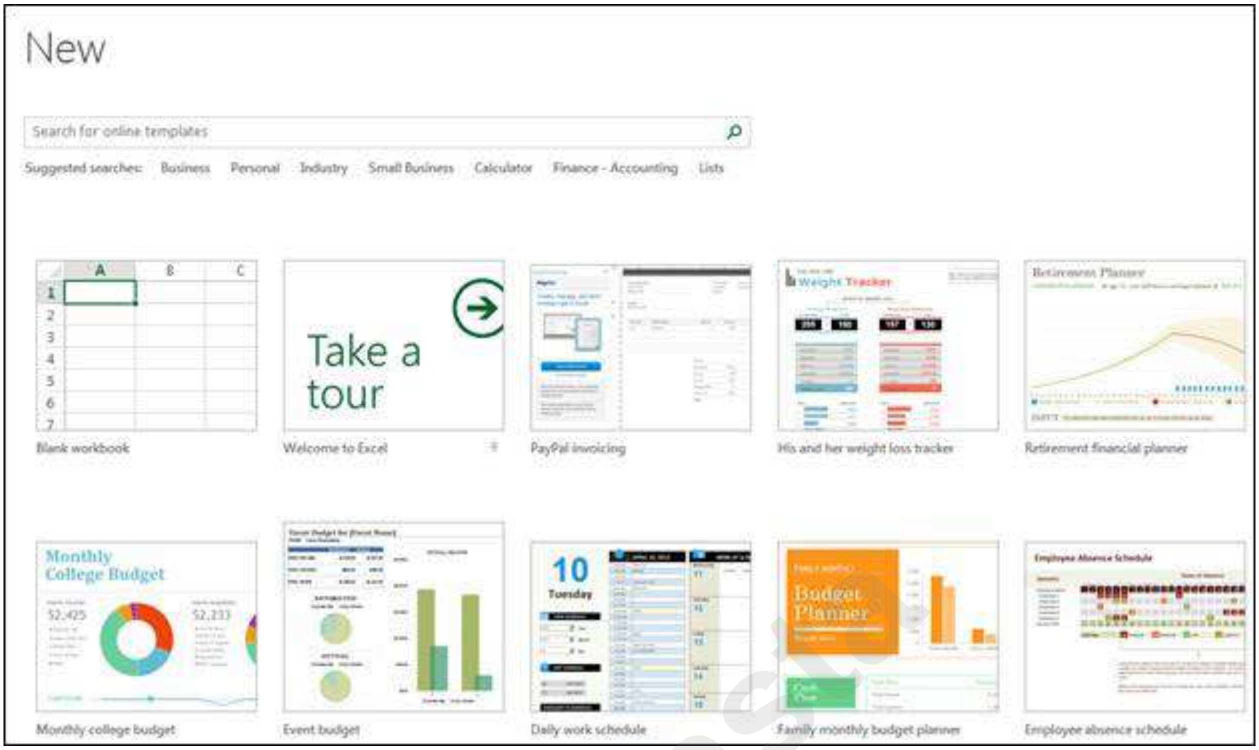
File tab contains the basic required options such as New, Open, Save, Save as, Print, Share, Export, and Close options. Other than the aforementioned options, we can find account and Excel options tab, too.





a) Info: - With this option, we can get the information about the particular Excel file. Created date, last modified date, Author name, Properties, versions etc.

b) New: – We use this option to open the new Excel file. We can open new file using shortcut key, Ctrl+N or by Clicking on File tab > New > Blank workbook. If Excel file is not opened, then Press Window+R and type Excel, New Excel file will open.



c) Open: – We use this option to open the existing file (shortcut Ctrl+O). "Open" option appears and you can choose to open the file.

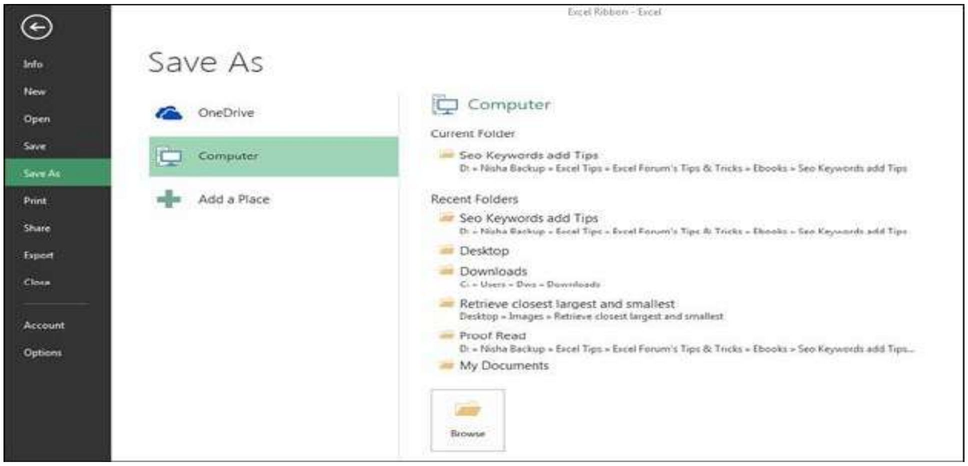
Alternatively, File tab > Open > choose the file

d) Save: – We use this option to save the current file.

Shortcut: – Ctrl+S

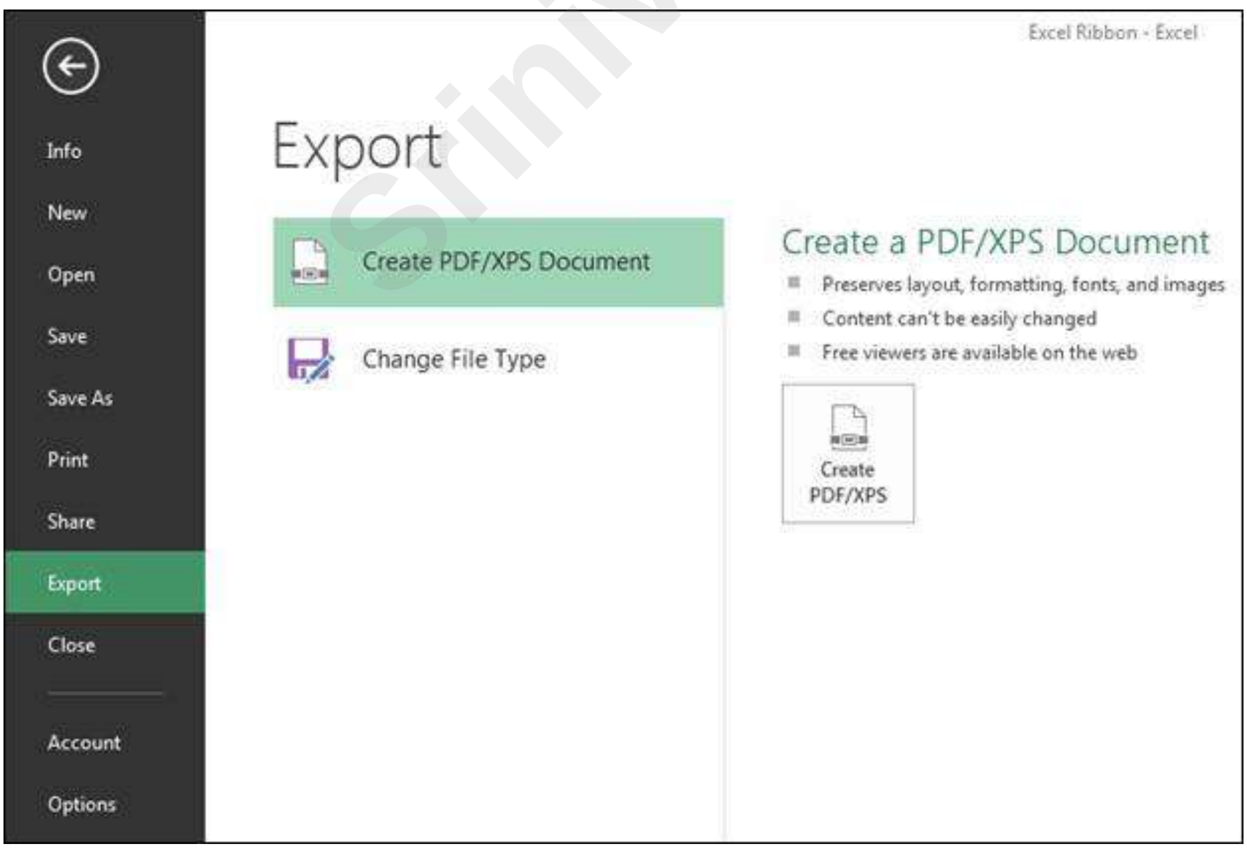
Path: – Click on File tab >Save

e) Save as: – We use this option to make another copy or save the file at another place. F12 is the shortcut key to save as the file or we can save the file following these steps: – Click on File tab > Save as and then choose the location.

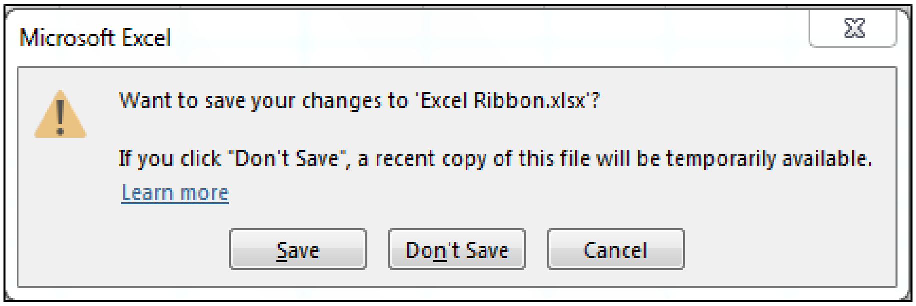


g) Share: – We use this option to share the file with multiple users and send it over email. To share the file we can follow the steps: – Click on File tab >Share.

h) Export: – We use this option to export the file in PDF or XPS document and we can change the file type as well. To Export the file, we can follow the steps: – Click on File tab >Export. And then we can export it as per our requirement.



i) Close: – We use this option to close the file. Ctrl+W is the shortcut key to close the workbook or we can follow the steps: – Click on File tab >Close, active file will be closed. When we close the file, we get the confirmation message to save the file or not or cancel the command.



j) Account: – We use this option to sign in to our office account and we can change office theme, too. We can follow the steps: – Click on File tab >Active, Active window will appear.

k) Options: – It was in the tool menu of the previous versions of 2007 MS. We use this option to add extra and advanced features, like Developer tab, Power pivot, Analysis toolpak and many more. Also, we can change default settings, like font size, font style, number of sheets etc. In Excel options, we have 10 categories:-

1) General

2) Formulas

3) Proofing

4) Save

5) Language

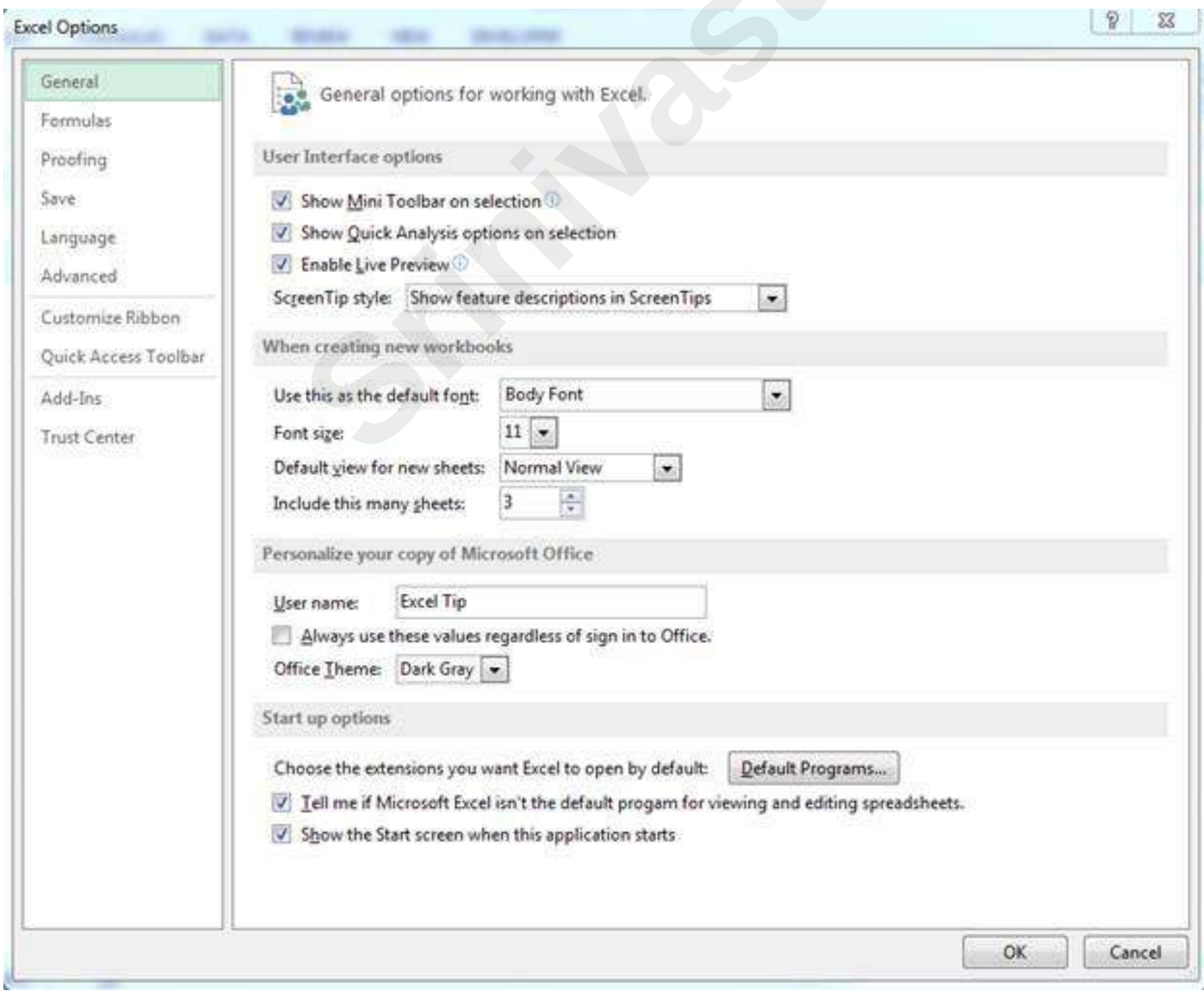
6) Advanced

7) Customize Ribbon

8) Quick Access toolbar

9) Add Ins

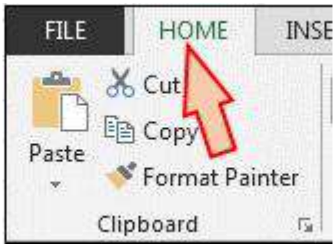
10) Trust Center



Home Tab in Microsoft Excel

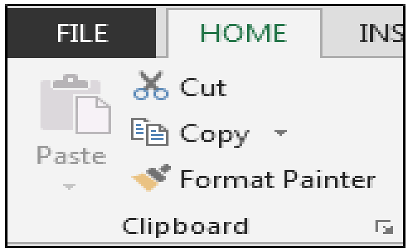
What is Home tab and its uses?

Home tab contains the most frequently used options such as cut-copy-paste, font formatting, alignment, Number, Conditional formatting, etc. All the options are used to format the data.

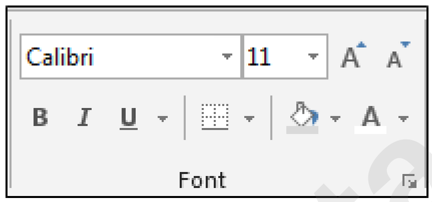


Under the Home tab, we have 7 groups: -

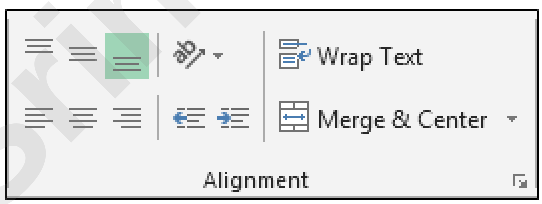
a) Clipboard: – This group contains frequently used commands: Cut, Copy, Paste and Format painter. Clipboard option allows us to collect text and graphic items and paste it.



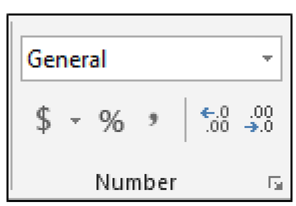
b) Font: – We use this option to change the font style and font-size. We can make it bold, italic and underline. Also, this group contains border styles, fill color, font color.



c) Alignment: – We use this option to change the alignment of cell's text to the right, left and middle. Also, we can subject the text to top, bottom, and middle alignment. In this group, we have Wrap text option to adjust and make the text visible within a cell, and we can also merge 2 or more cells, using merge option.

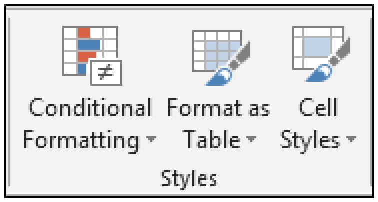


d) Number: – We use this option to change the number formatting into General, Percentage, Currency, Date, Time, Fraction etc. We can increase and decrease the decimal and convert the number into accounting number.

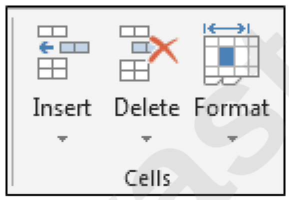


e) Styles: – In this option, we have Conditional Formatting, Format as Table and Cell Styles. Conditional formatting is used to highlight the cell or range on the basis of conditions. Format as table is having

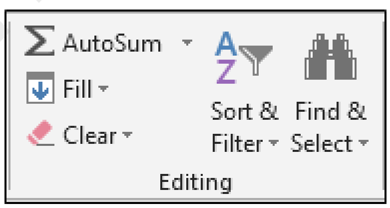
Ready made table format and Cell styles feature different types of built-in styles that are a combination of Font style, Font colour and Fill colour.



f) Cells: – We use this option to insert or delete cells, rows, columns and sheets. Also, we have format option to adjust the height, width of cells or range. Using this option, we can hide or unhide the range, protect the workbook, rename the sheet name, fill the tab color, move or copy to sheets, lock the cells.



g) Editing: – This option has Auto Sum feature to return the total of numbers and move the text to right, left, up and down, Clear the format, content, comments and hyperlink; sort the data and find and select option.



Insert Tab in Microsoft Excel

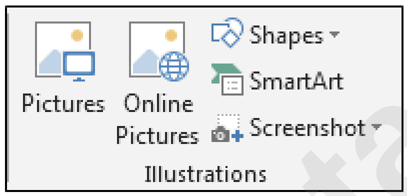
What is Insert tab and its uses?

We use Insert tab to insert the picture, charts, filter, hyperlink etc. We use this option to insert the objects in Excel. To open the insert tab, press shortcut keys Alt + N.

Under the Insert tab, we have 10 groups:-

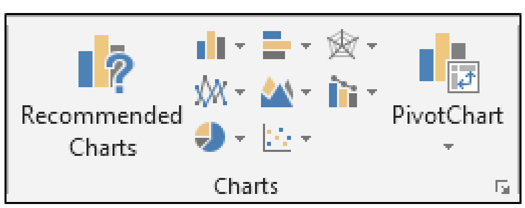
a) Tables: - We use this option to insert the dynamic table, Pivot table and recommended table. Pivot table is used to create the summary of report with the built-in calculation, and we have option to make our own calculation. Tables make it easy to sort, filter and format the data within a sheet. This option is also having recommended table that means on the basis of data, we can just insert the table as per the Excel's recommendation.

b) Illustration: - We use this option to insert the Pictures, Online Pictures, Shapes, SmartArt and Screenshot. It means if we want to insert any image, we can use Illustration feature.

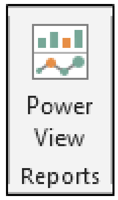


c) Apps: - We use this option to insert an app into the document and, in order to enhance the functionality, we can use web option.

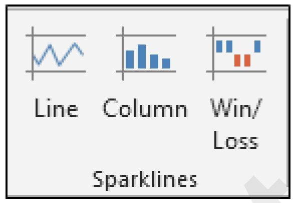
d) Charts: -Charts is very important and useful function in Excel. In excel, we have different and good numbers of readymade chart options. We have 8 types of different charts in Excel :- Column, Bar, Radar, Line, Area, Combo, Pie and Bubbles chart. We can insert Pivot chart as well as recommended chart, and if we don't know which chart we should insert for the data, we can use this option to fulfil the requirement.



e) Reports: -We use this option to create a better report on the basis of the decisions we take for business. It makes the report more interactive and decipherable.



f) Sparkline's: -Sparkline is a very tricky and useful option added by Microsoft Excel. On the basis of a range, it can visualize the trends in a single cell as charts. We have 3 different types of cell charts: - Line, Column and Win/Loss chart.

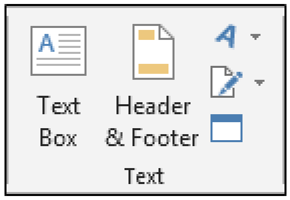


g) Filters: -We use this option to filter data visually and filter dates interactively. We have 2 options: Slicers and Timeline. We use Slicer to make the fast and easier to filter tables, Pivot tables, Pivot Charts and cube functions. Timeline makes it faster and easier to select time periods in order to filter Pivot Tables, Pivot Charts and Cube function.

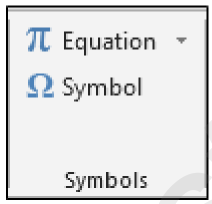
h) Links: -

- We use this option to create the link in the document for the quick access to webpage and files.
- We can also use it to access different locations in the document.

i) Text: -We use this option to insert the Text box, Header and Footer, Word art, Signature and objects. We insert Text box to write something in the image format. We use Header and Footer options to place the content on the top and bottom of the page. Word art makes the text stylish. Insert the Add Signature Lines that specify the individual who is supposed to sign it. And object option works for embedded objects, like documents or other files we have inserted into the document.



j) Symbols: - We use this option to insert the symbols and equation. Equation is used to insert the common mathematical equations to your document and also we can add equation by using the mathematical symbols. We use Symbols to insert the symbols which are not on the keyboard and, to create the equation, we use the symbols from here.



DATE AND TIME FUNCTIONS LEARNING OBJECTIVES

- Dates in Excels
- Times in Excels
- Useful Date/Time Shortcuts
- Custom Formatting
- Simple Date/Time Math
- Date Worksheet Functions
- Time Functions
- Dated if Function

Dates in Excel

If you've ever lost the date formatting on a cell, you have seen it turn into a strange number. For example, according to Excel these cells are the same number:

	A	B
1	10/17/2013	41564

The serial number 41, 564 tells us how many days it has been since January 1st, 1900*. The date serial numbers are sequential, one day at a time:

	A	B
1	10/17/2013	41564
2	10/18/2013	41565
3	10/19/2013	41566

*Note: The default date system for the Macintosh begins with January 1, 1904. This may cause some confusion if you try to use the same file in both a Mac and a PC. The setting can be changed in the Excel Options on the File menu.

Times in Excel

One day has 24 hours, so in Microsoft Excel, 1 is equivalent to 24 hours, 0.5 is equivalent to 12 hours. If we take our October 17th date, and add in a time of 12:00pm, it translates into 41564.5, the 0.5 representing the half- way point of the day.

1=24 hours, 0.5=12 hours, 0.25=6 hours…

	A	B
1	10/17/13 12:00 PM	41564.5

If you leave the date off a time, Excel will default to 1/0/1900 as the 'understood' date. You can ignore it, but realize that is what happens if you change a time format into a date/time format. All three of these cells contain 12:00 PM, they are just displayed with different formats:

	A	B	C
1	12:00 PM	1/0/1900	1/0/00 12:00 PM

<table>
<tr><th colspan="2">Date</th><th colspan="2">Month</th><th colspan="2">Year</th></tr>
<tr><td>d</td><td>3</td><td>m</td><td>2</td><td rowspan="2">YY</td><td rowspan="2">04</td></tr>
<tr><td>dd</td><td>03</td><td>mm</td><td>02</td></tr>
<tr><td>ddd</td><td>Tue</td><td>mmm</td><td>Feb</td><td rowspan="2">YYYY</td><td rowspan="2">2004</td></tr>
<tr><td>dddd</td><td>Tuesday</td><td>mmmm</td><td>February</td></tr>
</table>

Custom Formatting

Days, Months, and Years

You can format a cell with a preset list of options in the

Format Cells Window.

1. Click the More button in the Number group

2. Right-click on a cell and choose Format cells

3. Select the cell and press Ctrl-1 to open this window

Excel has a pretty extensive list of date and time formats, but it is possible to custom build a date format, using simple abbreviations.

Hours, minutes, and seconds

If you use "m" immediately after the "h" or "hh" code or immediately before the "ss" code, Excel displays minutes instead of the month.

Times for 1:02:05

Shortcut	Result	Note
Ctrl-;	Current Date	Control Semicolon
Ctrl-;	Current Time	Control Colon

Appropriate designator at the end. Access will accept any of the following: AM/PM; am/pm; A/P; a/p; AMPM.

Times for 17:02:05

Hours		24 hour clock		12 hour clock	
H	17	h:m	17:2	h:m am/pm	5:2 PM
hh	17	h:mm	17:02	h:mm am/pm	5:02 PM
h AM/PM	5 PM	hh:mm	17:02	hh:mm am/pm	05:02 PM

If you need to go smaller, adding .00 after the second format (ss.00) will give the fraction of a second.

Totaling Time

The time formats in Excel stay within the defaults of 24 hours, 60 minutes, 60 seconds. If we do math we may want it to display the times beyond these boundaries. For this, we use the brackets [ ] around the abbreviation.

Beyond the Boundaries

36 Hours		75 Minutes		75 Seconds	
h	**12**	**m**	**(assumes month)**	**s**	**15**
hh	**12**	**mm**	**(assumes month)**	**ss**	**15**
[h]	**36**	**[m]**	**75**	**[s]**	**75**

Simple Date/Time Math

Because dates and times are stored as numbers you can do simple math with them

Times are a fraction of a day. As mentioned earlier, 0.5=12 hours. If we multiply a time by 24 we should get the numeric value.

Time	Time * 24
8:45 AM	6:00 PM

Why doesn't it look right? You have to format the result as a true number field.

Time	Time * 24 (formatted as a number)
8:45 AM	8.75

Remember if you are going beyond the 24 hour clock, you will need to set up a custom format.

Date Worksheet Functions

Adapted from Excel Help

DATE - Returns the sequential serial number that represents a particular date

Syntax: DATE(year, month, day)

Year The value of the year argument can include one to four digits.

Month A positive or negative integer representing the month of the year from 1 to 12. If month is greater than 12, month adds that number of months to the first month in the year specified

Day A positive or negative integer representing the day of the month from 1 to 31.

DATEVAULE - Returns the serial number of a date

Converts a date that is stored as text to a serial number that Excel recognizes as a date. To view a date serial number as a date, you must apply a date format to the cell.

Syntax: DATE VALUE (date_text)

Date text represents a date between 01/01/1900 and 12/31/9999 saved as text

Equation	Result	Notes
=DATEVALUE(3/15/2009)	#VALUE!	Date not in Text format
=DATEVALUE("10/15/1905")	2115	
=DATEVALUE("3/20/2011")	40622	
=DATEVALUE(A1)	#VALUE!	A 1 = 12/5/1976
=DATEVALUE(A1)	28099	A 1 = "12/5/1976"

DAY - Returns the numeric value of the day in a valid date

Syntax: =DAY(serial number)

Serial Number is the date of the day you are trying to find.

Equation	Result	Notes
=DAY(3/15/2009)	0	*Serial Number* not a Date (Excel sees 3÷15÷2009)
=DAY("10/15/1905")	15	
=DAY(40622)	20	Equivalent to 3/20/2011
=DAY(A1)	5	A1 = 12/5/1976

DAYS360 - returns the number of days between two dates based on a 360-day year

360 day year assumes twelve 30-day months, this is often used with accounting calculations. If start date occurs after end date, the DAYS360 function returns a negative number.

Syntax: DAYS 360 (start date, end date, [method])

Start Date and End Date are valid dates which represent the starting and ending dates. Method Optional logical value that specifies whether to use the U.S. (NASD) or European method in the calculation. False or omitted will give us the U.S. method, True will use the European method.

Remarks: If *start_date* is the last day of the month, both methods set the start_date to the 30th of the month, but if the end_date is the last day of the month the US method will change the date to the 1st of the next month, the European will change the date to the 30th.

EDATE - Returns serial number of the date that is a number of months away from a date

Syntax: EDATE (start date, months)

Start Date is a valid date that represents the starting date

Months is the number of months before or after start date. A positive value for months yields a future date; a negative value yields a past date

EDATE - Returns serial number of the date that is a number of months away from a date

Equation	Result	Notes
=EDATE(A1, 1)	39899 03/27/2009	A1 = 02/27/2009
=EDATE(A1, 10)	40174 12/27/2009	A1 = 02/27/2009
=EDATE(A 1, 100)	42913 06/27/2017	A1 = 02/27/2009
=EDATE(A 1, -1)	39840 01/27/2009	A1 = 02/27/2009

EOMONTH - Returns serial number for the last day *of the month* EOMonth => End of month

Syntax: EOMONTH(start date, months)

Months number of months before or after start_date. A positive value for months yields a future date; a negative value yields a past date.

MONTH- Returns the numeric value of the month in a valid date

Syntax: =MONTH (serial number)

Serial Number is the date of the month you are trying to find.

Syntax: NETWORK DAYS (start_date, end_date, holidays)

Start Date and *End Date* are valid dates.

Holidays is an optional range of one or more dates to exclude from the working calendar. The list can be either a range of cells that contains the dates or an array constant of the serial numbers that represent the dates

Equation	**Result**	**Notes**
=NETWORKDAYS(A1, A2)	5	A1 = 02/27/2009, A2 = 03/05/2009
=NETWORKDAYS(B1, B2)	1	B1 = 01/30/2009, B2 = 02/01/2009
=NETWORKDAYS(C1, C2)	12	C = 07/01/2009, C2 = 07/15/2009 1

=NETWORKDAYS(C1, C2, C3)	11	C1 = 07/01/2009, C2 = 07/15/2009, C3 = 07/04/1999
=NETWORKDAYS(D1, D2, D3:D4)	259	D1 = 11/15/2009, D2 = 11/15/2010, D3 = 01/01/2010, D4 = 07/04/2010

WEEKDAY - Returns the day of the week corresponding to a date

Syntax: =WEEKDAY (serial number, return_type)

Serial Number is the date of the day you are trying to find.

Return Type is a number that determines the type of return value. 1 or omitted sees the week as 1- Sunday, 7-Saturday. 2 sees the week as 1-Monday, 7-Sunday. 3 sees the week as 0-Monday, 6- Sunday.

Equation	**Result**	**Notes**
=WEEKDAY("12/15/1976")	4	Wednesday, December 15, 1976
=WEEKDAY(40622)	1	Sunday, March 20, 2011
=WEEKDAY(A1)	7	A1 = Saturday, October 28, 1905

WEEKNUM - Returns the day of the week corresponding to a date

Syntax: =WEEKNUM (serial number, return_type)

Serial Number is the date of the day you are trying to find.

Return Type is a number that determines the type of return value. **1 -** Week begins on Sunday. Weekdays are numbered 1 through 7; **2 -** Week begins on Monday. Weekdays are numbered 1 through 7.

Remarks: The WEEKNUM function considers the week containing January 1 to be the first week of the year.

Equation	Result	Notes
=WEEKNUM ("12/15/1976")	51	Wednesday, December 15, 1976
=sWEEKNUM (40622)	13	Sunday, March 20, 2011
=WEEKNUM(A1)	43	A1 = Saturday, October 28, 1905

WORKDAY - Returns a date that is a number of working days before or after a date

Syntax: WORKDAY (start_date, days, holidays)

Start Date is a valid date that represents the starting date.

Days is the number of non-weekend and non-holiday days before or after start_date. A positive value yields a future date; a negative value, a past date.

Holidays is an optional range of one or more dates to exclude from the working calendar. The list can be either a range of cells that contains the dates or an array constant of the serial numbers that represent the dates

YEAR- Returns the numeric value of the year in a valid date

Syntax: =YEAR (serial number)

Serial Number is the date of the year you are trying to find.

Equation	Result	Notes
=YEAR("10/15/1905")	5	
=YEAR(40622)	2011	Equivalent to 3/20/2011
=YEAR(A1)	1976	A1 = 12/5/1976

YEARFRAC - Returns fraction of the year of the number of whole days *between two dates* Syntax: YEARFRAC (start_date, end date, basis)

Start Date and *End Date* are valid dates which represent the starting and ending dates

Basis is the type of day count basis to use

0-US/NASD 30/360; **1-**Actual/Actual; **2-**Actual/360; **3-**Actual/365; **4-**European 30/360

Equation	Result	Notes	
=YEARFRAC (A1, A2)	1	A1 = 1/1/2009	A2 = 1/1/2010
=YEARFRAC (A1, A3)	1.013888889	A1 = 1/1/2009, A3	= 1/1/2010
=YEARFRAC (A1, A4)	0.455555556	A1 = 1/1/2009, A4	= 6/15/2009
=YEARFRAC (A1, A5)	0.538888889	A1 = 1/1/2009, A5	= 7/15/2009
=YEARFRAC (A1, A6)	13.53888889	A1 = 1/1/2009, A6	= 7/15/2022

HOUR - Returns the hour of a time value

Syntax: HOUR (serial number)

Serial Number the time that contains the hour you want to find.

Remarks: Times may be entered as text strings within quotation marks (for example, "6:45 PM"), as decimal numbers (for example, 0.78125, which represents 6:45 PM), or as results of other formulas or functions (for example, TIMEVALUE ("6:45 PM")).

Equation	Result	Notes
=HOUR(A1)	20	A1 = 8:28 PM
=HOUR(B1)	8	B1 = 8:28 AM
=HOUR(C1)	15	C1 = 15:43:12
=HOUR(D1)	17	D1 = 1/2/2003 17:52

MINUTE- Returns the hour of a time value
Syntax: MINUTE (serial number)

Serial Number the time that contains the hour you want to find.

Remarks: Times may be entered as text strings within quotation marks (for example, "6:45 PM"), as decimal numbers (for example, 0.78125, which represents 6:45 PM), or as results of other formulas or functions (for example, TIMEVALUE ("6:45 PM")).

Equation	Result	Notes
=MINUTE(A1)	28	A1 = 8:28 PM
=MINUTE(C1)	43	C1 = 15:43:12
=MINUTE(D1)	52	D1 = 1/2/2003 17:52

NOW - the serial number of the current date *and time*

Syntax: NOW()

SECOND - Returns the Seconds of a time value

Syntax: SECOND (serial number)

Serial Number the time that contains the hour you want to find.

Remarks: Times may be entered as text strings within quotation marks (for example, "6:45 PM"), as decimal numbers (for example, 0.78125, which represents 6:45 PM), or as results of other formulas or functions (for example, TIMEVALUE("6:45 PM")).

Hour is a number from 0 (zero) to 32767 representing the hour. Any value greater than 23 will be divided by 24 and the remainder will be treated as the hour value

Minute is a number from 0 to 32767 representing the minute. Any value greater than 59 will be converted to hours and minutes.

Second is a number from 0 to 32767 representing the second. Any value greater than 59 will be converted to hours, minutes, and seconds.

Equation	Result	Notes
=TIME(15, 3, 15)	3:03:20 PM	
=TIME(0, 0, 2000)	12:33:20 AM	2000 seconds = 33 min, 20 sec
=TIME(C1, C2, C3)	6:12:09 AM	

TIMEVAULE - Returns the serial number of a time

Converts a time that is stored as text to a serial number that Excel recognizes as a time. To view a time serial number as a time, you must apply a time format to the cell.

Syntax: TIMEVALUE (time text)

time text represents a time between 01/01/1900 and 12/31/9999 saved as text

Datedif Function:

The DATEDIF function in Excel is a versatile and powerful tool for calculating the difference between two dates in various units such as days, months, or years. This function, though not listed in Excel's function library, offers a straightforward way to measure time spans, making it invaluable for tasks such as determining age, tenure, or the duration between events.

Interval	Description
D	Number of Days
M	Number of Months
Y	Number of Years
YM	Number of months, not counting years
YD	Number of days, not counting years
MD	Number of days, not counting years and months

Learning Objectives

- Fill Handle
- Mathematical Operations
- Building an Equation
- Order of Operations
- Formatting Dates
- Formula View
- Absolute/Relative
- Functions.
- Naming Cells

Fill Handle Options

When you use the Fill Handle, you will notice a symbol appear in the right hand bottom corner of your newly filled cells. This icon () represents your AutoFill Options. If you put your mouse over the icon you will see a drop down arrow that will give you a list of your fill options.

The four basic Fill Options are:

- Copy Cells – Repeat the cells along the selection
- Fill Series – Follow pattern along the selection
- Fill Format Only – Repeat the format of the cells along the selection
- Fill without Formatting – Follow the pattern along the selection, but not the format
- Flash Fill – Fills based on a pattern you establish in the same column

If you use the **Fill Handle** on cells with dates you will notice even more options:

- Fill Days
- Fill Weekdays
- Fill Months
- Fill Years

Building an Equation

You can directly type in values, but that data stays constant. If you want to have the answers to your equations update as you change your data, you should use the cell addresses. You will see the cell addresses change colors so you can tell which ones are used in your equation.

Type in the exact cell address

Cells are labeled by their row and column headings. Rows are numbered and go horizontally across (rows of chairs) and columns are lettered and go vertically top to bottom (columns of a building). When we refer to the address of a cell, we use the column letter then the row number such as A1.

- Click in the cell where the answer will appear
- Press the Equal sign (=)
- Type in the cell address you want to use in your equation
- Accept the answer or press the next math operator (+, -, *, /, ^)

	A	B	C
1	**1**	**2**	**=a1+b1**

Use the keyboard to point to the cell address

When you first enter a character into a cell, you will be in "ENTER" mode. This includes when you type an equal sign. But if you press an arrow key on the keyboard after the equal sign, Excel does not move out of the cell. Instead you will be put into a "POINT" mode, and Excel will color coordinate the cell you are selecting.

- Click in the cell where the answer will appear
- Press the Equal sign (=)
- Press the arrow keys until you are on the cell you want to use in your equation
- Accept the answer or press the next math operator (+, -, *, /, ^)

Use the mouse to point to the cell address

The mouse and arrow keys are both "pointers". If you press the equal sign and then use the mouse to click on another cell, Excel will put you into a "POINT" mode, and place the address of the cell you clicked on in your equation.

Click in the cell where the answer will appear

- Press the Equal sign (=)
- Use the mouse to click on the cell you want to use in your equation
- Accept the answer or press the next math operator (+, -, *, /, ^)

Order of Operations

Microsoft Excel respects the Order of Operations.

(1) Parenthesis
(2) Exponents (raised to a power)
(3) Multiplication and Division
(4) Addition and Subtraction

Parentheses	*Exponents*	*Multiply*	*Divide*	*Add*	*Subtract*
Please	*Excuse*	*My*	*Dear*	*Aun t*	*Sally*

This means with an equation such as

= 5 + 3 * 2

Excel will do the multiplication before it does the addition. Would be 11. If you wanted the addition to happen first, you have to use parentheses: = (5 + 3) * 2

Giving us a result of 16.

In math, we use the brackets, such as {[(5+3) *(4-2)] / 2}

In Excel we ONLY use parentheses

= ((5+3)*(4-2))/2) result 8.

Remember to let Excel know you want it to calculate something; you have to start with an equal sign if you get stuck inside an equation, or confused by a mistake, press **Esc** to cancel and try again.

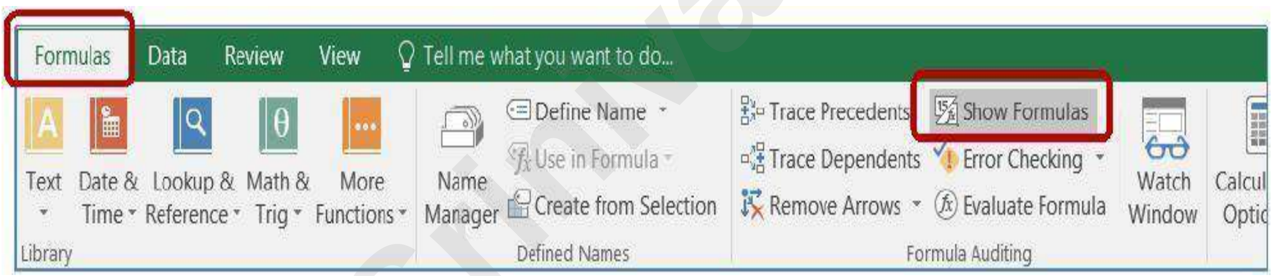
Formatting Dates

Dates and Times are technically numbers because they can be used in equations. Excel is able to do this because it has a "serial" number equivalent. Both columns of this table at the right of this page show the same value, but the dates in the second column have lost the date format. For a date, 0 = 1/1/1900. The numeric values in the second column represent how many days there are between 1/1/1900 and the date listed.

If your date loses its format you can modify it through the Number group or through the Format Cells Window. If you don't like the choices offered, you can custom build your date from the Custom section of the Number page in the Format Cells Window. Use M for months, D for days and Y for years. Excel is not case sensitive, upper or lowercase letters will work for these formats.

1993	3.00
/1995	6.00
002	9.00
2008	3.00
/2010	7.00
013	3.00
/2018	4.00

Our cells display the formula results. If we want to see the equation, we have to look at the formula bar or edit the cell. Sometimes it helps to see all the equations on a worksheet. The button for this option is on the Formula tab, in the Format Auditing group. The keyboard shortcut is Control-Tilde, ctrl ~ (the wavy line above Tab, below Esc on your



You can also turn this on through the Excel Options. From the File Menu, choose **Options**. On the **Advanced** tab, under the **Display Options for this Worksheet** group, check the box next to **Show formulas in cell instead of their calculated results**.

	A	B	C	D	E	F	G	H	I
1	Big City Store		Date:	10/8/2016					
2	Sales Report								
3									
4	Items	Price	Qty	SubTotal					
5	AAA	$ 123.00	987	$ 121,401.00					
6	BB	$ 456.00	654	$ 298,224.00					
7	C	$ 789.00	321	$ 253,269.00					
8				$ 672,894.00					

Calculated Results (normal view)

Formula View

	A	B	C	D	
1	Big City Store		Date:	42651	
2	Sales Report				
3					
4	**Items**	**Price**	**Qty**	**SubTotal**	
5	AAA	123	987	=B5*C5	
6	BB	456	654	=B6*C6	
7	C	789	321	=B7*C7	
8				=SUM(D5:D7)	

The formula view will stretch out the columns and all the numbers appear to have lost their format, including the dates and some alignments. Be careful adjusting the column widths, if you shrink them in the formula view, it will proportionally shrink in the calculated results view.

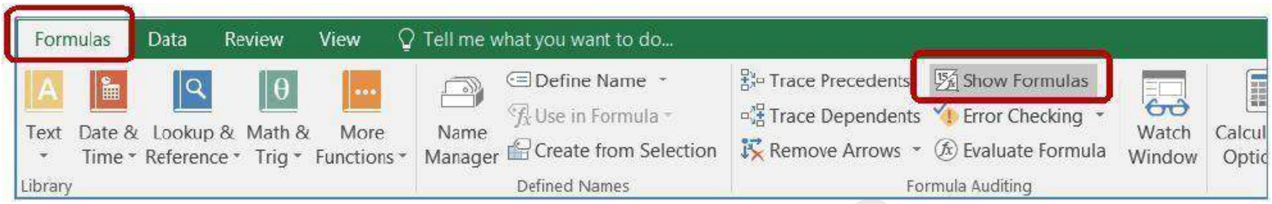
You can continue working in this view and print out the sheet with all the formulas showing. If you're going to print this view, I recommend turning on the print headings from the Page Layout tab.

Friday, February 1, 2002

dddd, mmmm d, yyyy

Formula View

Our cells display the formula results. If we want to see the equation, we have to look at the formula bar or edit the cell. Sometimes it helps to see all the equations on a worksheet. The button for this option is on the Formula tab, in the Format Auditing group. The keyboard shortcut is Control-Tilde, ctrl ~ (the wavy line above Tab, below Esc on your keyboard).



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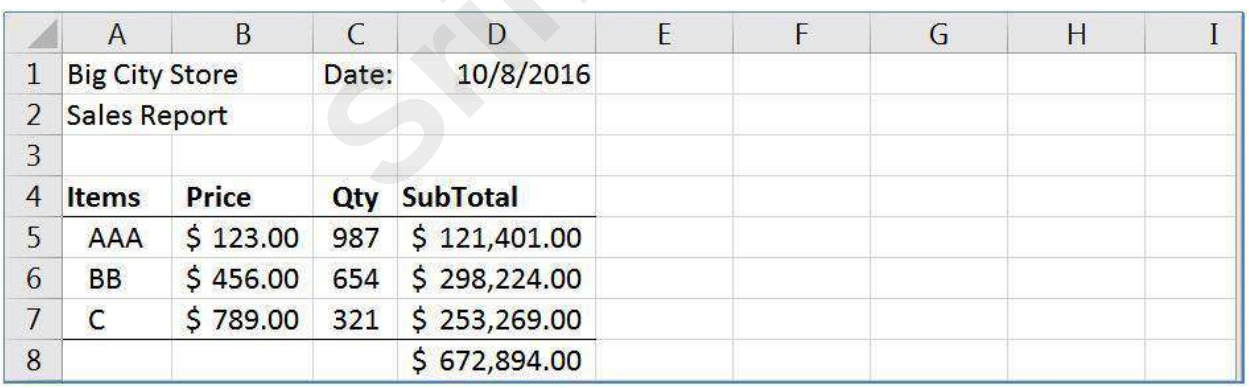
Calculated Results (normal view)



	A	B	C	D	E	F	G	H	I
1	Big City Store		Date:	10/8/2016					
2	Sales Report								
3									
4	**Items**	**Price**	**Qty**	**SubTotal**					
5	AAA	$ 123.00	987	$ 121,401.00					
6	BB	$ 456.00	654	$ 298,224.00					
7	C	$ 789.00	321	$ 253,269.00					
8				$ 672,894.00					

The formula view will stretch out the columns and all the numbers appear to have lost their format, including the dates and some alignments. Be careful adjusting the column widths, if you shrink them in the formula view, it will proportionally shrink in the calculated results view.

You can continue working in this view and print out the sheet with all the formulas showing. If you're going to print this view, I recommend turning on the print headings from the Page Layout tab.

Absolute/Relative

When you create an equation in Excel using cell addresses, Excel sets up the equation to have a **relative** reference. When you are using the Fill Handle or the Copy and Paste features the equation result is relative If this equation is copied into cell C2, or the Fill Handle is used to drag the equation down to C2, Excel will give you this result:

1	5	6	=A1+B1
2	12	4	=A2+B2

Since the equation was moved down, between rows, only the row number changes. If instead we moved the equation across, the row numbers will remain the same, but the column numbers will change:

The addresses in the equation are **relative** to where the answer is positioned. The equation in cell C1 of the table above states 'take the value of the cell that is two to the left from this cell and add it to the value of the cell that is one to the left from this cell'. When we fill or copy the cell over or down, the basic equation stays the same.

If you do not want a number to move relatively you can make it **absolute** by using dollar signs ($) in the equation. The **F4** button on the keyboard will place the dollar sign characters in for you while you are in Enter, Edit, or Point mode. (Think F4 **->** FORCE)

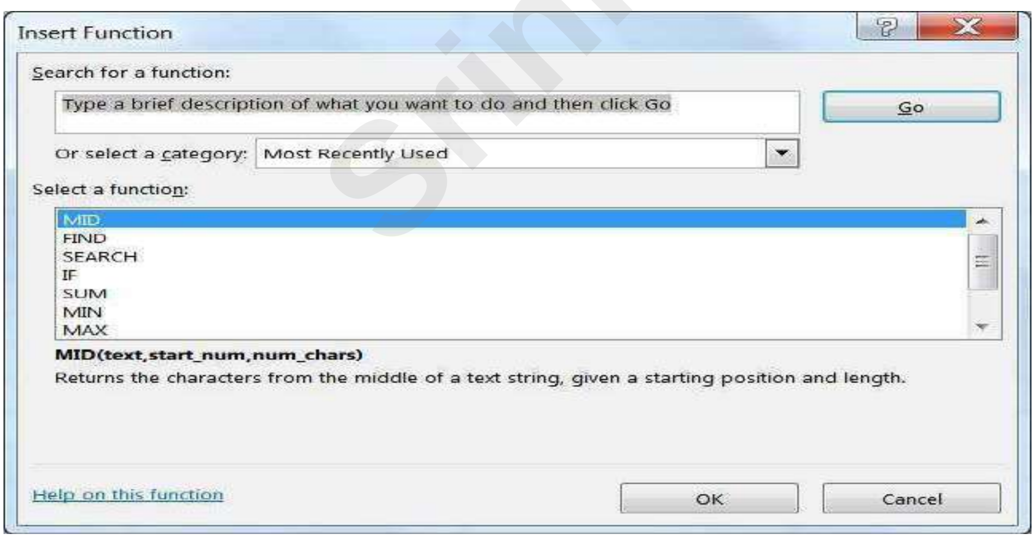
=$A$1 **-** Locks the reference into Cell A1

=$A1 **-** Locks the reference into Column A, but will allow the row number to change =A$1 **-** Locks the reference into Row 1, but will allow the row number to change

	A	B	C	D	E	F	G	H	I
1	5	6	=$A$1+B1		SubTotals:	123	456	789	
2	12	4	=$A$1+B2		TaxRate:	6.5%			
3					Taxes:	=F1*$F$2	=G1*$F$2	=H1*$F$2	

Functions

Microsoft Excel has several built in functions. To insert a function, click the **Insert Function** button  on the Formula Bar, or the **Insert Function** option from the **Formulas Tab**.

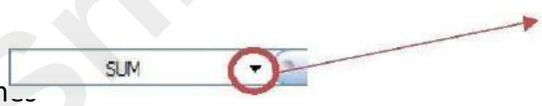


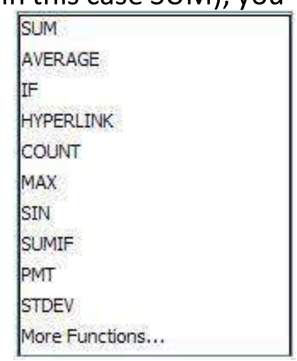
From here you can request a function to perform a particular task and Excel will make suggestions for you. If you **Search for a function:** Excel will return a **Recommended** category, offering all the functions it thinks might help in your search.

By default, the first category is a list of **Most Recently Used** functions. To see all the functions built into Excel, you can choose **All** from the **Select a category**: list.

The bottom of this window displays a description of the selected function. Each choice will show an example arrangement of the function, the arguments, and a description of what that function should do. If you need more information, click on the Help on this function option in the bottom left corner. If you have found the function you would like to use, select it and click **OK**.

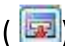
An easier way to access the list of **Most Recently Used** functions is to press the equal sign on the keyboard, as if you were going to type an equation. The name box, that displays which cell you are in, changes to the last function that was used. When you click on the arrow next to the listed function (in this case SUM), you will see a list of list of **Most Recently Used** functions.

becomes




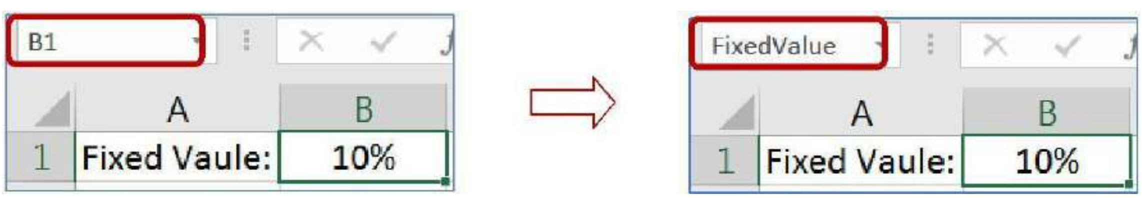
If the function you desire is not on the Most Recently Used list, chose the More Functions… option at the bottom of the list and you will get the above Insert Function dialog box.

Either selection method will open the Function Arguments window

The function name is listed in the top left corner and the description is across the bottom. There are blanks in the middle of the screen for the arguments of the statement. You can type in the cell addresses, move the window out of the way to try to select the addresses or let Excel help you move the window by using the **Collapse** or "go out and get it button" (). This button will *collapse* your Function Arguments window so you can select the data you wish to use as an argument in this function. Once you have chosen your desired data either press **Enter** or click on the **Expand** button () to return to the full window.

In the sample above, you can see we can **Sum** more than one number/set of numbers. As soon as you click into **Number 2** a **Number 3** will appear. The description tells us this will allow up to 255 arguments (number ranges) to sum.

Across the bottom of this window we can see a **Formula Result =**. This will show us the running total as we add in each part of the equation. Notice there is also an **=number** at the end of each argument line. This will give you a piece-by-piece result for each argument. This is especially helpful when using the logic functions, such as **If**.



Once a name is defined, you can use it in your equations. =B12*$B$1 means the same as

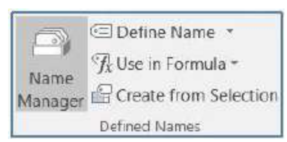
=B12*FixedValue

When you use the fill handle or the Copy/Paste feature, the B12 will change to be relative to the new location, but FixedValue will remain throughout, always pointing to cell B1.

This same method can be used for a range of cells. Select the desired range, click within the name box, erase the current cell address, type the name of the range, press Enter.

NOTE There are some limitations in naming. You cannot use many special characters such as the hyphen (**-**), and the name must be all one word, no spaces. In the example above we used capitalization to show multiple words, you can also use the underscore character (_).

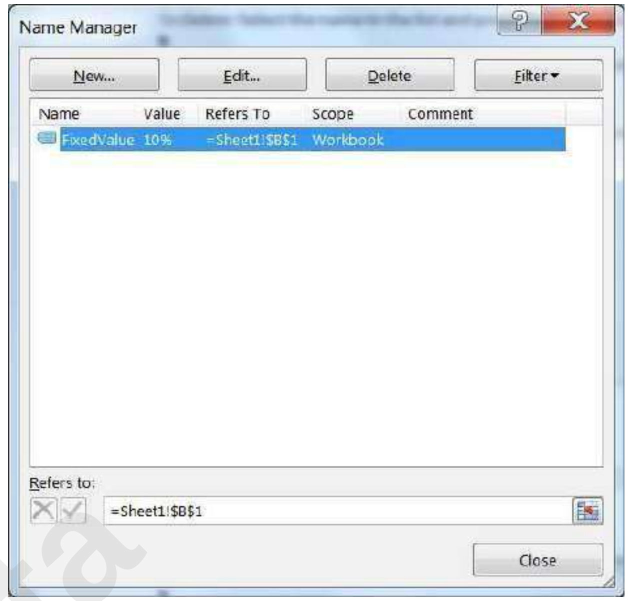
Defined Names group on the **Formula** Tab:



- **Define a Name** – create a new name
- **Use in Formula –** choose from a list of existing names, this can be used to begin a new formula or to add a name to a formula you are building.
- **Create from Selection –** make a new name based on a group of selected cells

Use the **Name Manager** to modify a Defined Name.

- *New:* Create a new Name
- *Edit…:* Rename, add a comment, change cell reference
- *Delete:* Select the name in the list and click **Delete** to remove it
- *Filter:* Filter the list of names to help you find the one you need.



To Redefine: Select the name in the list, erase the Refers to blank and choose your new range of cells. Click the Check to accept.

F3 is the keyboard shortcut to let you pick a name to use in your equation.

Creating Equations with Cell addresses (pg 3)

- Clear cell C1, type =a1+b1 O Not case sensitive

➢ Notice the color coding

➢ Accept data, should get a 3

➢ Change A1 to 5, accept to see answer in C1 change

-**USING KEYBOARD**	-**USING MOUSE**
[redacted]	
➢Go to C2	➢ Go to C3
➢press = sign	➢ press = sign
➢press arrow two times	➢ click on cell A3
➢press + sign	➢ press the + sign

- press ⍰ arrow once
- enter to accept

- click on cell B3
- click on check to accept

Basic math symbols (pg 3)

- Add +, Subtract -, Multiply *, Divide / O All found on the Num Pad
- Build these equations
  - In Cell C4 add A4 and B4: =A4+B4
  - In Cell C5 subtract A5 and B5: =A5-B5
  - In Cell C6 multiply A6 and B6: =A6*B6
  - In Cell C7 divide A7 and B7: =A7/B7 Remember the Order of Operations (pg 4)

- Parenthesis
- Exponents/Powers
- Multiplication/Division
- Addition/Subtractions
- Build these equations

	A	B	C
1	1	2	3
2	2	4	6
3	3	6	9
4	4	8	12
5	5	10	-5
6	6	12	72
7	7	14	0.5
8	8	16	17
9	9	18	25
10	10	20	

O In Cell C8 type =2+3*5 O In Cell C9 type =(2+3)*5

Date not Math?

- Put today's date in cell C10 O i.e. 10/5

O Note Excel sees it as a date not as an equation (10 divided by 5), Why? Because there is no Equal sign! 10/5 is October 5th, =10/5 is an equation with a result of 2

Totals Preview

- Select Column A

- Look in the status bar in the bottom right side of the window. When you select more than one cell, Excel will automatically calculate the Average, Count, and Sum of your selection.

Formatting

- Format column A with dollar signs

    O Select Column A, from the right-click or Home tab, choose the $ format

- Format column B to be centered
    - Select Column B, from the right-click or Home tab, choose Align Center
- Edit Cell C1 (double-click in cell)
    - When you accept, Excel recalculates answer and the format changes
- Format Column C with dollar signs

Note format in cell C10, no longer a date

Reformat Date

- Click on Cell C10 (we don't want the whole column)
- Choose **Short date** from the list of number formats, in the Number group on the Home Tab
- Choose "More Numbers" to view different date formats
- Choose the 14-Mar format to return to the original date formatting To custom build a data

  format see pg 4

Formula View (pg 5)

- Toggle cell data with ctrl-` (Ctrl-~… Wavy line above tab, below Esc)
- We can see all the formulas but NO formatting

- Toggle back to answer view

Create Data Table

- Click on the Plus sign next to Sheet 2 to crea

- Zoom to 150%

- Type in the table shown here

	A	B	C
1	Big City Store		
2	Sales Report		
3			
4	Items	Price	Qty
5	A	123	987
6	BB	456	654
7	CCC	789	321

Format Data Table

	A	B	C	D
1	Big City Store			
2	Sales Report			
3				
4	**Items**	**Price**	**Qty**	**SubTotal**
5	A	$ 123.00	987	
6	BB	$ 456.00	654	
7	CCC	$ 789.00	321	

- Bold Row 4

- Center Column C

- AutoFit Column C

- Dollar Format the Prices, B5 through B7

- In Cell D4 type: SubTotal O Should be bold already

Calculate SubTotals

- Enter First SubTotal
  - In Cell D5 Type: = B5*C5

$121,401.00

- Each total is going to be the price of the item, times the Qty sold of the item. If we had 3,000 records, we would not want to type the equation 3,000 times

  - Our equation is a pattern, "the cell two away, times the cell next to me"
  - Anywhere you copy or fill the equation, the pattern will follow

- Use Fill handle to pull down this "pattern"
- Switch to Formula view (Ctrl ~) to see results and switch back to normal view

	A	B	C	D
1	Big City Store			
2	Sales Report			
3				
4	**Items**	**Price**	**Qty**	**SubTotal**
5	A	123	987	=B5*C5
6	BB	456	654	=B6*C6
7	CCC	789	321	=B7*C7

Calculate Taxes

- In Cell D2 type: Tax Rate
- In Cell E2 type: 10%
- In Cell E4 type: Taxes
- In Cell E5 type: =D5*E2
  - Our pattern is Subtotal times Tax Rate Use fill handle to pull down equation
  - **Don't Panic,** these answers are supposed to look weird

Absolute vs Relative (pg 6)

What Happened?

Pattern "Cell next to me * Cell three above me"

As we move DOWN the pattern continues but it's wrong for this equation Erase the answers

In Cell E5 type the same equation: =D5*E2

Before you accept, press the **F4** button on the keyboard to lock cell address E2

=D5*$E$2

You can type in the $ signs, but **F4** is often faster Think **FORCE** for the F4 button

D5 is relative (always cell next to me)

$E$2 is absolute (always cell E2)

Use fill handle to pull down the equation

Calculate Totals

- In Cell F4 type: Total

In Cell F5 type: = D5+E5

- Use fill handle to pull down the equation

Calculate Grand Total

In Cell E8 Type: Big City Store Grand Total O From the Home tab, Align Right

If it won't let you, make sure you have accepted your entry (enter or ▯)

Click in Cell F8

Use AutoSum button Sigma **- ∑** in the **Editing** group on far right of the **Home** tab

O =SUM(F5:F7)

O In English, this reads "Calculate the Sum of F5 through F7"

	A	B	C	D	E	F
1	Big City Store					
2	Grand Total			Tax Rate:	10%	
3						
4	**Items**	**Price**	**Qty**	**SubTotal**	**Taxes**	**Total**
5	AAA	$ 123.00	987	$ 121,401.00	$ 12,140.10	$133,541.10
6	BB	$ 456.00	654	$ 298,224.00	$ 29,822.40	$328,046.40
7	C	$ 789.00	321	$ 253,269.00	$ 25,326.90	$278,595.90
8					Big City Store Grand Total	=SUM(F5:F7)

Note: When you click the AutoSum button Excel looks for numbers above the current cell. If it can't find a number, it will look to the left. The AutoSum button has a drop down menu to do quick calculations for Sum, Average, Max, Min, and Count.

Confirm your answer

Select cells F5:F8

Look at the status bar in the bottom right of the window to see the Average, Count, and Sum
O Sum should be $740,183.40

Change the fill color for cells F5:F8 to yellow These are now our "Yellow Numbers"

Set up for Functions

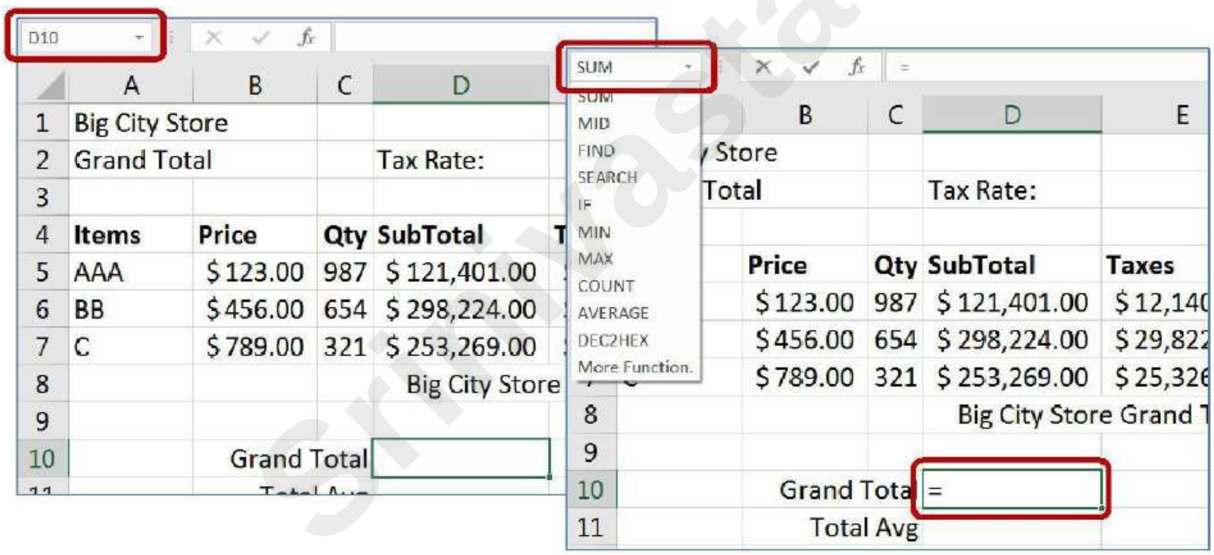
- In Cell C9 type: Grand Total
- In Cell C10 type: Total Avg
- In Cell C11 type: # of Items
- In Cell C12 type: Largest Sale
- In Cell C13 type: Smallest Sale
- Right align all titles in cells C9:C13
- click in cell D10

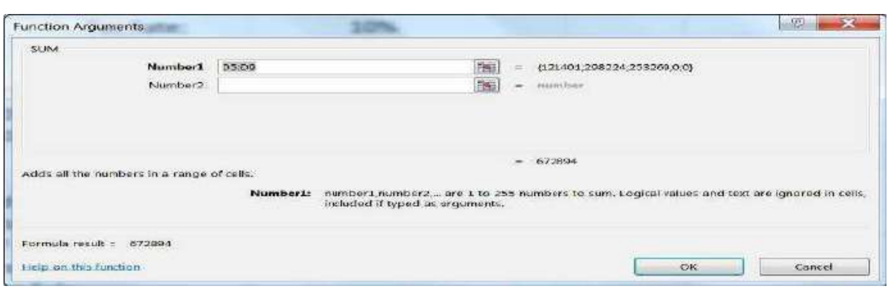
O We can tell we are in cell D10 because of the Name Box

- Press the Equal Sign =

O The Name Box has changed to a list of Most Recently Used Functions O Since the last function used on this computer is SUM, it's listed first

O This will open the **Function Arguments** window





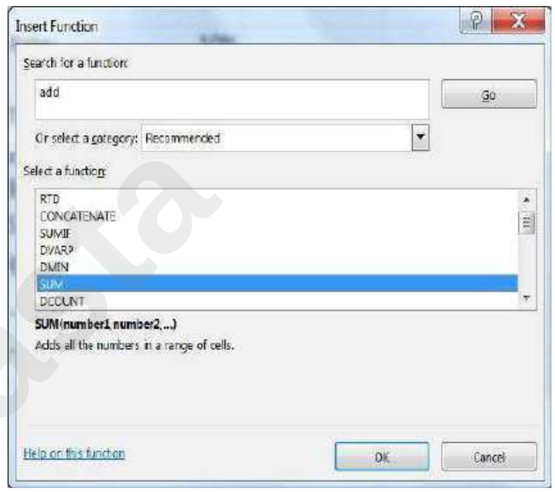
➢ Clear the contents in the Number1 box

➢ Move window so you can see the yellow numbers

➢ To move: click in the title bar or any grey blank space and drag

➢ Once you can see the yellow numbers, click in the first one, cell F5

➢ Drag to the last yellow number, F7



➢ Selection should be F5:F7

- Sum should be $740,183.40

- Clear Cell D10

- In Cell D10 Type: =

- Open the List of Recent Functions from the Name Box

- Click on the last option, More Functions… O If you click on a function name, Excel gives a brief description.

O Click on the Help on this function link for a Help article about this function, usually with examples.

Erase the "Type a brief description of what you want to do and then click Go" from the Search box. Type in ADD and click the GO button.

- Select Sum from the list and click OK
- This opens the same Function Arguments window
- Choose your Yellow Numbers and Click OK
- Sum should be $740,183.40
- Clear Cell D10

Editing a function (pg 8)

Press the Equal Sign

Choose SUM from function list Click OK

Answer should be WRONG

Why? Because we don’t have the yellow numbers

Click on the Edit Function box in front of the formula bar 

This opens the **Function Arguments** window Choose the Yellow Numbers and Click OK

Other Basic Functions

- Click in Cell D11
- Press Equal Sign =
- Click on List of Functions
- Choose Average (aka Arithmetic Mean)
- Choose Yellow Numbers and Click OK
- Using Yellow numbers for each O In cell D12 calculate the Count O In cell D13 calculate the Max

O In cell D14 calculate the Min

Final Result

	A	B	C	D	E	F
1	Big City Store					
2	Grand Total			Tax Rate:	10%	
3						
4	Items	Price	Qty	SubTotal	Taxes	Total
5	AAA	$ 123.00	987	$ 121,401.00	$ 12,140.10	$133,541.10
6	BB	$ 456.00	654	$ 298,224.00	$ 29,822.40	$328,046.40
7	C	$ 789.00	321	$ 253,269.00	$ 25,326.90	$278,595.90
8				Big City Store Grand Total		$740,183.40
9						
10		Grand Total		$ 740,183.40		
11		Total Avg		$ 246,727.80		
12		# of Items		3		
13		Largest Sale		$ 328,046.40		
14		Smallest Sale		$ 133,541.10		

	A	B	C	D	E	F
1	Big Cit					
2	Grand			Tax Rate:	0.1	
3						
4	Items	Price	Qty	SubTotal	Taxes	Total
5	AAA	123	987	=B5*C5	=D5*$E$2	=D5+E5
6	BB	456	654	=B6*C6	=D6*$E$2	=D6+E6
7	C	789	321	=B7*C7	=D7*$E$2	=D7+E7
8					e Grand Total	=SUM(F5:F7)
9						
10			Grand Total	=SUM(F5:F7)		
11			Total Avg	=AVERAGE(F5:F7)		
12			# of Items	=COUNT(F5:F7)		
13			Largest Sale	=MAX(F5:F7)		
14			Smallest Sale	=MIN(F5:F7)		

Reset for next lesson

Clear the calculated taxes, Cells E5:E7

Notice how it carries through to all the answers Clear the Grand Total in Cell F8

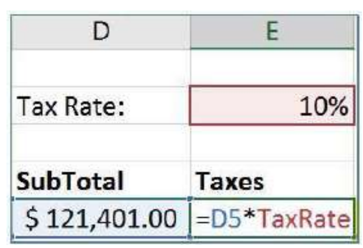
Clear the five functions you created in Cells D10:D14 Press Ctrl-Home to return to the top of the worksheet

Naming a Cell (pg 9)

- Click in cell E2 (10%)
- Click in Name box
- Erase E2 in Name box
- Type **TaxRate**
- No Spaces, No Hyphens, capitalization doesn't matter

- Press Enter and TaxRate should still be in the box
- Click anywhere else, see cell address in name box
- Click on E2 (10%), see "TaxRate" in Name Box
- Click anywhere else
- Click arrow next to Name box and choose TaxRate
- Should jump to cell E2
- Go to Sheet 1, Click on Name box menu, Choose TaxRate
- Should jump to Worksheet 3, Cell E2
- We have made a "bookmark" in our workbook

Using named cell in equations



- In Cell E5 type =D5*
- Click on cell E2
- Instead of E2 we see **TaxRate**
- TaxRate is always TaxRate, no ($) locks needed

➢ Accept and fill down for E6 and E7

Naming Ranges

- Select Yellow numbers

- Type **Total** in the Name box and press Enter

Using named cell in functions

- Click in cell F8

- Use AutoSum button Sigma **-** ∑ O
  =Sum (Total)

- In Cell D10 type: =

- Choose Sum from the List of Recently Used Functions

- Select your Yellow numbers, you should get Total

- Click OK

Typing in Functions

Clear cell D10

In Cell D10 Type: =SU

See list of functions that start with an SU Once you see the word **SUM** Select it Double-click or press

Tab (to GRAB!) Continue to type: TO

See list of all functions and Names that start with TO Double-click **Total** *or tab*

Press Enter to accept

Try the same for the other four functions. Type an equal sign, start typing the name of the function. Once you see it, double-click, or tab to grab the selection. Then type the first few letters of total. Double-click or tab to grab. Enter to accept.

Cell D11 the Average Cell D12 the Count

Cell D13 the Max
Cell D14 the Min

Answers will be the same as the ones on the top of Page 19 of this handout.

	A	B	C	D	E	F
1	Big City					
2	Grand '			Tax Rate:	0.1	
3						
4	**Items**	**Price**	**Qty**	**SubTotal**	**Taxes**	**Total**
5	AAA	123	987	=B5*C5	=D5*TaxRate	=D5+E5
6	BB	456	654	=B6*C6	=D6*TaxRate	=D6+E6
7	C	789	321	=B7*C7	=D7*TaxRate	=D7+E7
8					re Grand Total	=SUM(Total)
9						
10			and Total	=SUM(Total)		
11			Total Avg	=AVERAGE(Total)		
12			‡ of Items	=COUNT(Total)		
13			rgest Sale	=MAX(Total)		
14			allest Sale	=MIN(Total)		

Working with Ranges of Data

If we insert new data it must go inside the range

Insert two rows, one above item "C", one below item "C" O Right-click on Row Number 7, INSERT

Right-click on Row Number 9, INSERT Enter in data

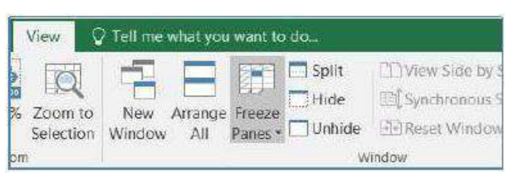
- In Cell A7 Type: D
- In Cell B7 Type: 159

- In Cell C7 Type: 951
- In Cell A9 Type: E
- In Cell B9 Type: 357
- In Cell C9 Type: 753

- Select Cells D5:F5
  - The first Sub Total through the First Total
  - The numbers should not change, just be selected

- Use fill handle to pull down through all the yellow numbers

4	Items	Price	Qty	SubTotal	Taxes	Total
5	AAA	$ 123.00	987	$ 121,401.00	$ 12,140.10	$ 133,541.10
6	BB	$ 456.00	654	$ 298,224.00	$ 29,822.40	$ 328,046.40
7	D	$ 159.00	951	$ 151,209.00	$ 15,120.90	$ 166,329.90
8	C	$ 789.00	321	$ 253,269.00	$ 25,326.90	$ 278,595.90
9	D	$ 357.00	753	$ 268,821.00	$ 26,882.10	$ 295,703.10
10				Big City Store Grand Total		$ 906,513.30
11						
12		Grand Total		$ 906,513.30		
13		Total Avg		$ 226,628.33		
14		# of Items		4		

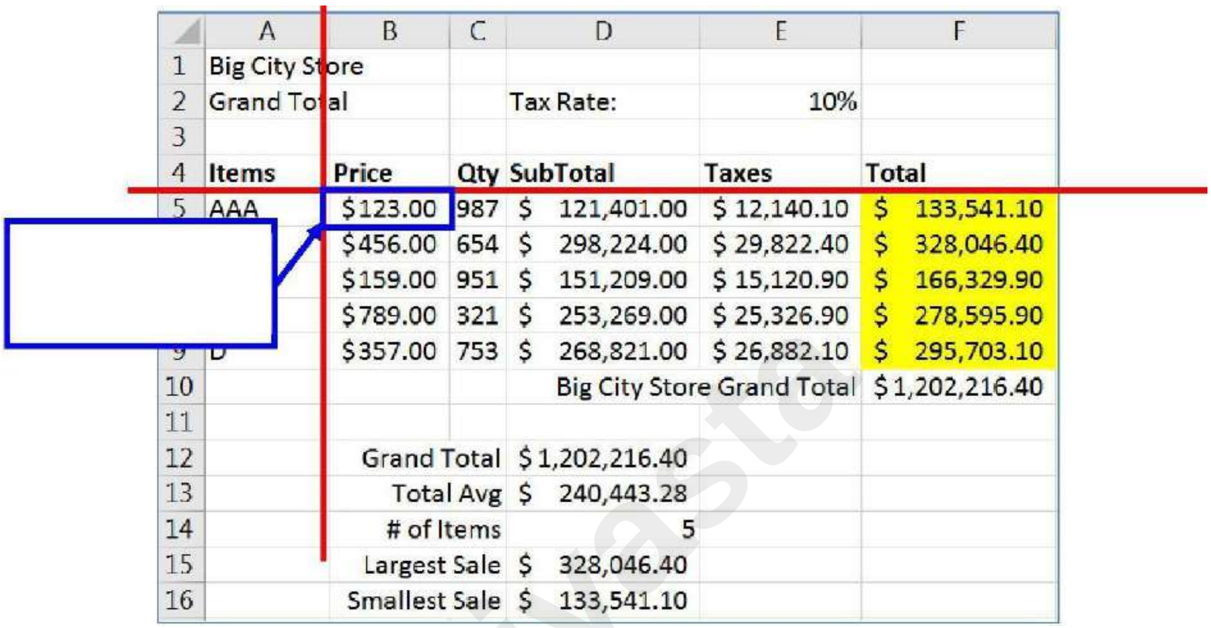


Freeze Panes

- Click inside Cell B5

- From the View tab in the Window group, choose Freeze Panes

  This will lock the first four rows and the first column into place, so as you scroll through the worksheet you can always cee the content in those cells.



	A	B	C	D	E	F
1	Big City Store					
2	Grand Total			Tax Rate:	10%	
3						
4	Items	Price	Qty	SubTotal	Taxes	Total
5	AAA	$123.00	987	$ 121,401.00	$ 12,140.10	$ 133,541.10
		$456.00	654	$ 298,224.00	$ 29,822.40	$ 328,046.40
		$159.00	951	$ 151,209.00	$ 15,120.90	$ 166,329.90
		$789.00	321	$ 253,269.00	$ 25,326.90	$ 278,595.90
9	D	$357.00	753	$ 268,821.00	$ 26,882.10	$ 295,703.10
10				Big City Store Grand Total		$1,202,216.40
11						
12		Grand Total		$1,202,216.40		
13		Total Avg		$ 240,443.28		
14		# of Items		5		
15		Largest Sale		$ 328,046.40		
16		Smallest Sale		$ 133,541.10		

Freeze panes above and to the left of THIS cell. (B5)

If Function (if there is time)

- In Cell G4 type: Continue
- Draw Logic Tree
- If our Total for this item is greater than $250,000, then yes we want to continue
- If our Total for this item is less than $250,000, then no we don’t want to continue

IF Item Total > 250000

T *F*

Yes *No*

- In Cell G5 type: =

O From the list of functions choose IF and fill in the parts from our logic tree

- Use fill handle to pull down the equation

- Edit the function in cell G4 to 280000

- DON'T FORGET TO: Use fill handle to fill down again

Charts

A chart helps you display your data into a graphical representation. There are many types of charts, but in this class we'll focus on simple column, line, and pie charts. There are examples of other charts near the end of this handouts.

The first thing to know is the data has to be organized so Excel can understand what you are trying to chart. Excel will chart your data selection or your connected data range. As long as there are no blank columns and no blank rows within your dataset, you can skip selecting the cells.

Value Axis

Chart Title

Here is a dataset we will use in class:

Item	1st Qtr	2nd Qtr	3rd Qtr	4th Qtr
Pants	456	489	423	468
Shoes	498	435	472	436
Socks	128	168	157	138
Blouses	579	498	531	589
Hats	126	129	123	119

This is a structured collection of related data set in a table format. When plotted onto a clustered column chart, like the one shown above, the titles in the first column of the dataset appear along our category axis. The titles in the first row appear within the legend. The values are represented by the height of each column.

Line charts are usually set up to go across a period of time, think *Time Line*. For this chart I've used the **Switch Row Column** tool so we can see the trend of the sales through the year. In this case our first column titles appear in the legend, and the first row of titles appears in our category axis.

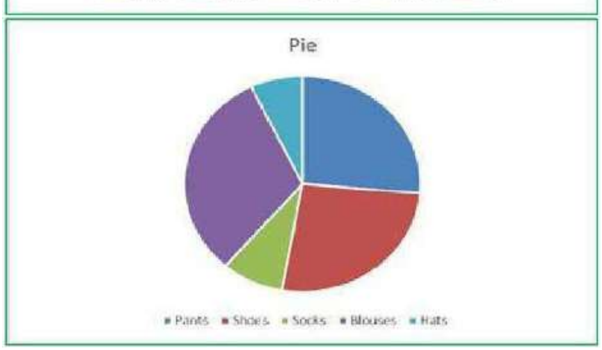
<u>Pie charts</u> are usually created to display the breakdown of the total values within the whole. Pie charts can only be based on one set of data. When you try to create one with the above dataset, you will only see the first value set

appear within the chart. If you want to go to an extreme and have all four quarters show, try using a *Doughnut* chart.

Creating a Chart

To create a chart make sure your cursor is in the dataset you would like to plot. If you want a subset of the dataset, select that portion. You can use your Ctrl key to add to a current selection.

You will find the **Charts** group on the **Insert** tab. Click on any small chart button to see a list of possible charts.



If you are unsure of the best chart option for your data use the **Recommend Charts** button. It will open the Insert Chart window shown here.

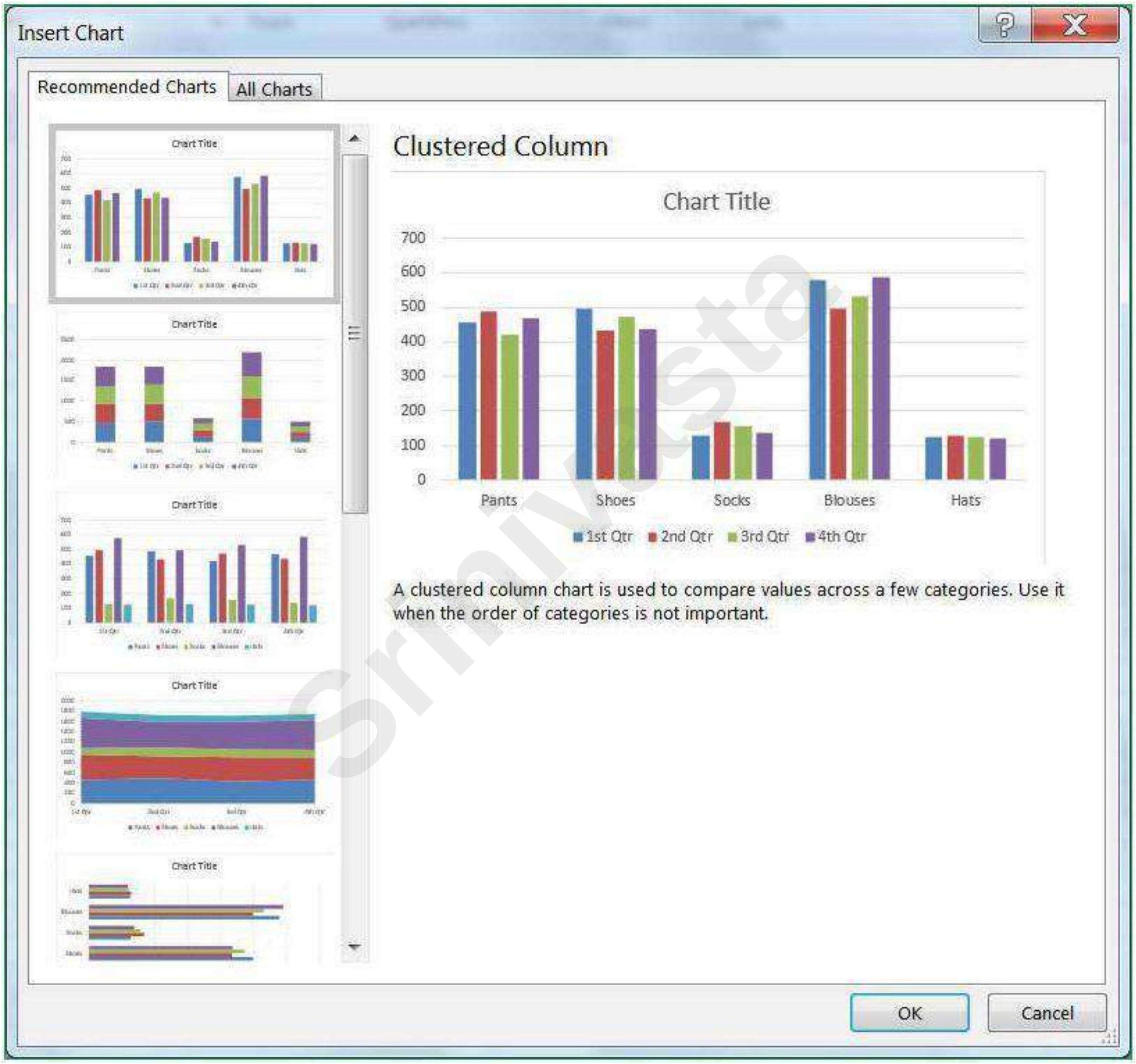
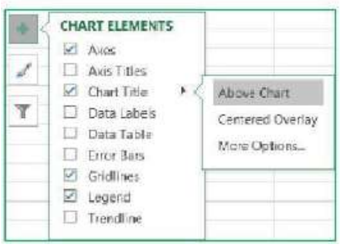


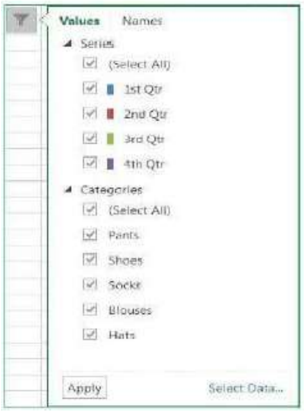
Chart Tools

When you select a chart, three buttons appear along the right side of the selection.



The plus sign is the Add Chart Element button. This option is used to add and remove different pieces of your chart. The list of options will vary depending on the type of chart. For example, a pie chart will not have a set of category axis titles. As you hover your mouse over each option, you will see a small arrow head pointing to the right. This will open another menu with more detailed choices. Each menu also has a "More Options..." button which will open a Format Pane on to customize each chart element.

There is a *Chart Style Gallery* and a *Colors* menu on the *Design* tab, but the **Chart Styles** button, the paint brush next to the chart, offers the same options.



If you are patient while you hover over each option, Excel will provide you with a Live Preview of the result. The **Color** options are available at the top of the menu.

The third button is a funnel. This is a Chart Filters button.

The Values group allows you to add and remove data points from the chart.

The Names page allows you to change the labels that appear in the legend (series) and axis titles (category).

The **Select Data…** option at the bottom of the window opens the same window as the *Select Data* button on the *Design* tab. From there you can change or adjust the range of cells used to create this chart.

Chart Tool Tabs

When a chart is selected two chart tool tabs appear at the end of the ribbon, **Design** and **Format**.

<u>Design Tab</u>

1. **<u>Add Chart Element</u>** – A menu of chart elements that can be added or removed to the chart. Each option will have a expand arrow at the end of the element name that will provide specifics and a **More**

Options button to open the Format Pane. This is the same as the Add Chart Element button that appears next to the selected chart.

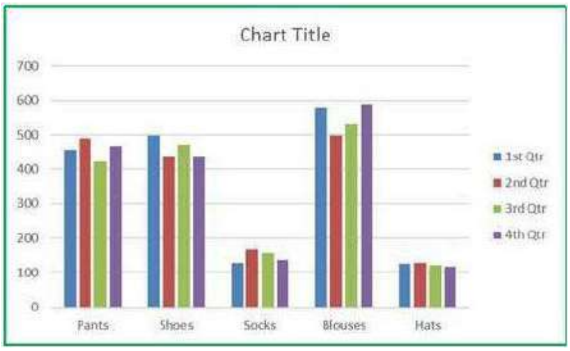
2. **Quick Layout** – A variety of layouts that offer suggested views and choices that adjust the chart elements such as adding a title, varying the space between columns, and moving the legend.

3. **Change Colors** – Different color that can be applied to your chart. Changing the Theme on the *Page Layout* tab will give you a different sets of colors.

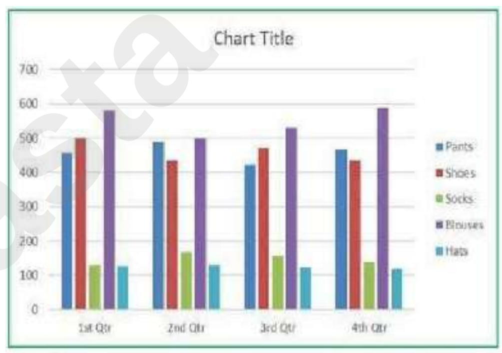
4. **Chart Style Gallery** – Different chart styles that can be applied to your chart. Because Excel automatically adjusts the Ribbon to fit on your screen, your copy of Excel may show less options than the picture above. Use the scroll arrows and open menu buttons at the right side of the gallery for more.

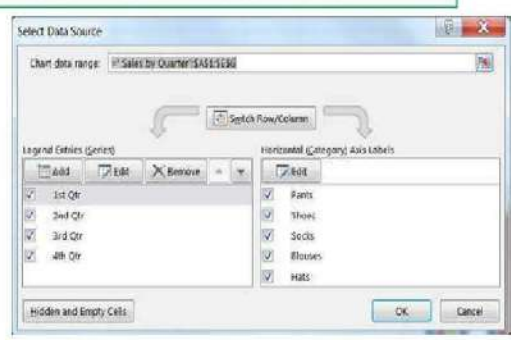
ITEM	1ST QTR	2ND QTR	3RD QTR	4TH QTR
PANTS	456	489	423	468
SHOES	498	435	472	436
SOCKS	128	168	157	138
BLOUSES	579	498	531	589
HATS	126	129	123	119

5. **Switch Row/Column** – Changes the direction the chart looks at the data. In our column chart, each column is plotted on the chart, when we **Switch** each row is plotted. We are swapping the category labels with the legend labels.



6. **Select Data** – Opens a Select Data Source window where you can customize the source of the chart data, even edit the labels. Use this window to reorder your legend and change how Line charts deal with blank cell values.





7. **Change Chart Type** – Opens Insert Chart window where you can change to other chart types. If you have multiple series you can change each to be different chart type by choosing the ***Combo*** chart type from the bottom of the left pane.

8. **Move Chart** – By default when you create a chart it is placed on the same worksheet as your data set. You can move the chart to its own worksheet or to any existing worksheet with the workbook.

Format Tab

Current Selection

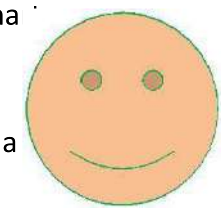
a. **Chart Elements** – This box shows the currently selected Chart Element, and the menu provides a list of the major chart elements. Choose an item from this list to select that element.

b. **Format Selection –** Opens the Format Pane based on the current selection shown in the Chart Elements box.

c. **Reset to Match Style** – Changes the current selection to match the original style of the chart.

2. Insert Shape

a. **Shape Gallery – Use** this gallery to find a shape such as a block arrow to add to your cha



b. **Change Shape** – Use this tool to change the current shape to a different one, perhaps a a rounded rectangle.

3. Shape Styles

Style Gallery – Different shape styles, options will vary based on the current selection.

Shape Fill – Menu of the most common fill colors and options, such as pictures and textures. For more options, open the Format Pane.

Shape Outline – Menu of the most common outline colors and options, such as dashes and arrows. For more options open the Format Pane.

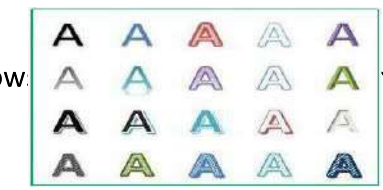
Shape Effects – Menu of the most common shape effects, such as shadows. For more options open the Format Pane.

4. WordArt Styles

WordArt Gallery – Different WordArt styles

Text Fill – Menu of the most common fill colors and options, such as p

Text Outline – Menu of the most common outline colo hes and line weight.



Text Effects – Menu of the most common Text effects, such as shadow ne Format Pane.

5. Arrange – Change the alignment and arrangement of multiple charts. Use the **Shift** key to select more than one chart at a time.

6. **Size** – Change the height and width of the chart.

Format Pane

There are multiple ways to open the Format Pane. Click on the **Format Selection** button in the *Format* tab

Click on **More** option from any menu

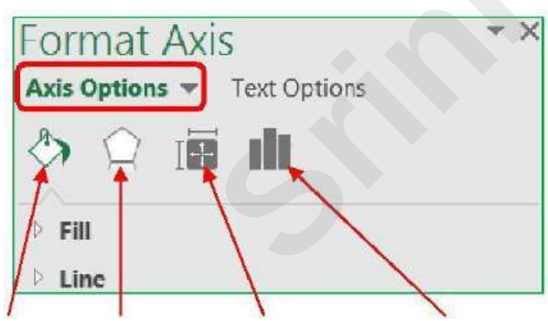
Right-click on a chart element and choose **Format…**

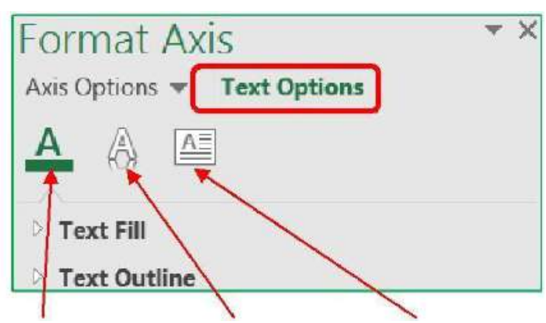
Double-click on a chart element

The format pane can remain open for as long as you need it. The properties shown change depending on the current selection. The current selection is shown on the Format tab and in the title of the Format Pane. The pane can be pulled free from the side by dragging the title toward the middle of the window. To return the pane to the side of the window drag it back into place or double-click the title of the Format Pane.

To close the pane, click on the X in the upper right hand corner. If you accidentally close the pane, use any method above to reopen it.

Within the Format Pane, click on each icon to see the subset of properties. Click on the expand arrow in front of the options to see the relevant properties.



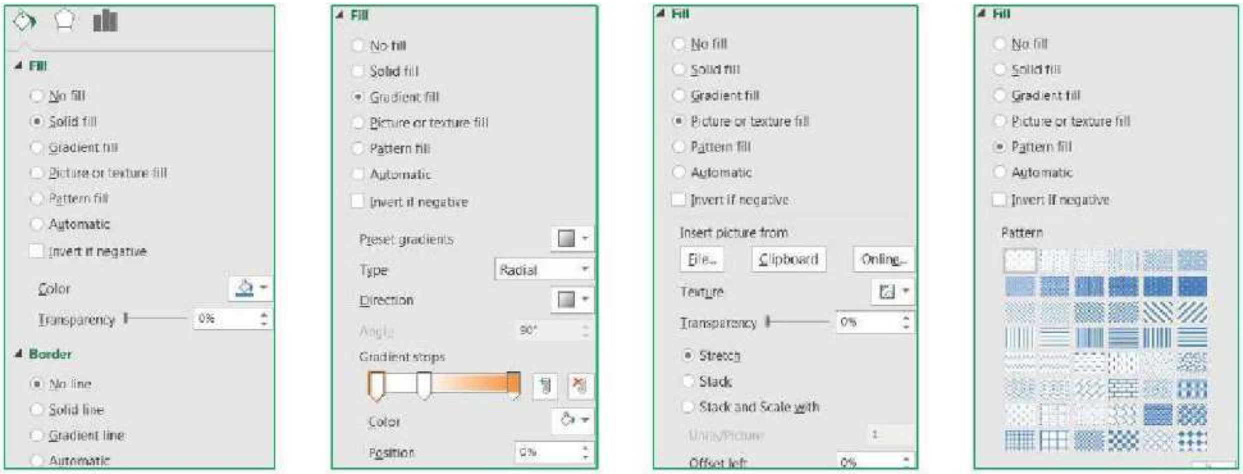


Fill & Line **Effects** **Size & Properties** **Options** **Text Fill & Text Outline** **T e x t**

Effects **Text Box Options**

The Fill & Line and Effects options are the same for all the of the chart elements. If an option cannot be applied to that chart element, Excel will disable (grey out) that option.

Below are the fill options for a Chart Element. Each **Fill** choice provides a new set of options.



How do I …?

Change Axis Numbers

Select the Axis by clicking on a number in the area. Open the Format pane, be sure the title says **Format Axis**. Click on the Options button. From here you can:

- Change the **Minimum** and **Maximum** numbers shown. These can be greater than the minimum and less than the maximum if you want.
- Change the **Major** unit, this is how the displayed number is chosen. If the major unit is 100 the chart axis will read 100, 200, 300. If it's 25 the chart will show 25, 50, 75.
- Change the **Display Units** to Thousands, Millions, Billions. This will change the unit
- shown in the labels and data tables as well.
- Change the **Format** of the numbers; number of decimals, include a dollar sign, etc.

Change Distance Between Columns

Select any column. Open the Format pane, be sure the title says **Format Data Series**. Click on the Options button.

From here you can:

- Change the selected series to be on a **secondary axis**
- Change the distance the series **overlap**
- Change the **width between** the each category grouping

Explode a Pie Chart

In the chart: Hover over a pie wedge. Click and drag the piece away from the center. To move one piece at a time, select the single pie wedge first, and then move it from the middle.

In the properties: Select a pie wedge. Open the Format pane, be sure the title says **Format Data Series**. Click on the Options button. 

From here you can:

- Change the rotation without changing the order of the data
- Change the explosion, how close the wedges are to each other

Add Trendlines and Error Bars

Select the chart. Click on the Add Chart Element button in the Design tab, or on the  button next to the chart.

You can add your own custom error bars, if needed, from the error bar options. You do have to format one series of error bars at a time.

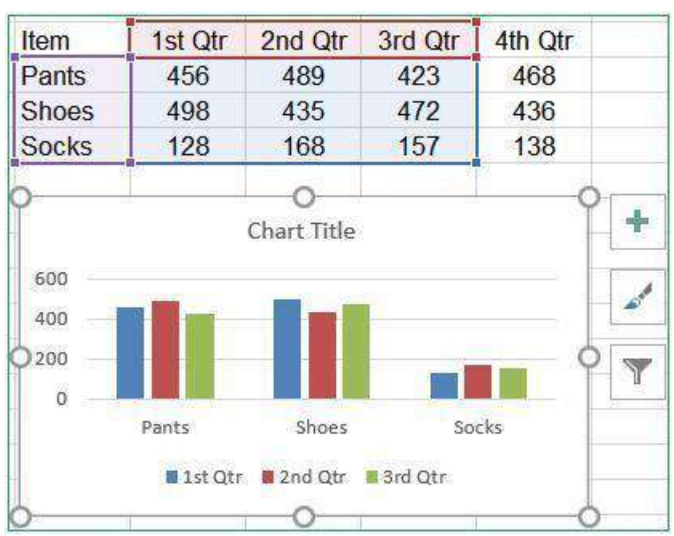
Make Charts the Same Size

Use the Height and Width properties found on the Format tab in the ribbon, or on the Format Pane for the Chart Area's Size & Options. You can use the alignment options on the format tab to make the charts line up.

Changing the Data Source From the Worksheet

When you select a chart you will see the Chart Tool tabs in the ribbon, and the three options buttons along the right side of the chart. If you can see the cells in the worksheet used for the chart you will also be able to see the data is selected and each section is shaded.

If you hover your mouse over the bottom right- hand corner of the data grouping you will get the two-way sizing arrow. If you click and drag the



selection you can manually change the chart data source.

If you know you will have more categories and series you can grow the data area beyond what's showing and Excel will assign new colors and make room in the chart for the new values.

From the Select Data Source Window

From the Chart Tools *Design* tab choose **Select Data**.

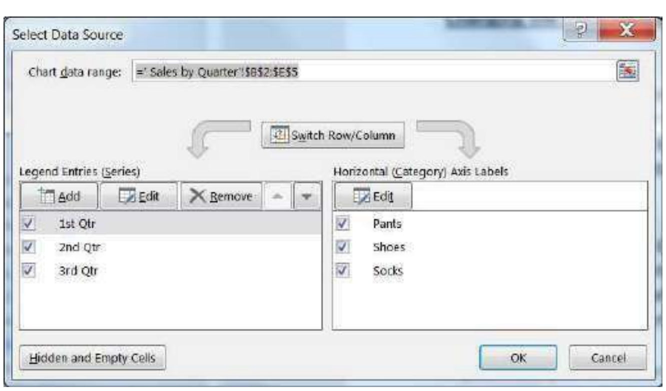
The **Chart data range** option can be a bit finicky so I recommend deleting the current range and selecting the new set from the worksheet.

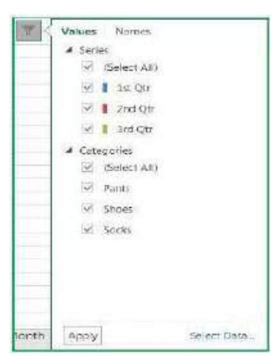
The chart is initially arranged to follow the order of the data, but if you would like the legend in a different order, you can rearrange the Legend Entries using the up and down arrows.

Removing data

Both of the above options will help you add and remove data. You can manually adjust the range in the worksheet or you can select a different range from the Select Data Source window. Both are great as long as you are using a consecutive range of data.

The Select Data Source window also had a **Remove** button to delete a series from the chart. Notice there is not one for the Category/Axis labels. To be able to remove one you will need to first **Switch Row/Column**. Once you've removed the categories, **Switch Row/Column** again.





From the chart itself you can click on the series you want to remove and press **Delete** on the keyboard. You can only delete the series, so the same actions apply in order to remove a category you will need to switch the row/columns first.

However, we now have a Filters button along the side of the selected chart. From here we can uncheck any of the values we do not want on the chart; Series and Categories. You must click the Apply button at the bottom of the menu for the filter to take effect.

Sorting and Filtering Sorting Data

On the far right side of the **Home** tab you will find a large Sort & Filter button. The menu you see when you click on the button is reflected in the Sort & Filter group of the **Data** tab.

If you make a selection of cells, Excel will think you only want to sort or filter by that selection. But if your dataset has no blank rows and no blank columns Excel will see the whole range as one data set.

You can have blank cells, but not completely blank columns/rows; if you are not sure that your dataset is consistent, click inside one cell, and press Ctrl-A. This will select all the cells within the dataset. A second "Ctrl-A", or pressing the shortcut in an empty cell, will select the entire sheet.

When you have completed a sort, you can click the Undo button (or Ctrl-Z). Excel will undo the sort and it will select the dataset it used in the sort. This is another way to see your dataset.

Ascending Sorts

	A	B	C
1	Apples	123	1/1/1971
2	Bananas	456	2/2/1982
3	Cherries	789	3/3/1993

1) **Text**: Sort alphabetically from A to Z
2) **Numbers**: Sorts from smallest number to largest number
3) **Dates**: Sorts from the newest date to the oldest date

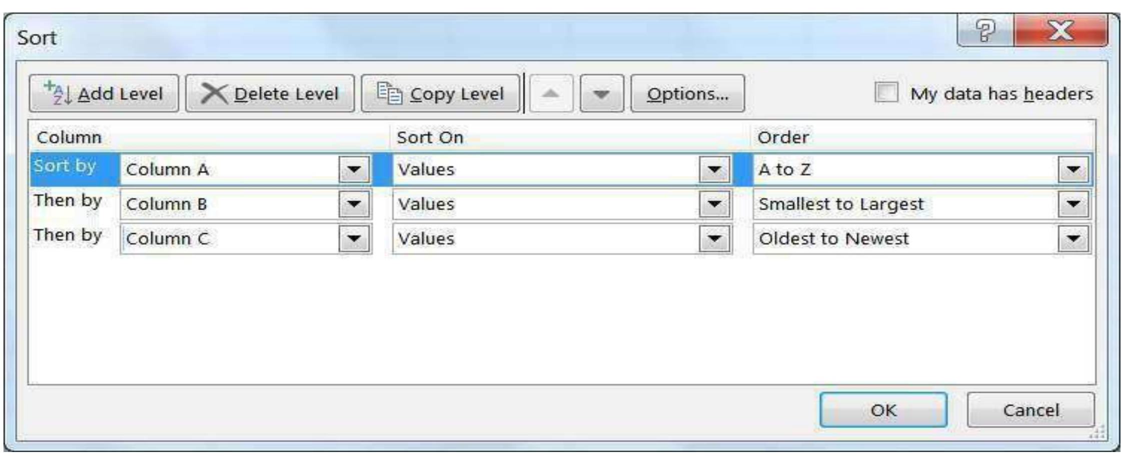
Descending Sorts

	A	B	C
1	Cherries	789	3/3/1993
2	Bananas	456	2/2/1982
3	Apples	123	1/1/1971

3) **Text**: Sort alphabetically from Z to A
4) **Numbers**: Sorts from largest number to smallest number
5) **Dates**: Sorts from the oldest date to the newest date

Custom Sorts

When you first open this window, Excel will show the most recent sort options. If you haven't created a sort yet, this window may be blank.

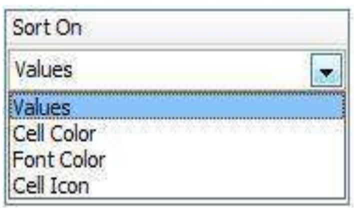


In Excel 2016, we can sort by 64 levels. From this sort window we can add levels, delete levels, copy levels, and even change the order of our sort using the up and down arrows in the toolbar.

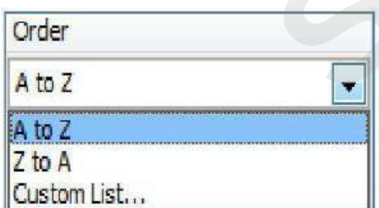
Column: The column drop-down menu will show the names of your columns, your 'fields'. If your data doesn't have titles Excel lists the column heading letters instead. If you were expecting titles, but is only showing the column letters, you can click on the check box in the upper right hand corner of the Sort window to let Excel know your data has headers.

Sort On: You can Sort on the values of the cells, the cell colors, the font colors, or the cell icons. Order:

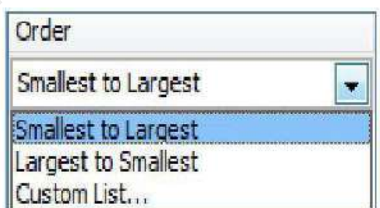
The order options change depending on the values in the cells.



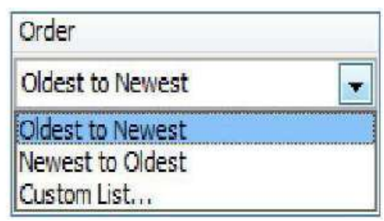
Text



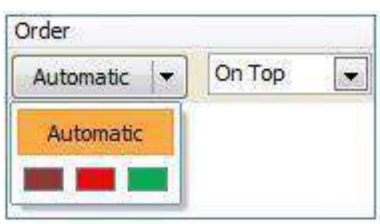
Number



Date



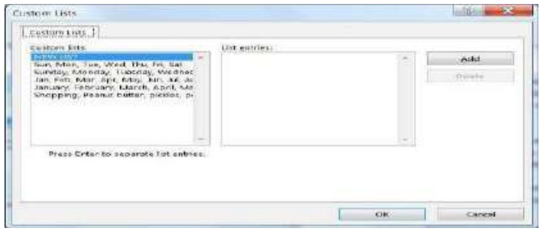
Font Color



Cell Color

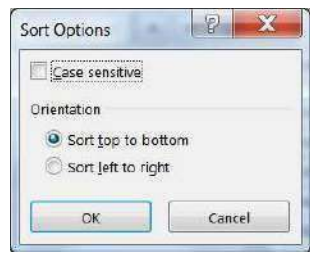
Custom Lists

Custom lists can be built through the Excel Options under the File menu in the Advanced section under General. Or by choosing Custom List… option at the bottom of each order box above.



If you choose this option, you will be able to select from one of these lists. Alphabetically, April comes before January. With the Custom List order, we can ensure January comes first.

These custom lists will work as patterns with the fill handle. Notice the "Shopping" list? Once I set this up, I can type any of the words in a cell and use the fill handle to follow this pattern.



Sort Options

- **Case sensitive:** Sort lowercase letters before uppercase letters

- **Orientation:** Sort vertically (top to bottom, sort rows) or horizontally (left to right, sort columns)

Filtering Data

Filters hide rows (records) based on criteria you set. You can turn the filter on and off by choosing Filter from the Sort & Filter button on the Home tab, or choosing the Filter button on the Data tab.

Excel will place a drop-down arrow at the end of each cell in the title row (the first row of the dataset). When you click on this arrow we see several options including our sort orders:

1) Sort Ascending, Descending, and by color

2) Clear the Filter

3) Filter by Color

4) Set a custom filter (text, number, date)

5) Search for a matching value in the column

6) List of values in the column (field). **Select All** will toggle between everything and nothing.

Once a filter has been set Excel will hide all the rows that don't match the criteria. The status bar will show how many records (rows) were found that matched. The row numbers of the original data will remain the same, but will appear blue. The dropdown arrows of the columns that are being filtered will show the filter icon (funnel). The double line between the row numbers indicate hidden rows.

Custom Filters

Depending on the data in the column you will have the option to set a custom filter based on text, numbers, and dates.

Text Filters **Number Filters** **Date Filters**

If you choose one of the options on the Filter List with the ellipsis (…), you will see a Custom Auto Filter window such as this. From here we can set up to two filters.

Be careful with the AND/OR relationships. If you ask Excel to show the rows where the City equals Jacksonville **AND** the City equals Gainesville, you will get no results, because one cell cannot equal both values. But if you ask for the same using the OR, Excel will show all the records for both cities.

Or's tend to work for exact matches (Equals This **OR** Equals That), whereas AND's tend to work for ranges (Greater than This **AND** Less than That).

You can use the "Wildcards"? and * to help you with your filter. ? is used for one character, * for multiple.

Equals Jacks* **->** Jacksonville, Jacksonville Beach, Jackson Heights

Some of the filter choices may work just as well. I could say Contains 'Jacks' or Begins with 'Jacks'.

SUBTOTAL Worksheet Function

We can do common mathematical functions with our filtered lists using the SUBTOTAL worksheet function. The syntax is for this function is "SUBTOTAL (function_num,ref1,ref2,...)".

Function_Num	**Function**	**Function_Num**	**Function**
1	AVERAGE	7	STDEV
2	COUNT	8	STDEVP
3	COUNTA	9	SUM
4	MAX	10	VAR
5	MIN	11	VARP
6	PRODUCT		

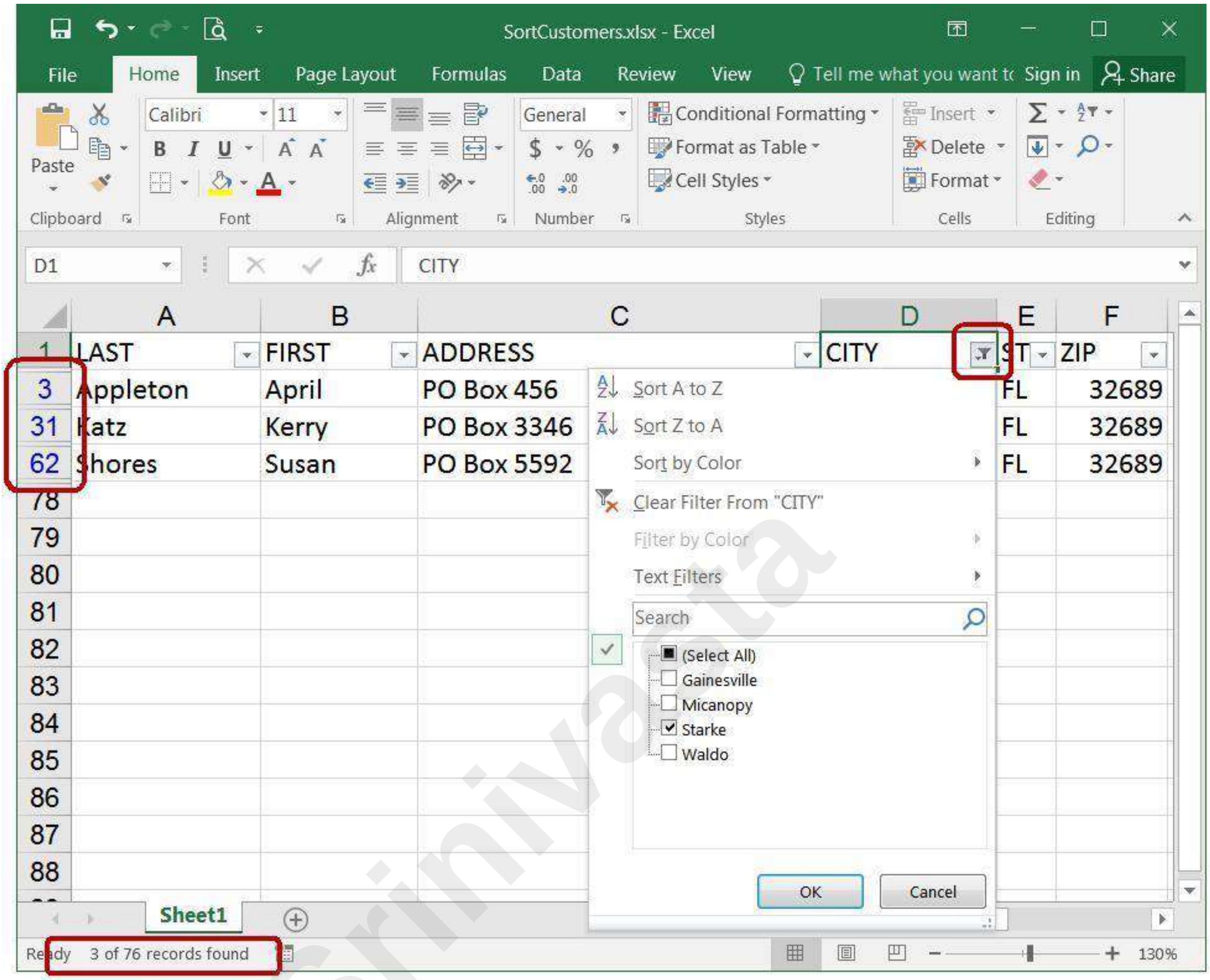
Function_num is the number 1 to 11 that specifies which 'function' to use in calculating subtotals within a list (see below). The ref1, ref2… are the ranges of data that should be used in the equation, there can be up to 29 different ranges used in this function.

Function numbers 1 through 11 will include manually-hidden rows, ones you have hidden yourself. Function numbers 101-111 will exclude your hidden rows from the function. Filtered- out rows are always excluded.

	A	B	C	D	E	F
1	Sum	1411		SubSum	1411	
2	Average	8.70987654		SubAvg	8.70987654	
3	Count	162		SubCount	162	
4						
5	Quarter	Item	Size	Color	# Sold	
6	1st Quarter	blouses	Large	Blue	14	
7	1st Quarter	blouses	Large	Red	6	
8	1st Quarter	blouses	Large	White	10	
9	1st Quarter	blouses	Medium	Blue	2	
10	1st Quarter	blouses	Medium	Red	4	

	A	B	C	D	E	F
1	Sum	1411		SubSum	24	
2	Average	8.70987654		SubAvg	8	
3	Count	162		SubCount	3	
4						
5	Quarter	Item	Size	Color	# Sold	
43	2nd Quarter	pants	Large	Red	8	
46	2nd Quarter	pants	Medium	Red	6	
49	2nd Quarter	pants	Small	Red	10	
168						
169						

	A	B	C	D	E
1	Sum	=SUM(E5:E168)		SubSum	=SUBTOTAL(9,E5:E168)
2	Average	=AVERAGE(E5:E168)		SubAvg	=SUBTOTAL(1,E5:E168)
3	Count	=COUNT(E5:E168)		SubCount	=SUBTOTAL(2,E5:E168)
4					
5	Quarter	Item	Size	Color	# Sold
43	2nd Quarter	pants	Large	Red	8
46	2nd Quarter	pants	Medium	Red	6
49	2nd Quarter	pants	Small	Red	10
168					
169					



Using the IF Function

The If function allows users to make logical comparisons between values, returning a value of either True, or False. Based on the result of the logical comparison, the if function can carry out a specific operation, based on a true or false value.

The syntax for an if function is: **=IF(logical_test, value_if_ture, value_if_false)**

Logical test: any value or expression that can return a TRUE or FALSE

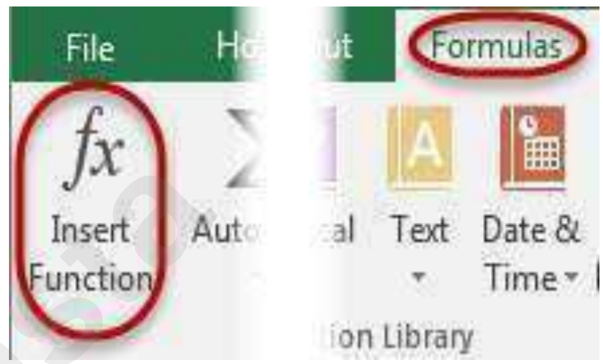
Value_if_true: The value to be returned, or operation to be performed, if the condition is TRUE. This can be text (must be in double quotes "text"), a number, or a function.

Value_if_false: The value to be returned, or operation to be performed, if the condition is FALSE.

Excel Comparison Operators

Comparison Operator

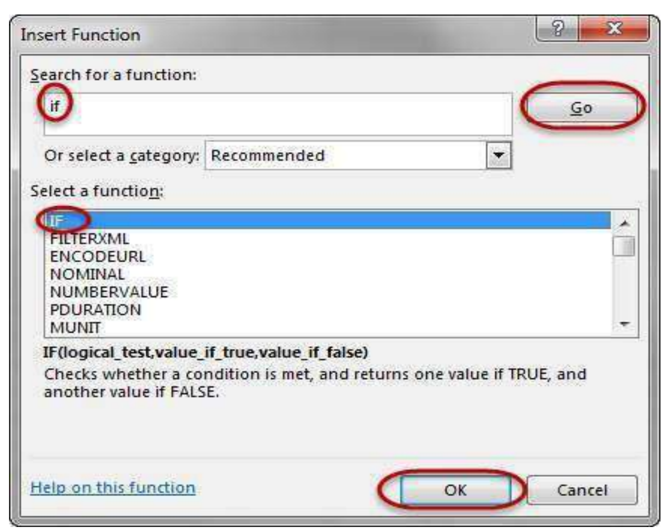
- = (equal sign)
- > (greater than sign)



- < (less than sign)
- >= (greater than or equal to sign)
- <= (less than or equal to sign)
- <> (not equal to sign)

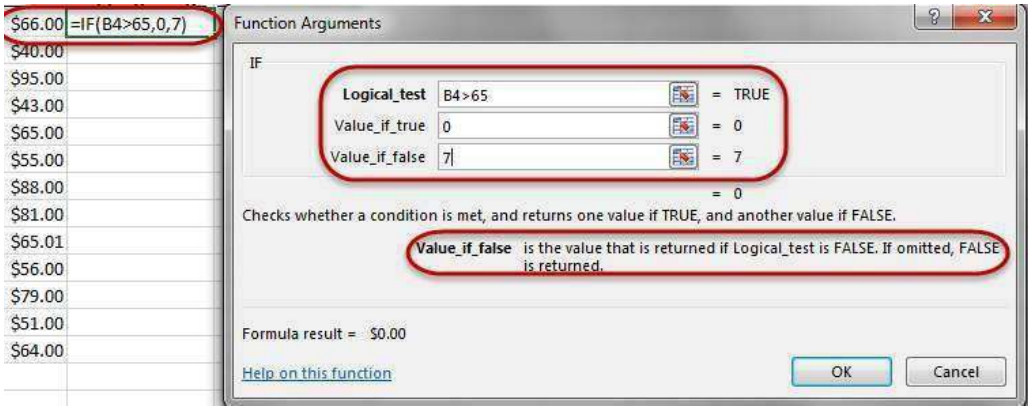
Steps for an If Statement:

1. Place cursor in the cell that will contain the If Statement.
2. Navigate to the Formulas tab, and click on the Insert Function icon.
3. Within the Insert Function window, search for If in the search textbox and hit the Go Button.



4. Select the if function from the select a function box. Select it by either double clicking on it or by selecting it and clicking the OK button.

5. In the Functions Argument window, we are able to see each argument of the if function. For the logical test, we want to select the Cell that we are comparing.



6. Value_if_true. What do we want to happen if our argument is True ?
    a. This can be as simple as a number, text (must be in quotes "", an arithmetic function, or even another function entirely.
7. Value_if_false. What do we want to happen if our argument is False?
    a. This can be as simple as a number, text (must be in quotes "", an arithmetic function, or even another function entirely.
8. When we have all of our arguments filled out, click on the OK button and our function is now complete.

9. To copy this function to the remaining cells, click back on the cell that contains the original function. Move your mouse into the lower right corner until your mouse turns into a dark plus sign.

10. Click and hold with your mouse as you drag to the bottom of the rest of the data. The function has now been copied to all of the other cells.

Best practices with if statements

Generally, it is not good practice to put static values (values that may need to change from time to time) directly into functions in Excel. The main reason for this is because if a change ever has to be made, the values can be hard to find and change. It is best to put these values into their own individual cell, and then reference the cell containing the value in the function. This way, any time the values has to change, the referenced cell's value is changed and all functions related to that function are now updated immediately. There isn't a need to search for each function to update them.

Display a formula with color coded cell references

When troubleshooting a function while in Excel, hit the F2 key. This will display all arguments in a function as color coded cell references and cells.

Cell References

When creating functions in Excel, users must know that there are two different cell reference types, Relative and Absolute.

Relative cell reference

It is important to understand how Excel identifies cells in formulas. By default, all cell references are relative references, which means the function is related to the cells around it. As a function is copied across multiple cells, it is updated based on the relative position of rows and columns.

The way Excel reads a function is by looking at the relationship of the cell references in relation to the cell containing the function. For example, in the

	A	B	C	
1	Dept A	Dept B	Totals	
2	100	300	400	=A2+B2
3	200	400	600	=A3+B3

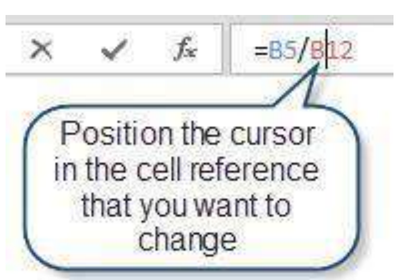
function contained in cell C3 to the right, Excel reads this as "take the number located two cells to the left and add that to the number in the cell located one column to the left". Therefore, when this function (which appears as =A2+B2) is copied to the cell below, it performs the calculation using the same pattern, "take the number located two cells to the left and add that to the number in the cell located one column to the left", but updates the formula to reflect the appropriate cell addresses. This is called a Relative Reference and is the feature that enables users to copy the same function to a different location within a worksheet.

Absolute cell reference

An absolute reference is a reference to a particular cell, or range of cells, that never updates as the function is copied to a new location. Absolute references can be made to an individual cell, or to keep a row or a column constant as the function is copied to a new location.

To make a cell reference absolute so that it will not adjust when a formula is copied, insert a dollar sign ($) in the appropriate position. By navigating into the formula in the Formula bar, users can cycle through the absolute reference options by hitting the F4 key on the keyboard. The first time the F4 key is pressed, a user will create an absolute reference to the cell reference in the function, the second time the reference will be to the row only, and the third time the reference will be to a particular column.

Press the F4 key one time



Two times

=B5/B$12

Three times

=B5/$B12

Example 1: $B$12 Both the Row and the Column are held Constant

Example 2: B$12 The Row is held constant; the Column will update as the function is copied.

Example3: $B12 The Column is constant; the Row will update as the function is copied.

Using OR & AND Functions

The And & Or functions are great functions to use within an if function because they will return a value of either True or False, which is what the first portion of an if statement needs in order to complete the function.

The And and Or functions will allow for multiple logical expressions to be created within the argument of the function. Each logical expression must be separated by a comma.

And Function

The And function is used to check multiple logical expressions and will return a True value if every one of the logical expressions is true.

The syntax is; **=AND(logical1,[logical2],…) OR Function**

The OR function is used to check multiple logical expressions and will return a True value as long as one of the logical expressions is true.

The syntax is; **=OR(logical1,[logical2],…)**

Nesting an AND or OR function into an If function

The easiest to incorporate an AND or OR function into an if function is to first start with the And or Or Function and then incorporate the function into the logical_test argument of an IF function.

Steps to create a nested AND or OR function within an IF function;

1. Navigating to the cell to contain the IF function and type in an equal sign, followed by AND. Select the AND function by double clicking on it, or using the arrow keys to select the function and then hit the tab key.
2. When the AND function is complete, finish the function by entering in a closed bracket).
3. Now, use the mouse and place the cursor between the equal sign and the A in the front of the AND function.

Users may also use the formula bar to do this, as it may be easier to access the first portion of the function in the formula bar.

With the cursor between the equal sign and the A, start typing if. Excel will populate a list of functions under the cell. Either double click on the If function, or use the arrow keys to highlight IF and then hit the tab key.

Excel will now display the arguments for the IF function underneath the cell, or the formula bar, depending on where the user is working. The argument that will be bolded will be the logical_test, which has already been completed

with the AND function. To enter in to the second portion, value_if_true, of the expression, move the cursor to the far right of the AND function, after the closed bracket and type in a comma.

Enter in what is to be done if the AND function is true, then type a comma to get to the last portion of the IF function, the value_if_false.

When all arguments have been fulfilled, hit the type in a closed bracket to complete the
Note: Users may hit the Enter key to complete the function, which may result in the following error;

This error is just notifying the user that the last closed bracket

is missing. By accepting this error, Excel will insert the closed bracket to complete the function.

Nested IF Statement

Sometimes there is more than one condition that needs to be tested in order to return a value. To test multiple conditions, users can use a nested IF function. A nested IF statement is and easy way to test for multiple conditions. Typically, the first condition will be tested and an answer will be provided if the condition is True. If the condition is false, a new IF statement will be used to test in the value_if_false portion of the first if statement.

The syntax would look similar to this;

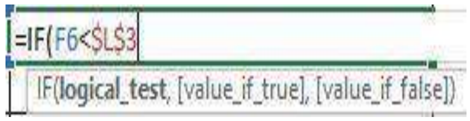
=IF(logical_test,value_if_true,IF(logical_test,value_if_true,value_if_false)) Note: Excel can nest up to 64 if statements.

Tip: When comparing values within a scale, make sure the values are sorted in either increasing or decreasing order to help make the nested if statement easier to assemble. If the scale is in Increasing order, the less than < sign will be used in the arguments and if the scale is in decreasing order, the greater than > sign will be used in the arguments.

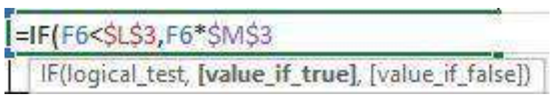
Steps for a Nested IT statement.

1. Position the cursor in the cell that will contain the formula. Type in an equal sign, followed by if. When if populates under the cell, either double click on if to select it,
or use the arrow keys to highlight it and then hit the tab key to select the IF function.

2. Enter in the first logical test within the If function, then hit the comma key to get to the value_if_true.



3. In the value_if_true portion of the argument, enter in what is to be done if the first logical test is true, and then hit the comma key to get to the value_if_false.



4. In the value_if_false, a new if statement will have to be started since we are comparing multiple logical tests. To start a new if statement, start typing IF within the value_if_false portion, the exact same way we would start a brand new IF function, but without the equal sign.

Excel will populate the if function underneath the cell that is being typed in. Select the if function, just as if this was a brand new if statement that was being started, by double clicking on the If, or by highlighting it and then hitting the tab key.

When this is done, users will notice that the argument below the cell changed from being in the value_if_false portion of the first IF function to now being the logical_test of the second if statement.

Tip: Treat this new IF function as its own, free standing if function by entering in all arguments within the IF function.

In the second IF functions logical_test, enter in the second comparison and then hit the comma key to get into the value_if_true.

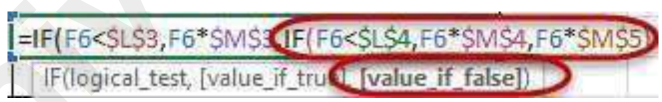
In the value_if_true portion of the argument, enter in what is to be done if the first logical test is true, and then hit the comma key to get to the value_if_false.

In this example, we are only comparing two values, so in the value_if_false portion, enter in what is to be performed if neither of the logical_tests are met.

Note: If more logical tests were being compared, another IF function would be entered into the value_if_false portion, following the same steps that were taken in step 4.ss

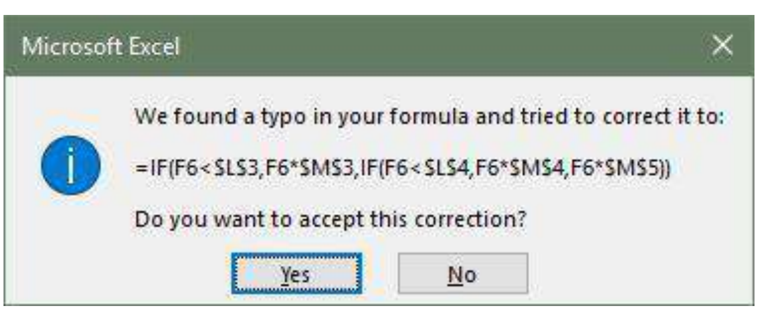
After the value_if_false portion is completed, complete the second IF function by typing in a closed bracket).

notice that the IF function argument below the cell will now display the value_if_false argument of the first IF function. Since the second IF statement has completed the value_if_false portion of the first IF function the nested if is now complete.



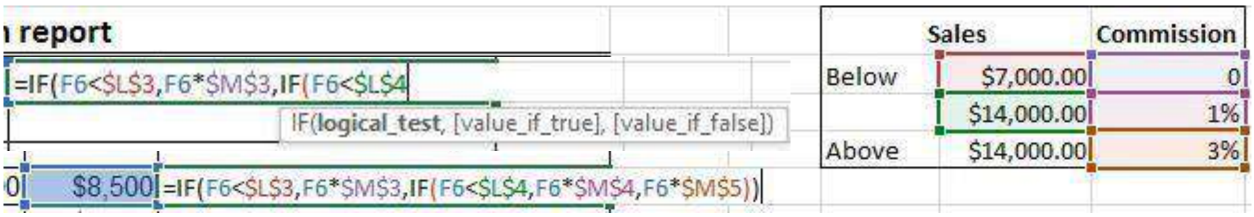
To complete the nested IF function, either type in another closed bracket), or hit the Enter key.

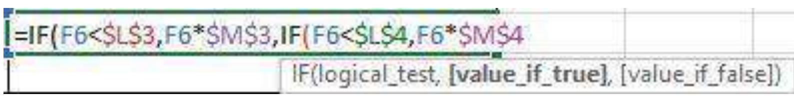
Note: Users may also hit the Enter key after completing the second IF function. If this is done, Excel will populate a window indicating that there is a typo in the formula, meaning there aren't any closed brackets to complete the function. To have Excel enter in the closed brackets, hit the Yes button.

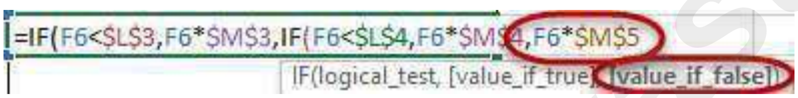


To verify the nested IF was entered in correctly, click on the cell containing the IF function and then hit the F2 key. This will display the function with colored cell references related to the cells on the worksheet.

s







PIVOT TABLES

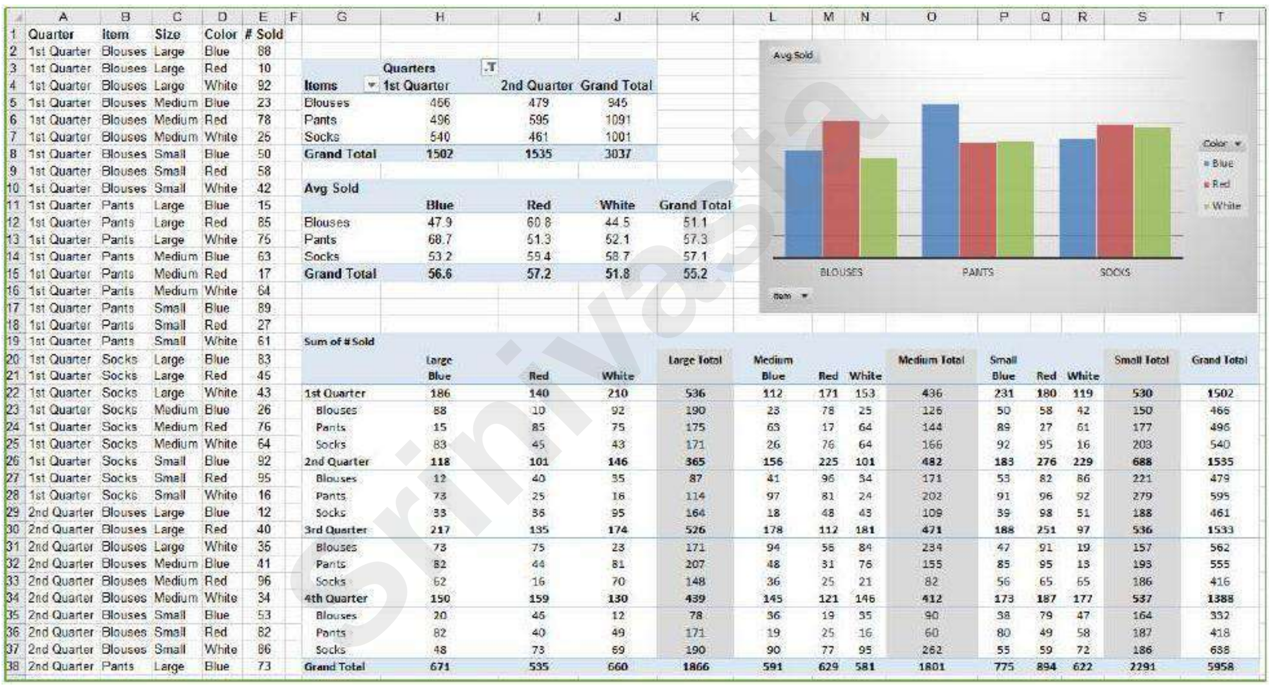
Learning Objectives

- PivotTables
- Formatting
- Pivot Tables - Excel 2016 Help File
- Exercises
- Things to Remember

PivotTables

PivotTables are summary reports. They give you the ability to take a large boring set of repetitive data and summarize it into a neat table that you can very easily rearrange, filter, format, and even chart.

Here's an example of a repetitive raw data set, three PivotTables, and a PivotChart.



	Quarter	Item	Size	Color	# Sold
2	1st Quarter	Blouses	Large	Blue	88
3	1st Quarter	Blouses	Large	Red	10
4	1st Quarter	Blouses	Large	White	92
5	1st Quarter	Blouses	Medium	Blue	23
6	1st Quarter	Blouses	Medium	Red	78
7	1st Quarter	Blouses	Medium	White	25
8	1st Quarter	Blouses	Small	Blue	50
9	1st Quarter	Blouses	Small	Red	58
10	1st Quarter	Blouses	Small	White	42
11	1st Quarter	Pants	Large	Blue	15
12	1st Quarter	Pants	Large	Red	85
13	1st Quarter	Pants	Large	White	75
14	1st Quarter	Pants	Medium	Blue	63
15	1st Quarter	Pants	Medium	Red	17
16	1st Quarter	Pants	Medium	White	64
17	1st Quarter	Pants	Small	Blue	89
18	1st Quarter	Pants	Small	Red	27
19	1st Quarter	Pants	Small	White	61
20	1st Quarter	Socks	Large	Blue	83
21	1st Quarter	Socks	Large	Red	45
22	1st Quarter	Socks	Large	White	43
23	1st Quarter	Socks	Medium	Blue	26
24	1st Quarter	Socks	Medium	Red	76
25	1st Quarter	Socks	Medium	White	64
26	1st Quarter	Socks	Small	Blue	92
27	1st Quarter	Socks	Small	Red	95
28	1st Quarter	Socks	Small	White	16
29	2nd Quarter	Blouses	Large	Blue	12
30	2nd Quarter	Blouses	Large	Red	40
31	2nd Quarter	Blouses	Large	White	35
32	2nd Quarter	Blouses	Medium	Blue	41
33	2nd Quarter	Blouses	Medium	Red	96
34	2nd Quarter	Blouses	Medium	White	34
35	2nd Quarter	Blouses	Small	Blue	53
36	2nd Quarter	Blouses	Small	Red	82
37	2nd Quarter	Blouses	Small	White	86
38	2nd Quarter	Pants	Large	Blue	73

	Quarters		
Items	1st Quarter	2nd Quarter	Grand Total
Blouses	466	479	945
Pants	496	595	1091
Socks	540	461	1001
Grand Total	**1502**	**1535**	**3037**

Avg Sold	Blue	Red	White	Grand Total
Blouses	47.9	60.8	44.5	51.1
Pants	68.7	51.3	52.1	57.3
Socks	53.2	59.4	58.7	57.1
Grand Total	**56.6**	**57.2**	**51.8**	**55.2**

Sum of # Sold	Large			Large Total	Medium			Medium Total	Small			Small Total	Grand Total
	Blue	Red	White		Blue	Red	White		Blue	Red	White		
1st Quarter	**186**	**140**	**210**	**536**	**112**	**171**	**153**	**436**	**231**	**180**	**119**	**530**	**1502**
Blouses	88	10	92	190	23	78	25	126	50	58	42	150	466
Pants	15	85	75	175	63	17	64	144	89	27	61	177	496
Socks	83	45	43	171	26	76	64	166	92	95	16	203	540
2nd Quarter	**118**	**101**	**146**	**365**	**156**	**225**	**101**	**482**	**183**	**276**	**229**	**688**	**1535**
Blouses	12	40	35	87	41	96	34	171	53	82	86	221	479
Pants	73	25	16	114	97	81	24	202	91	96	92	279	595
Socks	33	36	95	164	18	48	43	109	39	98	51	188	461
3rd Quarter	**217**	**135**	**174**	**526**	**178**	**112**	**181**	**471**	**188**	**251**	**97**	**536**	**1533**
Blouses	73	75	23	171	94	56	84	234	47	91	19	157	562
Pants	82	44	81	207	48	31	76	155	85	95	13	193	555
Socks	62	16	70	148	36	25	21	82	56	65	65	186	416
4th Quarter	**150**	**159**	**130**	**439**	**145**	**121**	**146**	**412**	**173**	**187**	**177**	**537**	**1388**
Blouses	20	45	12	78	36	19	35	90	38	79	47	164	332
Pants	82	40	49	171	19	25	16	60	80	49	58	187	418
Socks	48	73	69	190	90	77	95	262	55	59	72	186	638
Grand Total	**671**	**535**	**660**	**1866**	**591**	**629**	**581**	**1801**	**775**	**894**	**622**	**2291**	**5958**

Planning

The most important part of building a PivotTable is planning. You have to remove yourself from the raw data and think about the final result. There's a learning curve, have patience with yourself and with Excel and you'll get there.

The Data

The data has to be repetitive. It looks wrong at first, but the more boring and repetitive your data, the more you can do with it. In the Large Data 1 class we learn about sorting and filtering in Excel. We don't need to keep each quarter, each department, each person on a different sheet. If we keep all of the data in the same place, we can look at each category, one at a time, using the filter, and beautifully because it is all in the same place, we can look across categories and quickly summarize with a PivotTable.

The data has to be consistent. We don't want to see entries like: 4th Qtr, 4th Quarter, Qtr 4. In the Large Data 3 (vLookups) class we'll learn about validation rules, and in Large Data 4 (report) we'll see how to use vLookups to help us cleanup inconsistent data entry.

The definition of a database is a structured collection of related data. Rows of a data table are **records**.

When we filter in Excel, the Status Bar tells us how many Records were found.

Columns of a data table are **fields**. The column titles of your original dataset will appear in the PivotTable Field List. You'll use these titles to control the structure of the PivotTable. If possible, **use clear, concise, and unique column titles**. If you use the same column title more than once, Excel will add a number after the subsequent titles. Example: *Home Address, City, State, Zip, Work Address, City, State, Zip* becomes *Home Address, City, State, Zip, Work Address, City(2), State(2), Zip(2).*

The Result

By all means, jump in and play with the tools, get comfortable with how to make a PivotTable. But when it comes down to needing a specific report, you have to "show your work". Think about what it is you actually want to see.

"I want to know how many items we ordered." Okay. Do you care what the items were? Do you care about the data across the year, or do you only want the total? What about the other details? Each versus boxes? Would you like to compare your orders to another departments?

"I want to know how many patients were admitted last month." Okay. Do you want to see the break down by time, perhaps by morning and afternoon? By shift? By Department? Or would you like to have the flexibility to change (filter) the shifts and departments?

The more you think about what you want out of the glorious summary report known as a PivotTable, the better it will come together for you.

Building

The PivotTable field list shows the column titles of our original dataset, these are our **Fields**. If you rename, add, or delete columns in the dataset you will not see the change here until you refresh the data. The **Refresh** button is on the Analyze tab of the PivotTable Tools, and can be found on the shortcut menu if you right-click inside the table.

- Fields in the **Filters** will appear above the table.
- Fields in the **Columns** will appear at the top of each column of the PivotTable.
- Fields in the **Rows** will appear at the left of each row of the PivotTable.
- Fields in the **Values** will be summarized. By default, text and date fields will be counted, number fields will be summed.

Arranging Fields

Adding

- Click the check box in front of the field name

    a. Text fields will go into the **Rows** showing each unique value from the dataset

    b. Date fields will go into the **Rows** grouping the values across time

    c. Number fields will go into the **Value** as a sum

2. Drag fieldname from the field list to an area

    a. You will *have* to drag to add a field to the value area multiple times

3. Right-click on the fieldname in the field list and choose an area

Moving

- Drag fieldname from an area to a new area
- Right-click on the fieldname in the field list and choose a new area
- Left-click on a field in an area and choose a new area

Deleting

- Drag fieldname out of the area section
- Left-click on a field in the area and choose **Remove Field**

Value Field Settings

Fields added to the value section are summarized within the grouping of the row and column headings set in the PivotTable. Numbers will sum, other values will be counted. To change how the data is summarized, left-click on the fieldname in the Values area and choose **Value Field Settings**.

You can reformat your numbers with the Excel formatting tools, but if you reformat from the Value Field Settings window you'll format the Table, this means if you change the structure of your table by adding fields you won't have to reformat any new cells occupied by the table.

Page 7 of this handout discusses the **Show Values As** options.

> It is possible to create other summary options, but it's beyond the scope of this class. If you would like to explore, look at the **Fields, Items, & Sets** option on the Analyze tab

Formatting

The PivotTable Design tab has lots of style options to make the PivotTable look good. Change the Options settings to see how they vary in the style you chose.

The style options help with the look of the data, but in my opinion the first set of buttons on the Design tab are way better, as they determine how the data is pulled together within the PivotTable.

Original

Sum of # Sold	Column Labels			
Row Labels	Large	Medium	Small	Grand Total
1st Quarter	**536**	**436**	**530**	**1502**
Blouses	190	126	150	466
Pants	175	144	177	496
Socks	171	166	203	540
2nd Quarter	**365**	**482**	**688**	**1535**
Blouses	87	171	221	479
Pants	114	202	279	595
Socks	164	109	188	461
Grand Total	**901**	**918**	**1218**	**3037**

Subtotals at **Bottom**, **Blank Rows** inserted

Sum of # Sold	Column Labels			
Row Labels	Large	Medium	Small	Grand Total
1st Quarter				
Blouses	190	126	150	466
Pants	175	144	177	496
Socks	171	166	203	540
1st Quarter Total	**536**	**436**	**530**	**1502**
2nd Quarter				
Blouses	87	171	221	479
Pants	114	202	279	595
Socks	164	109	188	461
2nd Quarter Total	**365**	**482**	**688**	**1535**
Grand Total	**901**	**918**	**1218**	**3037**

No **Subtotals**, no **Grand Totals**

Report Layout: **Outline**

Report Layout: **Tabular**, no **Subtotals**, **Blank Rows** inserted

Sum of # Sold	Column Labels		
Row Labels	Large	Medium	Small
1st Quarter			
Blouses	190	126	150
Pants	175	144	177
Socks	171	166	203
2nd Quarter			
Blouses	87	171	221
Pants	114	202	279
Socks	164	109	188

Sum of # Sold		Size			
Quarter	Item	Large	Medium	Small	Grand Total
1st Quarter		**536**	**436**	**530**	**1502**
	Blouses	190	126	150	466
	Pants	175	144	177	496
	Socks	171	166	203	540
2nd Quarter		**365**	**482**	**688**	**1535**
	Blouses	87	171	221	479
	Pants	114	202	279	595
	Socks	164	109	188	461
Grand Total		**901**	**918**	**1218**	**3037**

Sum of # Sold		Size			
Quarter	Item	Large	Medium	Small	Grand Total
1st Quarter	Blouses	190	126	150	466
	Pants	175	144	177	496
	Socks	171	166	203	540
2nd Quarter	Blouses	87	171	221	479
	Pants	114	202	279	595
	Socks	164	109	188	461
Grand Total		**901**	**918**	**1218**	**3037**

Notice with the **Outline** and **Tabular** Report Layouts the **Row Labels** are no longer grouped, instead you see a heading for each column and row label.

Create a PivotTable

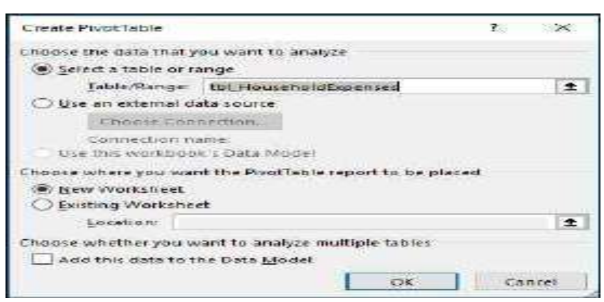
If you have limited experience with PivotTables, or are not sure how to get started, a **Recommended PivotTable** is a good choice. When you use this feature, Excel determines a meaningful layout by matching the data with the most suitable areas in the PivotTable. This helps give you a starting point for additional experimentation. After a recommended PivotTable is created, you can explore different orientations and rearrange fields to achieve your specific results.

1) Select the PivotTable that looks best to you and press **OK**. Excel will create a PivotTable on a new sheet, and display the **PivotTable Fields** List.

1) Click a cell in the source data or table range.
2) Go to **Insert** > **Tables** > **PivotTable**.

If you're using Excel for Mac 2011 and earlier, the PivotTable button is on the **Data** tab in the **Analysis** group.

3) Excel will display the **Create PivotTable** dialog with your range or table name selected. In this case, we're using a table called "tbl_HouseholdExpenses".



In the **Choose where you want the PivotTable report to be placed** section, select **New Worksheet**, or **Existing Worksheet**. For **Existing Worksheet**, you'll need to select both the worksheet and the cell where you want the PivotTable placed.

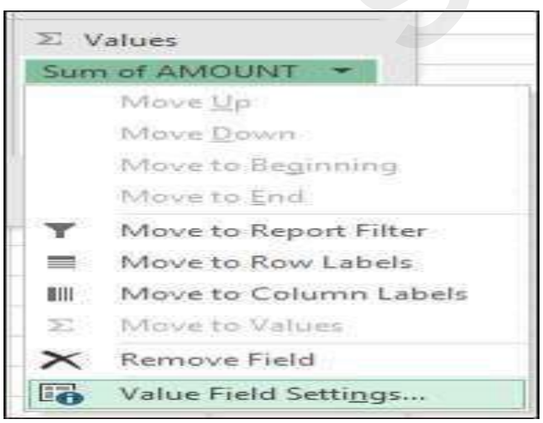
1) If you want to include multiple tables or data sources in your PivotTable, click the **Add this data to the Data Model** check box.

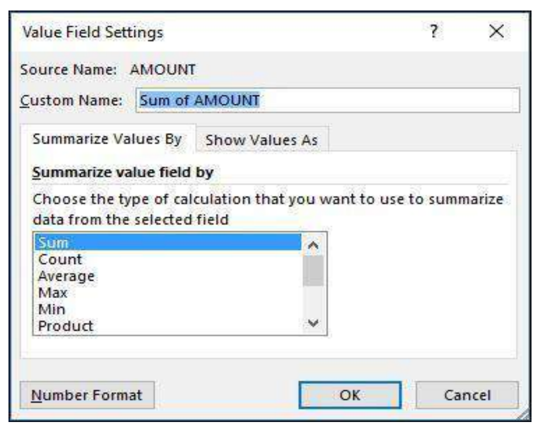
2) Click **OK**, and Excel will create a blank PivotTable, and display the **PivotTable Fields** list. Working with the PivotTable Fields list

In the Field Name area at the top, select the check box for any field you want to add to your PivotTable. By default, non-numeric fields are added to the Row area, date and time fields are added to the Column area, and numeric fields are added to the Values area. You can also manually drag-and-drop any available item into any of the PivotTable fields, or if you no longer want an item in your PivotTable, simply drag it out of the Fields list or uncheck it. Being able to rearrange Field items is one of the PivotTable features that makes it so easy to quickly change its appearance.

PivotTable Values Summarize Values By

By default, PivotTable fields that are placed in the Values area will be displayed as a SUM. If Excel interprets your data as text, it will be displayed as a COUNT. This is why it's so important to make sure you don't mix data types for value fields. You can change the default calculation by first clicking on the arrow to the right of the field name, then select the Value Field Settings option.





Next, change the calculation in the Summarize Values By section. Note that when you change the calculation method, Excel will automatically append it in the Custom Name section, like "Sum of FieldName", but you can change it. If you click the Number Format button, you can change the number format for the entire field.

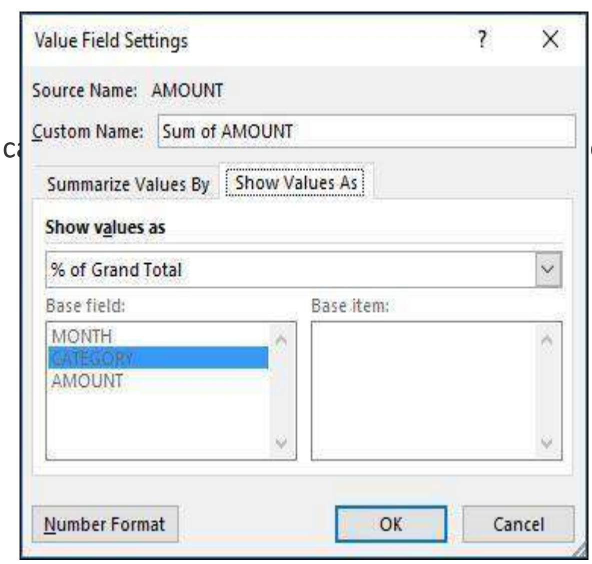
Tip: Since the changing the calculation in the Summarize Values By section will change the PivotTable field name, it's best not to rename your PivotTable fields until you're done setting up your PivotTable. One trick is to use Find & Replace (Ctrl+H) >Find what > "Sum of", then Replace with > leave blank to replace everything at once instead of manually retyping.

1) Show Values As

Instead of using a calculation to summarize the data, you can also display it as a percentage of a field. In the following example, we changed our household expense amounts to display as a **% of Grand Total** instead of the sum of the values.

Sum of AMOUNT	Column La			
Row Labels	January	February	March	Grand Total
Entertainment	5.10%	6.38%	6.13%	17.61%
Grocery	12.00%	12.25%	13.27%	37.52%
Household	8.93%	11.49%	10.21%	30.63%
Transportation	3.78%	5.87%	4.59%	14.24%
Grand Total	29.81%	35.99%	34.20%	100.00%

Once you've opened the **Value Field Setting** dialog, you c**es As** tab.



3) Display a value as both a calculation and percentage.

Simply drag the item into the **Values** section twice, then set the **Summarize Values By** and **Show Values As** options for each one.

Refreshing PivotTables

If you add new data to your PivotTable data source, any PivotTables that were built on that data source need to be refreshed.

To refresh just one PivotTable, you can right-click anywhere in the PivotTable range, then select Refresh.

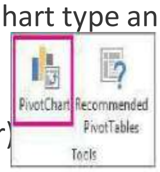
If you have multiple PivotTables, first select any cell in any PivotTable, then on the Ribbon go to PivotTable Tools >

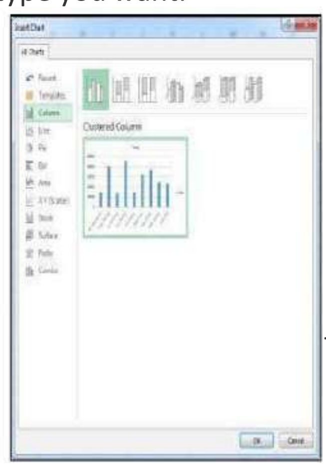
Analyze > **Data** > **Click** the arrow under the **Refresh** button and select **Refresh All**.

Create a PivotChart

1) Click anywhere in the PivotTable to show the PivotTable Tools on the ribbon.
2) Click **Analyze** > **PivotChart**.
3) In the **Insert Chart** dialog box, click the chart type and chart subtype you want.

You can use any chart type except an XY (scatter)





4) **Click OK**.

5) In the **PivotChart** that appears, click any tering options.

After you create a PivotChart, you can customize it, much like you'd do with any standard charts.

When you select the PivotChart two buttons appea r change chart elements such as titles or data labels, r PivotChart the same way you would in a standard cl

The **PivotChart Tools** are shown on the ribbon.

On the **Analyze**, **Design**, and **Format** tabs, you can r PivotChart.

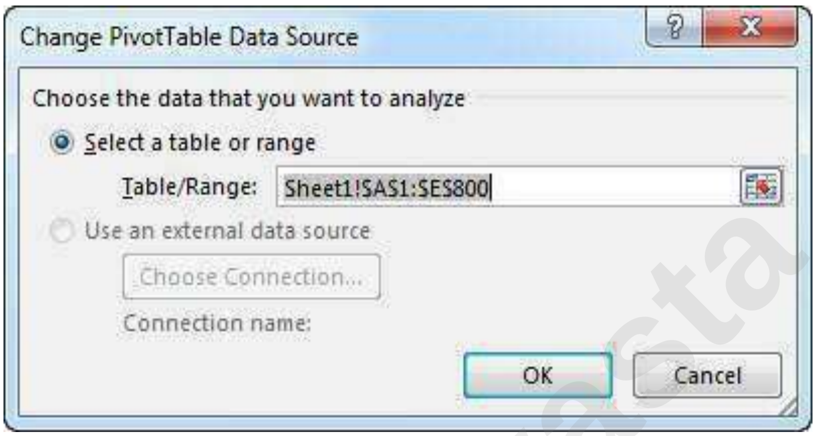
Change the source data

After you create a PivotTable, you can change the range of its source data. For example, you can expand the source data to include more rows of data. However, if the source data has been changed substantially—such as having more or fewer columns, consider creating a new PivotTable.

To change the data source of a PivotTable if it's a range of cells or an Excel table, do the following:

- Click anywhere in the PivotTable to show the **PivotTable Tools** on the ribbon.

- In the **Table/Range** box, enter the range you want to use.



Tip: Leave the dialog box open, and then select the table or range on your worksheet. If the data you want to include is on a different worksheet, click that worksheet, and then select the table or range.

Delete a PivotTable

When you no longer need a PivotTable, select the entire PivotTable, and press the Delete key to remove it. If you get a "Cannot change this part of a PivotTable report" message, smake sure the entire PivotTable is selected. Press Ctrl+A, and press Delete again.

If you're using a device that doesn't have a keyboard, try removing the PivotTable like this:

5) Pick a cell anywhere in the PivotTable to show the **PivotTable Tools** on the ribbon.

6) Click **Analyze** > **Select**, and then pick **Entire PivotTable**.

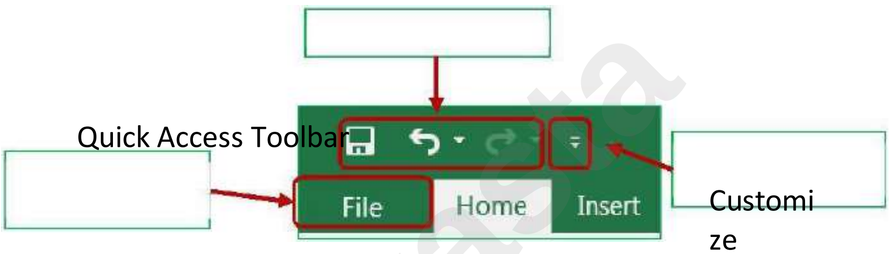
7) Press Delete.

Tip: If your PivotTable is on a separate sheet that has no other data you want to keep, deleting that sheet is a fast way to remove the PivotTable.

PAGE SETUP

Customizing the Quick Access Toolbar

Quick Access Toolbar



Customize Quick Access Toolbar

File Menu (Backstage view)

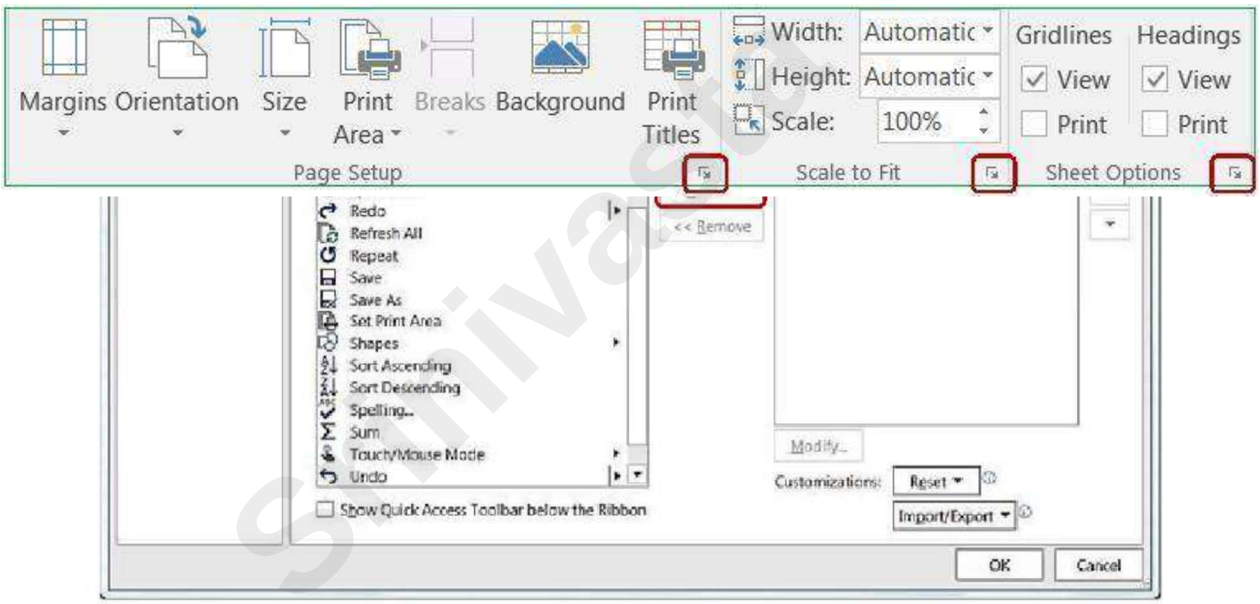
The upper left hand corner of the Excel Window has a Quick Access Toolbar. This is a very convenient location to place commonly used buttons. By default, this toolbar has Save, Undo, and Redo. For this Printing and Setup workshop we would like to add the Print Preview button to the toolbar.

There is a drop down button ( ) at the end of the Quick Access Toolbar that will help you decide which buttons to display. This drop down list displays several common choices, including our Print Preview and Print. When you choose a list item, Excel will place the button on the toolbar.

If you don’t see the option you would like, you can choose **More Commands**. This will open the Excel Options window. From the Customize Section, choose the command you would like to see and choose the **Add > >** button.

Page Setup

There are several of ways to customize your printouts in Microsoft Excel. Many of these options can be found on



The Page Layout tab in the ribbon.

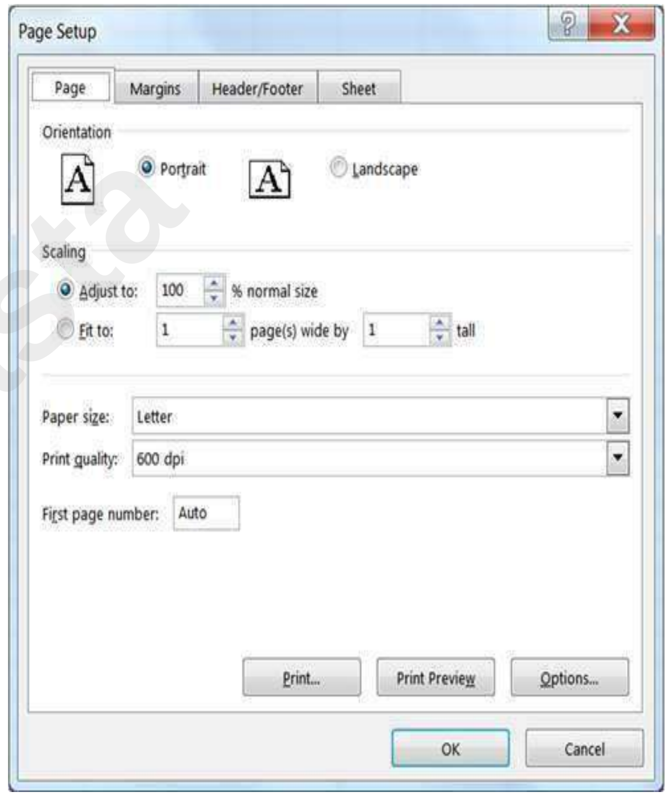
All of the page setup options can be found through the **Page Setup** window. You can view this window by clicking any of the launch dialog box ("More") buttons in the bottom right of any of the sections shown here. You can also open the window by clicking on the **Page Setup** button in the Print Preview.

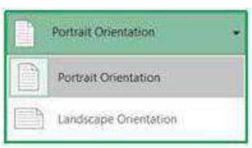
Page Tab

The first tab in the Page Setup window allows you to change some general page options.

__Orientation__

This option lets you set your print out to Portrait or Landscape. The image portrays the actual direction of the paper.

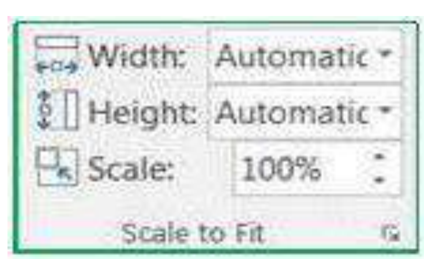




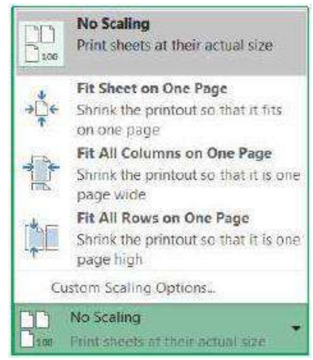
This is also a button on the **Page Layout** Tab and in the **Print Settings** in the print preview.

Scaling

This option can be used to change the "zoom" of the printout with the ***Adjust to:*** option or force the number of pages the printout must fit within using the ***Fit to:*** option. Like the zoom in the worksheet, you can adjust this option from 10% to 400%.



Scaling is also available on the **Page Layout** tab in the **Scale to Fit** section. The ***Width*** and ***Height*** options are the same as the *Fit to:* option allows you to specify how many pages wide or tall you want your printout to be. The **Scale** option is the same as to the *Adjust to:* option, it allows you to change the zoom percentage of the printout.



The **Scaling** option in the print preview offers four options.

- No Scaling – Scale to 100%
- Fit Sheet on One Page – Scale Width of only One Page
- Fit All Columns on One Page – Scale Height of only One Page
- Fit all rows on One Page – Scale whole printout to fit everything on one page.

Paper size

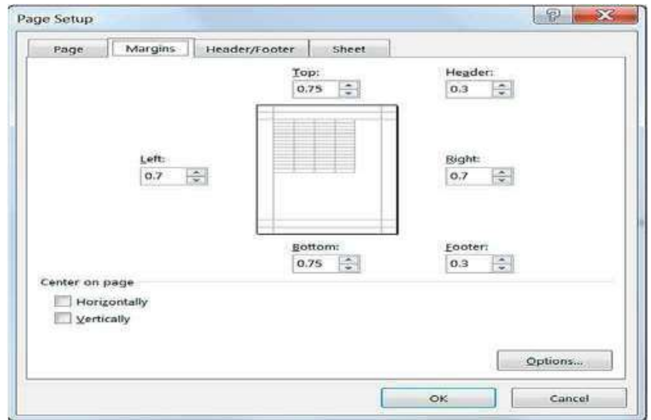
This option allows you to choose paper sizes such as Letter, and Legal. This option can also be changed by using the **Size** button on the **Page Setup** section of the **Print Layout** tab, and in the **Print Settings** in the print preview.

Print quality

This option allows you to specify the print quality of your worksheet. The higher the resolution (dots per inch – dpi), the better the quality of your printout. This option can only be changed in the **Page Setup** window.

First page number

By default, a printout starts on page one, but this option allows you to start on a different number. You might do this if you have a cover page, or if you're adding it to another report. This option can only be changed in the **Page Setup** window.



Margins Tab

Margins are the distance between your data and the edge of the page. Here on the second tab of the Page Setup window you can adjust the Top, Bottom, Left, and Right margins to an accuracy of a hundredth of an inch.

A preset list of Margins can also be found on the Page Setup section of the Page Layout tab and in the print preview menu. If you choose Custom Margins…, Excel will open the Margins page of the Page Setup window.

Header is the text that appears at the top of each printed page, and Footer is the text that appears at the bottom of every page. These do not show in the Normal view of the worksheet, but can be seen in the Print Preview and the Page Layout View.

These do not have to be the first row or column; any column can be chosen as your titles and will be repeated on each page of the printout. If you click on this button in the Page Layout tab, it will open the in the **Page Setup** window to the **Sheet** page.

Print Titles

This option allows you to set which **Rows to repeat at the top** of each page, and which **Columns to repeat on the left** of each page.

As with the **Print Area**, to set the rows and columns to repeat, click inside the blank in the **Page Setup** window and use your mouse to select the rows/columns you would like repeated on every page.

Work sheet, but can be seen in the Print Preview and the Page Layout View.

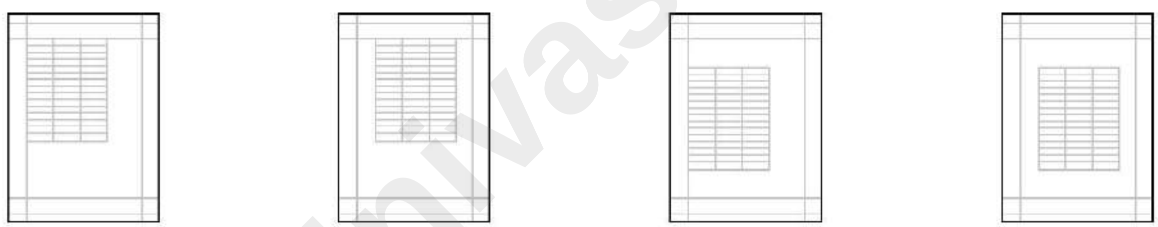
Keep the distance of the Header: and Footer: margins smaller than the Top:

and Bottom : margins to prevent overlapping the data.

<u>Center on page</u>

This option allows you to adjust how the data will be placed within the set margins.

No Centering *Centered Horizontally* *Centered Vertically* *Horizontally & Vertically*



Header/Footer Tab

The third tab of the Page Setup window is Header/Footer. Remember, Headers appear at the top, and Footers appear at the bottom of every page.

The Header and Footer drop down lists offer predefined options. The items listed pull the information such as Author from the Document Properties. These can be modified through the File tab, Info, Properties. Other information is derived from the Worksheet names, Workbook names, computer date and time settings, and calculated page numbers. Remember you can set the beginning page number on the Page tab.

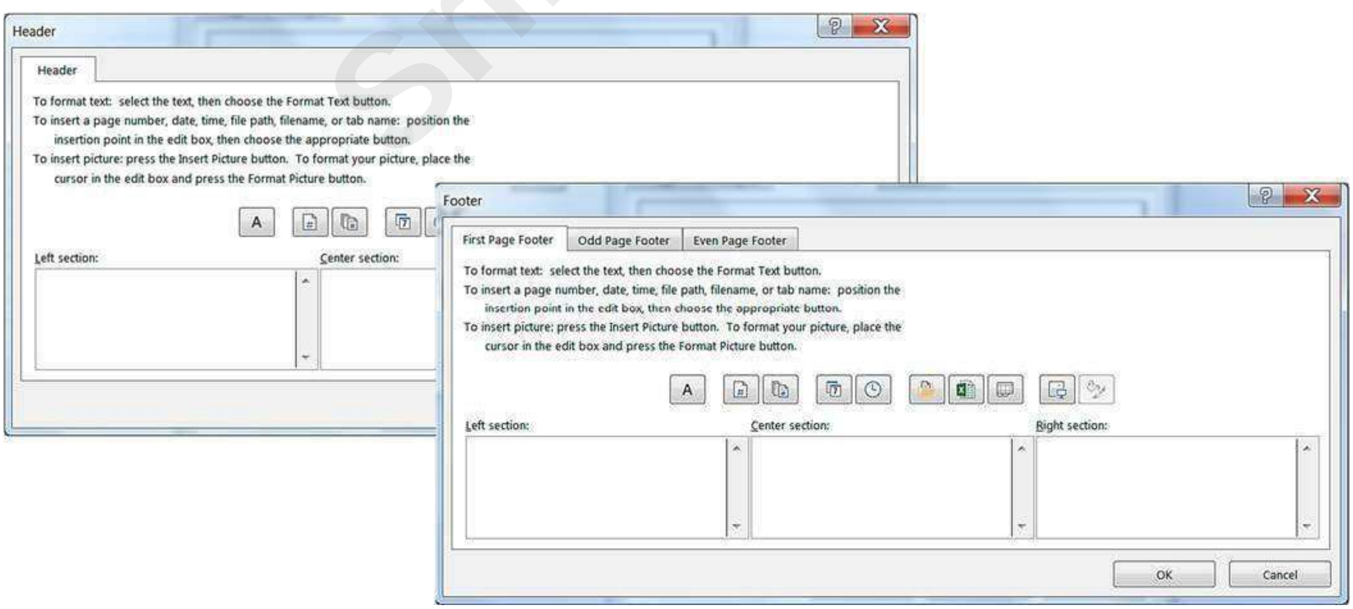
There are several Header and Footer options at the bottom of the window.

- **Different Odd and Even Pages** – Custom Headers and Footers can be created for odd vs. even pages.

- **Different First Page** – Custom Headers and Footers can be created for the first page to be different from the rest of the pages.

- **Scale with Document** – Keeps the same ***scaling*** as the worksheet. If we set the scale on the Page tab to print at 200%, this check box will ensure the header and footer also scale to 200%.

- **Align with Page Margins** – Aligns the Header and Footer with the left and right margins of the worksheet instead of the page itself.

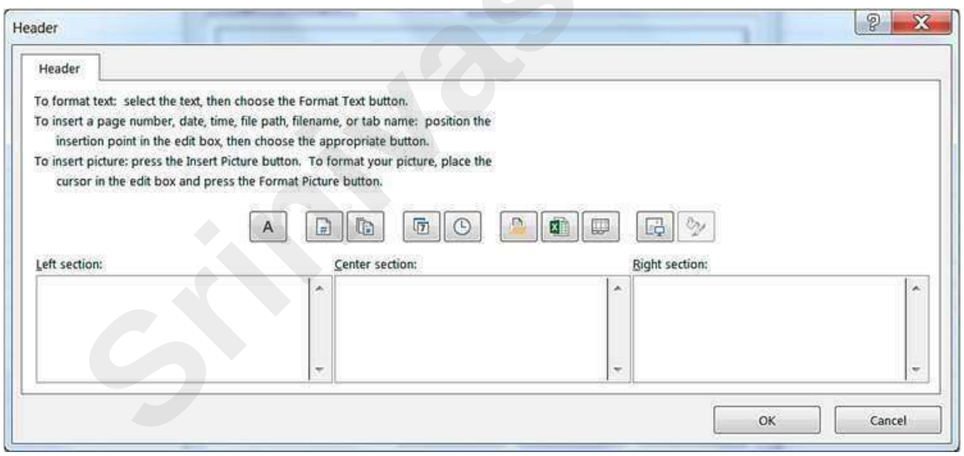


Custom Header/Footer

If you choose the options for *Odd*, *Even* and *First Page*, Excel will show new tabs here so you can set each section to have its own custom header and footer.

Header/Footer Elements (Custom Toolbar)

There are three sections you can enter text. The **Left Section** aligns its contents on the left side of the margin, the **Center Section** aligns its contents in the middle of the margin, and the **Right Section** aligns its contents to the right side of the margin. If you would like the Header/ Footer to align with the page instead of the margin,



1 2 3 4 5 6 7 8 9 10

turn off the **Align with Page Margins** option on the **Page Setup**.

1. **Font** – You can have several different fonts in the same section. You can set the format before or after you type your text. If you format after, you will need to select the text before you go into the Font window.

2. **Page number** – Inserts the text "&[Page]" which represents the current page number.

3. **Total number of pages** – Inserts the text "&[Pages]" which represents the total number of printed pages. If we wanted to see the phrase Page 1 of 6, we would need to type in the "Page" and " of " (don't forget the

spaces!) So our code would look like **- Page &[Page] of &[Pages]**

4. **Current date** – Inserts the text "&[Date]". This is the code to display the day the worksheet is printed. This button is automatically updated. If you want it to display the day the worksheet was created, you will need to type the actual date.

5. **Current time** – Inserts the text "&[Time]". Like the Date button, this will automatically update each time the file is printed.

6. **File path** – Inserts the text "&[Path]&[File]". This is the code to show the full file path. Like the Date and Time buttons, this will automatically update. If you move the file, or Save As into another location or with another name, this option will automatically change.

7. **Filename** – Inserts the text "&[File]". This is the code to display the current name of the file.

8. **Worksheet name** – Inserts the text "&[Tab]". This is the code to display the printed worksheet name.

9. **Insert Picture** – Inserts a graphic in the header/footer. This is most often used for logos and watermarks. The image will show in actual size behind the data. Very light images, such as a large graphic of the word DRAFT or CONFIDENTIAL can be inserted into a header and will appear on every page in the printout.

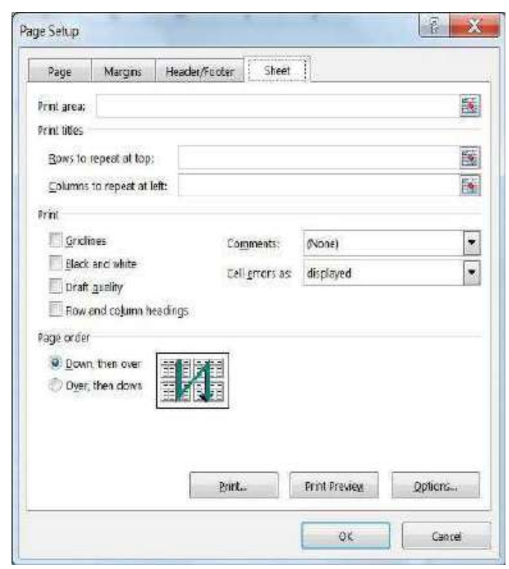
10. **Format Picture** – This button allows you to format the picture you have inserted. If there is no picture, this button will be disabled (grayed out).

The Page Layout View (found on the View tab), allows you to view and edit your Header/Footer and still see the data in your worksheet. From this view you will see the edges of the page and the page numbers in the status bar.

When you click in the Header/Footer area you will get a Header & Footer Tools – Design tab. From here you will see many of the options we have available when setting up our Header/Footer in the Page Setup window.

You may find it useful to create or edit your Header/Footer in this view, where you can see the changes as you make them. To switch views or work within the sheet, double-click back in the cells to leave the Header/Footer design tab.

Sheet Tab

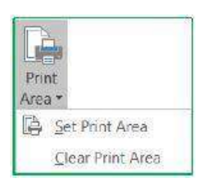


The final tab in the Page Setup window allows us to take things from the worksheet to add to our printout.

Print area

Page Layout tab. Select the desired

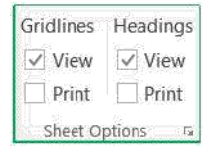
This option allows you to choose the range of cells to print. This option is not available if you enter **Page Setup** from the print preview, because you cannot "choose" cells.



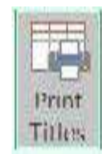
The Print area can also be set from the Page Setup group of the range of cells and choose, Set Print Area.

To set the print area from the Page Setup window, click inside Print area box and use the mouse to point to the cells you want. You can move the dialog box by clicking and dragging the blue title bar.

Print Titles



This option allows you to set which **Rows to repeat at the top** of each page, and which **Columns to repeat on the left** of each page. These do not have to be the first row or column; any column can be chosen as your titles and will be repeated on each page of the printout. If you click on this button in the Page Layout tab, it will open the in the **Page Setup** window to the **Sheet** page.



As with the **Print Area**, to set the rows and columns to repeat, click inside the blank in the **Page Setup** window and use your mouse to select the rows/columns you would like repeated on every page.

Gridlines

This option will print the gridlines around all the cells within the print area. The alternative is to place "border" around your cells. Gridlines can be turned on and off from the Page Layout tab, in the Sheet options section.

No Options Set

Jack	123
Jill	456
John	789

Gridlines

Jack	123
Jill	456
John	789

Black and white

This option will print your data in simple black and white. This does not include shades of gray. Excel will remove all color formatting from the printout.

No Options Set

Jack	123
Jill	456
John	789

Black and white

Jack	123
Jill	456
John	789

Draft quality

This option is the ideal quick printout. Depending on your printer it may reduce printing time. This option will not print gridlines or graphics.

<u>**Row and column headings**</u>

This option will print out the row headings (the row numbers) and the column headings (the column letters). Row and Column Headings can also be turned on and off from the Page Layout tab, in the Sheet options section.

No Options Set

Jack	123
Jill	456
John	789

Row and column headings

	A	B
1	Jack	123
2	Jill	456
3	John	789

Comments

These can be inserted through the **Review** tab in the **Comments** section. By default, comments are not printed, but here on the **Sheet** tab of the **Page Setup** window we have three options that will print the comments: *At end of sheet*, *As displayed on sheet*, and the default **-** *(none)*.

Cell errors as

This option allows you to choose how you want the errors to print. The default is to print the errors as you see them on the screen. But you can change this option to leave those cells blank or fill them all in with N/A or dashes (**----**).

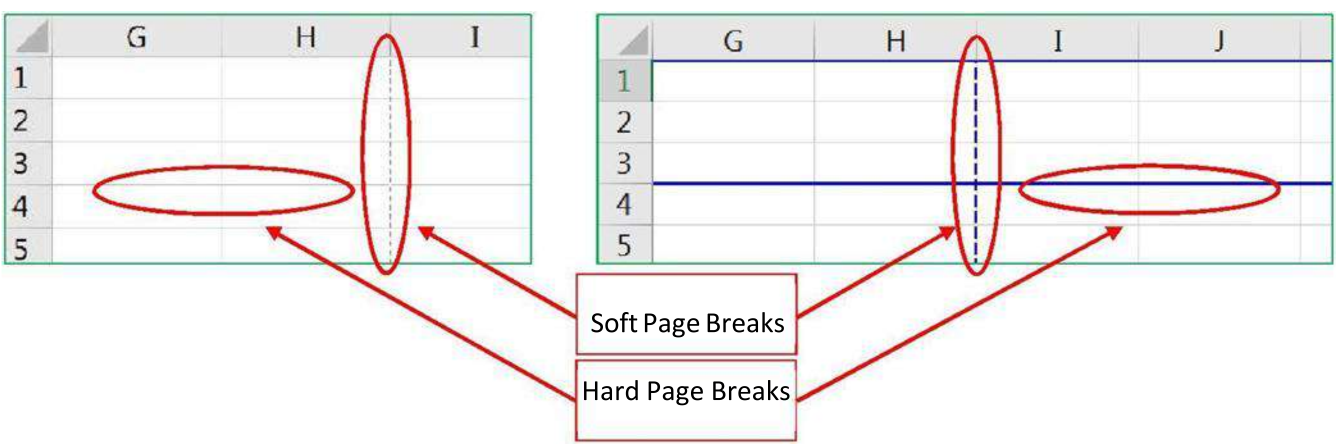
Page Order

This option allows you to decide how multiple pages will print out. By default, Excel will print data from the upper left corner straight down to the end, and then over to the next page full of data and down. If you want to change the print order, such that Excel reads across and then down, change this option.

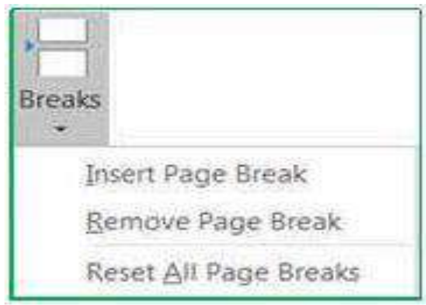
Page Breaks

You can tell Excel where to start a new page by choosing **Insert Page Break** from the Page Setup section of the **Print Layout** tab. The page break will be set above and to the left of the selected cell. In **Normal** view the page break will appear slightly thicker gridline. In **Page Break Preview** the page break will appear as a solid blue line.

Normal View *Page Break Preview*



Soft Page Breaks (natural breaks) Hard Page Breaks (inserted breaks)



You can remove page breaks that have been inserted. Put yourself in the first cell of that page (below and to the right of the break) and choose **Remove Page Break** from the Page Setup section of the Print Layout tab.

The **Reset All Page Breaks** option will remove all of the inserted page breaks and return the worksheet to the default soft page breaks (natural breaks).

Page Break Preview

The **Page Break Preview** can be accessed through the View tab, in the Workbook Views group. As with the normal view, you cannot see the headers and footers.

The Page Break Preview shows the order of the pages and allows you to manually move the page breaks. The dashed lines are soft (natural) page breaks. As you change the margins, scale, and orientation of the page, these page lines will automatically update. In this view only, you can drag a page break (a blue line) to another location. When you change where a page break is located, it changes for the entire worksheet. For the whole row and the whole column.

LEARNING OBJECTIVES S

Printing sounds like a simple concept, until that Excel data set just WON'T fit on that printed page! In this workshop we will work with the print preview; set paper size, page orientation, margins, headers and footers; learn how to print titles, gridlines, and column headings; work with the views Page Layout and Page Break Preview; change the document scaling to force the printout into a certain number of pages; and use print areas and page breaks to adjust what goes on each page. This **basic workshop** assumes some experience with **Microsoft Excel**.

How to use VLOOKUP in Excel 2016

VLOOKUP is a powerful function in Excel. In this tutorial, I'm going to show you how to use VLOOKUP in Excel 2016.

Specifically, I'm going to review an example with the steps needed to apply a VLOOKUP. But before we begin, let's first review the elements of the VLOOKUP function.

Elements of the VLOOKUP in Excel 2016

The VLOOKUP function in Excel has the following structure:

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

Let's now explain each element within this function:

1. **lookup_value** is the value that is present in your first table, which you wish to find in your second table
2. **table_array** is the range of cells for your second table
3. **col_index_num** is the column position in your second table. For example, the value of '2' represents the second column in your table
4. **[range_lookup]** can assume either a 'TRUE' value for an approximate match, or a 'FALSE' value for an exact match

In the next section, we'll review an example to demonstrate how to use VLOOKUP in Excel 2016.

Example of using the VLOOKUP function

To start, let's review an example, where you have two tables in Excel:

- The first table contains **Client Data** with 3 columns: Client First Name, Client Last Name and Country Code. Let's call this table the 'Client' table
- The second table contains **Country Data** with 2 columns: Country Code and Country Name. Let's call this table the 'Country' tables

For each client name, under the first Client table, the goal is to display the Country Name (to be taken from the second Country table). You can then use the VLOOKUP function to automate this task.

Here is the Client data that can be copied into Excel (under the range of cells A1 to C26). You'll need to rename the first sheet to **Client Data**:

Client First-Name	Client Last-Name	Country Code
Elden	Stanhope	107
Emile	Nation	103
Velda	Labriola	105
Vanesa	Rakowski	109
Sharron	Gorby	120
Cecilia	Swearingen	101
Doretha	Saffell	106
Mariah	Heilman	103
Rana	Bigby	107
In	Vallecillo	116
Librada	Taylor	119
Elane	Mayes	103
Alix	Haskett	109
Anastacia	Goldblatt	115
Melania	Whitlock	108
Dario P	arker	111
Rochel	Witherspoon	106
Lucrecia	Tarrance	114
Jermaine	Lamprecht	102
Regine	Bascom	109
Bryanna	Stansberry	107
Naomi	Arai	114
Darin	Blumstein	108
Greg	Ricketts	102
Wilburn	Guse	118

This is how the client data would look like in Excel:

Next, copy the Country data into a second sheet under the range of cells A1 to B21. You'll need to rename the second sheet to **Country Data**:

Country Code	Country Name	
101		Australia
102	Austria	
103	Belgium	
104	Belize	
105	Canada	
106	China	
107	Finland	
108	France	
109		Germany
110	Greece	
111	Italy	
112	Japan	
113	Mexico	
114		Morocco
115	Nigeria	
116	Peru	
117	Russia	
118	Sweden	
119	United Kingdom	
120		Uruguay

And this is how the country data would look like in Excel:

Notice that both of the tables have the column *Country Code*, which contains common values under both tables. We will use the values under the Country Code column to connect between the two tables.

Once you connect the two tables via the Country Code, you'll be able to achieve the *goal* of displaying the Country Name for each client under the Client table.

Steps to apply the VLOOKUP in Excel 2016

Let's now look at the steps to apply the VLOOKUP function:

(1) First, go to the 'Client Data' sheet

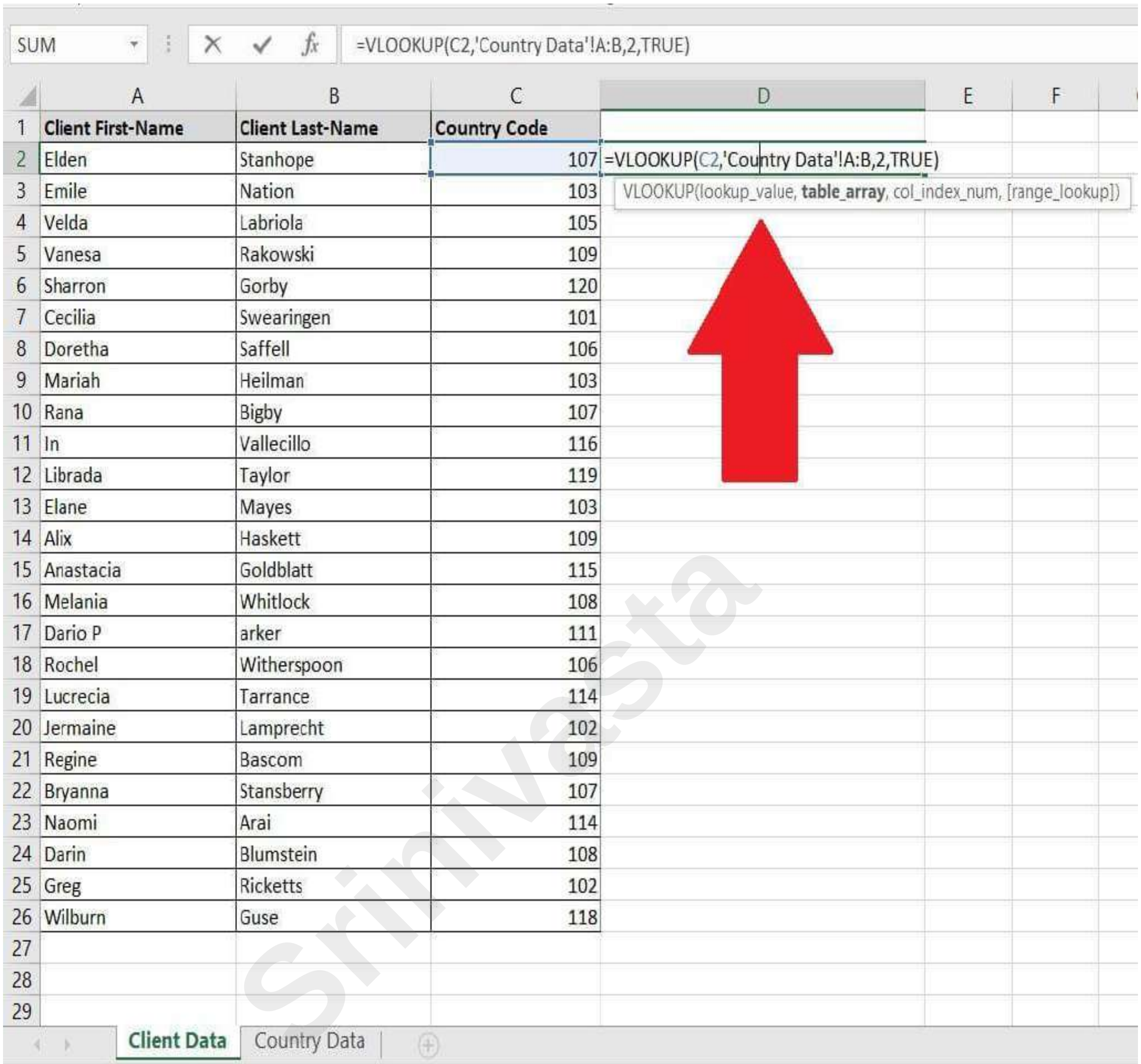
(2) Next, double-click on cell D2

(3) Then, type/copy:

```
=VLOOKUP(C2,'Country Data'!A:B,2,TRUE)
```

- **C2** is your *lookup_value*. It represents the value that is present in your first Client table, which you wish to find in your second Country table
- **'Country Data'!A:B** is your *table_array*. It represents the range of cells for your second Country table, where you want to find your lookup_value
- The value **2** is your *col_index_num*. It reflects the column position in your Country table. Here, '2' means the second column under your Country table. The values under this second column (i.e., the Country Name column) will then be displayed when you apply the VLOOKUP
- **TRUE** is your *[range_lookup]* which reflects an approximate match This is how your VLOOKUP

function would look like in Excel:

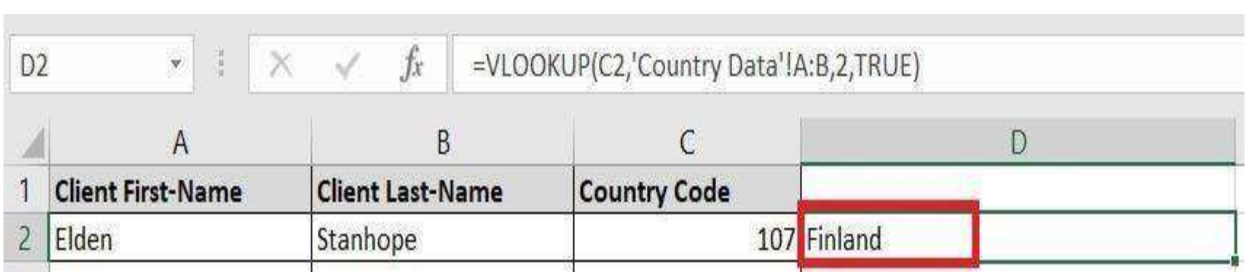


SUM | =VLOOKUP(C2,'Country Data'!A:B,2,TRUE)

	A	B	C	D
1	Client First-Name	Client Last-Name	Country Code	
2	Elden	Stanhope	107	=VLOOKUP(C2,'Country Data'!A:B,2,TRUE)
3	Emile	Nation	103	VLOOKUP(lookup_value, **table_array**, col_index_num, [range_lookup])
4	Velda	Labriola	105	
5	Vanesa	Rakowski	109	
6	Sharron	Gorby	120	
7	Cecilia	Swearingen	101	
8	Doretha	Saffell	106	
9	Mariah	Heilman	103	
10	Rana	Bigby	107	
11	In	Vallecillo	116	
12	Librada	Taylor	119	
13	Elane	Mayes	103	
14	Alix	Haskett	109	
15	Anastacia	Goldblatt	115	
16	Melania	Whitlock	108	
17	Dario P	arker	111	
18	Rochel	Witherspoon	106	
19	Lucrecia	Tarrance	114	
20	Jermaine	Lamprecht	102	
21	Regine	Bascom	109	
22	Bryanna	Stansberry	107	
23	Naomi	Arai	114	
24	Darin	Blumstein	108	
25	Greg	Ricketts	102	
26	Wilburn	Guse	118	
27				
28				
29				

Client Data | Country Data

(4) Press Enter, and you'll now see the result of your VLOOKUP function in cell D2. In our case, the country name associated with the country code of 107 is "Finland":



D2 | =VLOOKUP(C2,'Country Data'!A:B,2,TRUE)

	A	B	C	D
1	Client First-Name	Client Last-Name	Country Code	
2	Elden	Stanhope	107	Finland

(5) Finally, drag-down the VLOOKUP function from cell D2 to cell D26, so that you can display the Country Name for each client under your first Client table:

D2 =VLOOKUP(C2,'Country Data'!A:B,2,TRUE)

	A	B	C	D
1	**Client First-Name**	**Client Last-Name**	**Country Code**	
2	Elden	Stanhope	107	Finland
3	Emile	Nation	103	Belgium
4	Velda	Labriola	105	Canada
5	Vanesa	Rakowski	109	Germany
6	Sharron	Gorby	120	Uruguay
7	Cecilia	Swearingen	101	Australia
8	Doretha	Saffell	106	China
9	Mariah	Heilman	103	Belgium
10	Rana	Bigby	107	Finland
11	In	Vallecillo	116	Peru
12	Librada	Taylor	119	United Kingdom
13	Elane	Mayes	103	Belgium
14	Alix	Haskett	109	Germany
15	Anastacia	Goldblatt	115	Nigeria
16	Melania	Whitlock	108	France
17	Dario P	arker	111	Italy
18	Rochel	Witherspoon	106	China
19	Lucrecia	Tarrance	114	Morocco
20	Jermaine	Lamprecht	102	Austria
21	Regine	Bascom	109	Germany
22	Bryanna	Stansberry	107	Finland
23	Naomi	Arai	114	Morocco
24	Darin	Blumstein	108	France
25	Greg	Ricketts	102	Austria
26	Wilburn	Guse	118	Sweden
27				
28				
29				

Client Data | Country Data

Microsoft Office

PowerPoint 2016 for Windows

Introduction to PowerPoint

Introductions

Microsoft Office PowerPoint 2016 is a presentation software application that aids users in the creation of professional, high-impact, dynamic presentations. Slides are the building blocks of a PowerPoint presentation. By using slides, the focus is not only on the speaker, but on the visuals (slides) as well.

Learning Objectives

After viewing this booklet, you will be able to:

1) Become familiar with Power Point Interface
2) Create a new presentation and save it
3) Add slides to a presentation
4) Delete and Rearrange slides
5) Apply a design theme
6) Work with themes and background styles
7) Use the various PowerPoint views
8) Enter and edit text
9) Insert graphics and other objects
10) Play the slideshow
11) Print handouts

Best Practices for Creating Presentations

Slide layout, font, color scheme, and content are the main components to developing a great presentation. Follow the guidelines below to create a good presentation:

1. Identify the critical information that needs to be presented and include it in your presentation.
2. Use no more than six bullet points per slide.
3. Keep bullet points short and to the point. Incomplete sentences are okay.

4. Minimize the number of font types used in your presentation.
5. Keep font sizes consistent.
6. Do not make all of the text uppercase.
7. For contrast, use a light-colored font on a dark background and vice versa.
8. Use bold formatting to make appropriate words stand out.
9. Minimize the use of italics. They are more difficult to read.
10. Do not vary the look of one slide greatly from the next. Consistency is key.
11. Identify text that can be represented pictorially and use appropriate graphics in its place.
12. Remove unnecessary graphics that are not relevant to the information presented.
13. Use consistent colors and font size on each slide.
14. Do not use unusually bright colors.
15. Do not clutter the slides with too many graphics.
16. Use graphics and transitions sparingly.

The Microsoft PowerPoint 2016 Interface

There are a number of prominent changes to the look and functionality of Microsoft PowerPoint 2016.

Let us have a look at its latest interface.

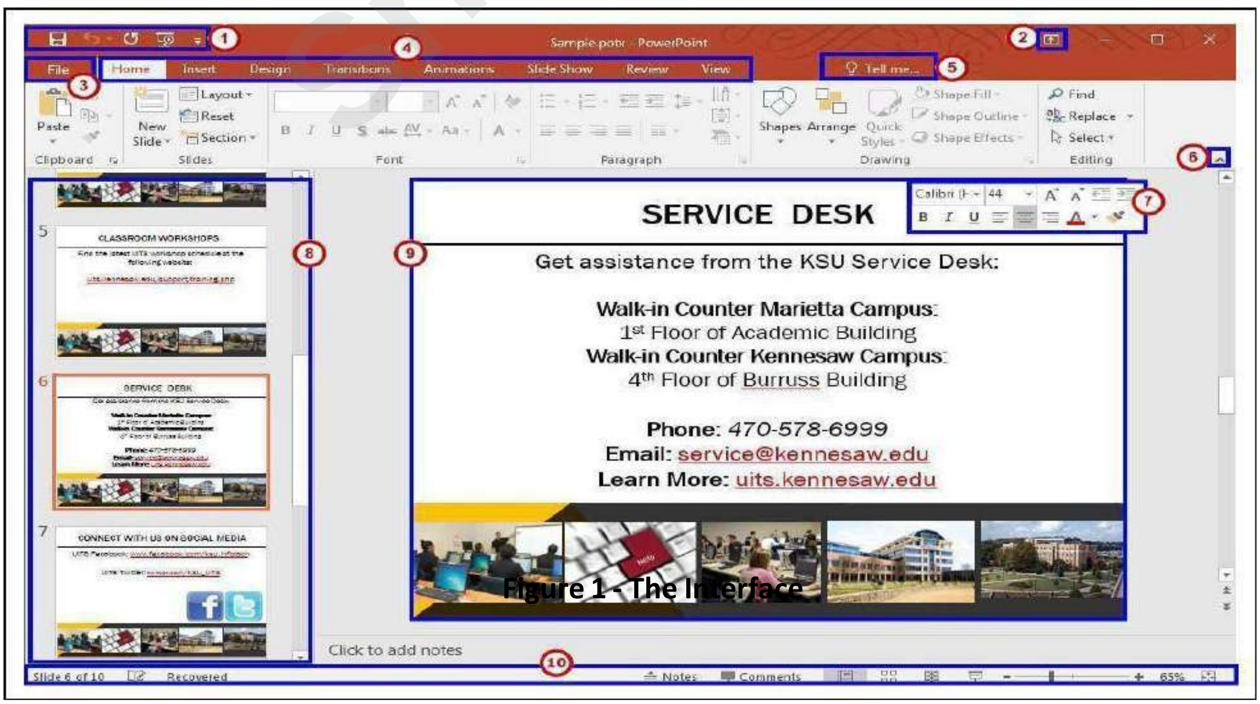


Figure 1 - The Interface

1. **Quick Access Toolbar** - Allows you to keep shortcuts to your favorite and frequently used tools.
2. **Ribbon Display Options** - You can collapse, auto hide, or show the whole ribbon.
3. **File Tab (Backstage View)** - The backstage view is where you manage your files and the information/properties about them (e.g. open, save, print, protect document, etc.).
4. **Ribbon** - Tabbed interface, where you can access the tools for formatting your presentation. The Home tab will be used more frequently than the others.
5. **Tell Me** - Look up PowerPoint tools, get help, or search the web.
6. **Collapse the Ribbon**- This button will collapse the ribbon. Click on the **pinned icon** to re-open the ribbon.
7. **Mini Toolbar** - Select or right-click text or objects to get a mini formatting toolbar.
8. **Slides Tab** - Use this to navigate through your slides within your presentation.
9. **Slide** - This is where you type, edit, insert content into your selected slide.
10. **Status Bar** - View which slide you are currently on, how many slides there are, speaker notes or comments in your presentation, change your views, or change your zoom level.

The Ribbon

The ribbon is a panel that contains functional groupings of buttons and drop-down lists organized by tabs (see Figure 2). The ribbon is designed to help you quickly find the commands that you need to complete a task.

Figure 2 - The Ribbon

The ribbon is made up of a set of tabs that pertain to the different functionalities of PowerPoint, such as designing slides, inserting media onto slides, or applying animations. Each tab is further divided into logical groups (of buttons), such as the Font group shown in Figure 1 above.

There are also contextual tabs that appear, depending on what you are working on.

For example, if you have inserted pictures, the Picture Tools tab appears whenever a picture is selected (See Figure 3).

Figure 3 - Contextual Tab

The Quick Access Toolbar

The Quick Access Toolbar is a small toolbar at the top left of the application window that you can customize to contain the buttons for the functions that you use most often.

Figure 4 - Quick Access Toolbar

To customize the Quick Access Toolbar:

1) Click the **drop-down arrow** on the far right (See Figure 5).
2) Click on any **listed command** to add it to the *Quick Access Toolbar* (See Figure 5).
3) Click **More Commands…** to choose from a comprehensive list of commands (See Figure 5).

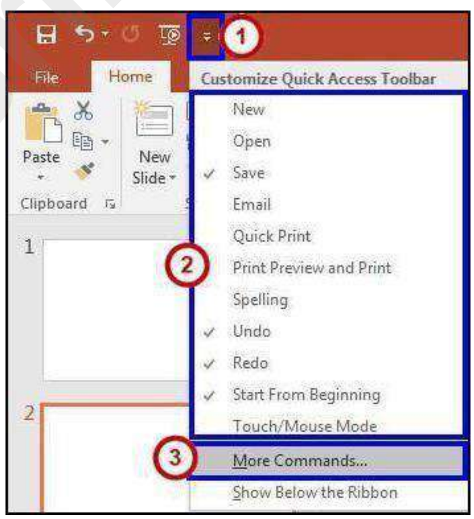


Figure 5 - Customize Quick Access Toolbar

4. Select a command from the list by clicking on it (See Figure 6).
5. Click the **Add** button (See Figure 6).
6. Repeat steps 4 & 5 to add additional commands (See Figure 6).

7. Click on the **OK** button to confirm your selection (See Figure 6).

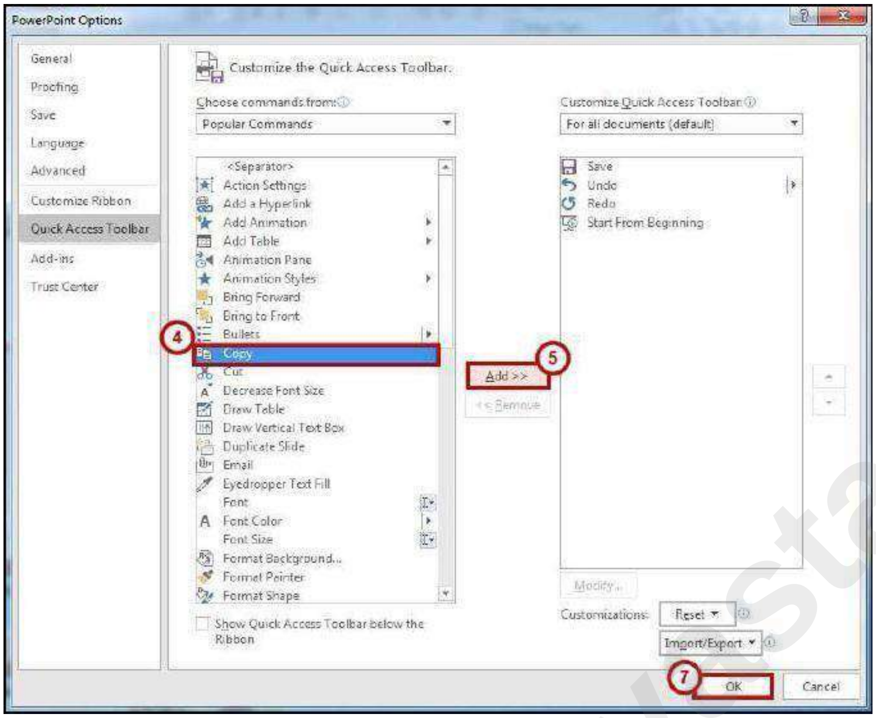


Figure 6 - Quick Access Toolbar Options

Tell Me

The Tell Me feature allows users to enter words and phrases related to what you want to do next to quickly access features or actions. It can also be used to look up helpful information related to the topic. It is located on the *Menu bar*, above the *Ribbon*.

Search for Features

1. Click in the **Tell Me** box.

Figure 7 - Tell Me

2. Type the **feature** you are looking for (See Figure 8).
3. In the *Tell Me* drop-down, you will receive a list of *features* based on your search. Click the **Feature** you were looking for (See Figure 8).

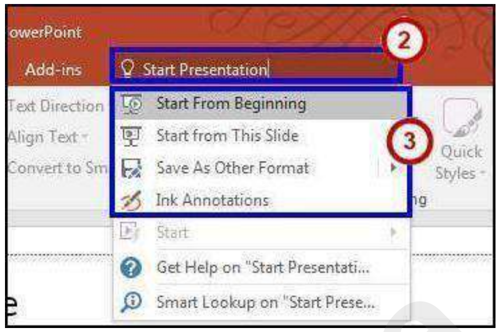


Figure 8 - Select a Feature

4. You will either be taken to the *feature* or a dialog box of that feature will *open*.

Get Help with PowerPoint

1. Click in the **Tell Me** box.

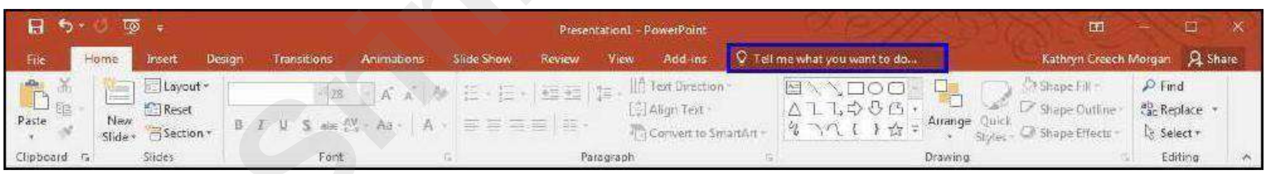


Figure 9 - Tell Me

2. Type the **feature** you want help with (See Figure 10).
3. In the *Tell Me* drop-down, click **Get Help on feature** (See Figure 10).

Figure 10 - Get Help on Feature

4. In the *PowerPoint 2016 Help* dialog box, you will get a list of help topics based on your search. Clickthe **Topic** you wanted help with.

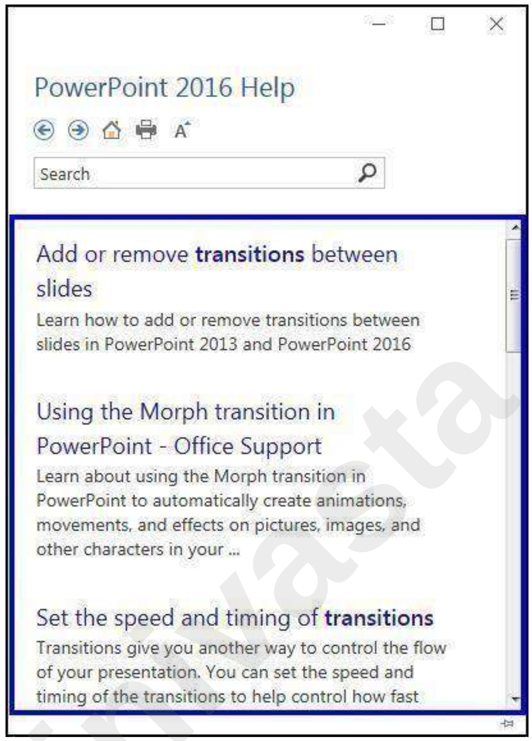


Figure 11 – PowerPoint 2016 Help

The Smart Lookup Tool

Smart Lookup uses *Bing* to provide you with search results for your selected term or phrase. It is located under the *Review* tab within the *Insights* section. The following explains how to use the *Smart Lookup* functionality.

1. Highlight the **word or phrase** you want to find information about (See Figure 12).
2. Right-click on the **word or phrase**.
3. Click **Smart Lookup** (See Figure 12).

Figure 12 - Smart Lookup

4. The *Insights* pane displays the information relevant to your selection. In the *Insights* pane, you receive the following information:

   a. **Explore** - Wiki articles, image search, and related searches from the internet (See Figure 13).
   b. **Define** - A list of definitions (See Figure 13).

Note: The *Insights* pane uses the Microsoft search engine Bing. For *Smart Lookup* to work you have to be connected to the internet.

5. To close the *Smart Lookup Insights* pane, click the **Exit** (**X**) button in the top right corner of the pane (See Figure 13).

Figure 13 - Insights Panel

Galleries

A Gallery is a collection of pre-defined formats which can be applied to various elements in Office applications, such as the Themes Gallery in PowerPoint (See Figure 14). A Gallery most often appears as a result of clicking on an item on one of the *Ribbon tabs*.

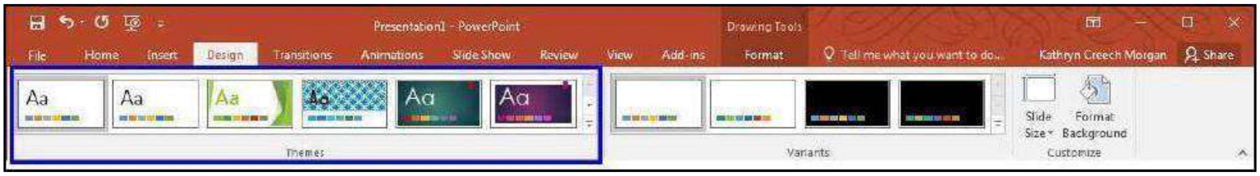


Figure 14 - Theme Gallery

The selections in a Gallery incorporate a feature called **Live Preview**. When the mouse cursor hovers over a selection in a Gallery, your document takes on the formatting attributes of that selection in order to give you a preview of how that selection will look when applied to your document.

Mini Toolbar

The Mini Toolbar is a semi-transparent toolbar that appears when you select text (See Figure 15). When the mouse cursor hovers over the Mini Toolbar, it becomes completely solid and can be used to format the selected text.

Figure 15 - Mini Toolbar

Status Bar

The Status Bar can be customized to display specific information. Below, in Figure 16, is the default Status Bar for PowerPoint:

Figure 16 - Status Bar

Right-clicking on the **Status Bar** brings up the menu to the right, which enables you to change the contents of the Status Bar by checking or un-checking an item (See **Error! Reference source not found.**).

Figure 17 - Status Bar Menu

Themes

A *Theme* is a set of formatting options that is applied to an entire presentation. A theme includes a set of colors, a set of fonts, and a set of effects. Using themes shortens formatting time and provides a unified, professional appearance.

The *Themes group* is located on the *Design tab*; it allows you to select a theme from the Themes Gallery, apply variants, and customize the colors, fonts, and effects of a theme (see Figure 18 on the next page).

Figure 18 - Design Tab: Themes

Applying a Theme to a Presentation

1. On the Ribbon, select the **Design tab** (See Figure 18).
2. In the *Themes* group, hover over a theme with your mouse to see a preview.
3. Click the **arrows** to scroll to additional themes.

Figure 19 - Themes

4. Select a theme by clicking on the **thumbnail** of your choice within the *Themes* group.

Applying a Theme Variant

1. On the Ribbon, select the **Design** tab (See Figure 18).
2. In the *Variants* group, hover over a **variant** with your mouse to see a preview.
3. Click the **down-arrow** to view any additional variants (See Figure 20).
4. Select a **variant** by clicking the thumbnail of your choice within the *Variants* group.

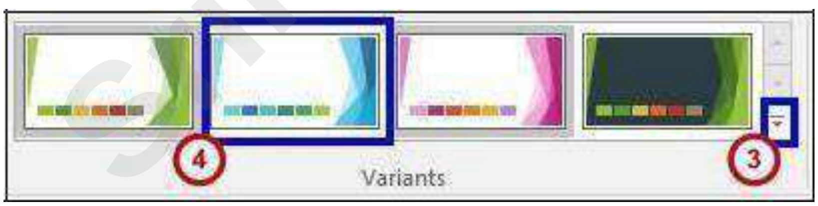


Figure 20 - Variants Gallery

Changing the Colour Scheme of a Theme

It may be necessary to change the color of a theme to better suit your presentation. In order to change the color scheme after applying a theme to your presentation:

From the Variants group, click the **down arrow** with the line above it, in the bottom right corner.

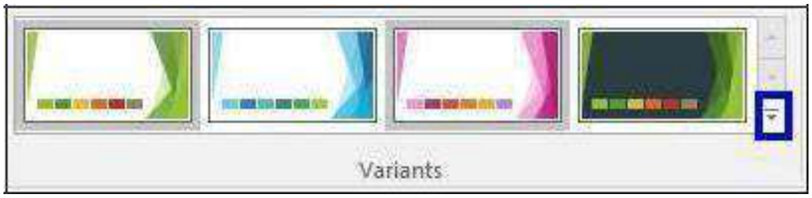


Figure 21 - Variants Drop-down

Select **colors** from the menu.

Figure 22 - Variants Menu

Select a color scheme from the list that appears.

Changing the Fonts of a Theme

In order to change the fonts of an applied theme:

1. From the *Variants* group, click the **down arrow** with the line above it, in the bottom right corner (See Figure 21).
2. Select **Fonts** from the menu.
3. Select your desired font from the list that appears.

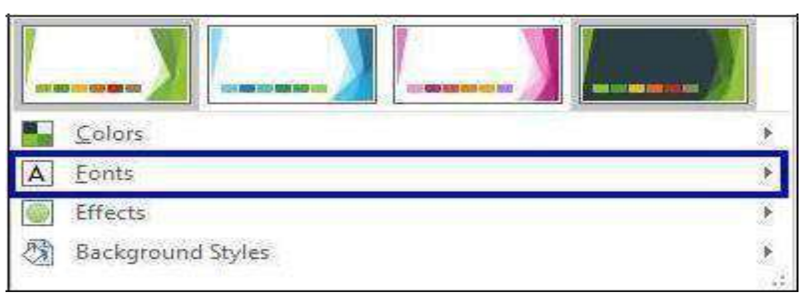


Figure 22 - Variants Menu

Changing the Effects of a Theme

In order to alter the visual effects of an applied theme:

1. From the *Variants* group, click the down arrow with the line above it, in the bottom right corner (See Figure 21).
2. Select **Effects** from the menu.

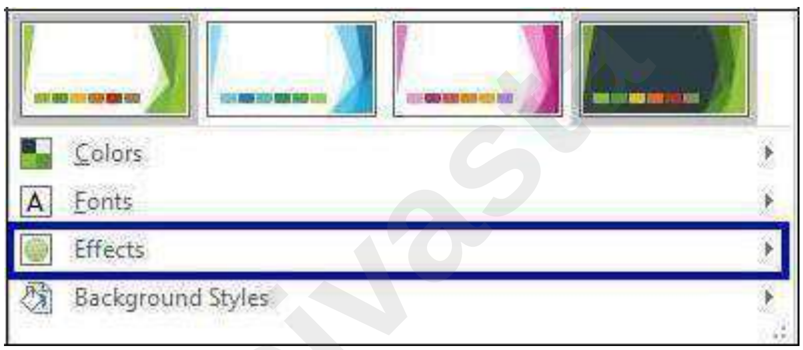


Figure 24 – Effects

3. Select your desired effect from the list that appears.

The File Tab

The **File** tab, shown below in Figure 25, provides a centralized location called the *Microsoft Office Backstage* view (see Figure 26). The *Backstage* view is used for all tasks related to PowerPoint file management: opening, creating, closing, sharing, saving, printing, converting to PDF, emailing, and publishing. The Backstage view also allows for viewing document properties, setting permissions, and managing different versions of the same document. (See the *PowerPoint 2016 Quick Guide* located at http://uits.kennesaw.edu/cdoc, for additional information on the Backstage View).

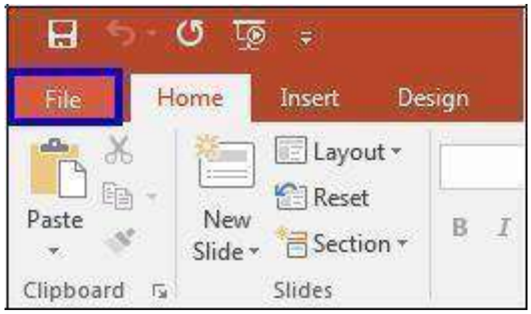


Figure 25 - File Tab

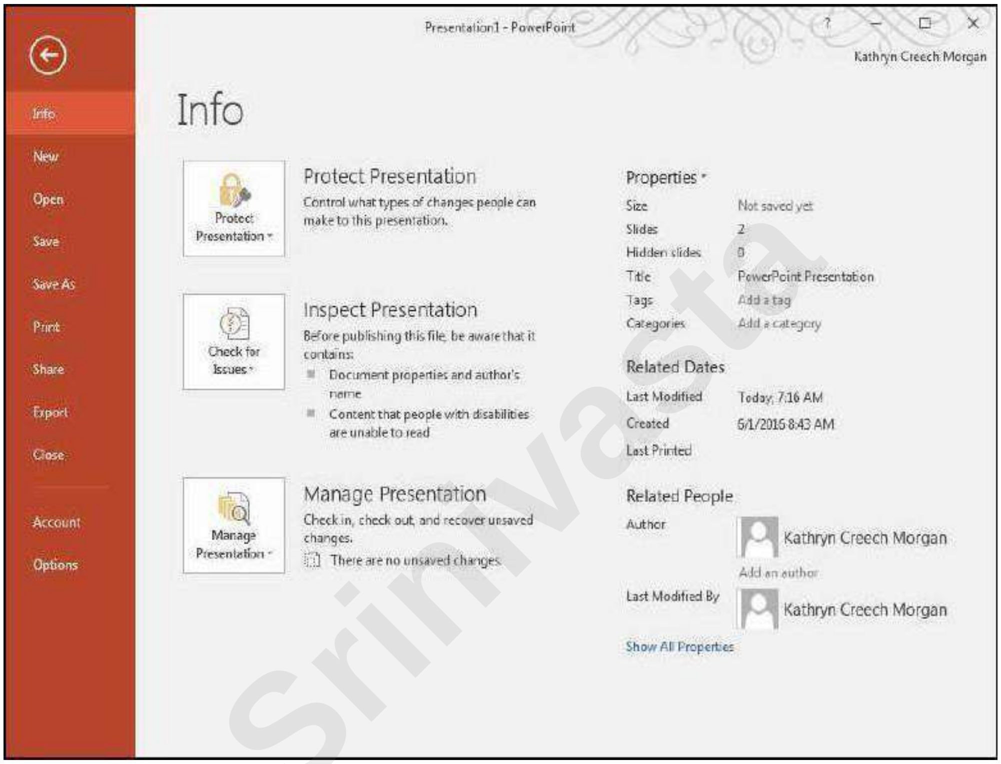


Figure 26 - Backstage View

Navigation

PowerPoint 2016 opens in the *Normal View* showing the *Slides Pane* on the left side of the window.

Slides Pane

The Slides pane shown on in Figure 27, displays all the slides available in a presentation and helps to navigate through the presentation. The slides are listed in sequence and you can shuffle the slides by dragging a slide from the current location and placing it in the preferred location.

Slide Preview

Select a slide in the slides pane to preview it in the Slide Preview window (See Figure 27). The slide preview all you to see how your text looks on each slide. You can add graphics, video and audio, create hyperlinks, and add animations to individual slides.

Notes Pane

It can be helpful to use the Notes Pane to remind yourself of speaking points for your presentation (see Figure 27). These personal notes can also be printed out for future referencing. Notes entered in the Notes Pane will not appear on the slide show.

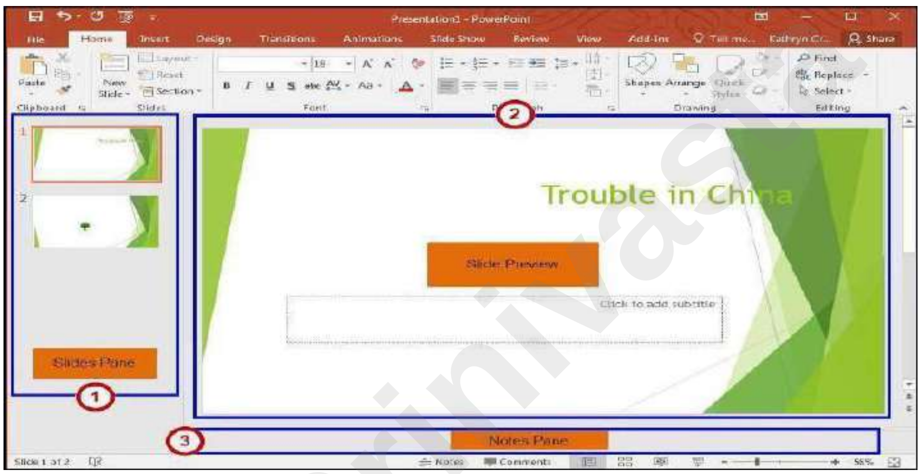


Figure 27 - Normal View

Changing Views

PowerPoint 2016 allows you to see and edit your slides in several views. To work with your presentation in a different view, click the **View** tab on the ribbon (see Figure 28) and select the appropriate view, or, click on the appropriate **Shortcut button** at the bottom right area of the Status Bar (see Figure 29).

Figure 28 - View Tab

Figure 29 - Status Bar

How to Create a New Presentation?

1. Click on the **File tab**.
2. Choose **New** (See Figure 30).
3. Double-click on **Blank presentation** (See Figure 30).

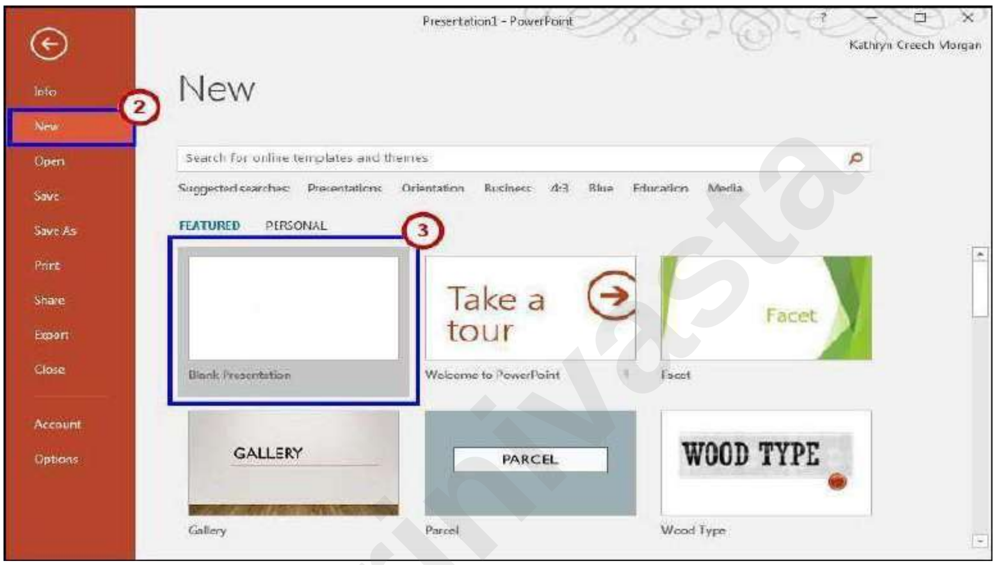


Figure 30 - Creating a New Presentation

4. A new blank presentation will be created.

Saving a Presentation

You created a new presentation. Now, you should save the presentation so that you can use it later. The *Save* command, available from the *File* tab, is used to save a newly created presentation or to save the changes made to an existing presentation. When saving a file for the first time, you are prompted to enter a file name for the presentation, and you are asked in which location you would like the file to be saved.

File Formats

PowerPoint 2016 uses PowerPoint Presentation (.pptx) as the default file format. Additional formats include PowerPoint 97-2003 (.ppt), PowerPoint Show (.ppsx), PowerPoint Show 97-2003 (.pps), Windows Media Video (.wmv), as well as GIF, JPEG, PNG, TIF and BMP. The PowerPoint Show is a presentation that always opens in Slide Show view rather than in Normal view.

How to Save a Presentation

1. Click the **File tab**.
2. Choose **Save As** to save the presentation with a new name in (See Figure 31).
3. Select **Computer** to save to the local drive (computer/laptop etc.) (See Figure 31).
4. Select your desired **folder** (See Figure 31).

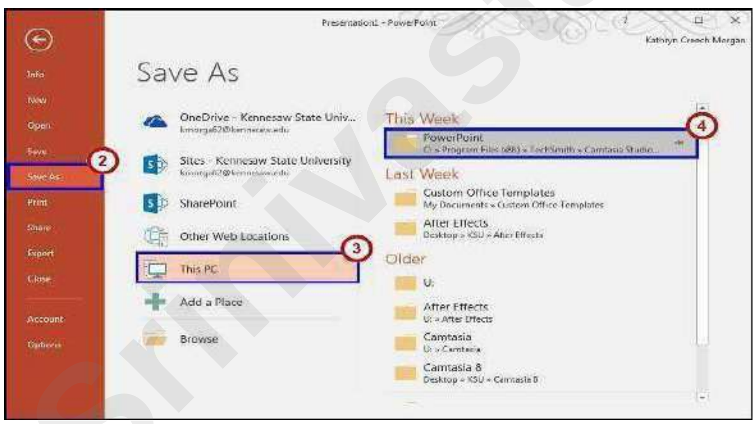


Figure 31 - Save As

5. In the **File Name** text box, type the name of the file (See Figure 32).
6. Click **Save as Type** and then do one of the following (See Figure 32):
   a. For a presentation that can be opened only in PowerPoint 2016 or in PowerPoint 2013, inthe *Save as Type* list, select **PowerPoint Presentation** (*.pptx).
   b. For a presentation that can be opened in either PowerPoint 2016 or earlier versions of PowerPoint, select **PowerPoint 97-2003 Presentation** (*.ppt).
7. Click **Save** (See Figure 32).

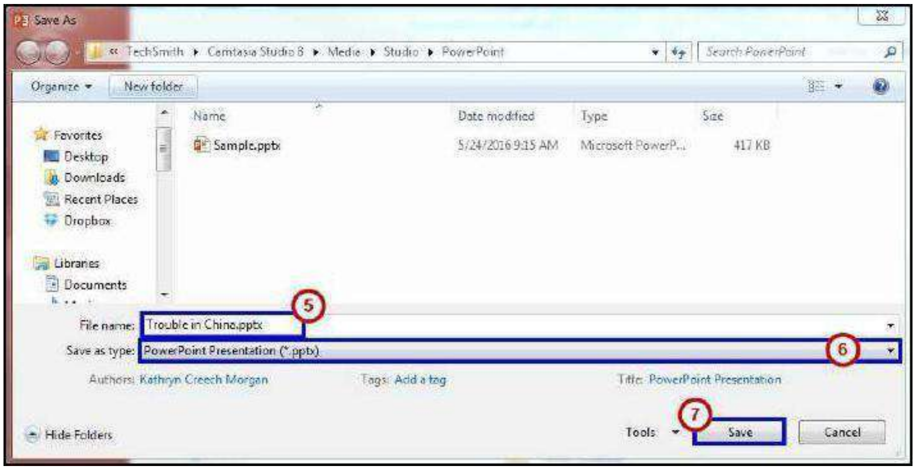


Figure 32 - File Name and Save as Type

Note: You can also press CTRL+S or click **Save** near the top of the screen to save your presentation quickly at any time.

Appearance

Slides

You are familiar with creating a new presentation and saving the changes you make to an existing presentation. You may need to add slides to the presentation to include more information.

Before you begin creating a presentation it is important that you decide on a design and layout. Slides and layouts are the basic building blocks of any presentation. For a presentation to be effective, care should be taken to apply the right slide layouts. Being able to add the appropriate slide layout to your presentation will enable you to present information more relevantly to your audience. PowerPoint offers several built-in slide layouts to deliver visually effective presentations.

Adding a New Slide

1. Within the slides pane, select the slide that you would like to insert a new slide after.
2. On the **Home tab** in the ribbon, click the drop-down arrow next to **New Slide**, within the Slides group, to display the default list of layouts (See Figure 33).
3. From the **New Slide** drop-down list, select a layout to insert (See Figure 33).

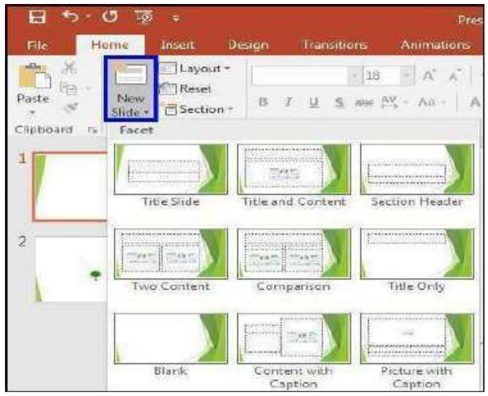


Figure 33 - New Slide

Types of Slide Layouts

Title Slide	This layout includes place holders for a main title and a subtitle
Title And Content	This layout includes a place holders to enter slide title and a place for text, charts, tables, pictures, clipart and Smart ART Graph
Section Header	This Layout allows space for section and sub-section titles.
Two Content	Much like the Title and Content slide layout, this layout offers a place for slide title text and two content places for text, charts, tables, pictures, clip art, and SmartArt graphics.
Comparison	This layout is just like the Two Content layout with the addition of two text placeholders to aid in compare and contrast slides.
Title Only	This layout offers a place to enter title text.
Blank	This is a blank slide with no placeholders.
Content With caption	With this slide you can enter a title, text, and content such as additional text, charts, tables, pictures, clip art, and SmartArt graphics.
Picture With Caption	This layout offers a place for a picture and caption text.

Table 2 - Slide Layouts

Text

A presentation is not all about pictures and background color—it depends mostly on the text. The message of your presentation is conveyed through the text. The visual aids are simply cosmetics to help support your message. The instructions below explain how to work with text.

Entering Text

Most slides contain one or more t*ext placeholders*. These placeholders are available for you to type text on the slide layout chosen. In order to add text to a slide, click in the **placeholder** and begin typing.

The placeholder is movable and you can position it anywhere on the slide. You can also resize a text placeholder by dragging the **sizing handles** (See Figure 34). Removing an unwanted text placeholder from a slide is as simple as selecting it and pressing Delete.

A text placeholder can contain multiple lines of text and will adjust the size of the text and the amount of space between the lines if the text exceeds the allowed space.

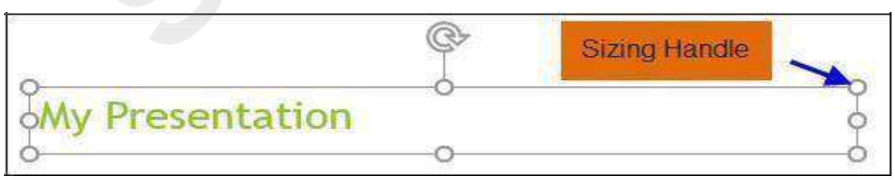


Figure 34 - Text Placeholders

Text Boxes can be added to any slide in order to provide additional room outside of the text placeholders.

Adding a Text Box

1. From the *Insert* tab, click on **Text Box**.

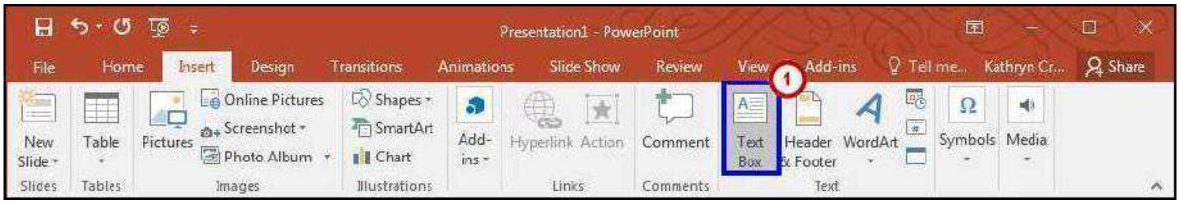


Figure 35 - Inserting a Text Box

2. Left-click on the **area of the slide** where you want to add text.
3. While maintaining the left-click, drag the **mouse cursor** down a bit and then to the right, then release. The *dashed text box* appears.
4. Left-click once inside the **text box** and start entering your text (See Figure 36).

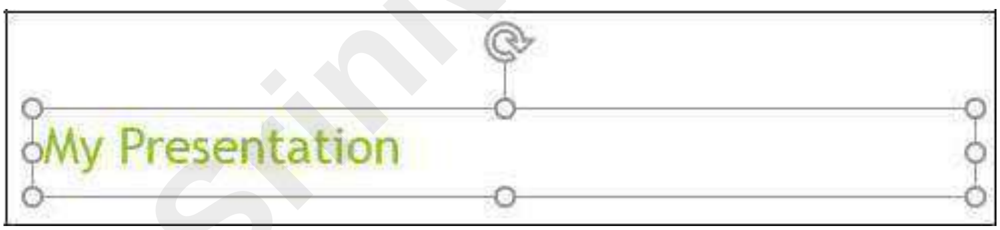


Figure 36 - Text Entry in a New Textbox

Formatting Text

When creating a presentation, it is likely that you will be doing some formatting. In order to edit text on a presentation:

1. Select the **text** (by left-clicking and maintaining the left-click while dragging the mouse cursor across the text).
2. Use the Font Group on the *Home* tab or right-click on the **selected text** and choose formatting options from the *Mini Toolbar* (See Figure 37).
3. Make necessary changes to the font and click **outside** of the text placeholder to accept the changes.

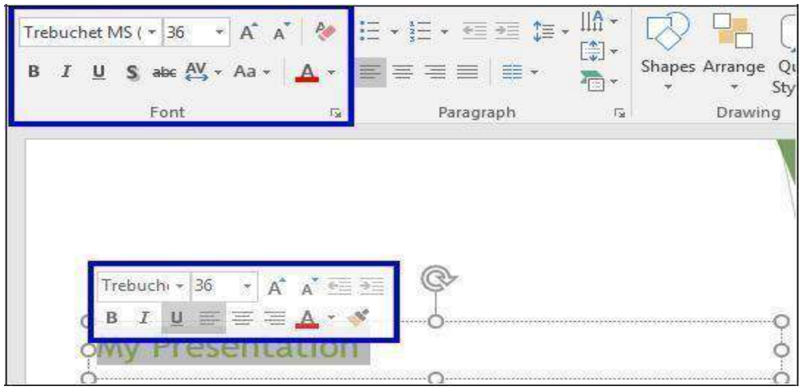


Figure 37 - Formatting Text

Graphics

Another way to add emphasis to your presentation is to have visual aids or graphics. Although we learned earlier that the most important part of your presentation is the message, it is always helpful to use graphics to get your point across more concisely and in a shorter period of time.

Inserting Pictures

When inserting clip art onto a preselected slide layout:

1. Go to the **Insert** tab.
2. Type in your **key word** or **phrase** of the object you are looking for into the search box.
3. This will open the Insert Pictures window.

Figure 38 - Search for Clipart

4. Scroll through the given results to find your desired clip art (See Figure 39).
5. Once found, click on the **image** (See Figure 39).
6. Select **Insert** to add the clip art to your slide (See Figure 39).

Figure 39 - Insert Clip Art

Note: You are responsible for respecting others' rights, including copyright, so be mindful when selecting your image(s).

Images from a File

Images from your own collection and experiences may also add value to your presentation. You must have the image saved prior to adding it to your presentation.

Inserting an Image from a file

Place your cursor where you would like the image to appear. Select the **Insert** tab. Click **Pictures** (See Figure 40). In the Insert Picture dialog box, navigate to find your image. Select your image, and click.

Insert

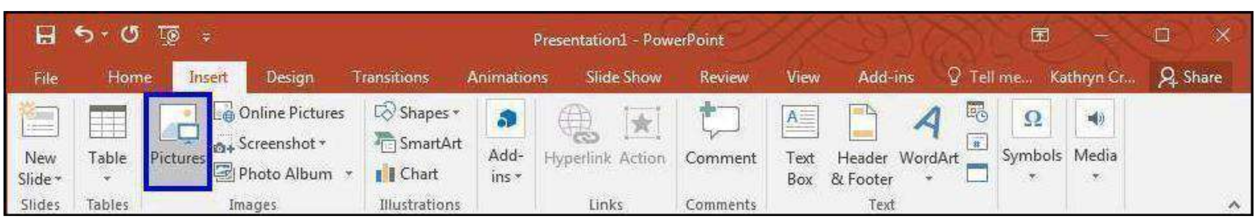


Figure 40 - Insert Pictures

Draw Shapes

Shapes are simple geometric objects that are pre-created by PowerPoint and can be modified. A shape can contain text or can appear without it. It can also be filled with color, and the outline of the shape can be given a different style and color.

Inserting a Shape

1. Select the **Insert** Tab.
2. Click **Shapes**.
3. Select the **shape** you wish to draw
4. Your cursor becomes a small black plus sign.
5. In the *Slide* pane, point the **crosshair** mouse pointer to the upper-left corner of the area where you want to draw the shape, hold the left mouse button down, and then drag diagonally down to the right to create the shape.

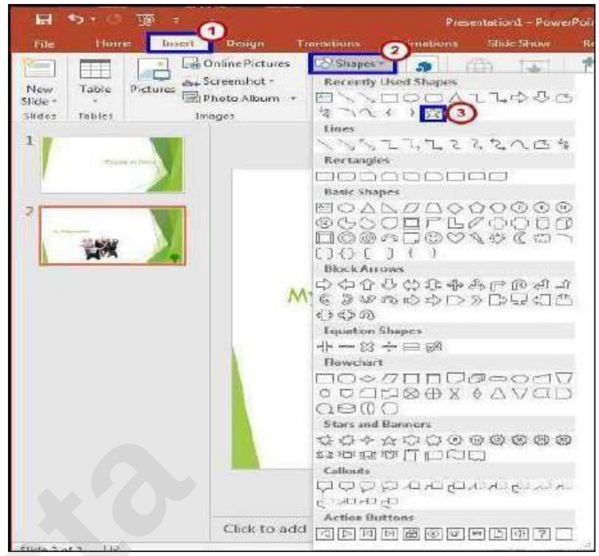


Figure 41 - Insert Shapes

Altering a Shape

1. You can now move your shape if necessary by clicking on the **object**, holding down the left mouse button, and dragging it to another location.
2. You may also alter the look of your shape by selecting the **shape** and clicking the **Format** tab (See Figure 42).
3. In the *Shape Styles* group, scroll through **additional styles** and click on the **desired style** to apply it to the shape (See Figure 42).

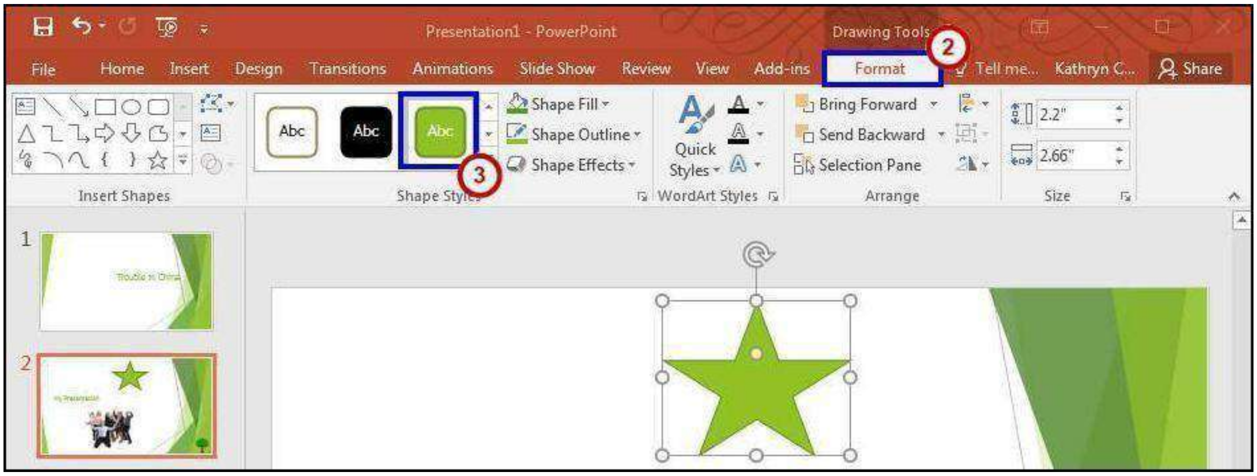


Figure 42 - Altering Shapes

Hyperlinks

You may have an additional document or a great website to enhance your presentation. Adding hyperlinks into your presentation can allow you to quickly jump to supplemental information while you are presenting.

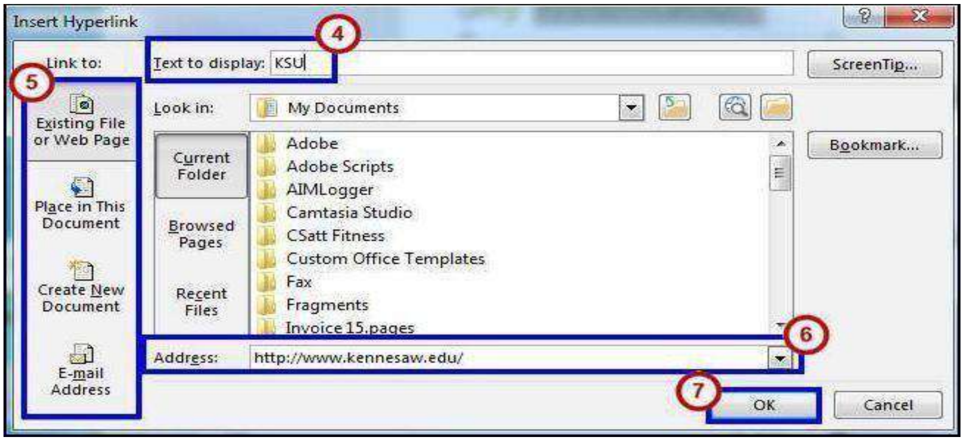
Adding a Hyperlink

1. Select the **text** that you would like to change to a hyperlink.
2. Select the **Insert tab** (See Figure 43).
3. Click **Hyperlink** (See Figure 43).

Figure 43 - Inserting a Hyperlink

4. The text you selected will appear in the *Text to Display* field at the top of the window (See Figure 44). You can change the text if you would like.
5. Select the location where you want to link to from the *Link To* column on the left (See Figure 44).
6. Type the address that you want to link to in the Address field (See Figure 44).

7. Click **OK** (See Figure 44). The text that you selected will now hyperlink to the web address.



Note: You can also type the hyperlink out and press **enter**, and *PowerPoint* will automatically create the hyperlink.

Header and Footer

Occasionally it is necessary to add information to the *Header* or *Footer* of a *PowerPoint* presentation, Just the printout or both. This may include information such as the author's name, date, and time. organization, class information, etc.

Adding a Header or Footer

1. Select the **Insert tab**.
2. Click **Header & Footer**.

Figure 45 - Inserting Header & Footer

3. Select whether you would like to add these settings to the *Slides* or *Notes and Handouts* by selecting the appropriate tab (See Figure 46).
4. Click inside the checkbox to add the **Date and Time** or **Slide Number** (See Figure 46).

5. Type additional information such as the author's name in the Footer box. (See Figure 46).
6. If you would like this information to appear on all slides, click **Apply to All** (See Figure 46).

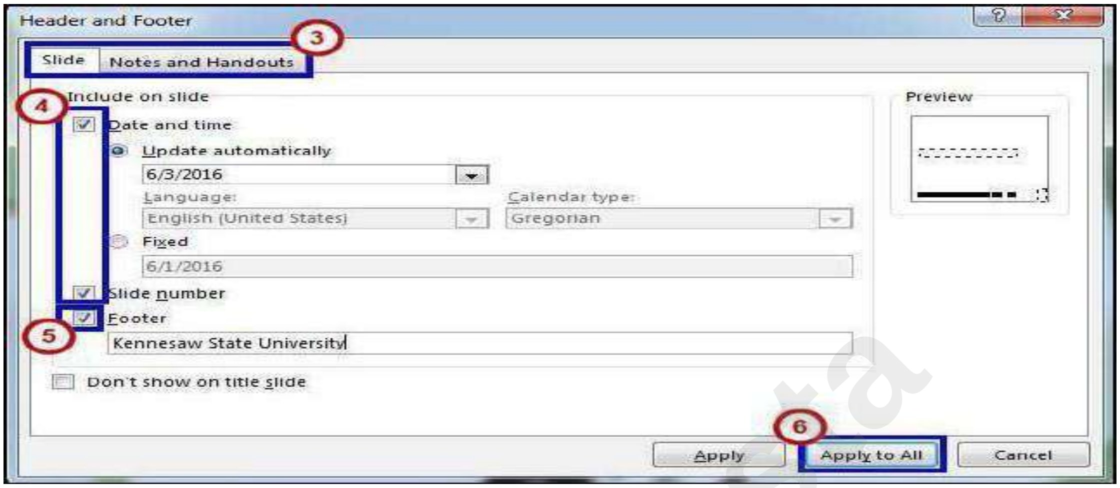


Figure 46 - Header and Footer Settings

Printing

PowerPoint 2016 allows you to print your presentation in order to aid you in presenting or to give your audience something to take notes on. There are multiple formats available when printing; you should choose a format which best suits your presentation and audience needs.

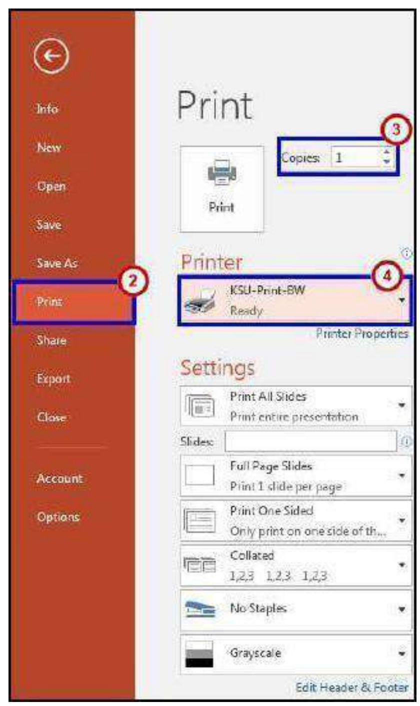


Figure 47 - Printer Settings

Printing Your Presentation

From the Backstage view:

1. Click the **File** tab.
2. Click on **Print** to view the printer settings

(See **Error! Reference source not found.**).

3. Indicate the **number of copies** you would like to print (See **Error! Reference source not found.**).
4. Select the **printer** (See **Error! Reference source not Found)**

From the Settings section:

1. Click on **Print All Slides**. If you do not need to print all slides, you may select another option in the window (See Figure 48).

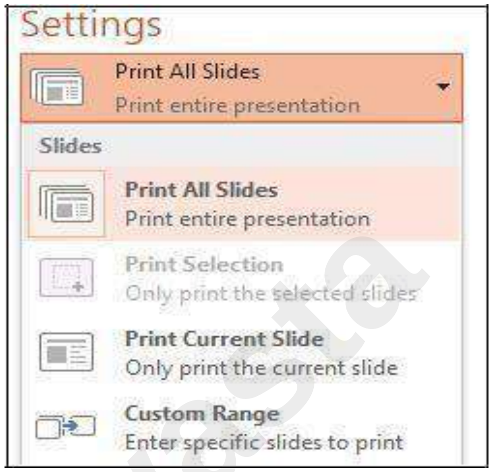
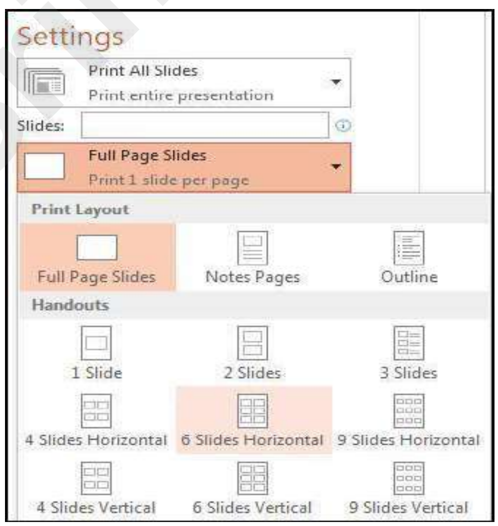


Figure 48 - Print All Slides

2. Click on **Full Page Slides** to access Print Layout options such as Handouts and Notes Pages (See Figure 49).



Note: If you select **Handouts**, choose the amount of slides that should appear on the printed page from the Handouts box. The *3 Slides* option allows for notes to be written on the handout.

3. Click on **Print One Sided** to access options to print on both sides of the paper (See Figure 50).

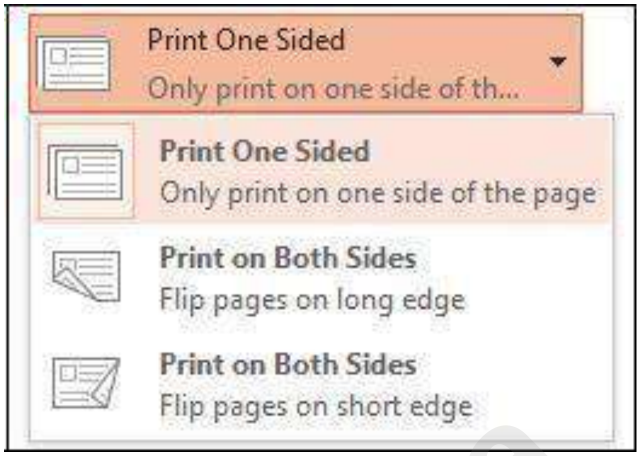


Figure 50 - Print One Sided

4. Click on **Grayscale** to access options to print in grayscale, color, or black & white

Figure 51 - Grayscale

5. Once you are ready to print, click on the **Print**  button.

Transitions & Animations; Timing the Presentation

Introduction

This booklet is the companion the *PowerPoint 2016: Transitions & Animations; Timing Your Presentation* workshop. It will explain how to add transitions and animations to presentation slides. Text, graphics, diagrams, charts, shapes, and other objects placed on your slides can all be animated to draw attention, emphasize important points, control the flow of information, and add interest to the presentation during the slideshow. You can rehearse your presentation to make sure that it fits within a certain time frame. This booklet will also explain how to time your presentation to record the amount of time needed to present each slide.

This booklet offers step-by-step instructions to creating dynamic presentations using transitions, animations, and timing. For other functionalities, please refer to the PowerPoint 2016: Intro to PowerPoint booklet.

Learning Objectives

After reading this booklet, you should be able to:

1. Apply transitions
2. Change the properties of a transition
3. Create one or more animations to a slide
4. Reorder animations
5. Use the animation painter
6. Remove an animation
7. Use the animation pane
8. Time the presentation

Transitions

Transitions are used to control the pace of your presentation and create a better flow between slides. Rather than simply changing from one slide to the next, you can apply special effects to the slides as they transition. The steps below explain how to apply transitions to your slides.

Adding a Slide Transition

1. Select the slide you wish to apply a transition to.
2. Select the *Transitions* tab (See Figure 1).
3. Select a **transition effect** from the *Transitions to this Slide* group (See Figure 1).
4. Select the **down-arrow** to view all of the available Transitions at once (See Figure 1).

Figure 1 - Apply Transitions

5. Once you select a transition type, click **Preview** to view the effect on the slide.

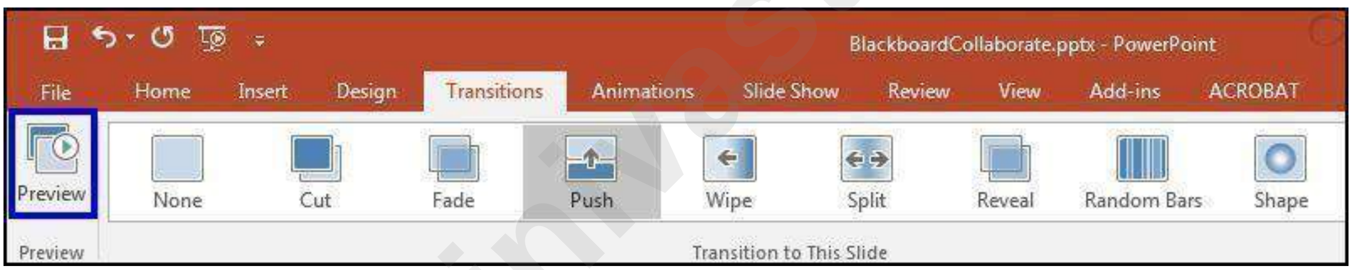


Figure 2 - Preview

6. If you would like the same transition for all slides in the presentation, click **Apply To All** in the *Timing* group.

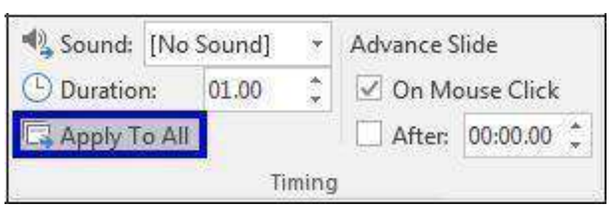


Figure 3 - Apply To All

Changing the Properties of a Transition

Occasionally you may experience the transition speed is too fast or too slow. You may also want to add sound to a transition effect to enhance your presentation as well as control how the slide advances. In order to change these settings, follow the steps below:

1. Select the **Transitions** tab.
2. In the *Timing* group, click the **down-arrow** for Sound to add a sound effect for the transition (See Figure 4).
3. Select the **up/down arrow** next to Duration to choose the amount of time the transition lasts. (See Figure 4).

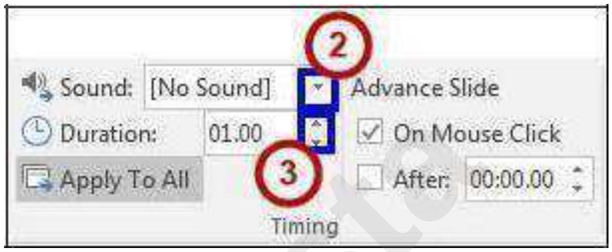


Figure 4 - Sound and Duration

a. Click the **Preview** button as shown in the Preview group as shown in Figure 2, to test yoursettings.

b. If you would like the same setting for all slides within the presentation, click **Apply to All**.

c. You can also set how the slide will advance; either by a mouse click or after a certain amount oftime has elapsed.

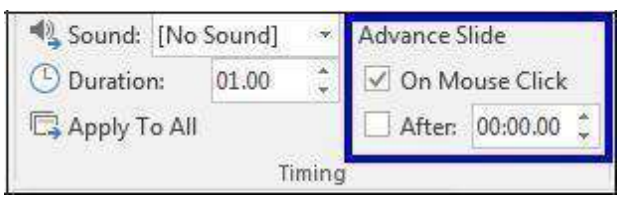


Figure 5 - Advance Slide

Creating One or More Custom Animations

1. Click the **Animations** tab.
2. Click the object (i.e.: clipart, textbox, shape) you would like to animate to select it
3. Click the **Add Animation** button.

Figure 6 - Add Animation

4. Select the animation from the drop-down list which offers four types of animation effects(See Figure 7):

Entrance: Objects can enter the slide via any of the entrance effects, including

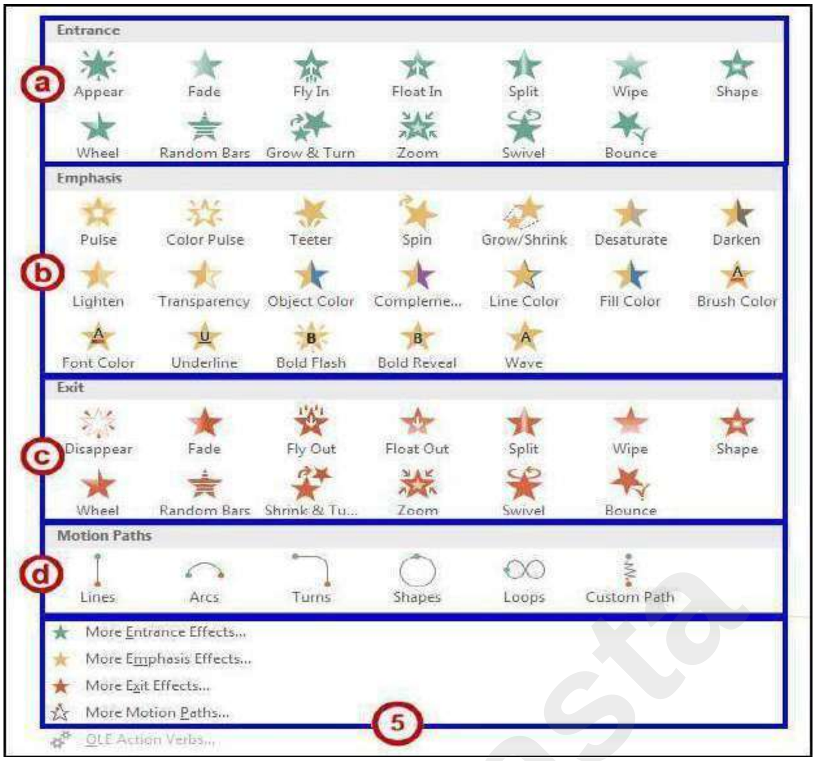
Fly In, *Dissolve In*, *Grow & Turn*, *Swish*, and *Crawl In* (See Figure 7).

Note: If you do not apply an entrance effect, the animated object starts at the position where you placed it on the slide.

5. **Emphasis**: This effect draws attention to an object that is already on the slide via any of the emphasis effects available, including *Flash Bulb*, *Spin*, *Grow & Shrink*, and *Color Wave* (See Figure 7).
6. **Exit**: Objects can leave the slide via any of the exit effects such as *Fly Out*, *Disappear*, *Spiral Out*, and *Sink Down* (See Figure 7).
7. **Motion Paths**: Objects can travel along a track which was created based on predefined motionpaths such as *Circle*, *Funnel*, *Stairs Down* or *Wave.* The object could also travel along one of four drawn custom paths: *Line*, *Curve*, *Freeform*, and *Scribble* (See **Error! Reference source not found.**Figure 7).

Note: To see a preview of the animation, select an animation and click the **Preview** button on the ribbon.

1) To view additional effects, select either **More Entrance Effects**, **More Emphasis Effects**, **More Exit Effects**, or **More Motion Paths** from the bottom of the *Add Animations* menu
(See Figure 7).



2) Click the option of your choice to apply that animation effect to the selected object.

3) The Effect Options button may become available (Some effects such as Appear do not have any effect options). Click the **Effect Options** button, and change the option if desired.

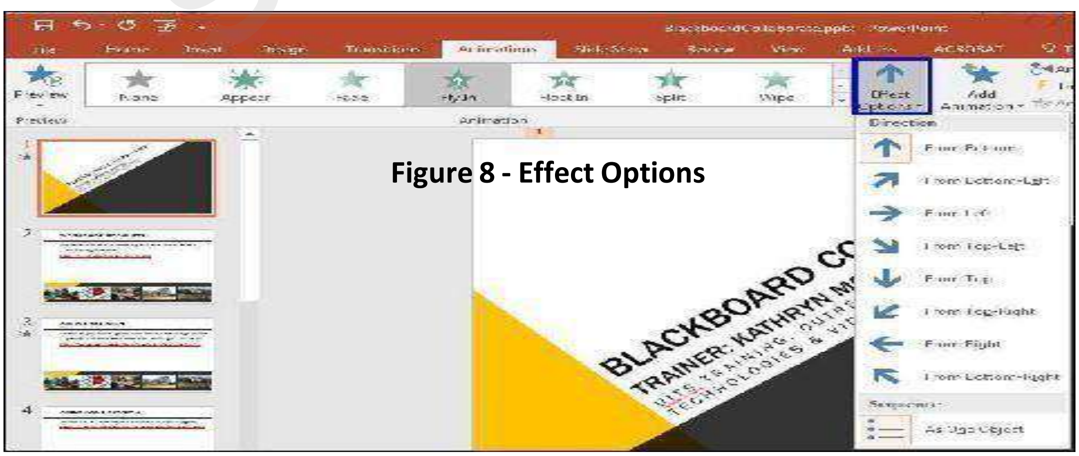


Figure 8 - Effect Options

Note: The effect options will vary depending on which animation is chosen.

a) Apply the timing options for your animation effect.

By default, the animation will play upon a mouse click. To make the animation play automatically when the slide loads, change the **Start** from *On Click* to *After Previous*.

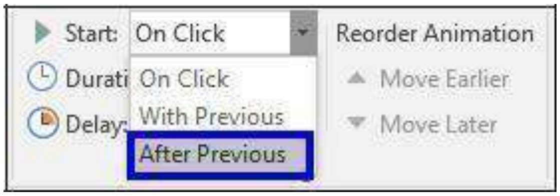


Figure 9 - Timing Options

b) You can specify the length of an animation by editing the **duration**.

Note: The longer the duration, the slower the effect.

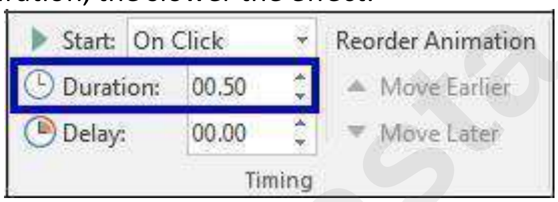


Figure 10 - Duration

c) You can set the animation to play after a certain number of seconds by specifying a **delay**.

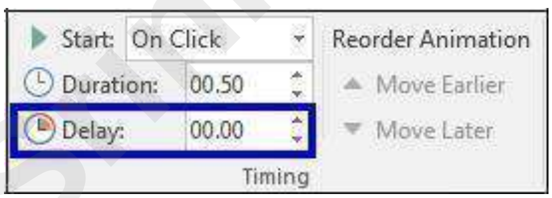


Figure 11 - Delay

d) Once the basic settings are in place, click the **Preview** button on the left side of theAnimations ribbon to visualize the animation.

Figure 12 - Preview

e) To create more than one animation for a given object, click the **Add Animation** button again and repeat steps 4 through 9. You can, for example, give an object an *entrance* effect, an *emphasis*effect, and an *exit* effect. This would let you bring an object onscreen, draw attention to it, and then have it leave the screen.

Reordering Animations

After applying a few animations to one or more objects on a slide, you may wish to change their order sequence.

1. Click the **object** that has the animations you wish to reorder.

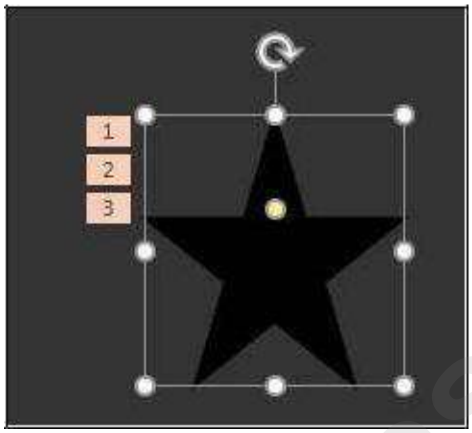


Figure 13 - Select Object

2. Click the **number** to the left of the object that is representing the effect you wish to move.

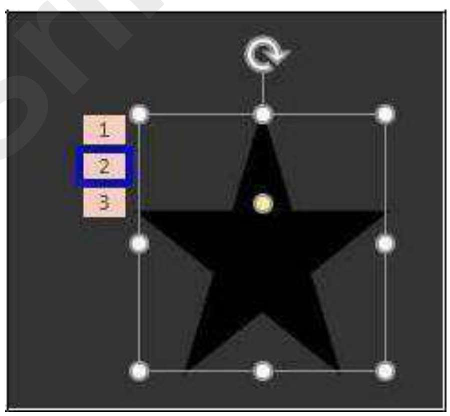


Figure 14 - Select Effects

3. At the right side of the ribbon, in the *Reorder Animation* group, click the button of your choice to **Move Earlier** or **Move Later**.

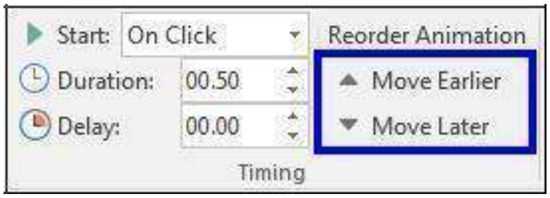


Figure 15 - Reorder Animation

4. Upon making these adjustments, the animation order will change.

The Animation Painter

The **Animation Painter** makes it easy to copy a complete animation effect from one object to another.

1. Click the **animated object**.

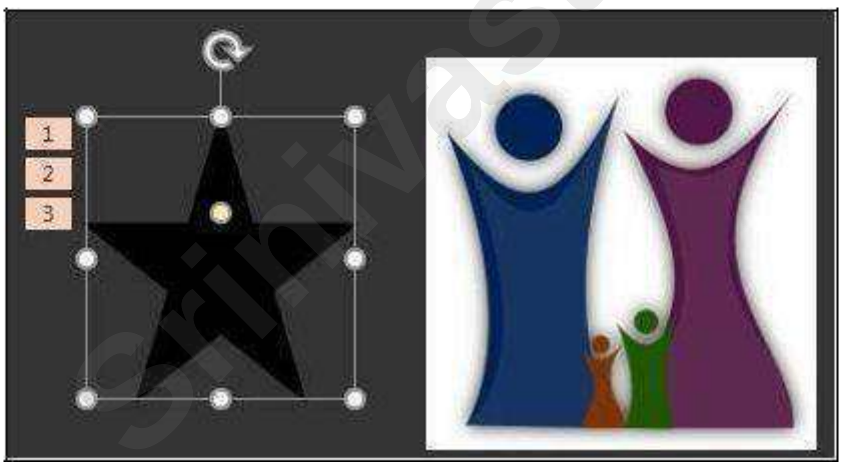


Figure 16 - Select Animated Object

2. Click the **Animation Painter** button on the ribbon. A small paint brush will appear next to the cursor.

Figure 17 - Animation Painter

3. Click the **object** where you wish to apply the animation(s). The animations created for the first object are applied to the second object selected.

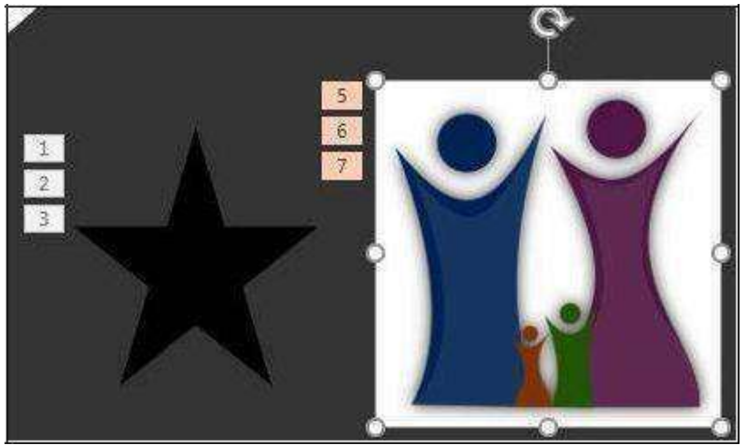


Figure 18 - Duplicated Animation Effects

Removing an Animation

1. Click the **object** that has an animation already applied and that you wish to remove.
2. Click the **number** to the left of the object that is representing the effect you wish to delete.

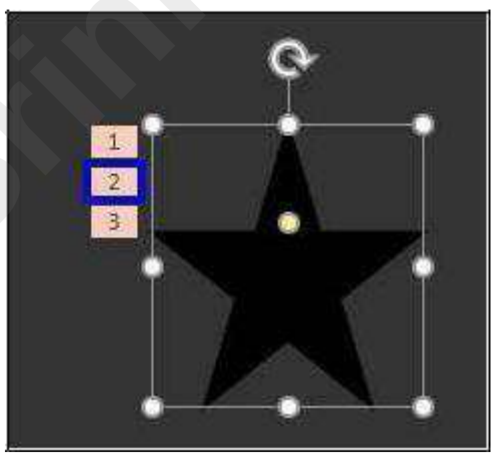


Figure 19 - Select Effect to be Deleted

3. Press the **Delete** key on the keyboard. This will delete the animation.

The Animation Pane

You may access additional and more advanced animation options such as timeline, sound, and timing by enabling the animation pane.

1. From the *Animations* tab, click the **Animation Pane** button on the ribbon.

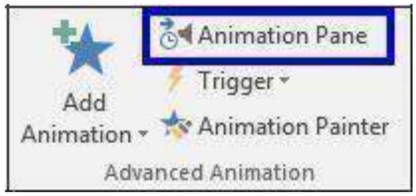


Figure 20 - Animation Pane Button

2. In the list in the *Animation Pane*, click the **animation** to be adjusted to select it.

3. Click the small **drop-down arrow** to the right of the selected animation.

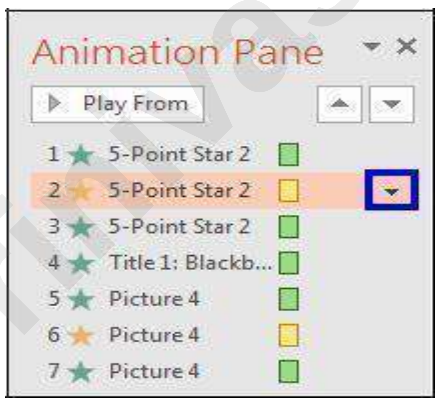


Figure 21 - Animation Pane

4. Select **Effect Options**.

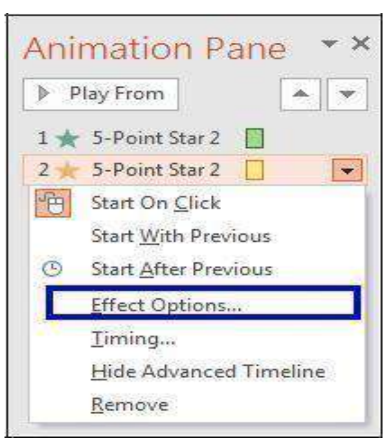


Figure 22 - Select Effect Options

5. Under the *Enhancements* section, you can add **sound** and set the object to **not dim** after the animation.

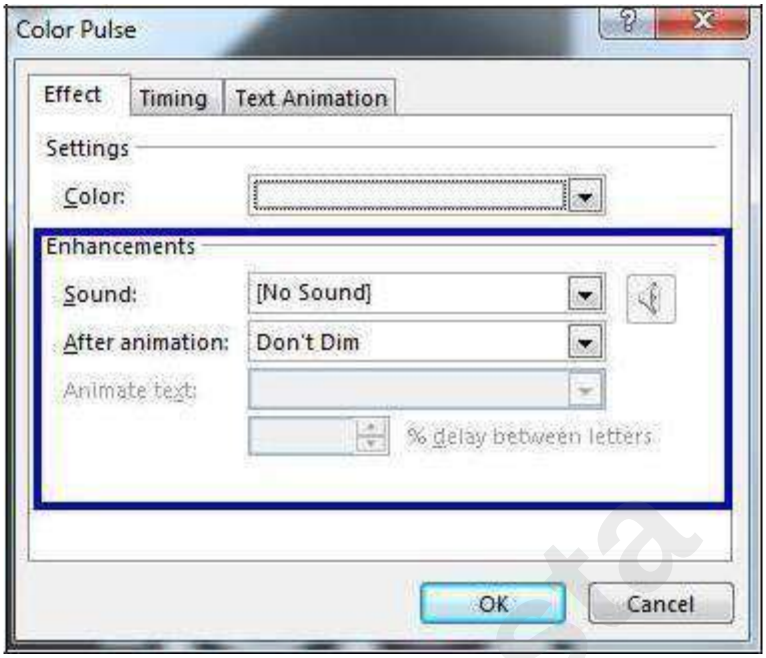


Figure 23 – Enhancements

6. Click the *Timing* tab. From the **Timing** tab, you may set the animation to repeat and/or rewind when done playing. The **Speed** setting can be adjusted as well. Some effects have an additional Property setting that allows you to control the range of an object's movement. (See Figure 24).

7. Click **OK** (See Figure 24).

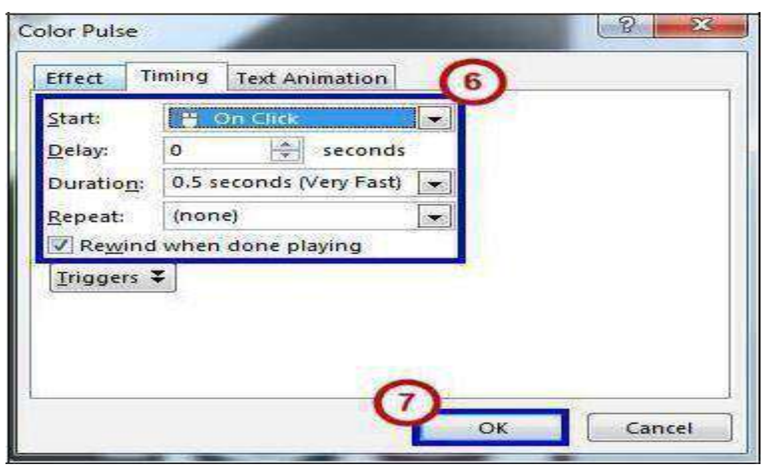


Figure 24 - Timing Tab

8. Click the **Play From** button on top of the Animation Pane to see your animation(s) in action.

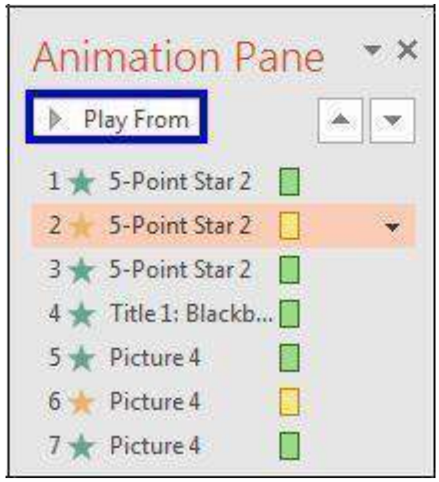


Figure 25 - Play From

Note: Click the **X** in the upper right corner of the *Animation Pane*, to close it.

Timing the Presentation

A good way to determine how long it will actually take you to do the presentation is to use the Rehearse Timing tool.

1. On the *Slide Show* tab, in the *Set Up* group, click **Rehearse Timings**. The Rehearsal toolbar
2. appears and the Slide Time box begins timing the presentation.

Figure 26 - Rehearse Timings

The Rehearsal Toolbar

1. Next (advance to next slide)
2. Pause
3. Slide Time
4. Repeat
5. Total presentation time

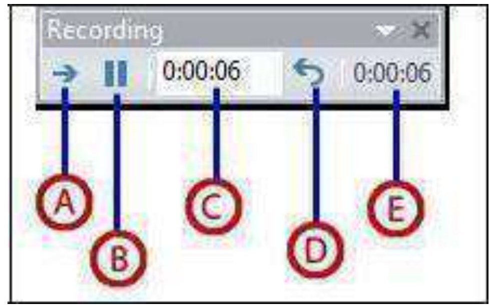


Figure 27 - Rehearsal Toolbar

a. While timing your presentation, do one or more of the following on the Rehearsal toolbar:

To move to the next slide, click **Next** (See Figure 27).

To temporarily stop recording the time, click **Pause** (See Figure 27).

To restart recording, click the **Resume Recording** button in the window that appears (See Figure 28).

Figure 28 - Resume Recording

To set an exact length of time for a slide to appear, type the length of time in the Slide Time box.

To restart the recording time for the current slide, click **Repeat**.

b. After you set the time for the last slide, a message box displays the total time for the presentation and prompts you to do one of the following:

To keep the recorded slide timings, click **Yes**.To discard the recorded slide timings, click **No**.

c. Slide Sorter view appears and displays the time of each slide in your presentation.**Advancing Slides Automatically**

You can set your slides to advance automatically so that you will not have to advance them manually.

1. Select the slide that you would like to apply a timing for.
2. Select the **Transitions** tab on the ribbon.
3. Within the *Timing* group, go to the **Advance Slide** section.
4. Insert a check next to **After**, and then enter the amount of time that you would like to display the slide.

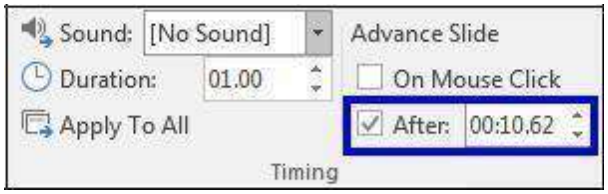


Figure 29 - Advance Slide

5. Select another slide and repeat the process.

Audio and video

Introduction

Adding audio and video to a PowerPoint presentation can be very useful, and can make your presentation more appealing to your audience. Audio can be added to slides in the form of background music, sound effects, or recordings. Video can also be added to a slide and played during your presentation. This is a great way to make your presentation more engaging for your audience.

Once your slideshow is complete, you'll need to present it. This booklet will teach you how to present.your presentation and explain how to use the presentation tools.

In PowerPoint 2016, you can insert and playback many different file formats.

Supported audio file formats:

FILE FORMAT	EXTENSION
AIFF Audio file	.aiff
AU Audio file	.au
MIDI file	.mid or .midi
MP3 Audio file	.mp3
Advanced Audio Coding - MPEG-4 Audio file	.m4a, .mp4
Windows Audio file	.wav
Windows Media Audio file	.wma

Supported video file formats:

FILE FORMAT	EXTENSION
Windows Media file	.asf
Windows Video file (Some .avi files may require additional codecs)	..avi
MP4 Video file	.mp4, .m4v, .mov
Movie file	.mpg or .mpeg
Adobe Flash Media	.swf
Windows Media Video file	.wmv

Learning Objectives

Upon completing this documentation, you will be able to:

1. Add audio to the slide show
2. Add video to the slide show
3. Edit an audio file
4. Edit a video
5. Set up a slide show
6. Start the slide show
7. Understand how to use the presentation tools and features

Audio

In PowerPoint, you can add audio to your presentation from clipart audio, from a file, or you can record audio. The instructions below explain how to add audio to your slide show using each of these methods.

To Insert Audio from a File

1. In Normal view, navigate to the slide that you want to add audio.
2. Click the **Insert** tab on the *Ribbon*, and then click the **Audio** button.

Figure 1 - Ribbon

3. From the drop-down list, within the Media group, select **Audio on My PC**.

Figure 2 - Audio on My PC

4. The *Insert Audio* window will appear. Navigate to the location of the audio file.

Figure 3 - Insert Audio

5. Select the audio file, and then click the **Insert** button.

6. The audio file will appear on the slide.

To Record Audio

1. In Normal view, navigate to the slide that you want to add audio.
2. Click the Insert tab on the Ribbon, and then select **Audio**.

Figure 4 - Ribbon

3. From the drop-down list, within the Media group, select **Record Audio**.

Figure 5 - Record Audio

4. The Recorder will appear. Type a **name** for the recording in the *Name* field.

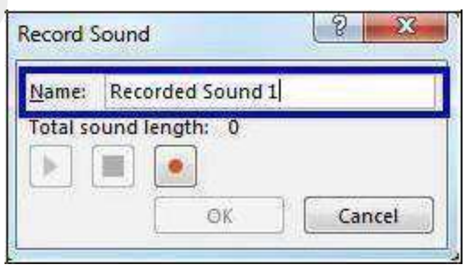
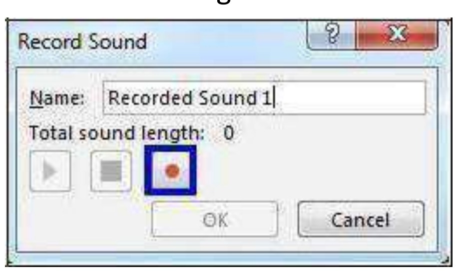


Figure 6 - Name the Recording

5. Click the **Record** button to start the recording.



6. To preview your recording, click the **Play** button.

Figure 9 - Play Button

Note: If you do not like the recording and want to re-record the audio, select the **Record** button again.

7. Once you are done, click **OK**. The audio file should appear on the slide.

8. If you do not want to keep the recording, click the **Cancel** button.

The Audio File

a. **Play/Pause –** Click the play button to play/pause button to play and pause the audio file (See Figure 10).
b. **Timeline –** The timeline will advance as the audio plays. Click anywhere on the timeline to go to a different place in the audio file (See Figure 10).
c. **Back –** Move back .25 seconds in the audio file (See Figure 10).
d. **Forward –** Move forward .25 seconds in the file (See Figure 10).
e. **Timer –** View the time for the audio file (See Figure 10).
f. **Volume –** Using your mouse, hover over the volume icon to access the slider. Scroll up to increase the volume. Scroll down to decrease the volume. Click the volume icon to Mute/Unmute the audio file (See Figure 10).

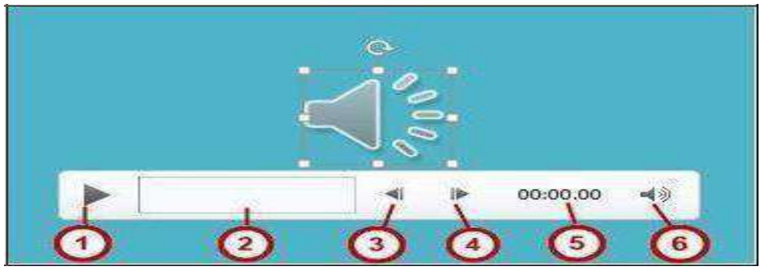


Figure 10 - Audio File

Deleting an Audio File

1. Navigate to the slide that contains the audio that you would like to delete.
2. Select the **audio file** icon.
3. Press the **Delete** or **Backspace** key on your keyboard.

Moving an Audio File

1) Navigate to the **slide** that contains the audio that you would like to move.
2) Click and drag the **audio file** to the desired location.

Playing Background Music Across all Slides

1. Select the **audio file** to be played across all slides.
2. On the *Audio Tools* contextual tab, select the **Playback** tab.
3. Click the **Play in Background** button. The audio file will now play across all slides until you reach the end of the presentation.

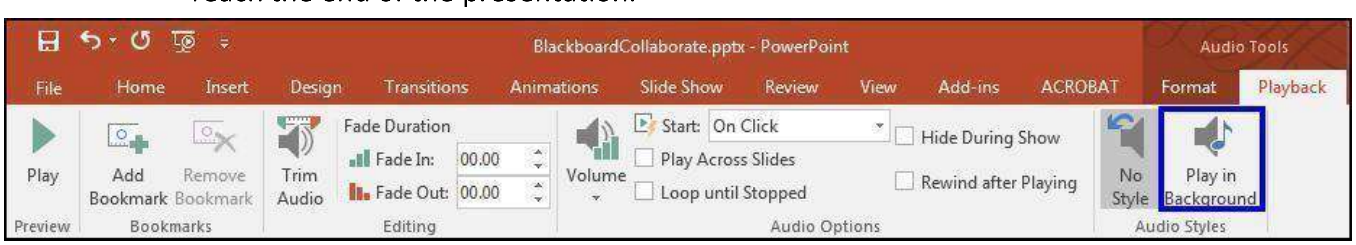


Figure 11 - Play in Background

Editing Audio

Sometimes you may need to make changes to your audio file. The following instructions explain how to edit the audio file.

To Trim an Audio File:

1. Select the audio file that you want to edit.
2. Click the **Playback** tab on the Ribbon.
3. Select **Trim Audio**.

Figure 12 - Trim Audio

4. The *Trim Audio* window will appear. Click and drag the green handle to set the start time (See Figure 13).
5. Click and drag the *red handle* to set the end time (See Figure 13).
6. Click the **Play** button to preview the changes (See Figure 13).
7. If necessary, adjust the handles again. Once you are done, click **OK** (See Figure 13).

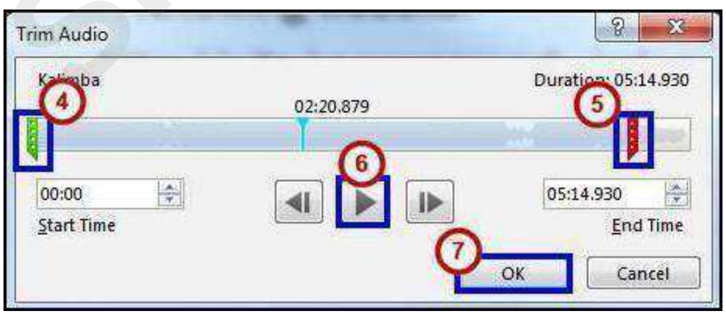


Figure 13 - Trim Audio

Note: You can also adjust the start or end time by selecting the green or red handles, and then clicking the **Previous Frame** or **Next Frame** buttons.

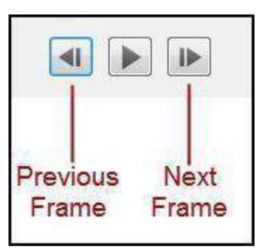


Figure 14 - Previous/Next Frame

To add a Fade In and Fade Out:

1. Select the **audio file** that you would like to edit.
2. Click the **Playback** tab on the Ribbon.
3. Under the Fade Duration section, either type in the desired times for the **Fade in** and **Fade Out**, or use the up and down arrows to adjust the times (See Figure 15).

Figure 15 - Fade In/Fade Out

Audio Options

Additional options that control how your audio file will play are found on the *Playback* tab, in the *Audio Options* group.

1. **Volume –** Adjust the volume for the audio file (See Figure 16).
2. **Start –** Controls whether the audio file starts when you click the mouse, or automatically (See Figure 16).
3. **Play Across Slide –** Select the checkbox to play the audio file across all slides (See Figure 16).
4. **Loop until Stopped –** Select the checkbox to replay the audio file until stopped (See Figure 16**Error! Reference source not found.**).
5. **Hide During Show –** Select the checkbox to hide the audio file icon while the slide show is playing (See Figure 16).
6. **Rewind after Playing –** Select this checkbox to return the audio file to the beginning when it is finished playing (See Figure 16).

Figure 16 - Audio Options

Applying an Audio Style

Once an audio file has been added to your slide, you can format it to make it look more presentable.

- Select the **audio file** that you would like to format.
- Select the **Format** tab (See Figure 17).
- Hover over the **picture styles** for a preview of each one (See Figure 17).
- Click the **style** of your choice. The style will be applied to the audio file. (See Figure 17).

Figure 17 - Picture Styles

Video

PowerPoint allows you to insert a video on to a slide and play it during your presentation. You can edit the video within PowerPoint using the trim feature, and the fade in/fade out feature. You can also format the appearance of the video.

Inserting a Video from a File

1. Navigate to the **slide** where you want to insert a video.
2. Select the **Insert** tab.
3. Click the **Video drop-down arrow**.

Figure 18 - Insert Video

- Select **Video on My PC**.

- Locate the **video file**, and then click **Insert**.

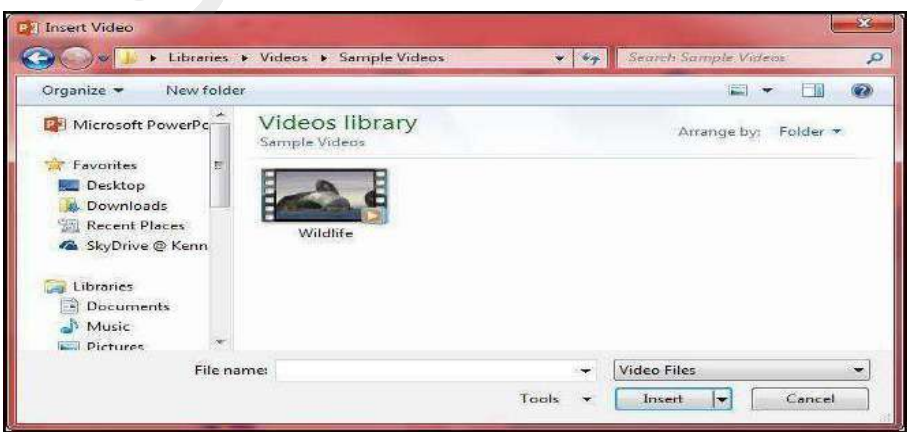


Figure 20 - Insert Video

- The video will be added to the slide. Click and drag the **sizing handles** to resize the video.

Figure 21 - Sizing Handles

Inserting an Online Video

If you choose to inserting an online video, you have the choice to select a video from YouTube or you can insert an embed code. You may also choose to insert a hyperlink to the video. For more information on how to insert a hyperlink, see the booklet PowerPoint 2016: Intro to PowerPoint, located on the UITS Documentation Center

Previewing a Video

Click the video to select it.

1. **Play/Pause –** Click the Play/Pause button to preview the video (See Figure 22).
2. **Timeline –** To go directly to a specific part of the video, click anywhere on the timeline (See Figure 22).
3. **Back –** Move back .25 seconds (See Figure 22).
4. **Forward –** Move forward .25 seconds (See Figure 22).

5. **Volume –** Using your mouse, hover over the volume icon to access the slider. Scroll up to increase the volume. Scroll down to decrease the volume. Click the volume icon to Mute/Unmute the video (See Figure 22).

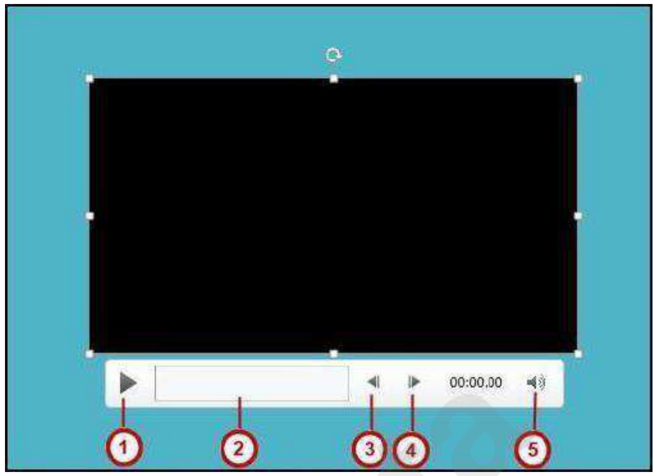


Figure 22 - Preview Video

Deleting a Video

1. Click the **video** to select it.
2. Press the **Delete** or **Backspace** key on your keyboard.
3. The video will be removed from the slide.

Moving a Video

1. Navigate to the **slide** that contains the video that you would like to move.
2. Click and drag the **video** to the desired location.

Editing a Video

Sometimes you may need to make changes to your video file. The following instructions explain how to edit a video file from your computer:

To Trim a Video

1. Select the video that you would like to trim.
2. Click the **Playback** tab on the Ribbon and select the **Trim Video** option in the Editing group.

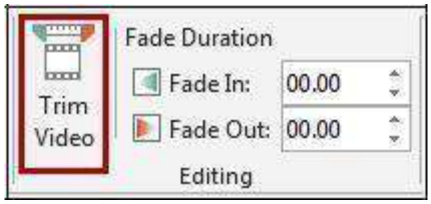


Figure 23 - Trim Video Button

3. The Trim Video window will appear. Use the **green handle** to set the start time (See Figure 24).
4. Use the **red handle** to set the end time (See Figure 24).
5. Click **Play** to preview the video (See Figure 24).
6. Click **OK** when you are done trimming the video (See Figure 24).

Note: You can also adjust the start or end time by clicking the up or down arrows in the Start Time or End Time fields (See Figure 24).

Note: You cannot edit video files that you acquire online.

To Add a Fade In and Fade Out

1. Select the **video** that you would like to apply fading to.
   Click the **Playback** tab on the *Ribbon*, and go to the Fade Duration section.
2. To adjust the *Fade In* or *Fade Out*, either type in the desired time, or use the up and down arrow keys.

Figure 30 – Fade In/Fade Out

Video Options

Additional options that control how your video will play are found on the *Playback* tab, in the *Audio Options* group. **Volume –** Change the volume for the video (See Figure 25).

1. **Start –** Control whether you want the video to start automatically or when you click your mouse (See Figure 25).
2. **Play Full Screen –** The video will fill your entire screen while playing (See Figure 25).
3. **Hide While Not Playing –** When the video is not playing, it will be hidden (See Figure 25).
4. **Loop until Stopped –** The video will replay until **it's** stopped (See Figure 25).
5. **Rewind after Playing –** When the video has finished playing, it will return to the beginning (See Figure 25).

Applying a Video Style

1. To apply a video style, select a video in your presentation.
2. Click the **Format** tab (See Figure 26).
3. Use your mouse to hover over a **video style** to see a preview of the style.
4. Select the **style** of your choice (See Figure 26).

5. The style will be applied to your video (See Figure 26).

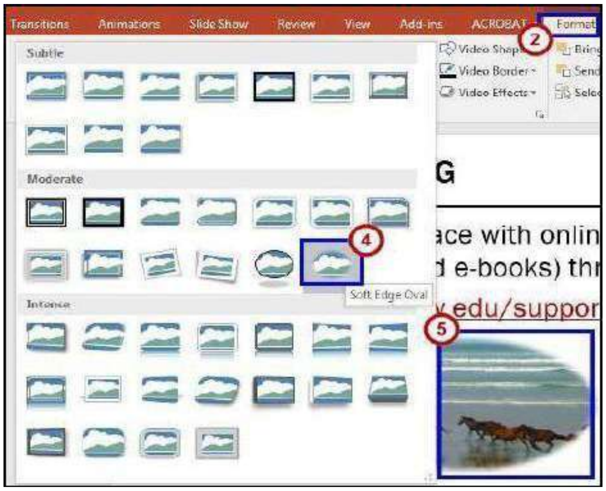


Figure 26 - Video Styles

Slide Show Setup Options

PowerPoint has several options to choose from to set up your slide show.

1. Select the **Slide Show** tab on the ribbon (See Figure 27).
2. Click **Set Up Slide Show** in the *Set Up* group (See Figure 27).

Figure 27 - Set Up Slide Show

3) The *Set Up Show* window will appear (See Figure 28).
4) Select your desired **options** for your presentation (See Figure 28).

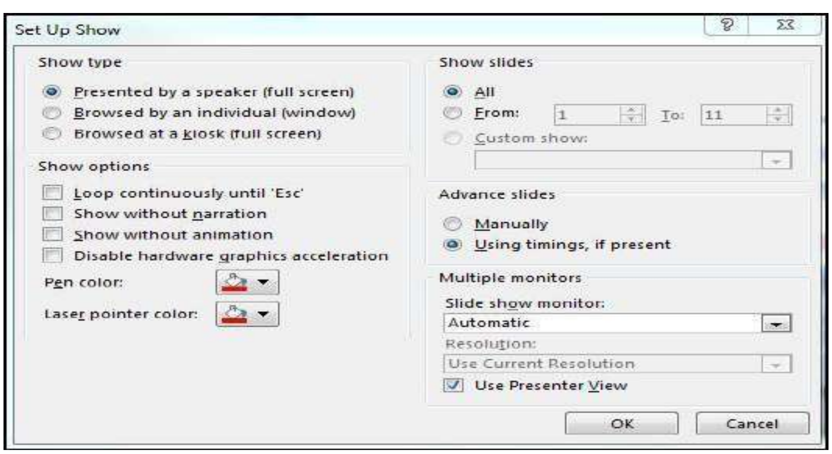


Figure 28 - Set Up Show

Hiding a Slide

If there is a slide that you do not want to show during the Slide Show, PowerPoint offers the option to hide a slide.

1. Select the **slide** that you wish to hide.
2. Click the **Slide Show** tab (See Figure 29).
3. Select **Hide Slide** within the *Set Up* group (See Figure 29)

Figure 29 - Hide Slide

4. When presenting your slideshow, the slide will not be seen.

Presenting Your Slide Show

PowerPoint presentations are meant to be supplemental information, not a script. Review your presentation for content and try to avoid reading directly from the slides. The following instructions explain how to present your slide show.

Starting the Slide Show

1. Select the **Slide Show** tab (See Figure 30).
2. Click **From Beginning** to begin the presentation from the first slide or click **From CurrentSlide** to begin the presentation from the slide which currently appears in the Slide Preview pane (See Figure 30).

Figure 30 - Start Slideshow

3. You can advance to the next slide by pressing **Enter**, the **Spacebar**, or by clicking the left mouse button. You will also notice left and right arrows in the bottom-left corner of theslide. Clicking one of these arrows will advance the presentation to the next slide or theprevious slide.
4. Pressing the **Escape (Esc)** key will end your presentation and return you back to the Normal View.

Note: You can also run the presentation from the current slide by clicking the slide show icon  located on the status bar.

Presentation Tools

PowerPoint provides tools that are available for use while presenting your slide show.

Once in Slide Show view, icons will appear in the bottom-left corner of your screen (see Figure 37).

Figure 31 - Presentation Tools

 **Back** - Select the **Back** button to move to the Previous slide in the slide show.

 **Forward** - Select the **Forward** button to move to the Next slide in the slide show.

 **Pen Tools** - Use the **pen tools** to make notations on your slides.

 **See All Slides** - To jump to a specific slide, select the **See All Slides** icon, and then select the slide that you would like to go to next.

 **Zoom** - Select the **Zoom** icon to zoom in on a specific area of the slide.

 **Black or unblack slide** - Allows you to black out a slide during your presentation.

 **More** - Select the **More** button for additional features available in PowerPoint. Select the More icon to access the taskbar on your computer. Having the ability to access the taskbar while in slide show view, will allow you to access the Internet or other files or programs during your presentation. Follow the instructions below to access the taskbar.

Pen Tool

1. Select the **Pen Tools** icon.
2. From the menu, select the **Pen** or **Highlighter**, and then click and drag the mouse to mark on your slides (See Figure 32).
3. Select **Laser Pointer** to draw attention to certain parts of the slide (See Figure 32).
4. Select **Eraser** to erase a marking on a slide (See Figure 32).
5. Select **Erase All Ink on Slide** to erase all of the markings on a slide (See Figure 32).

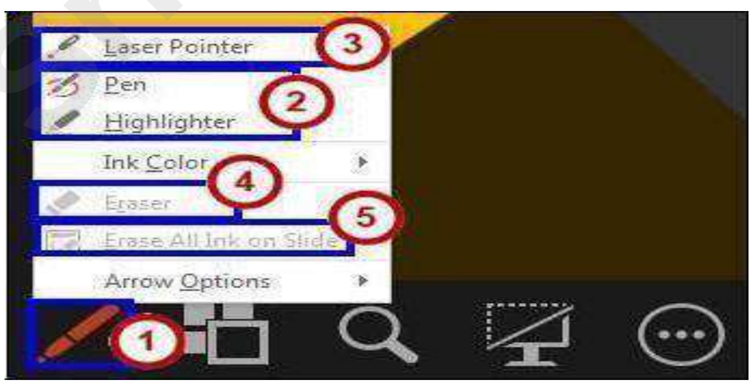


Figure 32 - Presentation Ink Markings Menu

More

1. Select the **More** button (See Figure 33).
2. In the menu that appears, select **Screen** (See Figure 33).
3. Click **Show Taskbar** (See Figure 33). The taskbar will appear at the bottom of the screen.

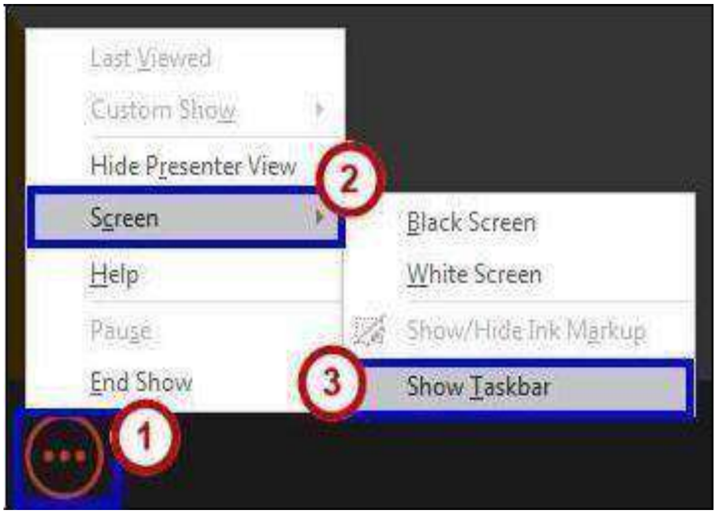


Figure 33 - More Menu: Show Taskbar

You can also white out your screen during your presentation.

1. Select the **More** icon (See Figure 34).
2. In the menu that appears, select **Screen** (See Figure 34).
3. Click **White Screen**. Your screen will become completely white (See Figure 34).

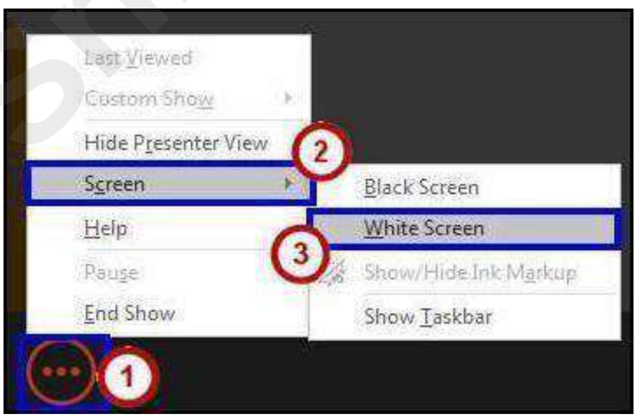


Figure 34 - More Menu: White Screen

Note: You can also access any of the presenting tools mentioned above, by right-clicking anywhere on the screen during your slideshow.

This material is for training purposes only and cannot be the basis for litigation. This protects you from any legal issues regarding the content of your training.